

COMPASS_Synoptic_MacrophyteFluxes_2025_CO2

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Set Up

Pull in Raw Datafiles

```
# Create a temporary directory
tmp_dir <- tempdir()

# Get list of files inside the zip
zip_contents <- unzip("7810_data/concentrations/Test_FW_Autochamber_Data.zip", list = TRUE)

# Keep only 8100 files from freshwater
CO2_files <- zip_contents$Name[
  grepl("data$", zip_contents$Name, ignore.case = TRUE)& #8100 files
  grepl("TG10", zip_contents$Name, ignore.case = TRUE)] #Freshwater autochamber test

CO2_files

## [1] "Test_FW_Autochamber_Data/Test_FW_Autochamber_7810-20_Data/TG10-01725-2026-05-12T090000.data"

# Extract the 8100 files
unzip(
  "7810_data/concentrations/Test_FW_Autochamber_Data.zip",
  files = CO2_files,
  exdir = tmp_dir
)

# List extracted 8100 files (full paths)
file.list <- list.files(
  path = tmp_dir,
  pattern = "TG10.*\\.data$",
  full.names = TRUE,
  recursive = TRUE
)

head(file.list)

## [1] "C:\\Users\\readz\\AppData\\Local\\Temp\\RtmpczclDg/Test_FW_Autochamber_Data/Test_FW_Autochamber_7810-20_Data\\TG10-01725-2026-05-12T090000.data"
```

```
#Read in the data
alldat1 <- lapply(file.list, ffi_read_LI7810)
```

```
## TG10-01725-2026-05-12T090000.data: read 422623 rows of TG10-01725 data, 2026-05-13 11:08:32 to 2026-
```

```
alldat <- data.table::rbindlist(alldat1, fill=TRUE)
```

Read in metadata file and match to raw data

```
#read in metadatfile
md <- "Processed_Data/COMPASS_TEMPESTX_SoilFluxes_8100_Metadata_processed.csv"
#metadat <- read_excel(md)
metadat <- read.csv(md)

##Make it so the plot column is unique for each run (Plot_obs#?)

##Separate Date and Time
metadat <- metadat %>%
  mutate(
    # Ensure it's a date-time object
    start_time = ymd_hms(start_time),
    # Extract date
    Date = as.character(date(start_time)),
    # Extract time
    Time = format(start_time, "%H:%M:%S"),
    #Parses the string as "minutes:seconds" and extracts the total duration in seconds from the period
    total_seconds = as.numeric(ms(obs_length)),
    #Add unique plot ID for each timestamp
    unique_ID = paste("Collar", label, "Timestamp", start_time, sep = "_"),
    Vtotal_m3 = Vtotal/(100^3))

head(metadat)
```

```
##          SN label chamber Vtotal obs_length DeadBand TimeClosing
## 1 81A-1767    41      101 6918.9      01:30      00:25          13
## 2 81A-1767    40      101 6926.9      01:30      00:25          14
## 3 81A-1767    40      101 6926.9      01:30      00:25          14
## 4 81A-1767    38      101 6926.9      01:30      00:25          14
## 5 81A-1767    45      101 6926.9      01:30      00:25          14
## 6 81A-1767    41      101 6918.9      01:30      00:25          13
##          start_time          end_time      Tcham Pressure  V1_avg
## 1 2026-05-18 00:10:01 2026-05-18 00:11:43  18.30179 101.1351 2.664151
## 2 2026-05-18 00:37:16 2026-05-18 00:38:59  18.09935 101.1842 2.771963
## 3 2026-05-17 15:36:57 2026-05-17 15:38:40  25.71168 101.0164 2.768224
## 4 2026-05-17 15:39:57 2026-05-17 15:41:40 -70.76159 101.0138 2.663551
## 5 2026-05-17 15:42:59 2026-05-17 15:44:42 -67.98766 101.0062 2.522150
## 6 2026-05-17 16:27:52 2026-05-17 16:29:34  25.75925 100.9553 2.649557
##          V2_avg  V3_avg  V4_avg      Date      Time total_seconds
## 1 -0.5314623 144.72676 0.03563208 2026-05-18 00:10:01          90
## 2  0.7748598 141.20145 0.13486916 2026-05-18 00:37:16          90
```

```
## 3 0.7743458 140.82571 0.14283178 2026-05-17 15:36:57 90
## 4 -0.5024393 72.46708 0.05562617 2026-05-17 15:39:57 90
## 5 0.8122430 15.50519 0.10640187 2026-05-17 15:42:59 90
## 6 -0.5321321 147.78287 0.03377358 2026-05-17 16:27:52 90
##
##          unique_ID Vtotal_m3
## 1 Collar_41_Timestamp_2026-05-18 00:10:01 0.0069189
## 2 Collar_40_Timestamp_2026-05-18 00:37:16 0.0069269
## 3 Collar_40_Timestamp_2026-05-17 15:36:57 0.0069269
## 4 Collar_38_Timestamp_2026-05-17 15:39:57 0.0069269
## 5 Collar_45_Timestamp_2026-05-17 15:42:59 0.0069269
## 6 Collar_41_Timestamp_2026-05-17 16:27:52 0.0069189
```

Match metadata to raw data

```
#match the metadata to the concentration data
alldat$metadat_row <- ffi_metadata_match(
  data_timestamps = alldat$TIMESTAMP,
  start_dates = metadat$Date,
  start_times = metadat$Time,
  obs_lengths = metadat$total_seconds)
```

```
## 8 entries had no timestamp matches!
```

```
head(alldat)
```

```
##      DATAH      SECONDS NANOSECONDS  NDX  DIAG REMARK  H2O  CO2  CH4 CAVITY_P
##      <char>      <int>      <int> <int> <int> <lgcl> <num> <num> <num> <num>
## 1: DATA 1778684912      7683992    0  256    NA    NA    NA    NA    101.513
## 2: DATA 1778684912    407668113    0  256    NA    NA    NA    NA    101.513
## 3: DATA 1778684913    257642984   17  256    NA    NA    NA    NA    101.935
## 4: DATA 1778684914    257601976   21  256    NA    NA    NA    NA    102.355
## 5: DATA 1778684915    257582902   25  256    NA    NA    NA    NA    102.354
## 6: DATA 1778684916    257550001   29  256    NA    NA    NA    NA    102.353
##      CAVITY_T LASER_PHASE_P LASER_T RESIDUAL RING_DOWN_TIME THERMAL_ENCLOSURE_T
##      <num>      <num>      <num>      <num>      <num>      <num>
## 1: 16.4309      101.188  24.3400      0      0      16.5229
## 2: 16.4309      101.188  24.3400      0      0      16.5229
## 3: 16.4309      101.608  24.3417      0      0      16.5229
## 4: 16.4309      102.028  28.0549      0      0      16.5233
## 5: 16.4309      102.027  33.6430      0      0      16.5237
## 6: 16.4313      102.026  34.3508      0      0      16.5241
##      PHASE_ERROR LASER_T_SHIFT INPUT_VOLTAGE  CHK      TIMESTAMP      TZ
##      <num>      <num>      <num> <int>      <POS>      <char>
## 1:      0      0      11.7513   97 2026-05-13 11:08:32 US/Eastern
## 2:      0      0      11.7513  185 2026-05-13 11:08:32 US/Eastern
## 3:    -8000      0      11.7054   48 2026-05-13 11:08:33 US/Eastern
## 4:   -16000      0      11.6077   88 2026-05-13 11:08:34 US/Eastern
## 5:   -16000      0      11.6104   27 2026-05-13 11:08:35 US/Eastern
## 6:   -16000      0      11.6159   68 2026-05-13 11:08:36 US/Eastern
##      SN  MODEL metadat_row
##      <char> <char>      <num>
```

```
## 1: TG10-01725 LI-7810      NA
## 2: TG10-01725 LI-7810      NA
## 3: TG10-01725 LI-7810      NA
## 4: TG10-01725 LI-7810      NA
## 5: TG10-01725 LI-7810      NA
## 6: TG10-01725 LI-7810      NA
```

```
#Set plot, temp, volume, etc. here
alldat$Plot <- metadat$unique_ID[alldat$metadat_row]
alldat$Volume <- metadat$Vtotal_m3[alldat$metadat_row]
alldat$final_temp <- metadat$Tcham[alldat$metadat_row] #from 8100 file
# alldat$L1_Temp <- metadat$L1temp$wx_tempavg15[alldat$metadat_row] #calculate temp for the measurement
# alldat$Site <- metadat$Site[alldat$metadat_row]
metadat$metadat_row <- seq_len(nrow(metadat))

#We need one more line that creates a column for final temp and takes the L1 data but if there isn't L1
# in this case we will use the by hand data i guess? but we could look to using some sort of NOAA data

# alldat$Hand_Temp <- as.numeric(alldat$Hand_Temp)
#
# alldat <- alldat %>%
#   mutate(final_temp = coalesce(L1_Temp, Hand_Temp))

#remove the rows that don't match to metadata
alldat <- alldat %>%
filter(!is.na(metadat_row))

#can remove any row that have NA's in the CO2 column
alldat <- alldat %>%
filter(!is.na(CO2))
```

Calculate fluxes

```
#change the units
alldat$CO2_umol <- ffi_ppm_to_umol(alldat$CO2,
                                   volume = alldat$Volume # m3
                                   )      # degrees C
```

```
## Assuming temp = 20 C
```

```
## Assuming atm = 101325 Pa
```

```
## Using R = 8.31446261815324 m3 Pa K-1 mol-1
```

```
#> Assuming atm = 101325 Pa
#> Using R = 8.31446261815324 m3 Pa K-1 mol-1
```

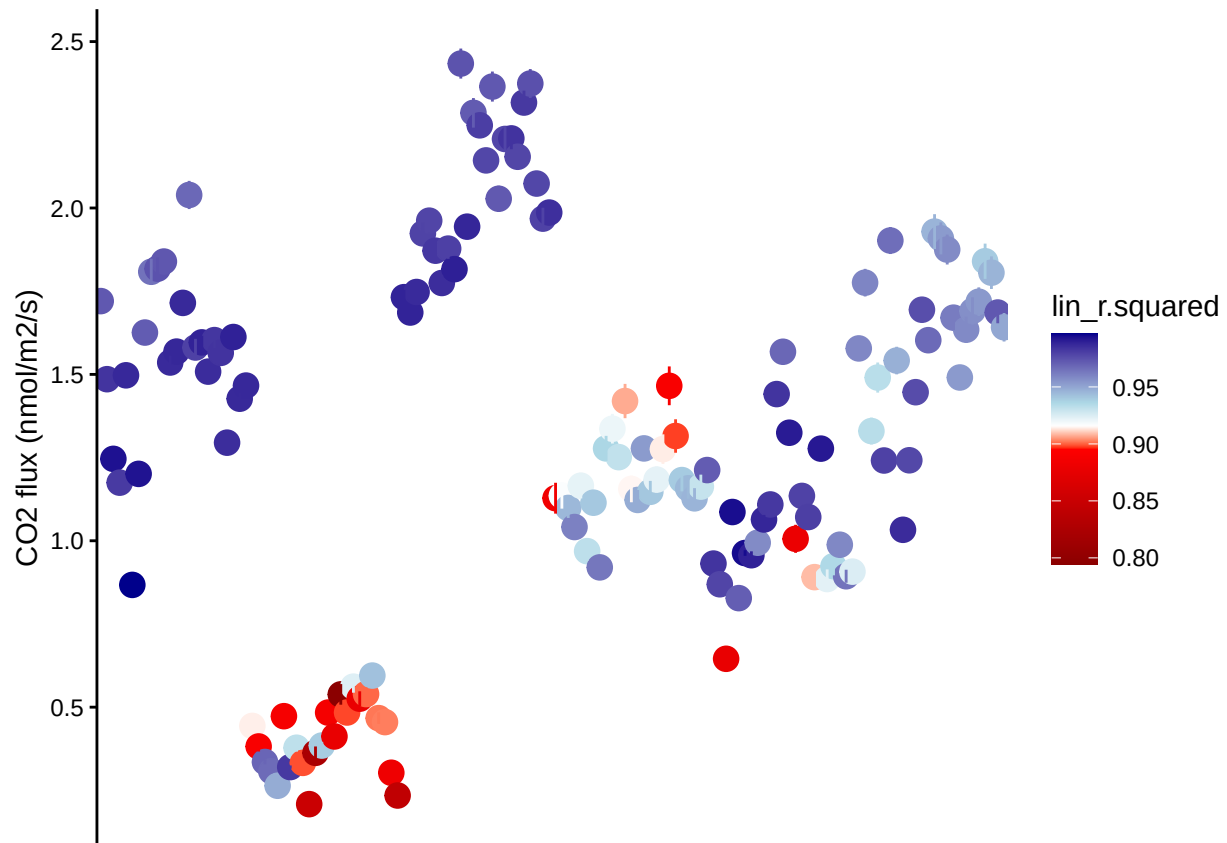
```
alldat$CO2_umol_m2 <- alldat$CO2_umol / 0.16
```

```
fluxes <- ffi_compute_fluxes(alldat,
```



```
## 5      1.4972049      0.019758196      854.5419      1.0792208
## 6      0.8673809      0.006030173      803.4086      0.3293767
##      lin_p.value lin_r.squared lin_RMSE poly_AIC poly_r.squared poly_RMSE
## 1 1.853387e-62      0.9722168 6.800692 383.3730      0.9961619 2.560723
## 2 7.106953e-71      0.9830940 4.555834 302.3000      0.9981110 1.542776
## 3 4.524456e-85      0.9926871 2.501438 273.3440      0.9981127 1.287379
## 4 4.253328e-69      0.9812243 3.800068 253.6873      0.9983578 1.138548
## 5 8.257915e-75      0.9865981 4.080917 235.7195      0.9991880 1.017608
## 6 2.336742e-96      0.9962442 1.245490 263.6382      0.9965369 1.211606
##      rob_AIC rob_converged rob_flux.estimate rob_flux.std.error rob_RMSE
## 1 541.5403      TRUE      1.666235      0.026188600 5.7298499
## 2 479.0675      TRUE      1.441454      0.017559603 3.6810128
## 3 378.2802      TRUE      1.239191      0.012815849 2.4143969
## 4 447.6384      TRUE      1.147217      0.015008772 3.6842487
## 5 458.3789      TRUE      1.471469      0.018187570 3.9043876
## 6 266.8394      TRUE      0.863090      0.005540475 0.9431363
##      TIMESTAMP_min      TIMESTAMP_max
## 1 2026-05-17 10:40:06 2026-05-17 10:41:25
## 2 2026-05-17 11:40:05 2026-05-17 11:41:24
## 3 2026-05-17 12:40:06 2026-05-17 12:41:25
## 4 2026-05-17 13:40:05 2026-05-17 13:41:24
## 5 2026-05-17 14:40:08 2026-05-17 14:41:27
## 6 2026-05-17 15:40:07 2026-05-17 15:41:26
```

```
#take a look at the fluxes we calculated with R2
ggplot(fluxes, aes(Plot, lin_flux.estimate, color = lin_r.squared)) +
  geom_point(size=4) + theme_classic() +
  geom_linerange(aes(ymin = lin_flux.estimate- lin_flux.std.error,
                    ymax = lin_flux.estimate+ lin_flux.std.error)) +
  scale_color_gradientn(colours = c("darkred", "red", "white", "lightblue", "darkblue"),
                        values = c(0, 0.5, 0.6, 0.7, 1)) +
  ylab("CO2 flux (nmol/m2/s)") +
  theme(axis.title.x = element_blank(),
        axis.text.x = element_blank(),
        axis.ticks.x = element_blank())
```

```
#sort fluxes by R2
fluxes_good <- as.data.frame(subset(fluxes, lin_r.squared >= .90, select = Plot:TIMESTAMP_max))
fluxes_weird <- as.data.frame(subset(fluxes, lin_r.squared < .90, select = Plot:TIMESTAMP_max))

#negative CO2 fluxes
fluxes_negative <- as.data.frame(subset(fluxes, lin_flux.estimate < 0, select = Plot:TIMESTAMP_max))

#find fluxes that represent the first and fourth quartile of the data
low <- quantile(fluxes$lin_flux.estimate, 0.05)
high <- quantile(fluxes$lin_flux.estimate, 0.95)

fluxes_low <- as.data.frame(subset(fluxes, lin_flux.estimate <= low, select = Plot:TIMESTAMP_max))
fluxes_high <- as.data.frame(subset(fluxes, lin_flux.estimate >= high, select = Plot:TIMESTAMP_max))

tails <- rbind(fluxes_low, fluxes_high)
```

Check the fluxes for which I changed the start time due to lags or ebullition (recorded in notes of metadata)

```
tobechecked <- c()

colnames <- c("Plot")
```

```

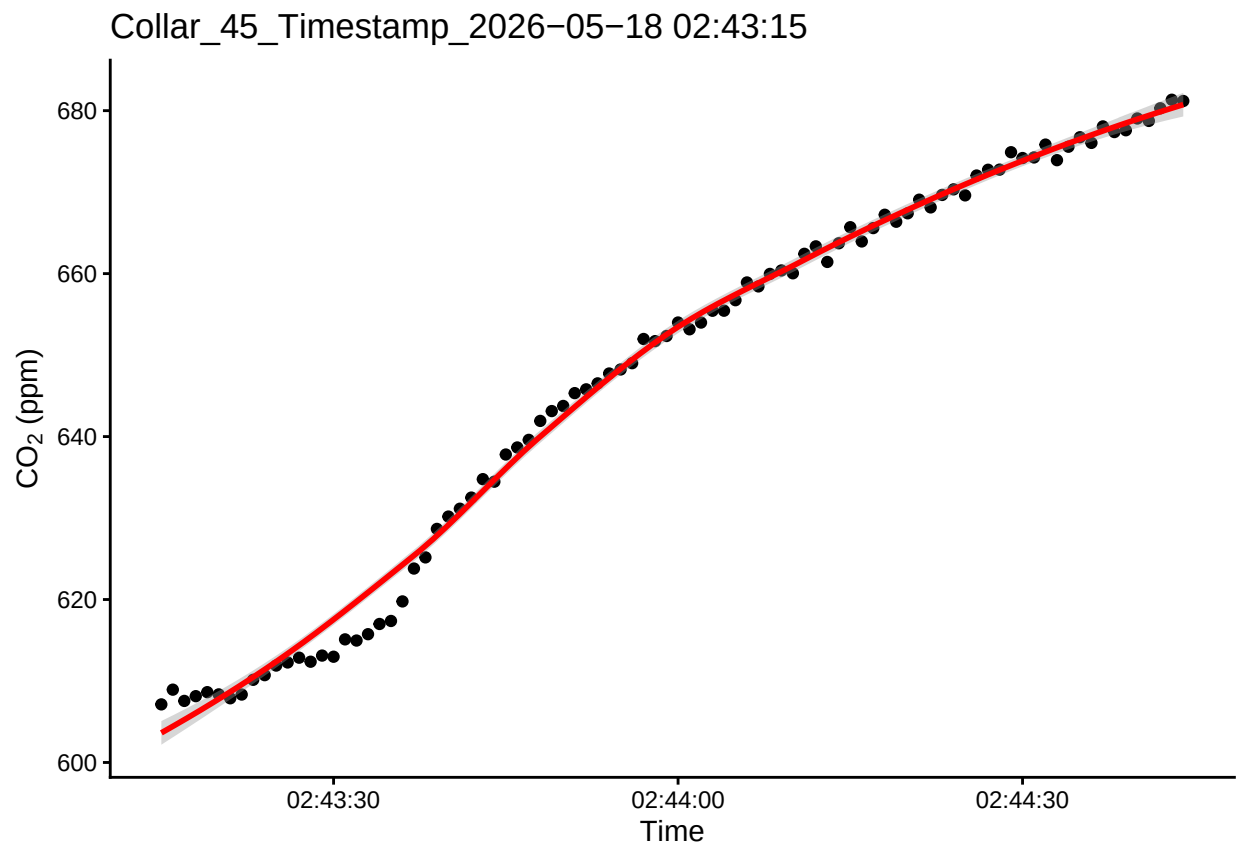
#Loop to run plots above
for(P in tobechecked){
  r1 <- subset(alldat, Plot == P)
  p1 <- ggplot(r1, aes(TIMESTAMP, CO2)) + geom_point() + theme_classic() +
  ylab(expression(paste( CO [2], " (ppm)"))) +
  xlab("Time") + labs(title = P) +
  geom_smooth(color="red")
  print(p1)
}

#list of the ones that I changed and checked and looked good?
fixedplots <- c()

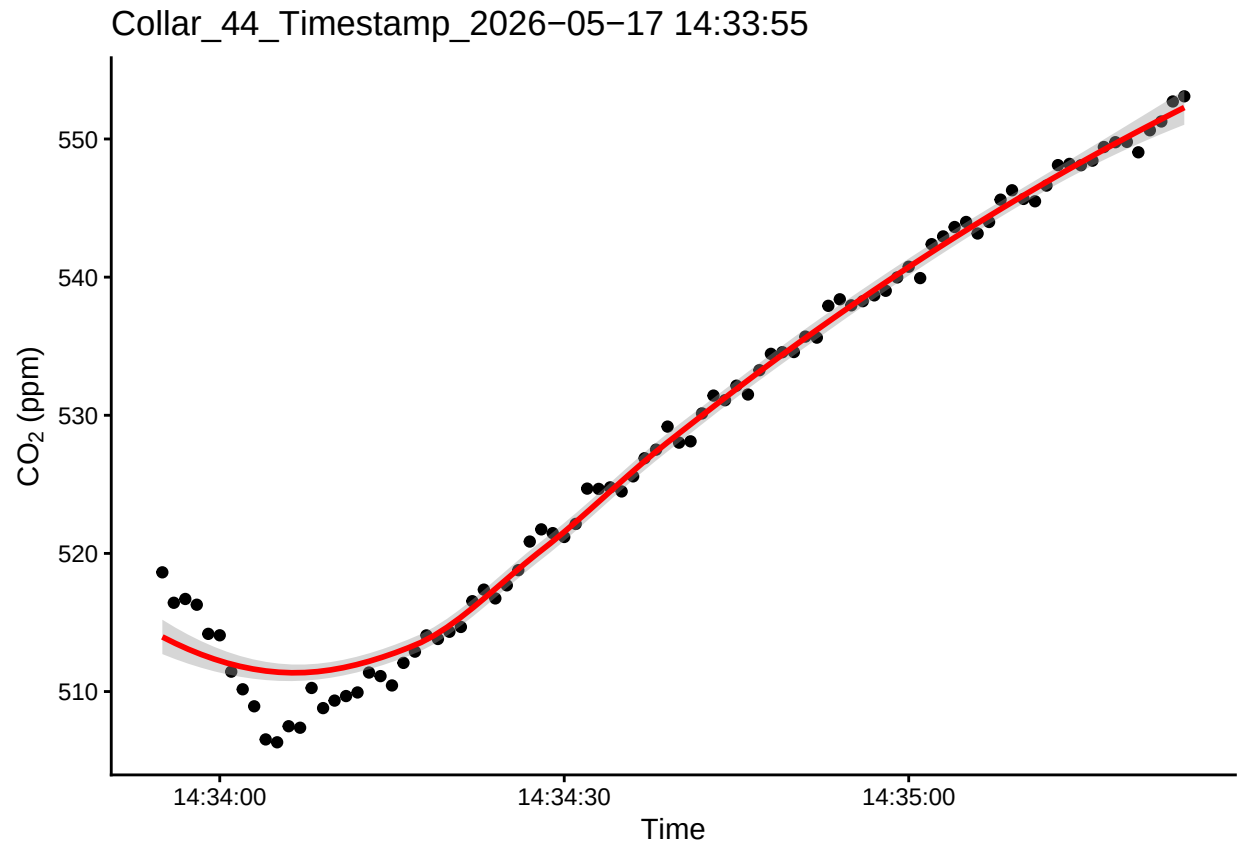
```

Spot check a few fluxes with $R^2 > 0.9$

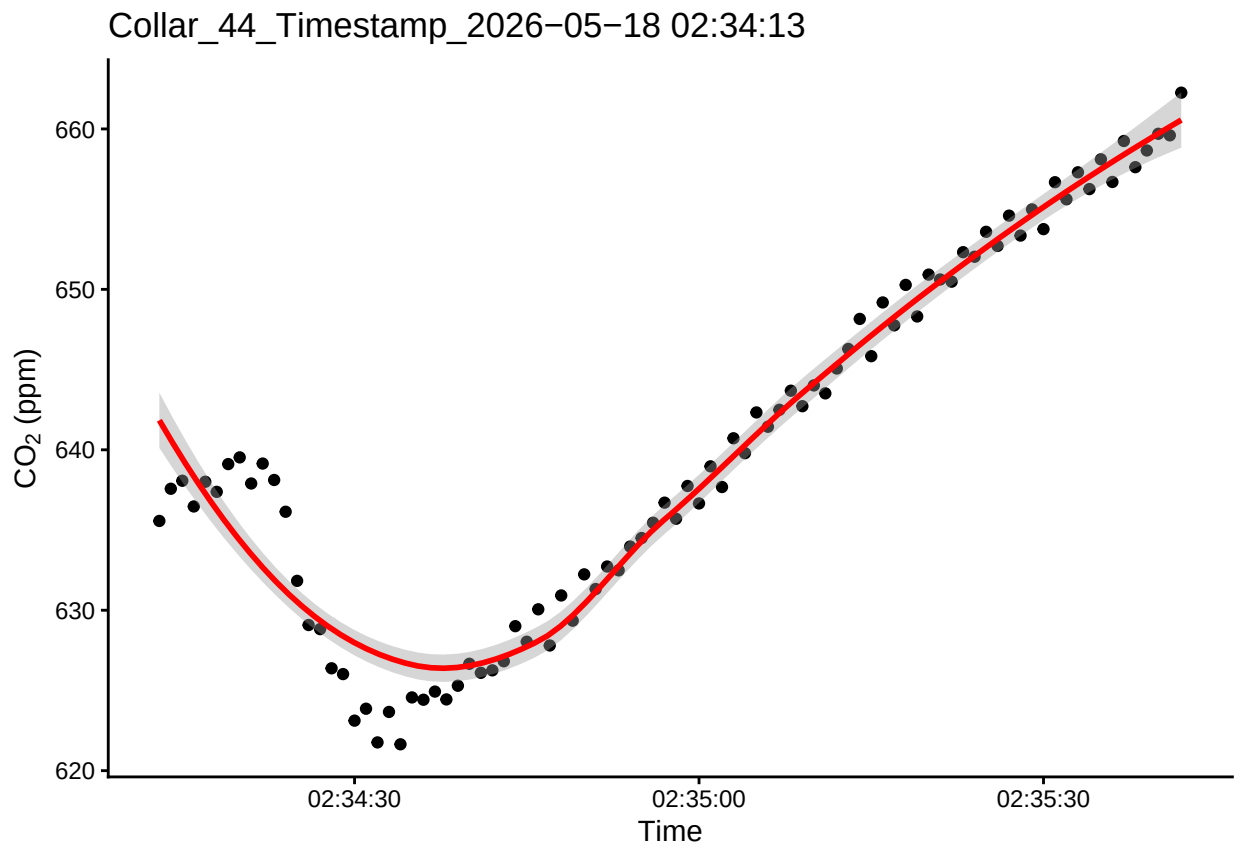
```
## 'geom_smooth()' using method = 'loess' and formula = 'y ~ x'
```



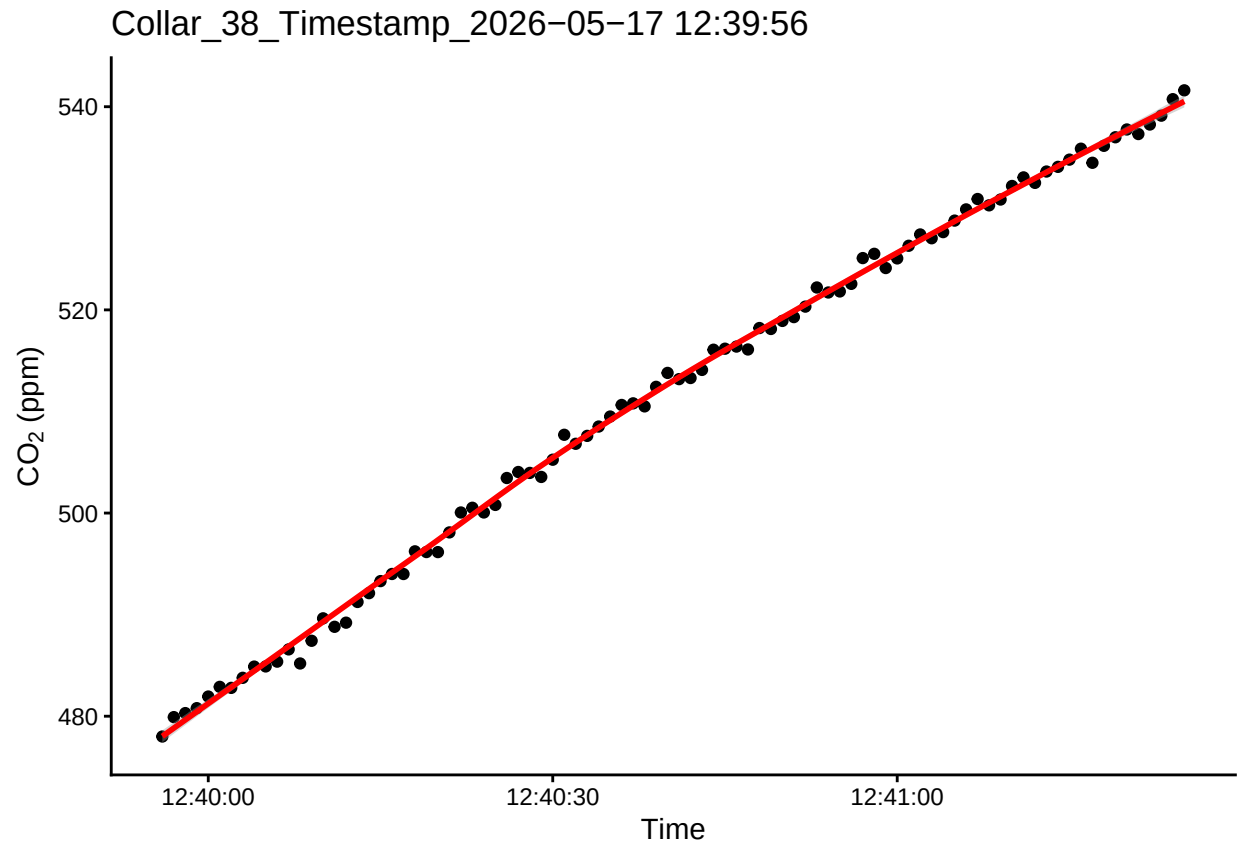
```
## 'geom_smooth()' using method = 'loess' and formula = 'y ~ x'
```



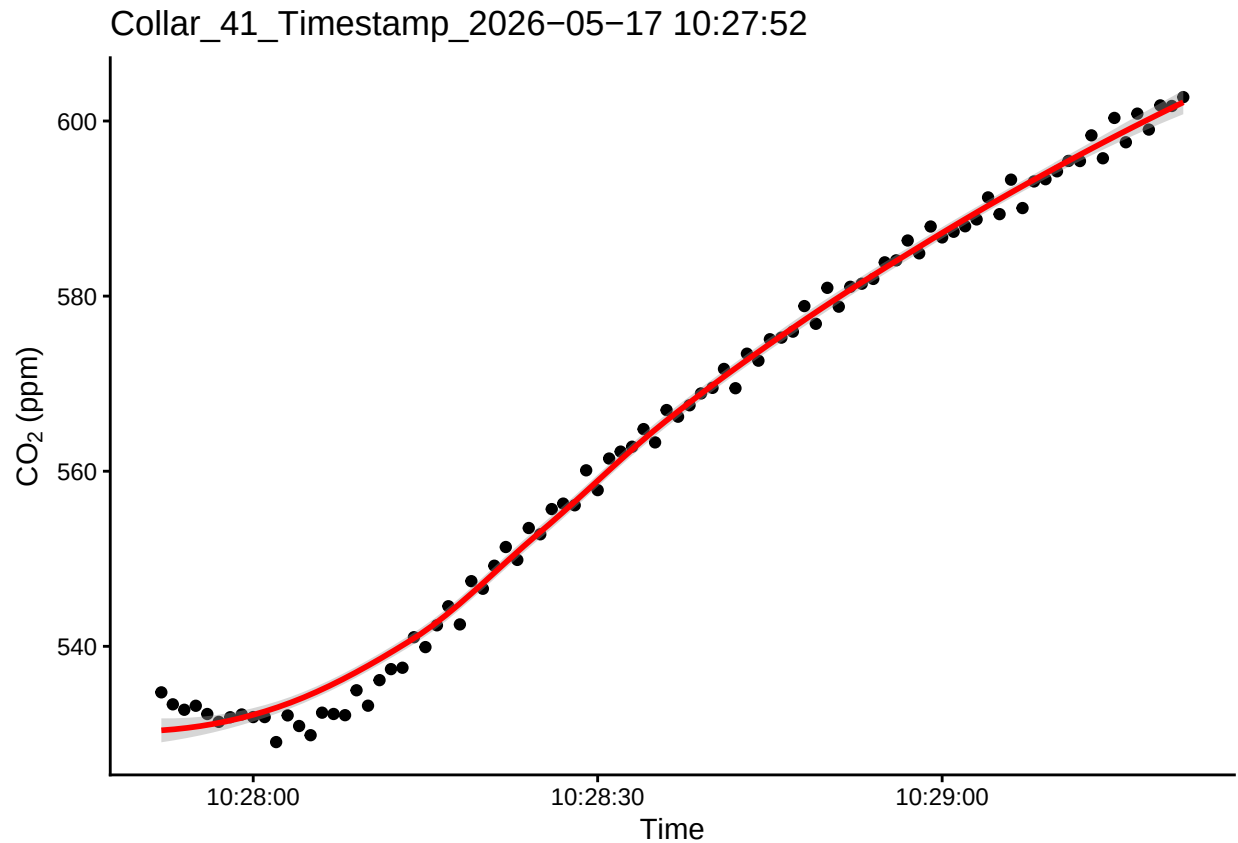
```
## 'geom_smooth()' using method = 'loess' and formula = 'y ~ x'
```



```
## 'geom_smooth()' using method = 'loess' and formula = 'y ~ x'
```



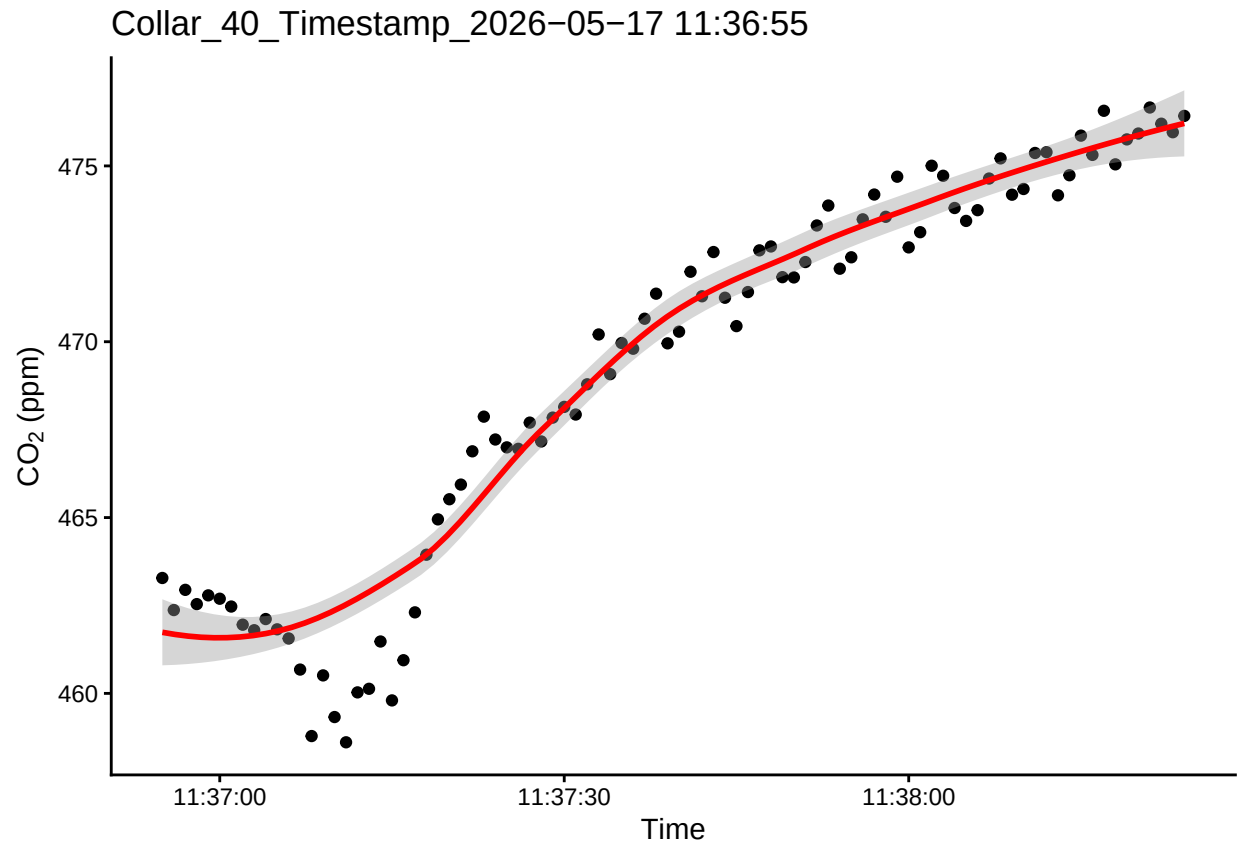
```
## 'geom_smooth()' using method = 'loess' and formula = 'y ~ x'
```



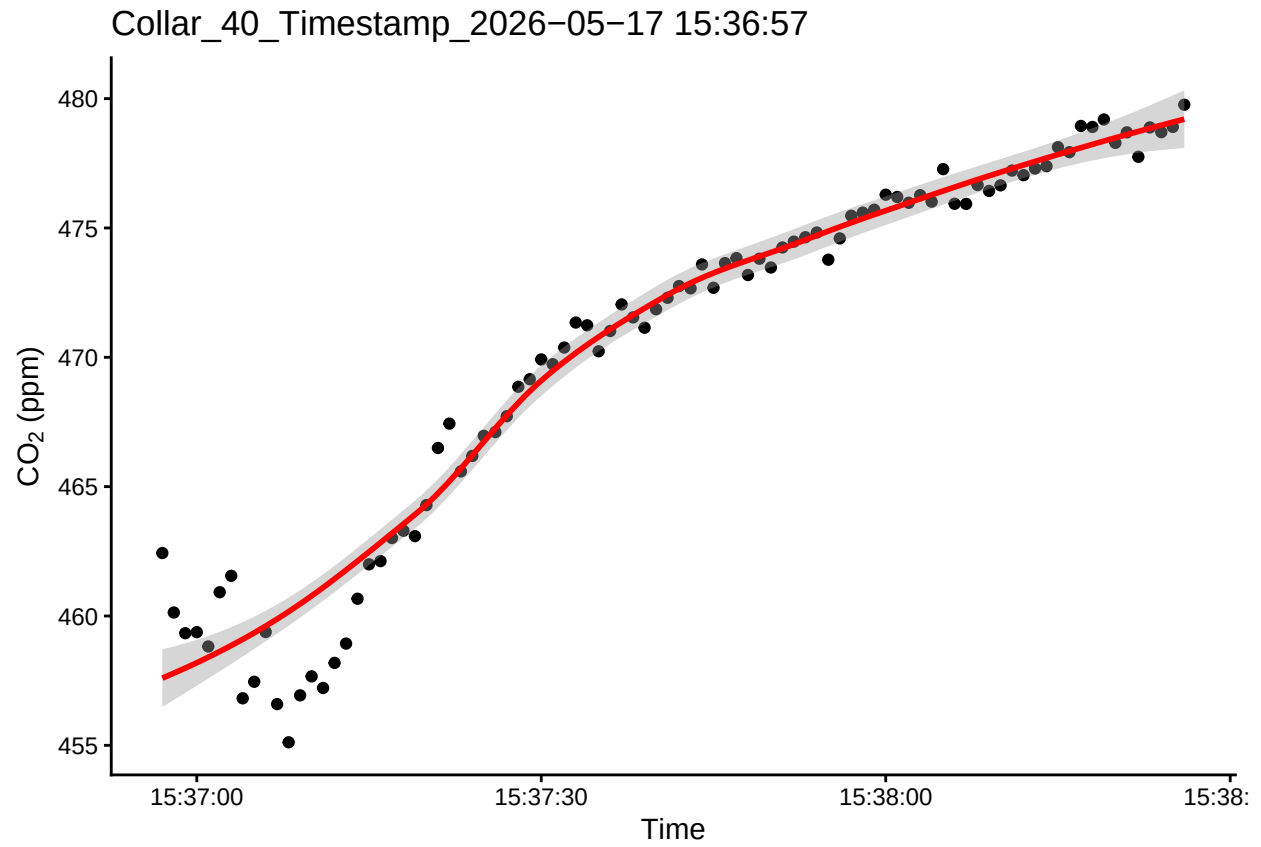
Take a look at the bad fluxes < 0.9

If a flux was started too early/late, change start time or Obs_length in metadata file and note in the note column

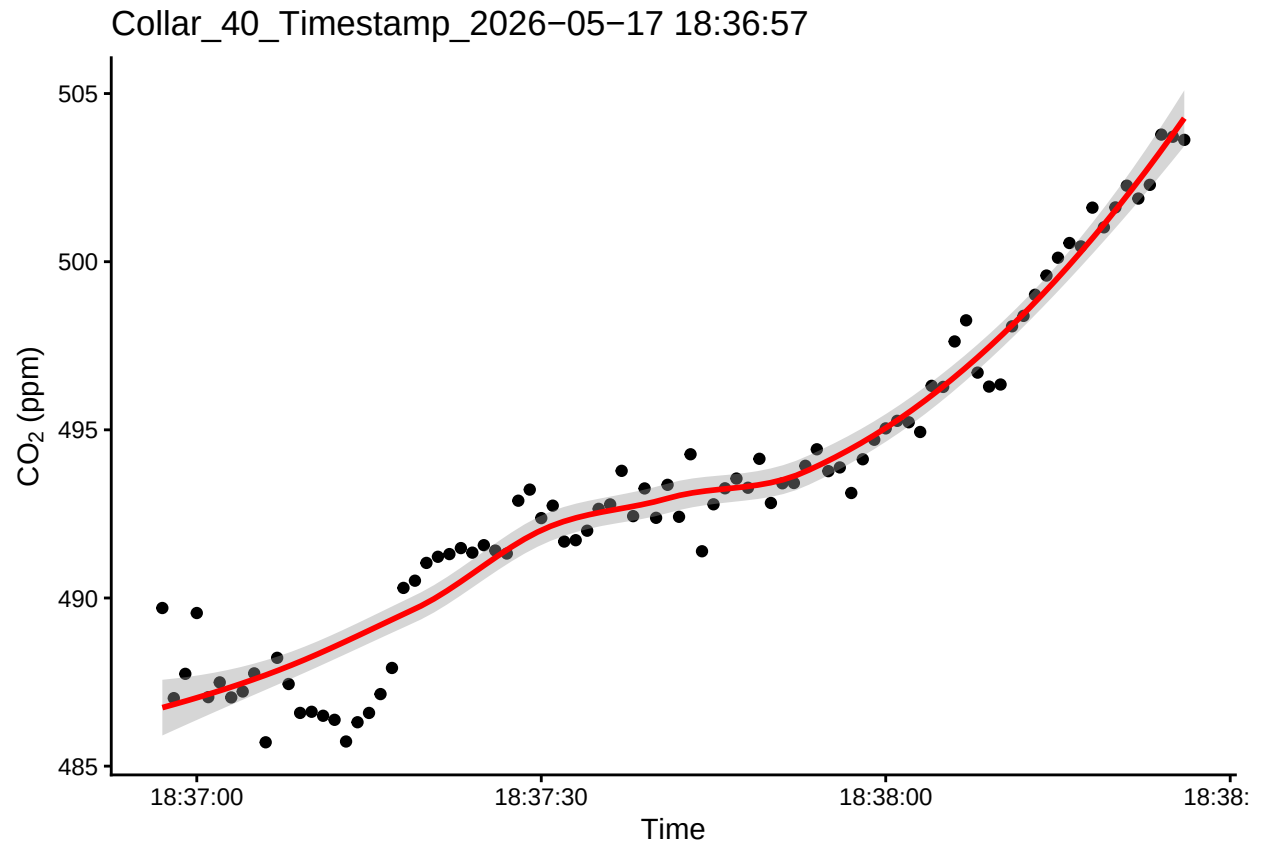
```
## 'geom_smooth()' using method = 'loess' and formula = 'y ~ x'
```



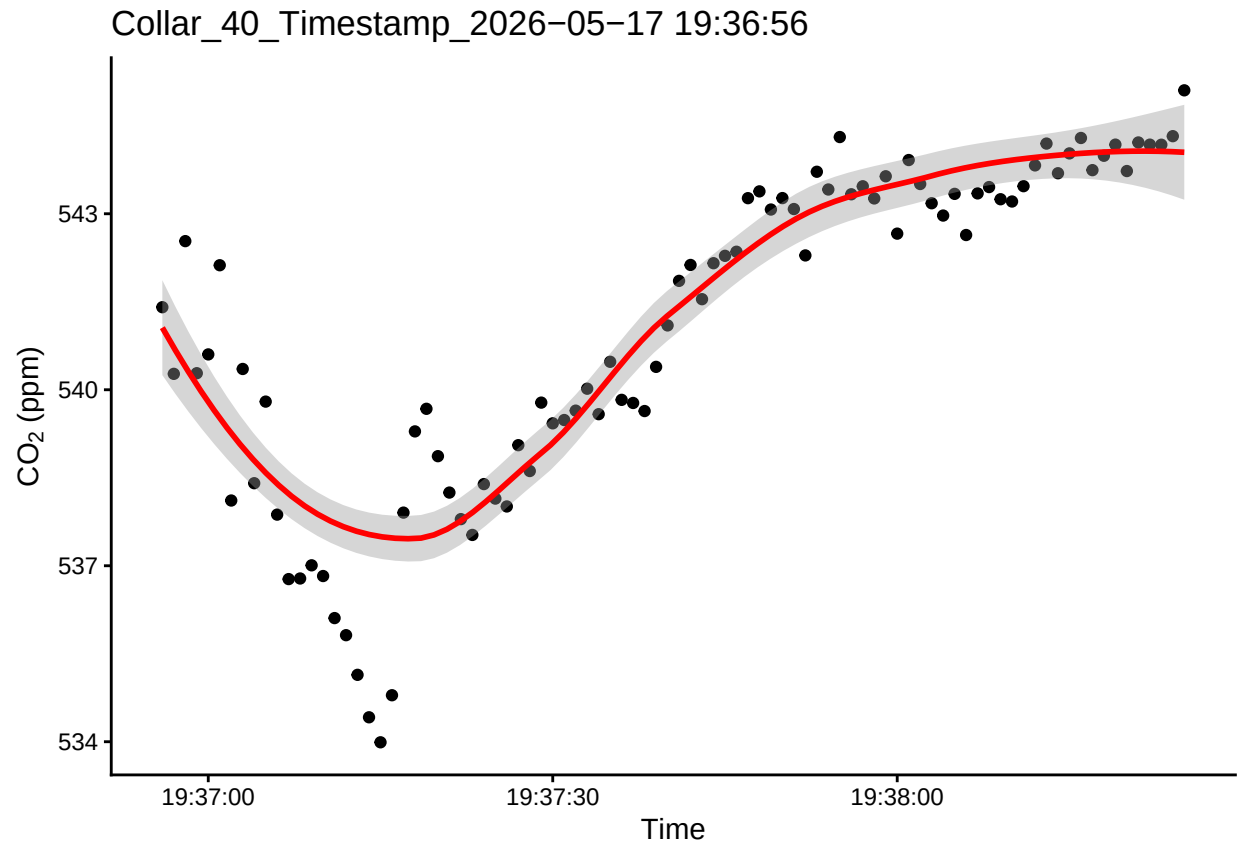
```
## 'geom_smooth()' using method = 'loess' and formula = 'y ~ x'
```



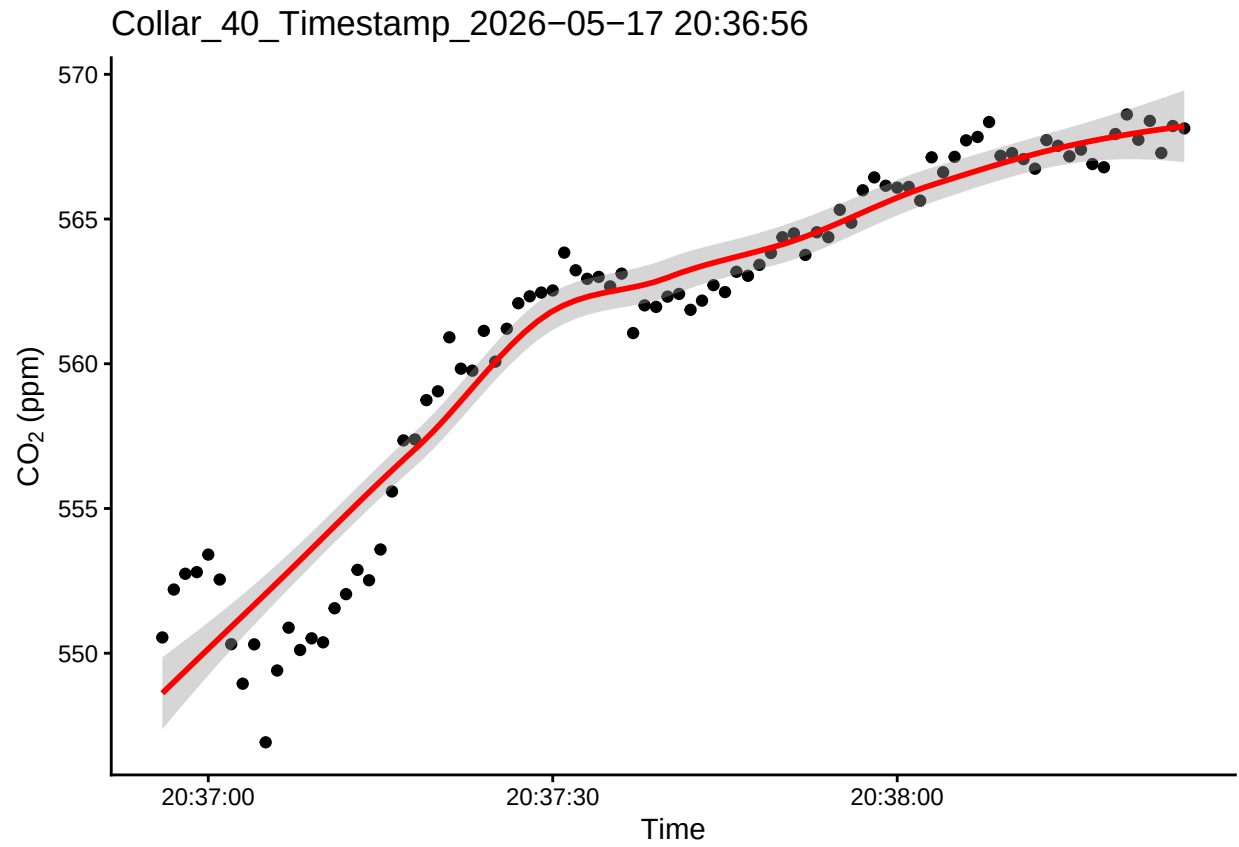
```
## 'geom_smooth()' using method = 'loess' and formula = 'y ~ x'
```

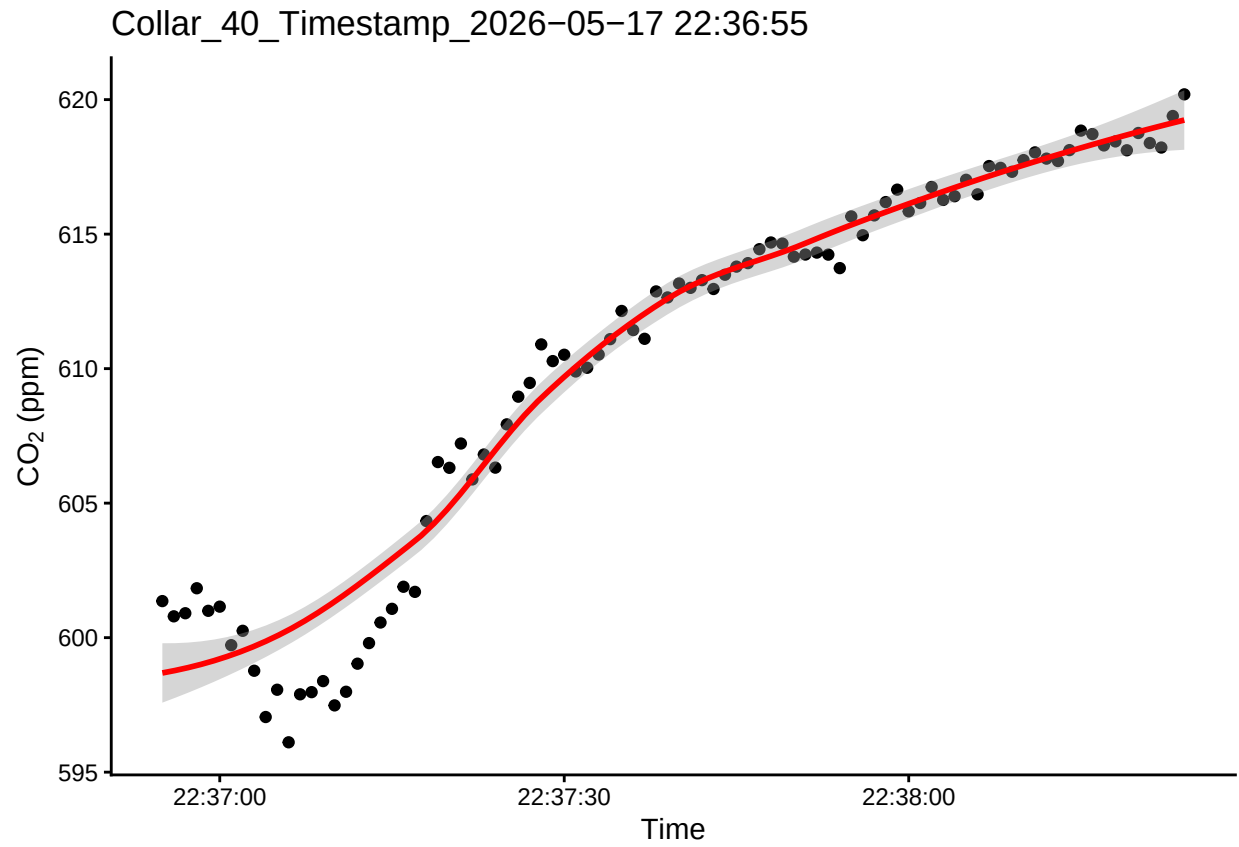
```
## 'geom_smooth()' using method = 'loess' and formula = 'y ~ x'
```



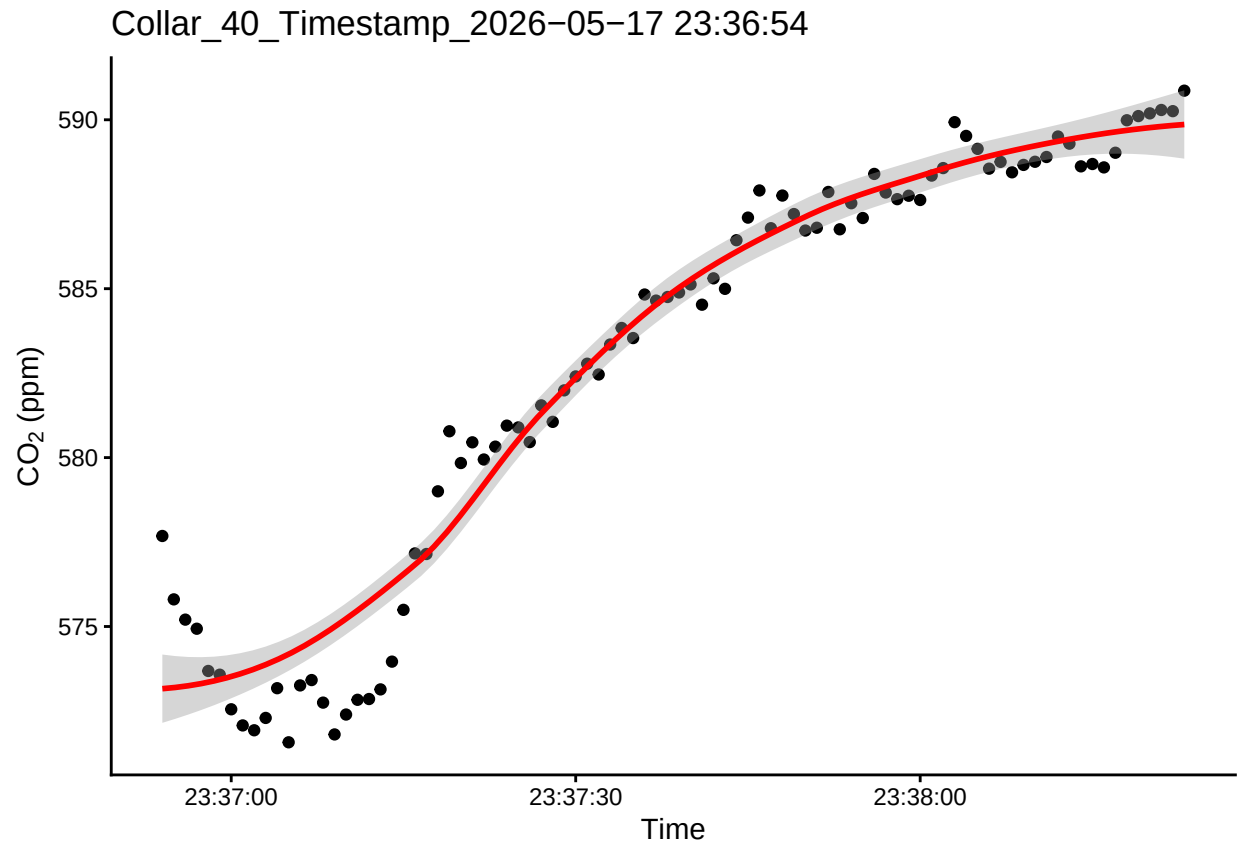
```
## 'geom_smooth()' using method = 'loess' and formula = 'y ~ x'
```



```
## 'geom_smooth()' using method = 'loess' and formula = 'y ~ x'
```

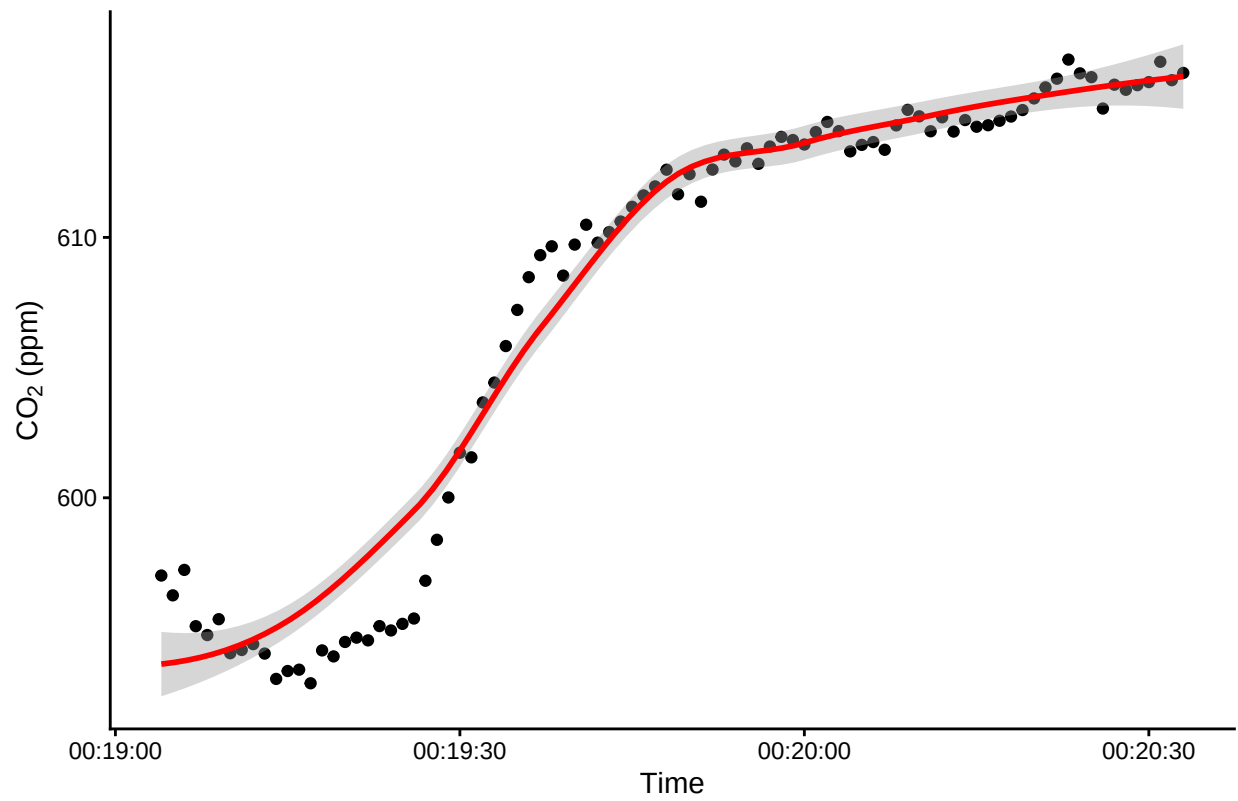


```
## 'geom_smooth()' using method = 'loess' and formula = 'y ~ x'
```

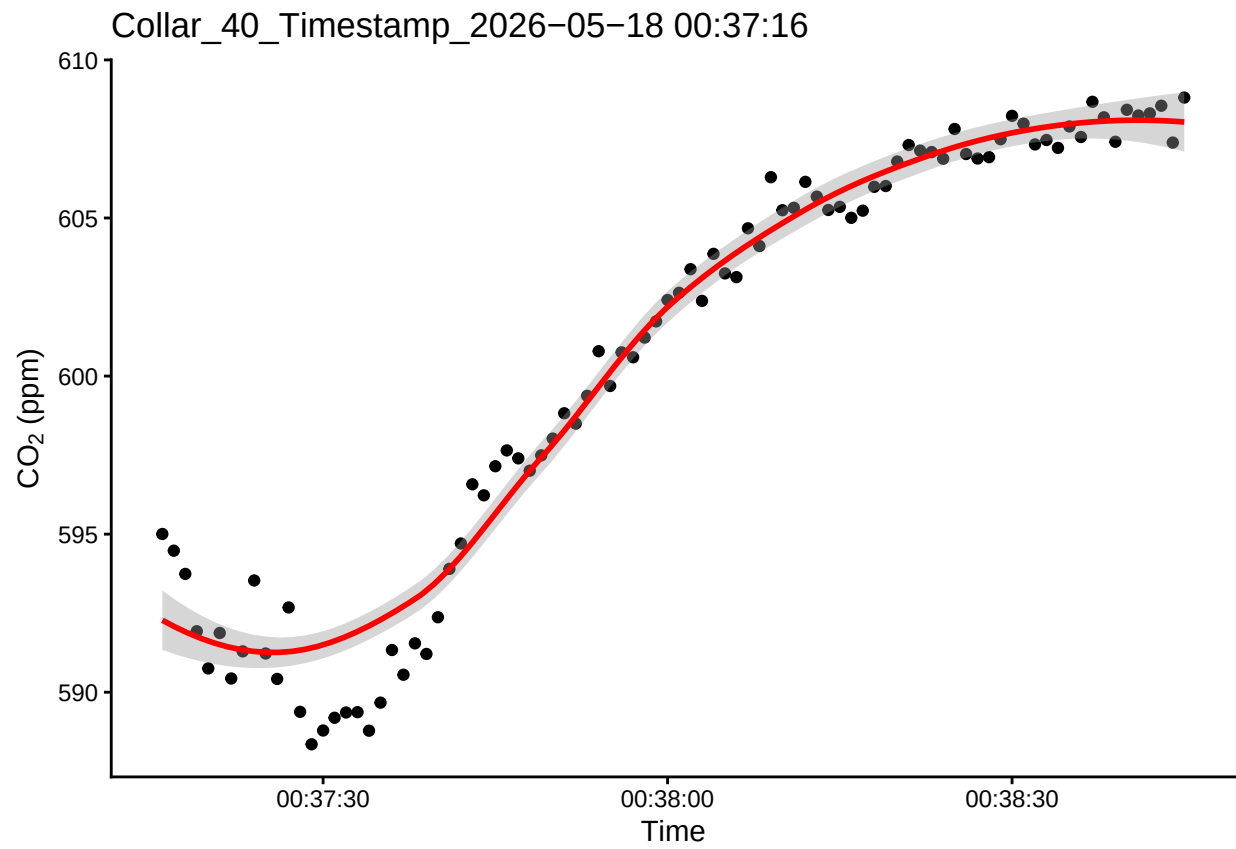


```
## 'geom_smooth()' using method = 'loess' and formula = 'y ~ x'
```

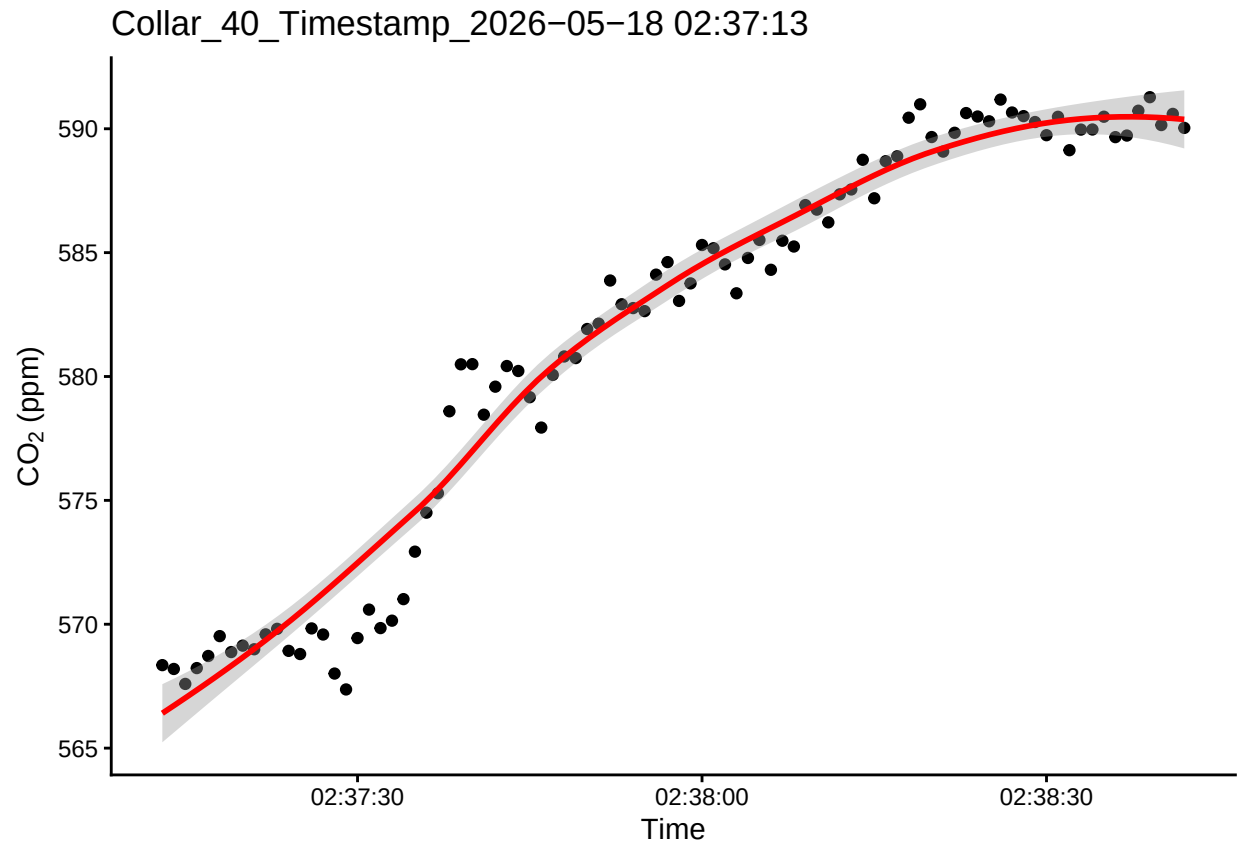
Collar_40_Timestamp_2026-05-18 00:19:04



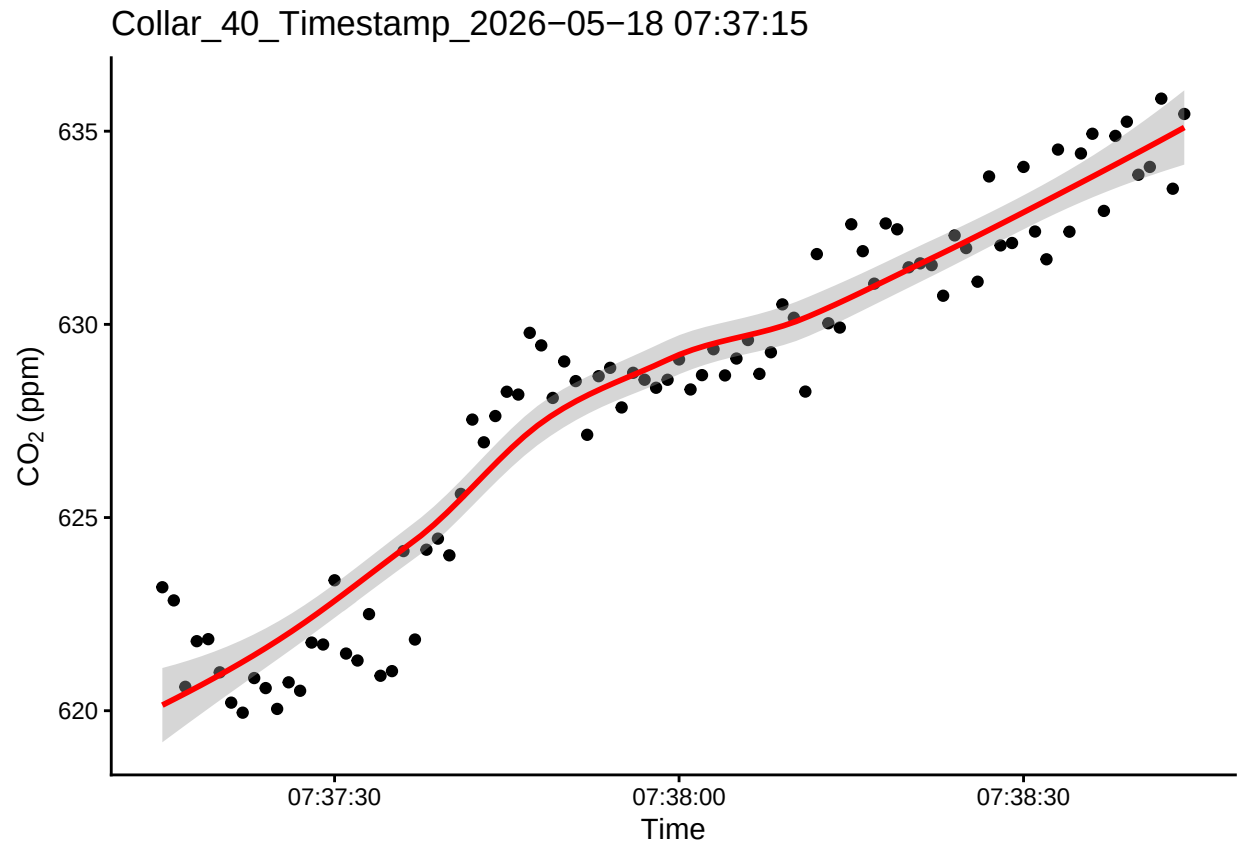
```
## 'geom_smooth()' using method = 'loess' and formula = 'y ~ x'
```



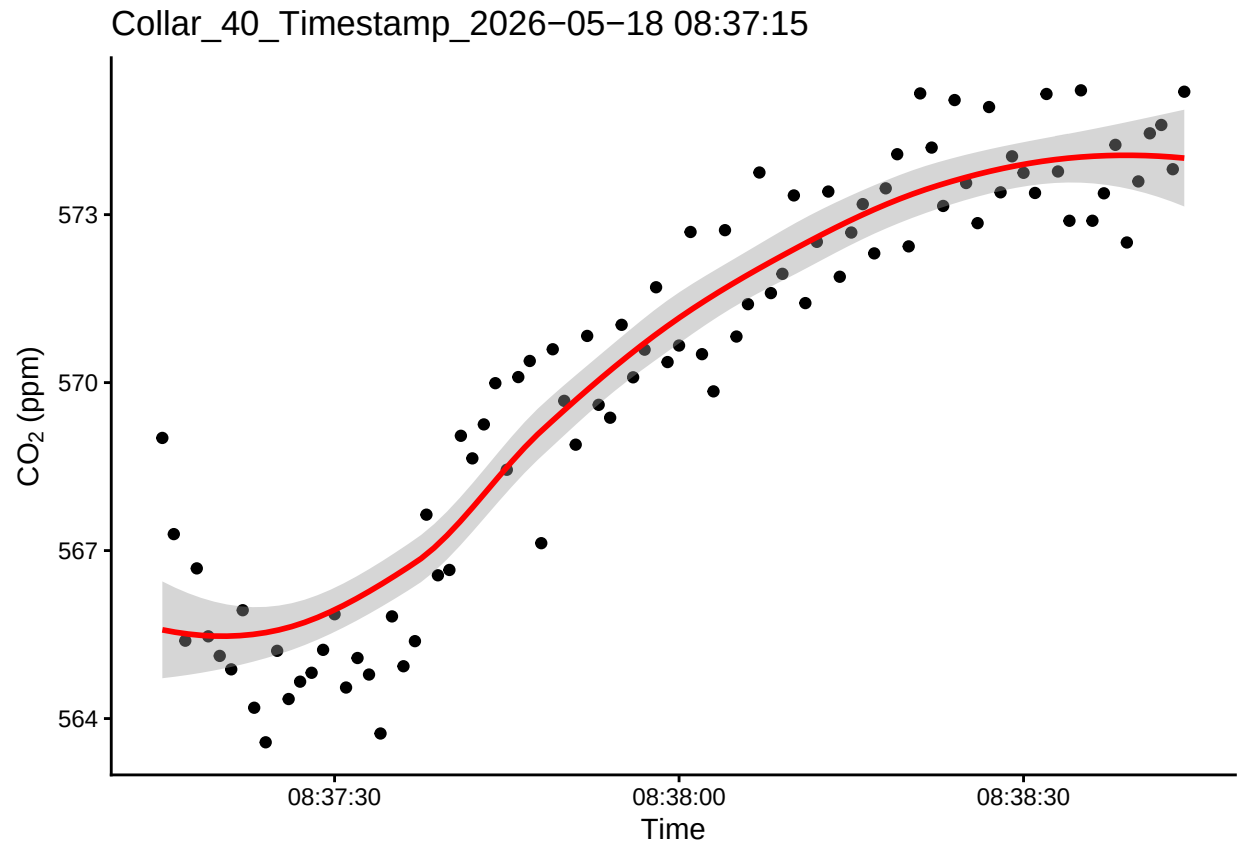
```
## 'geom_smooth()' using method = 'loess' and formula = 'y ~ x'
```



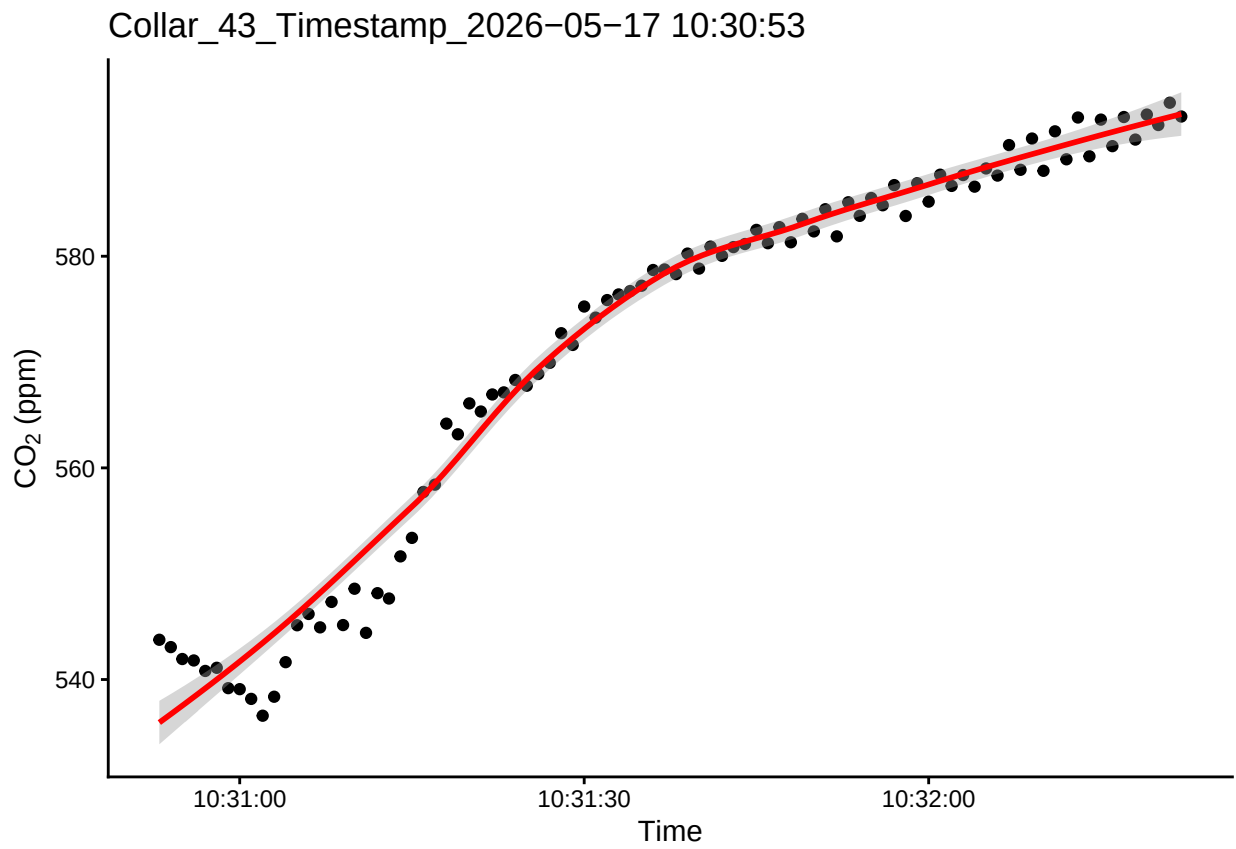
```
## 'geom_smooth()' using method = 'loess' and formula = 'y ~ x'
```

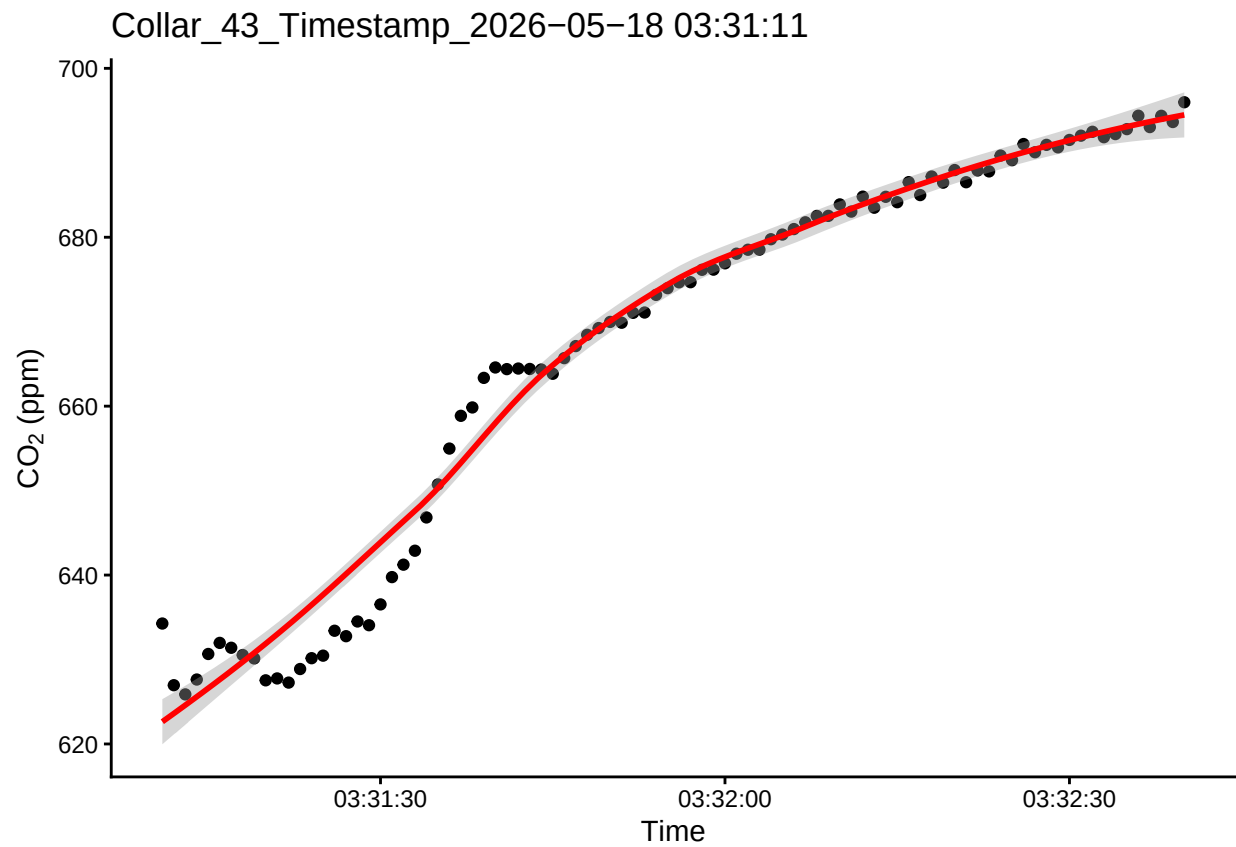
```
## 'geom_smooth()' using method = 'loess' and formula = 'y ~ x'
```



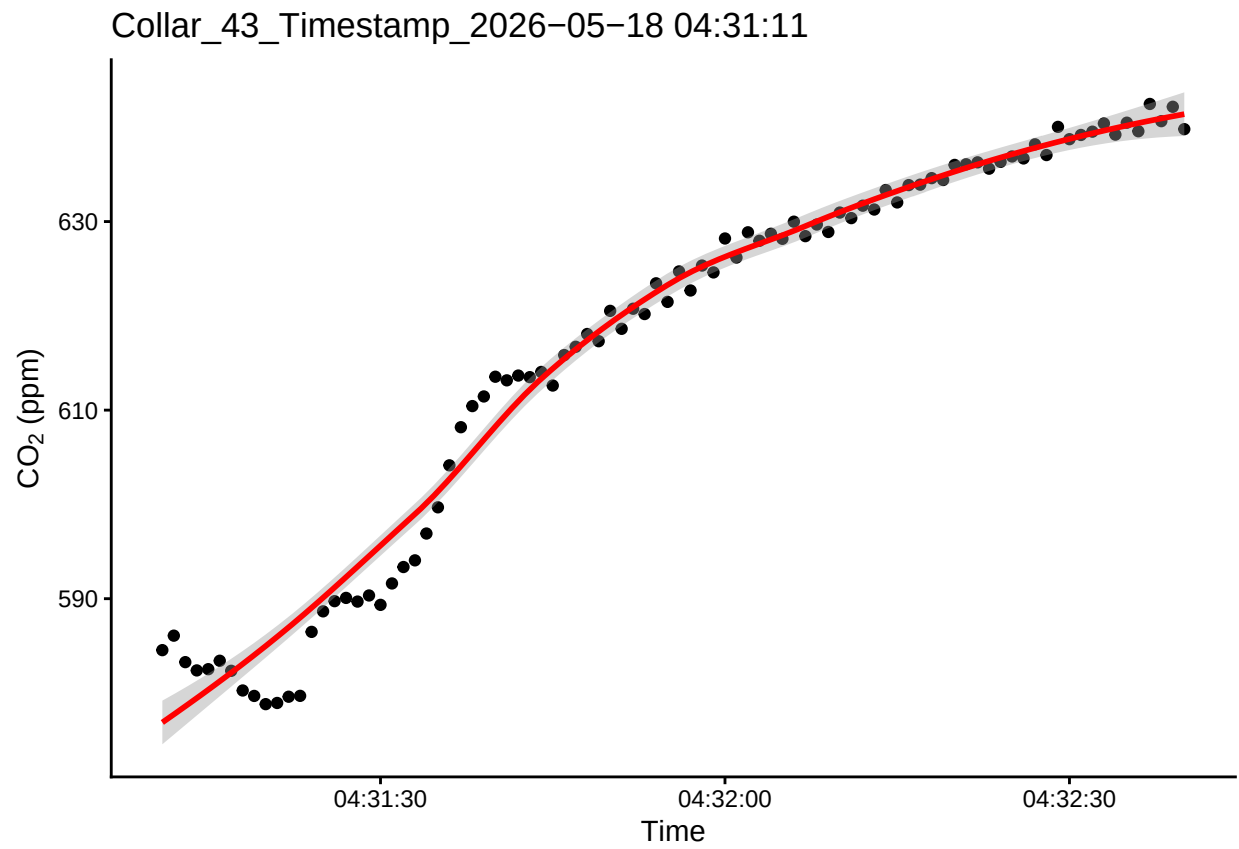
```
## 'geom_smooth()' using method = 'loess' and formula = 'y ~ x'
```



```
## 'geom_smooth()' using method = 'loess' and formula = 'y ~ x'
```

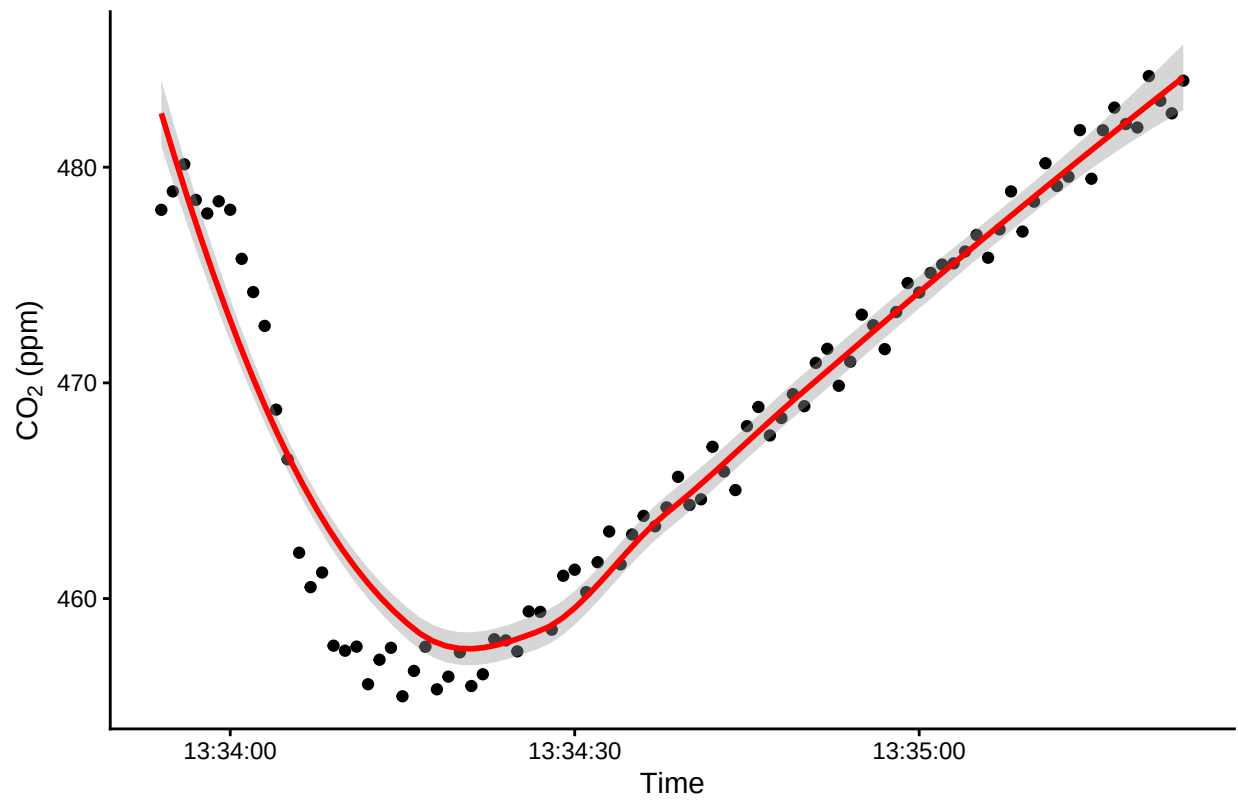


```
## 'geom_smooth()' using method = 'loess' and formula = 'y ~ x'
```

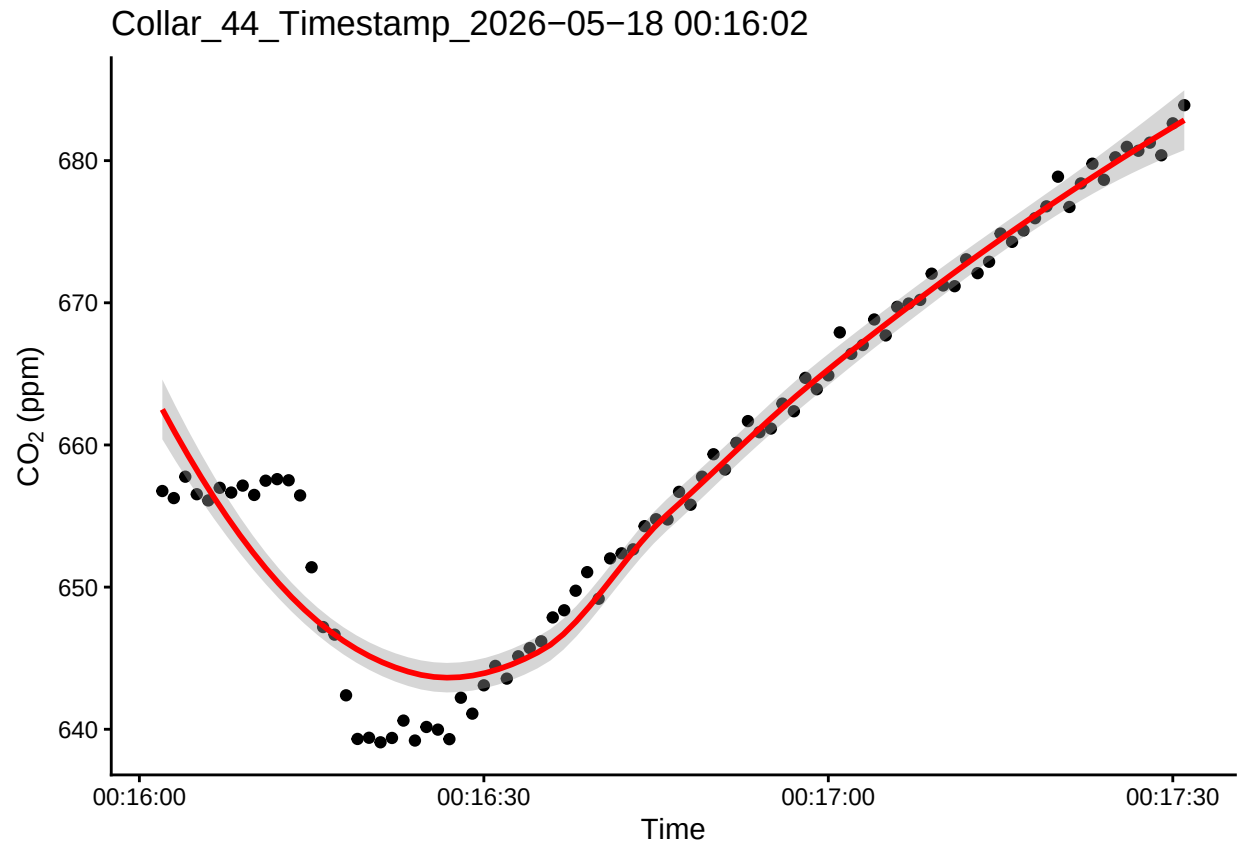


```
## 'geom_smooth()' using method = 'loess' and formula = 'y ~ x'
```

Collar_44_Timestamp_2026-05-17 13:33:54

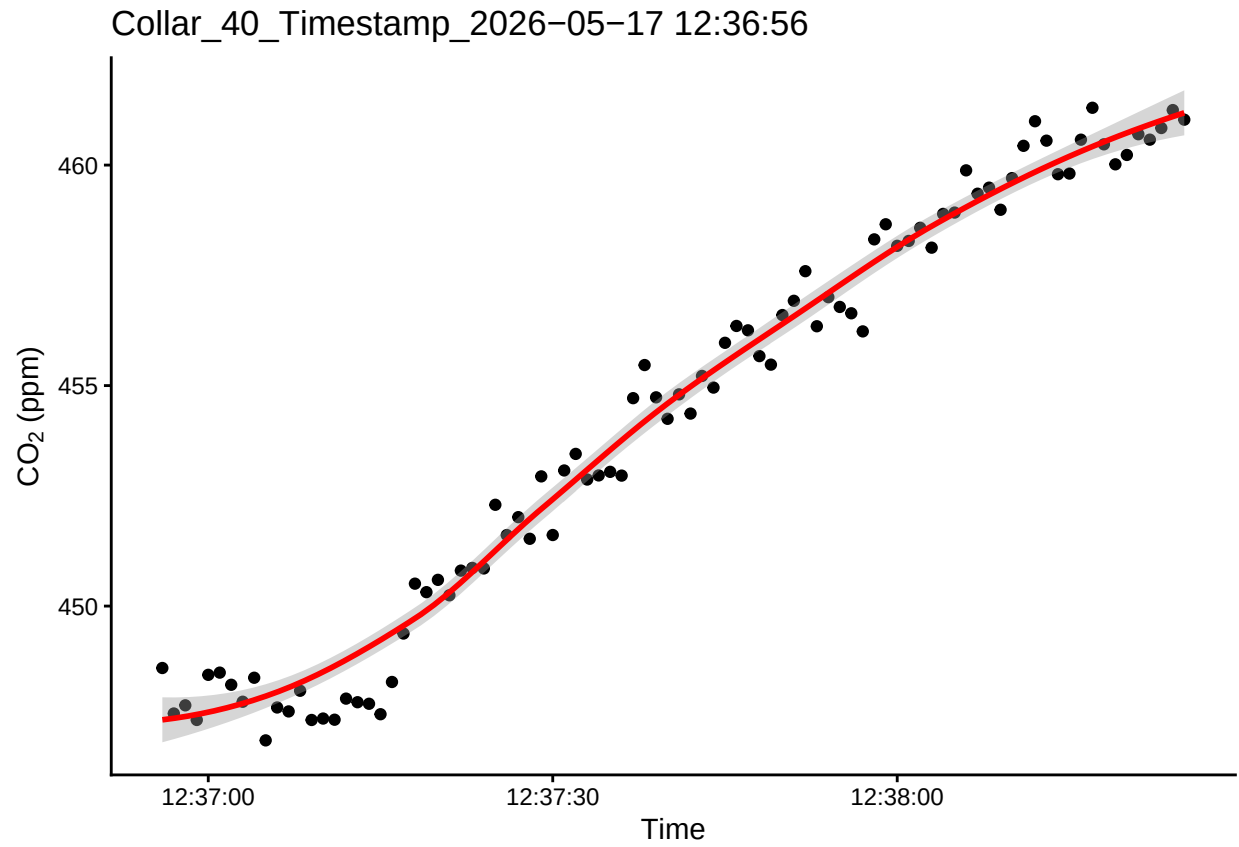


```
## 'geom_smooth()' using method = 'loess' and formula = 'y ~ x'
```

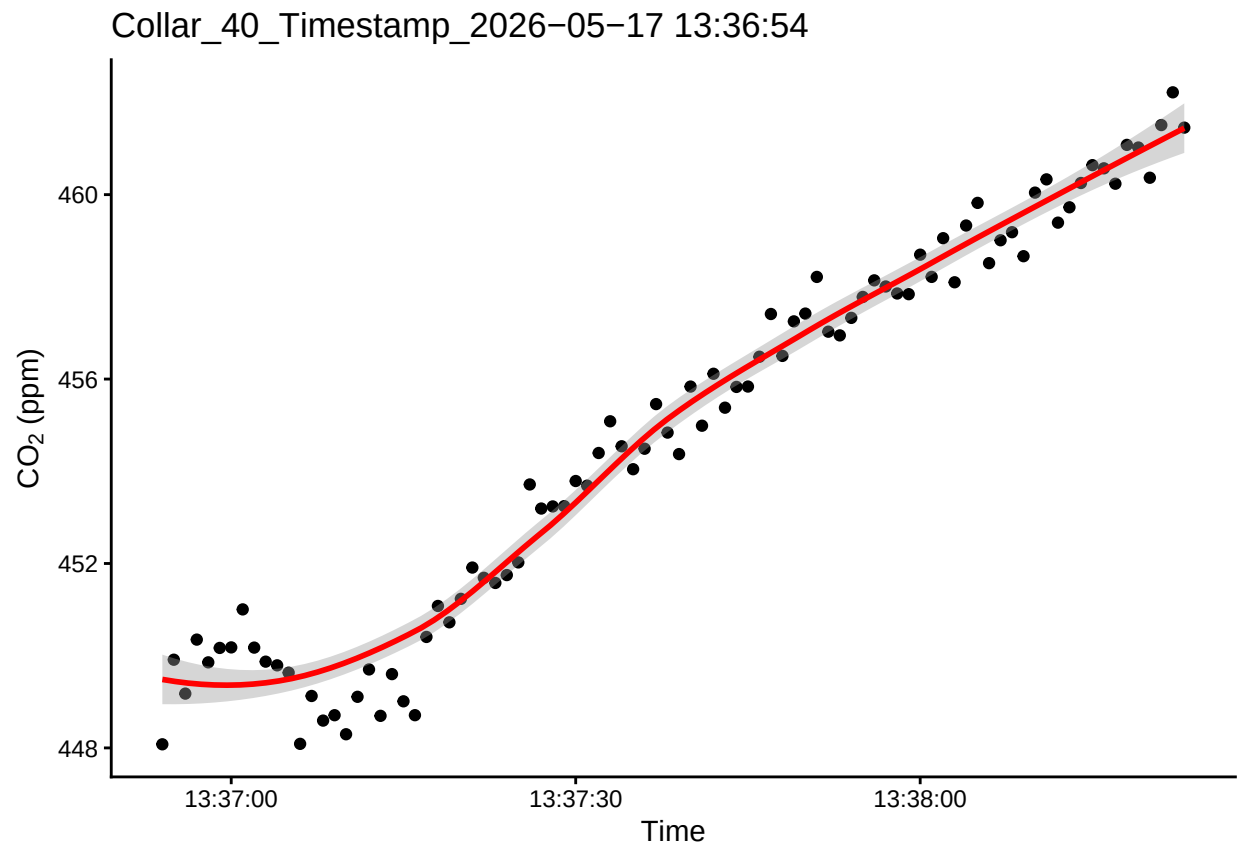


Look at the really high or really low ones to see if there is a timing issue or something

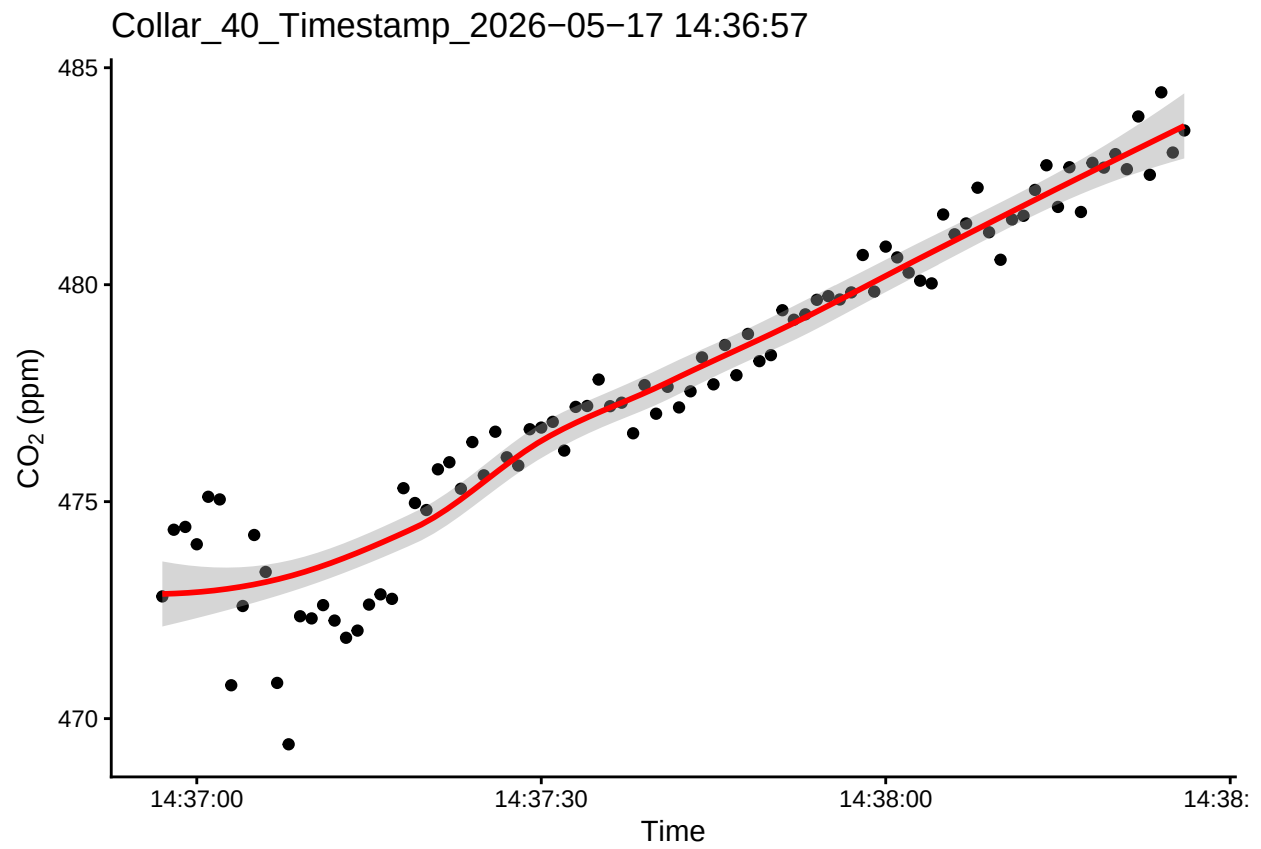
```
## 'geom_smooth()' using method = 'loess' and formula = 'y ~ x'
```



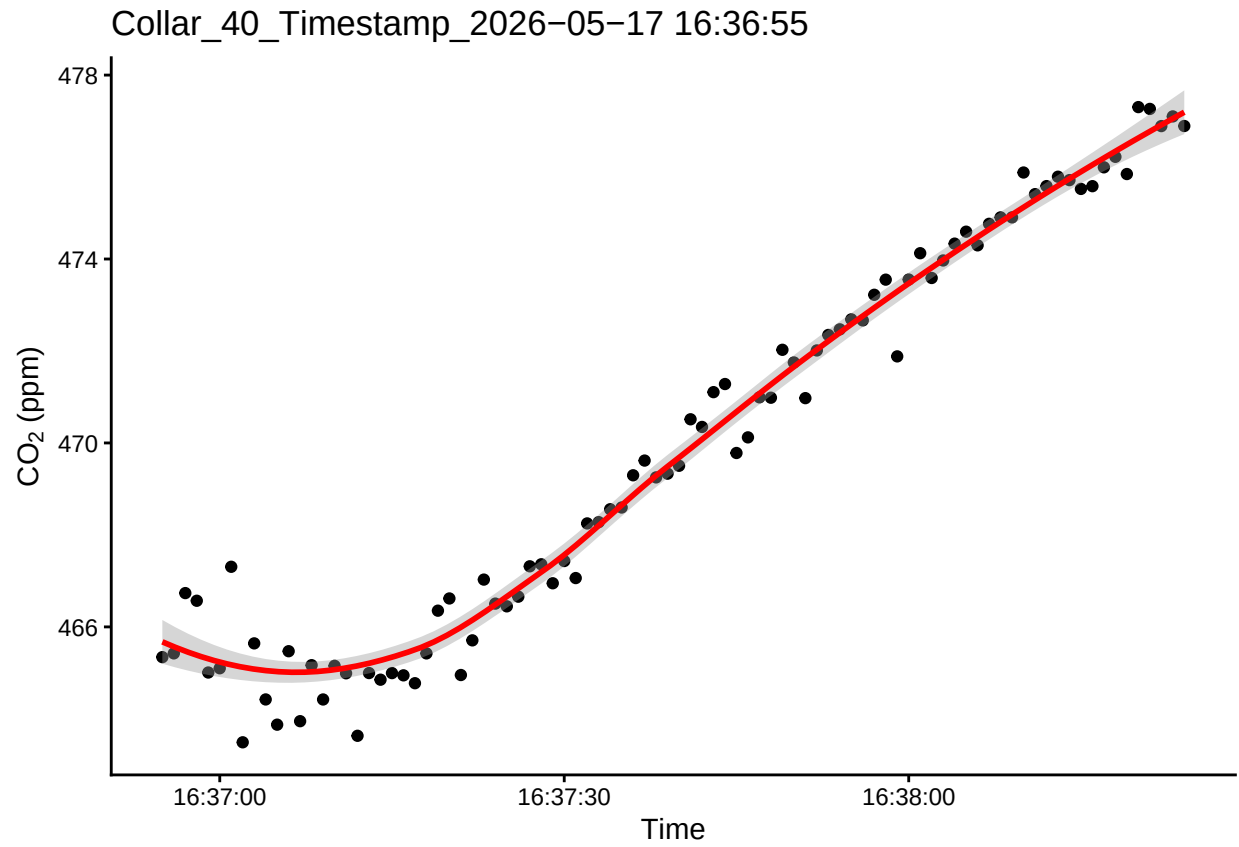
```
## 'geom_smooth()' using method = 'loess' and formula = 'y ~ x'
```

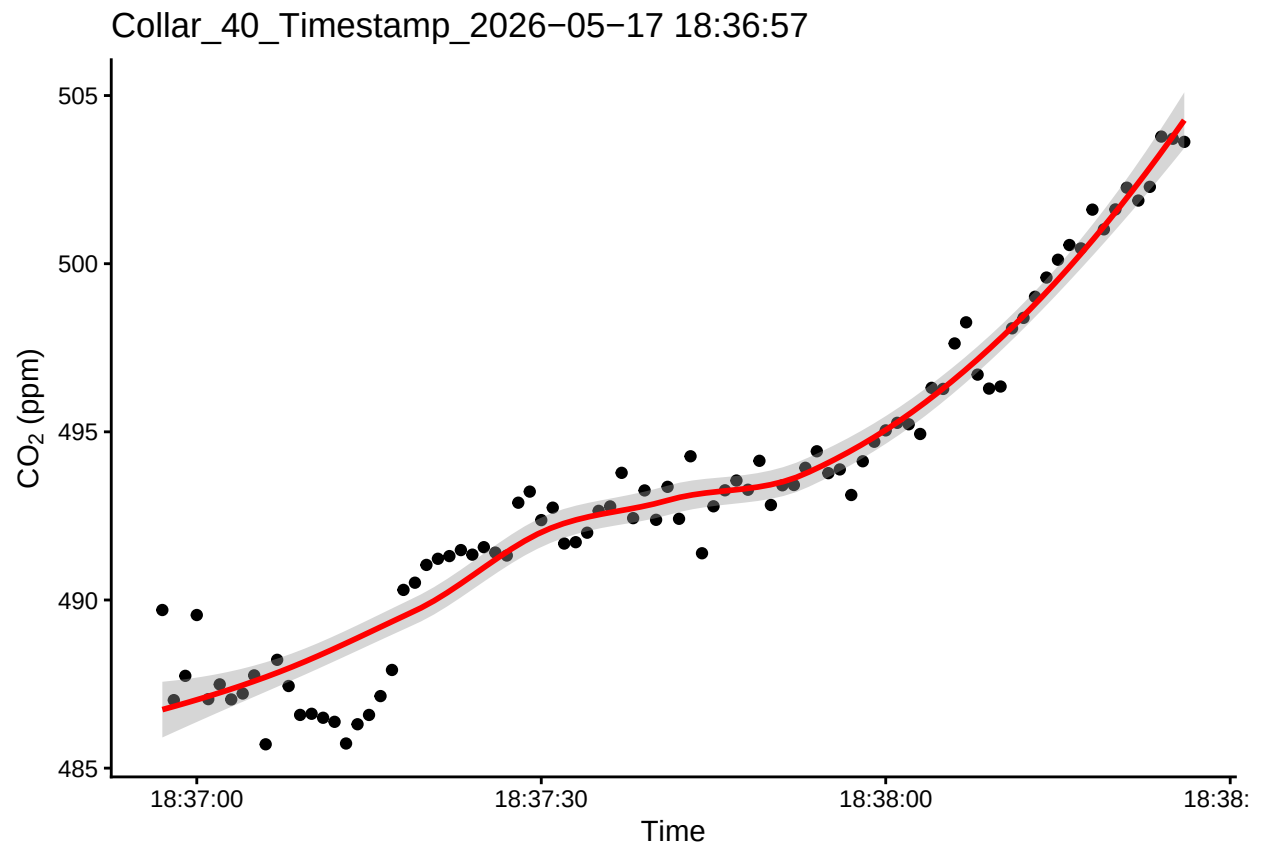
```
## 'geom_smooth()' using method = 'loess' and formula = 'y ~ x'
```



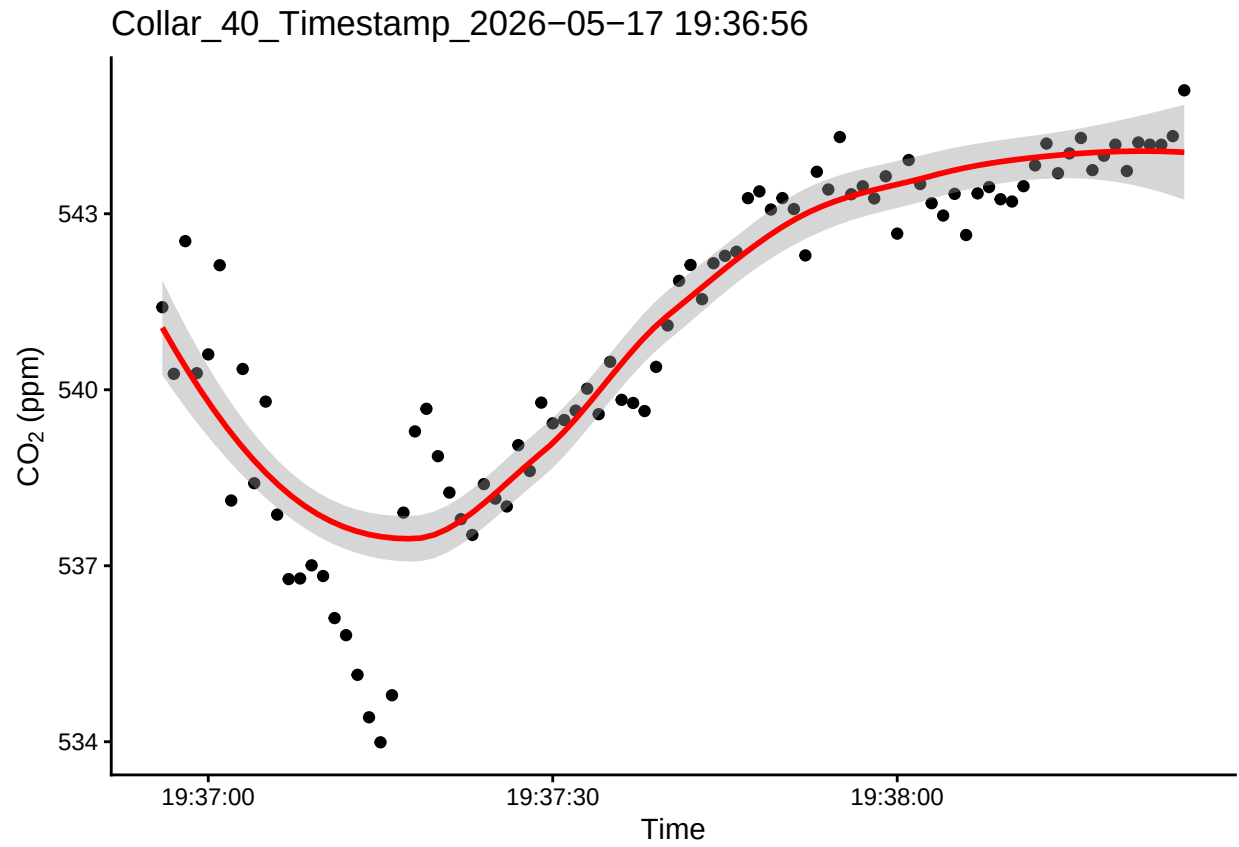
```
## 'geom_smooth()' using method = 'loess' and formula = 'y ~ x'
```



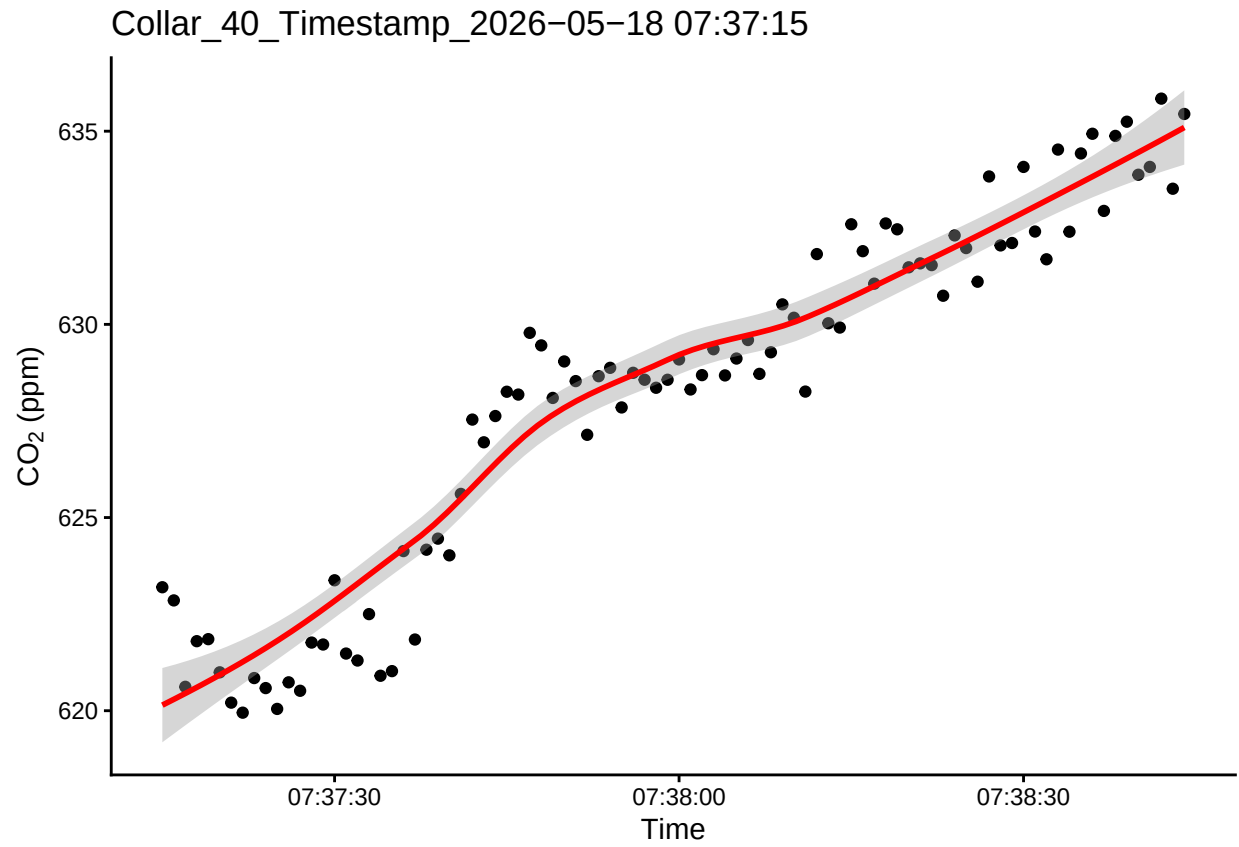
```
## 'geom_smooth()' using method = 'loess' and formula = 'y ~ x'
```



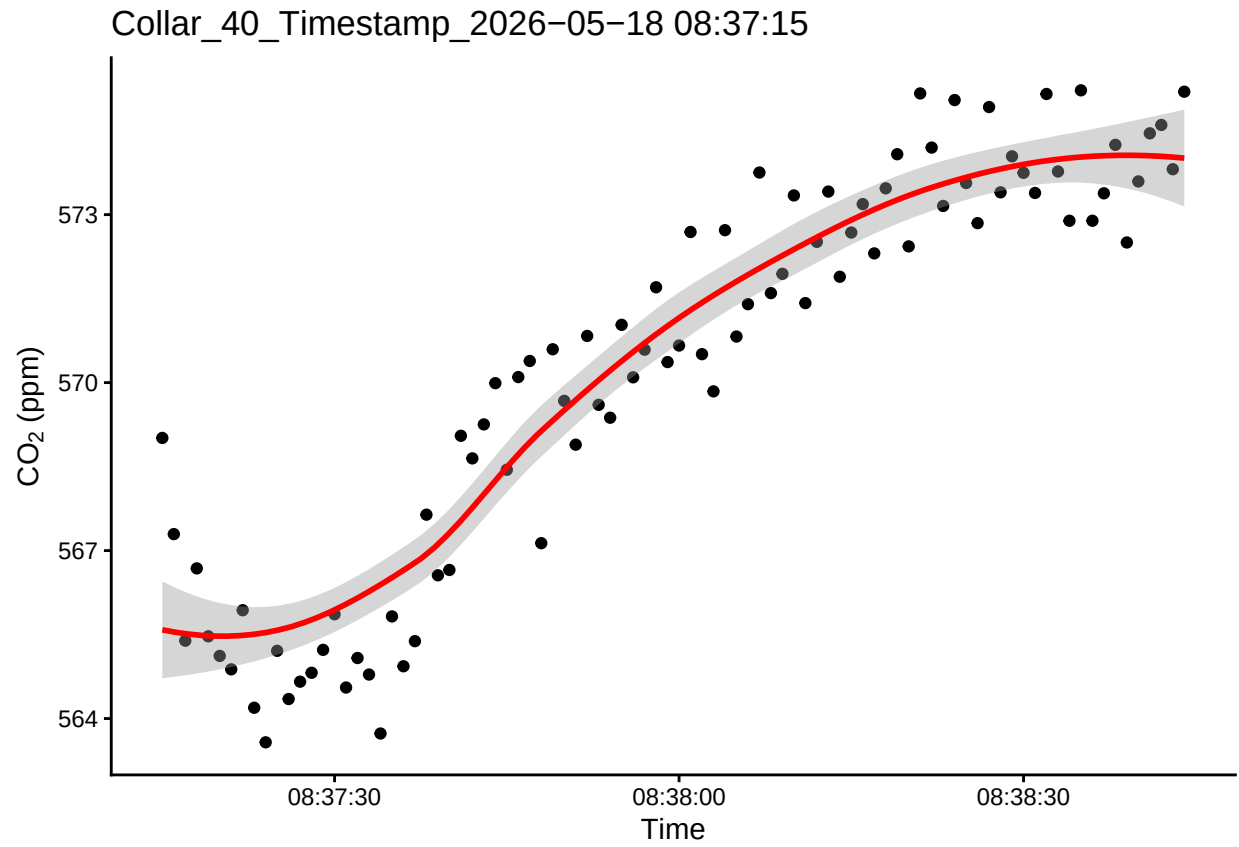
```
## 'geom_smooth()' using method = 'loess' and formula = 'y ~ x'
```



```
## 'geom_smooth()' using method = 'loess' and formula = 'y ~ x'
```

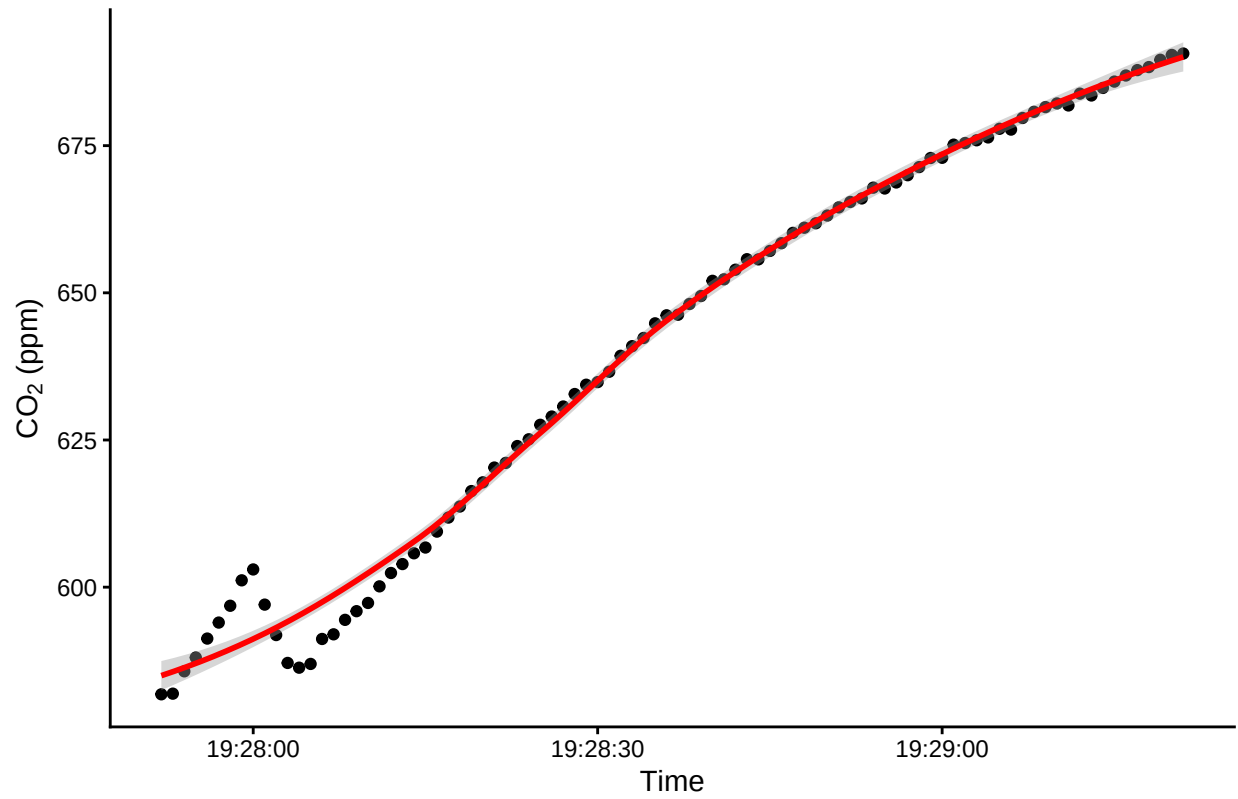


```
## 'geom_smooth()' using method = 'loess' and formula = 'y ~ x'
```



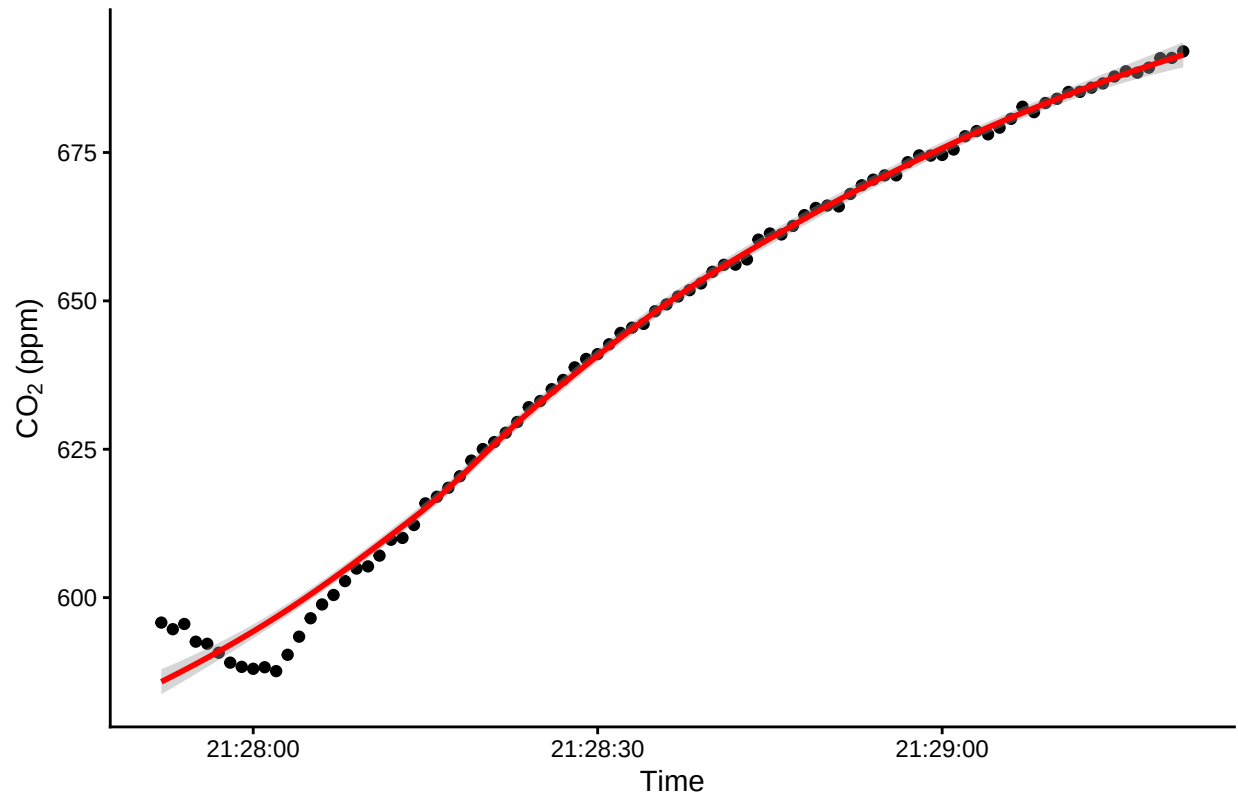
```
## 'geom_smooth()' using method = 'loess' and formula = 'y ~ x'
```

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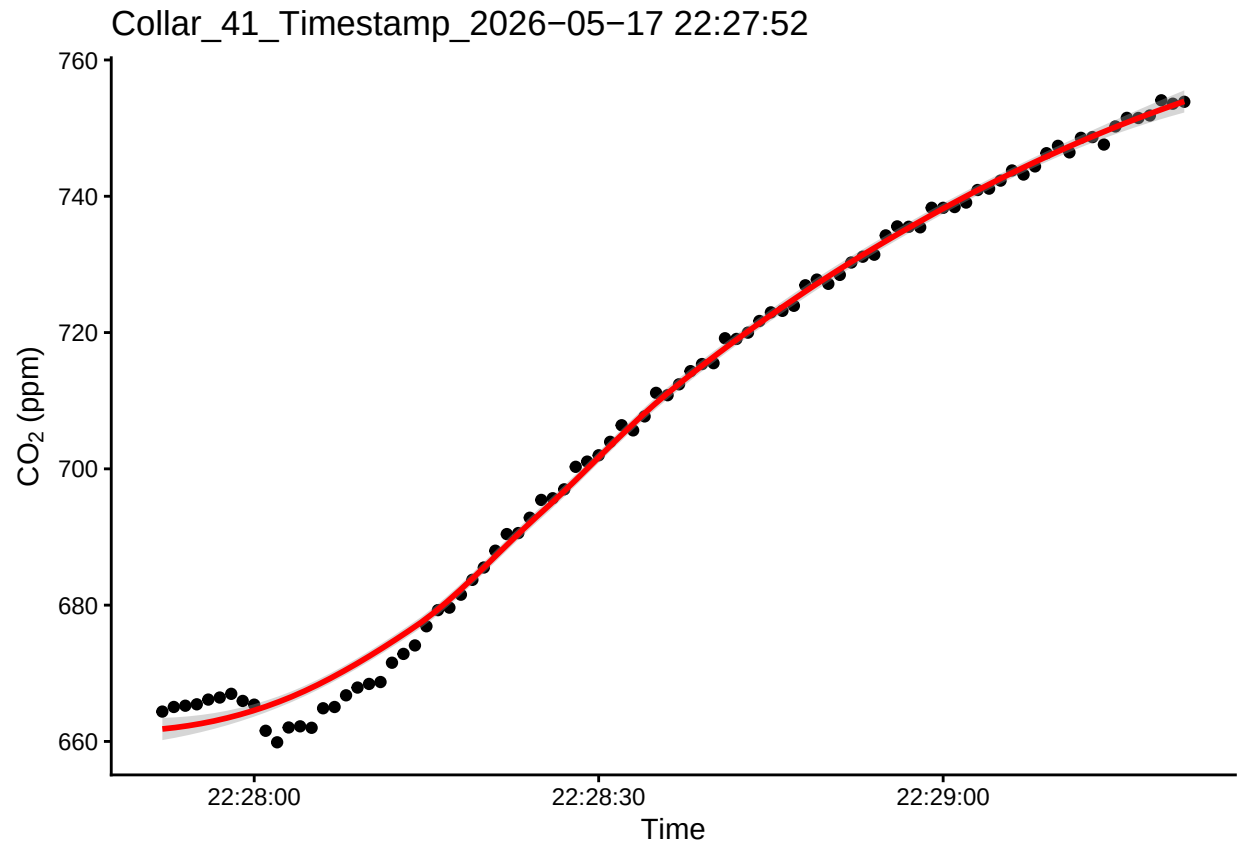


```
## 'geom_smooth()' using method = 'loess' and formula = 'y ~ x'
```

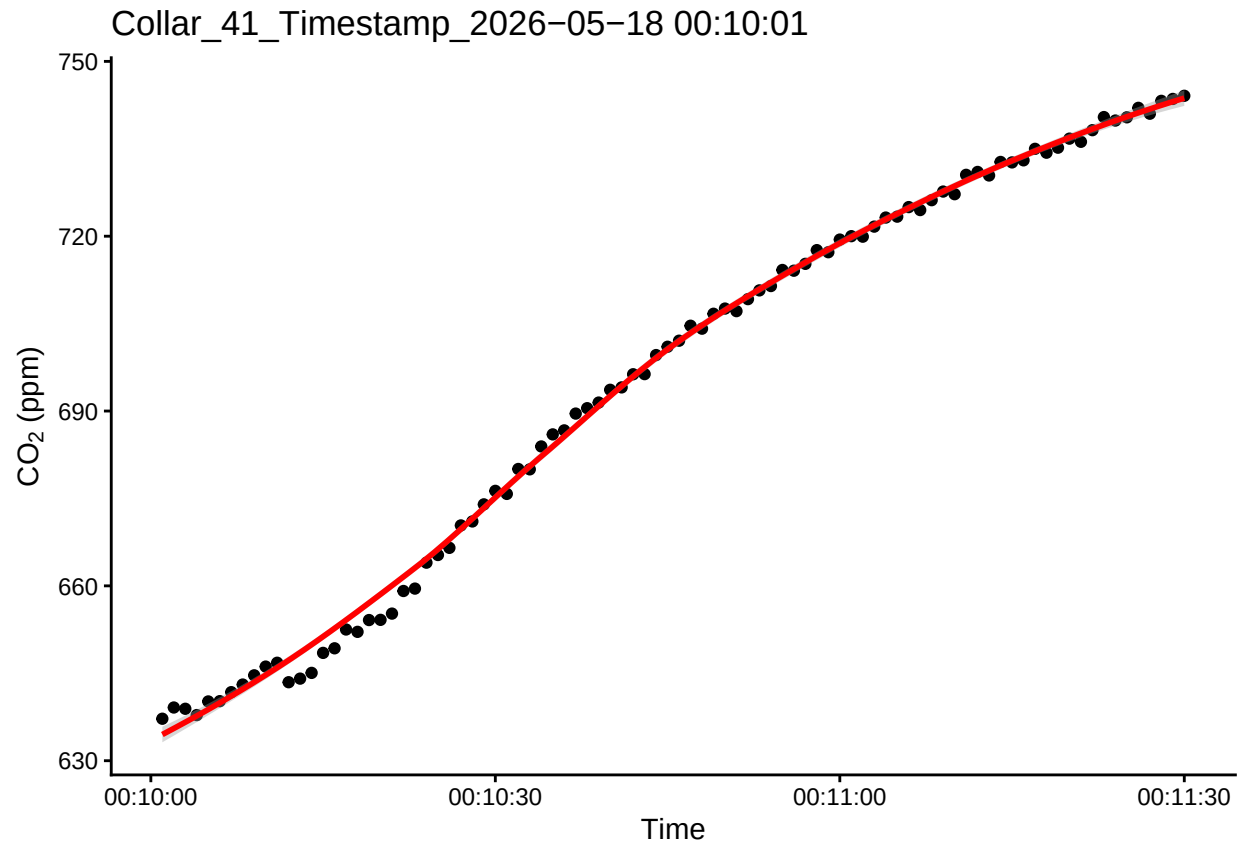

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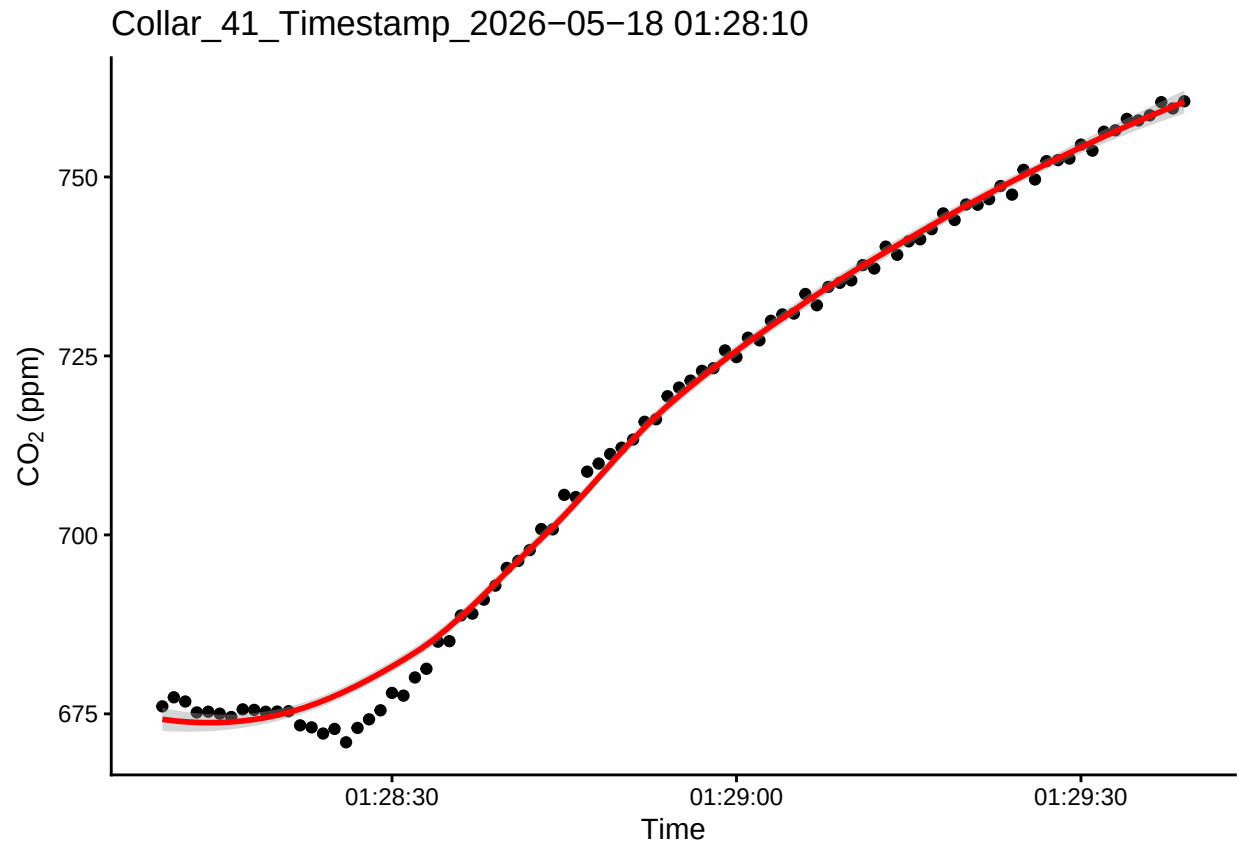
```
## 'geom_smooth()' using method = 'loess' and formula = 'y ~ x'
```



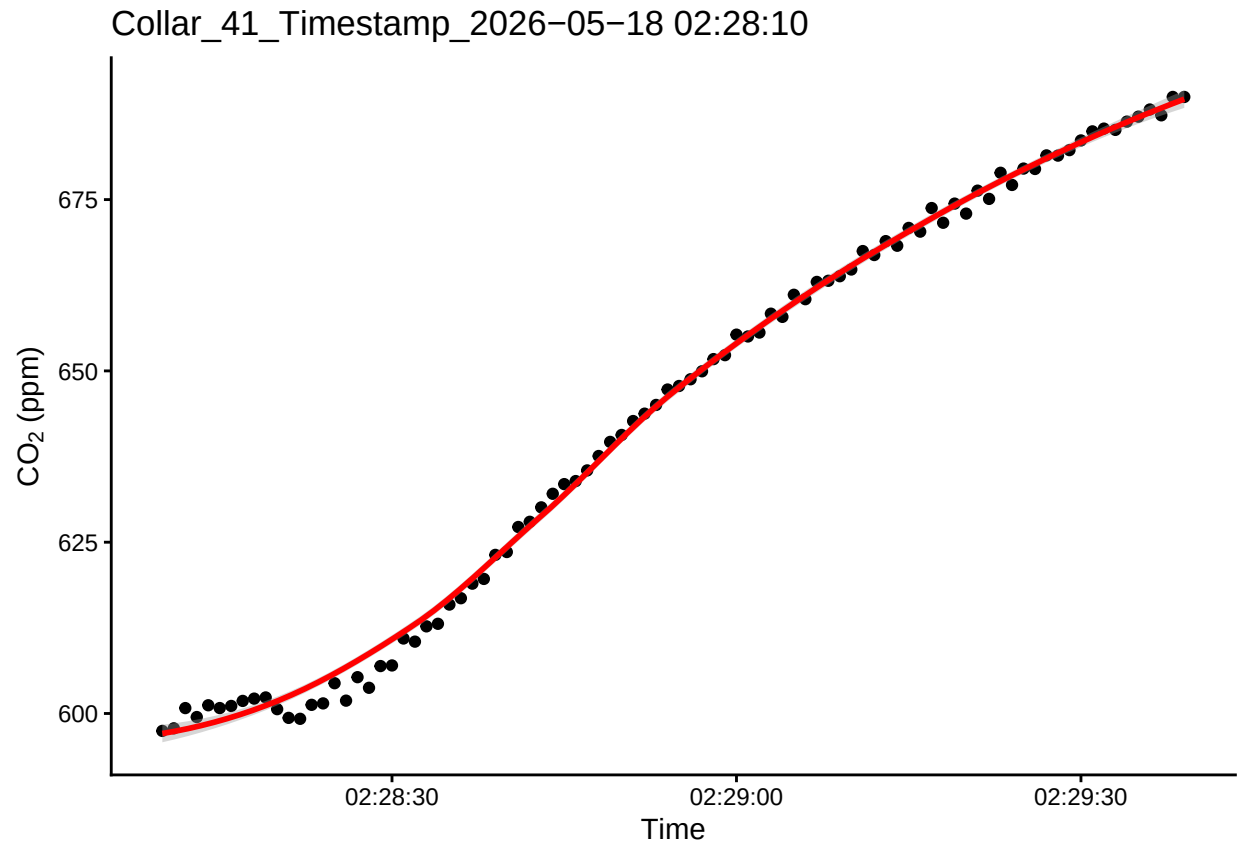
```
## 'geom_smooth()' using method = 'loess' and formula = 'y ~ x'
```



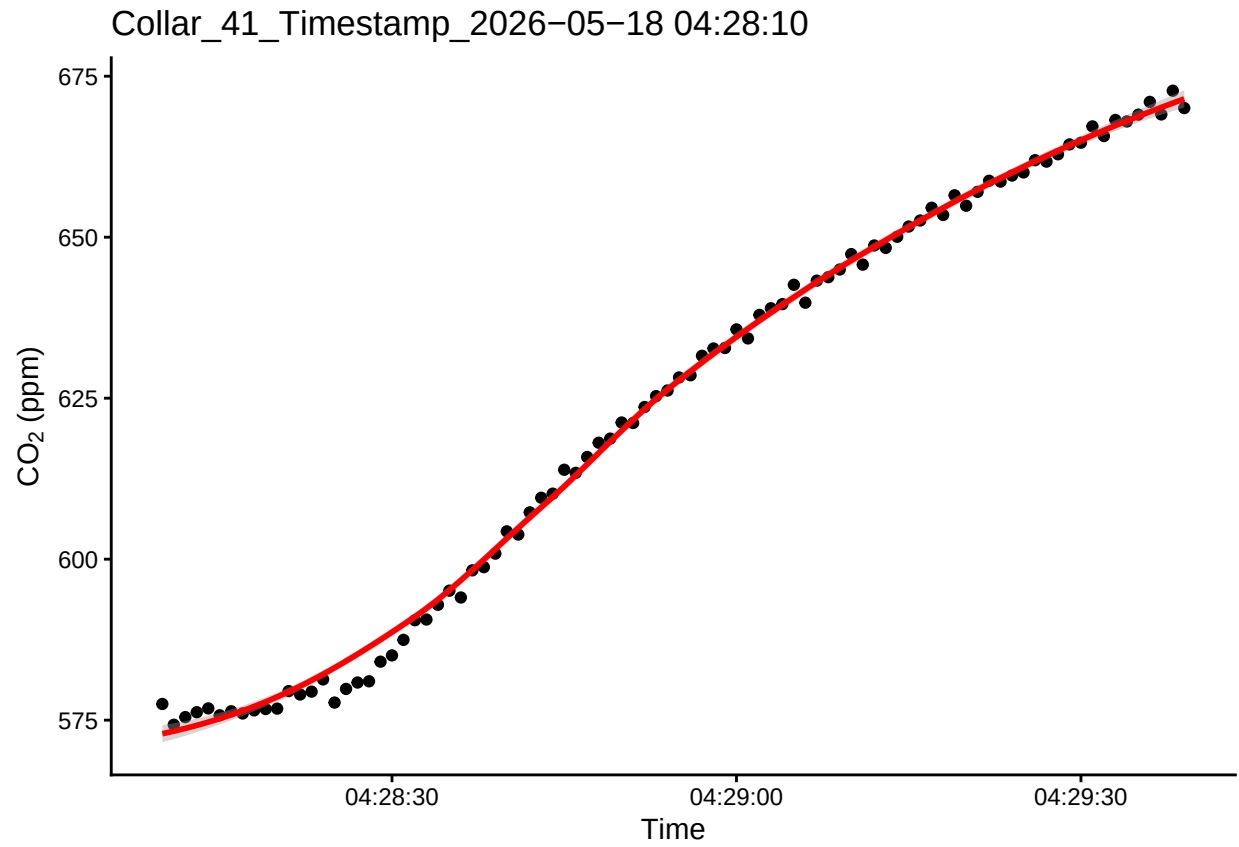
```
## 'geom_smooth()' using method = 'loess' and formula = 'y ~ x'
```



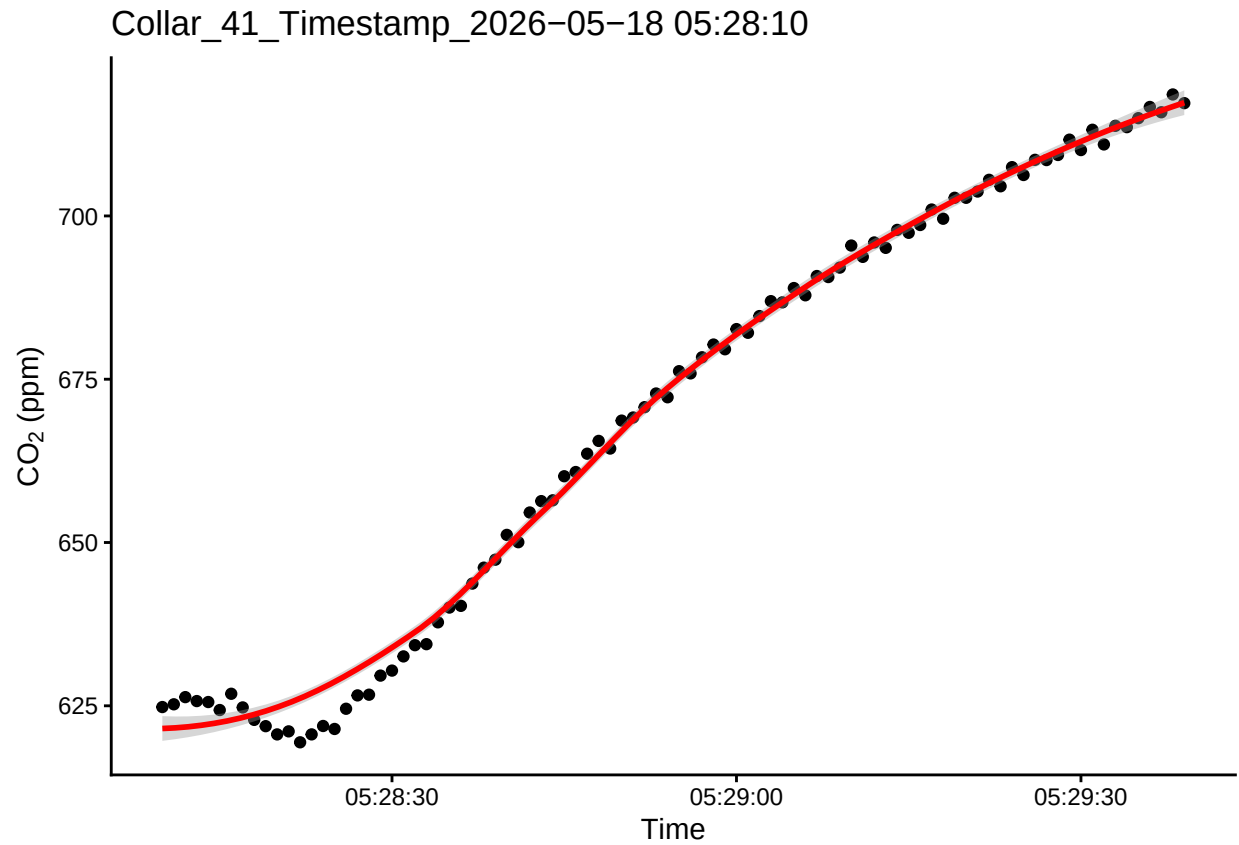
```
## 'geom_smooth()' using method = 'loess' and formula = 'y ~ x'
```



```
## 'geom_smooth()' using method = 'loess' and formula = 'y ~ x'
```



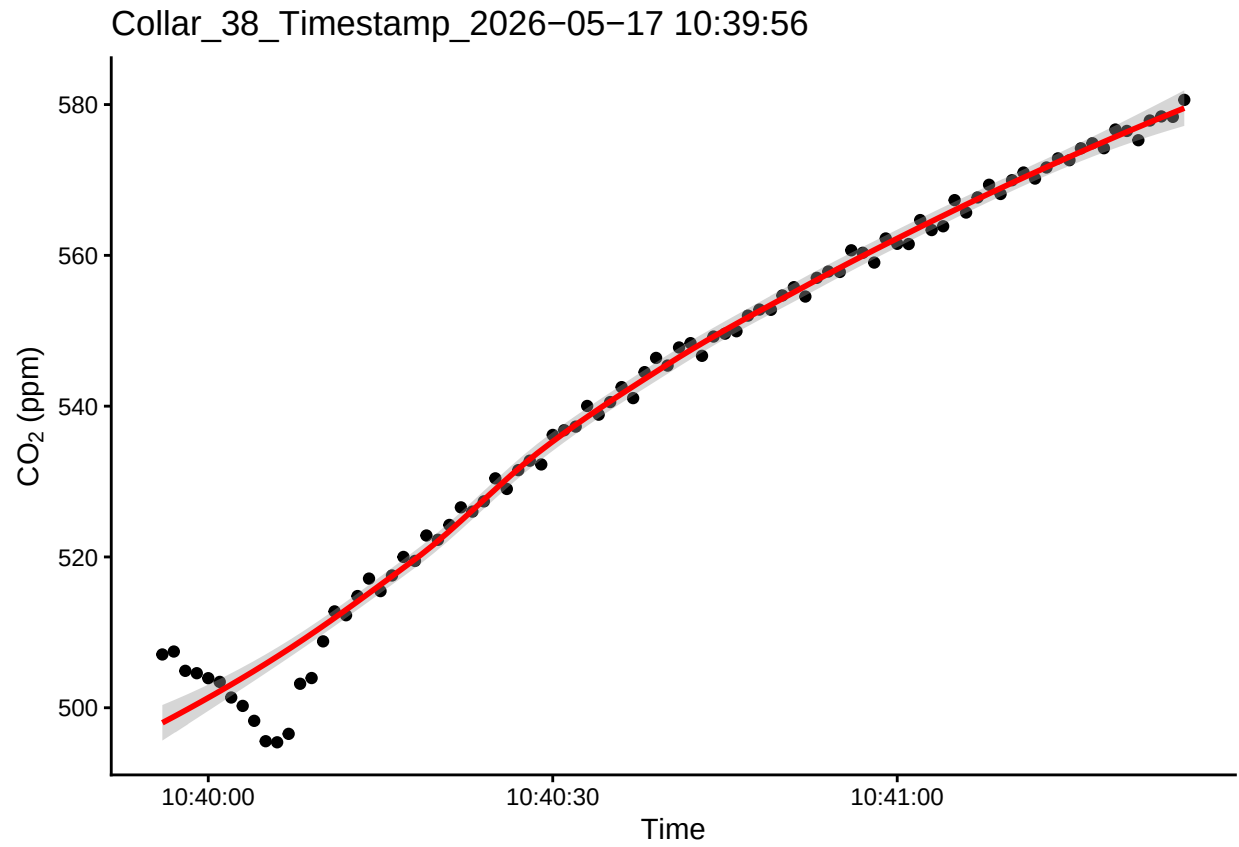
```
## 'geom_smooth()' using method = 'loess' and formula = 'y ~ x'
```



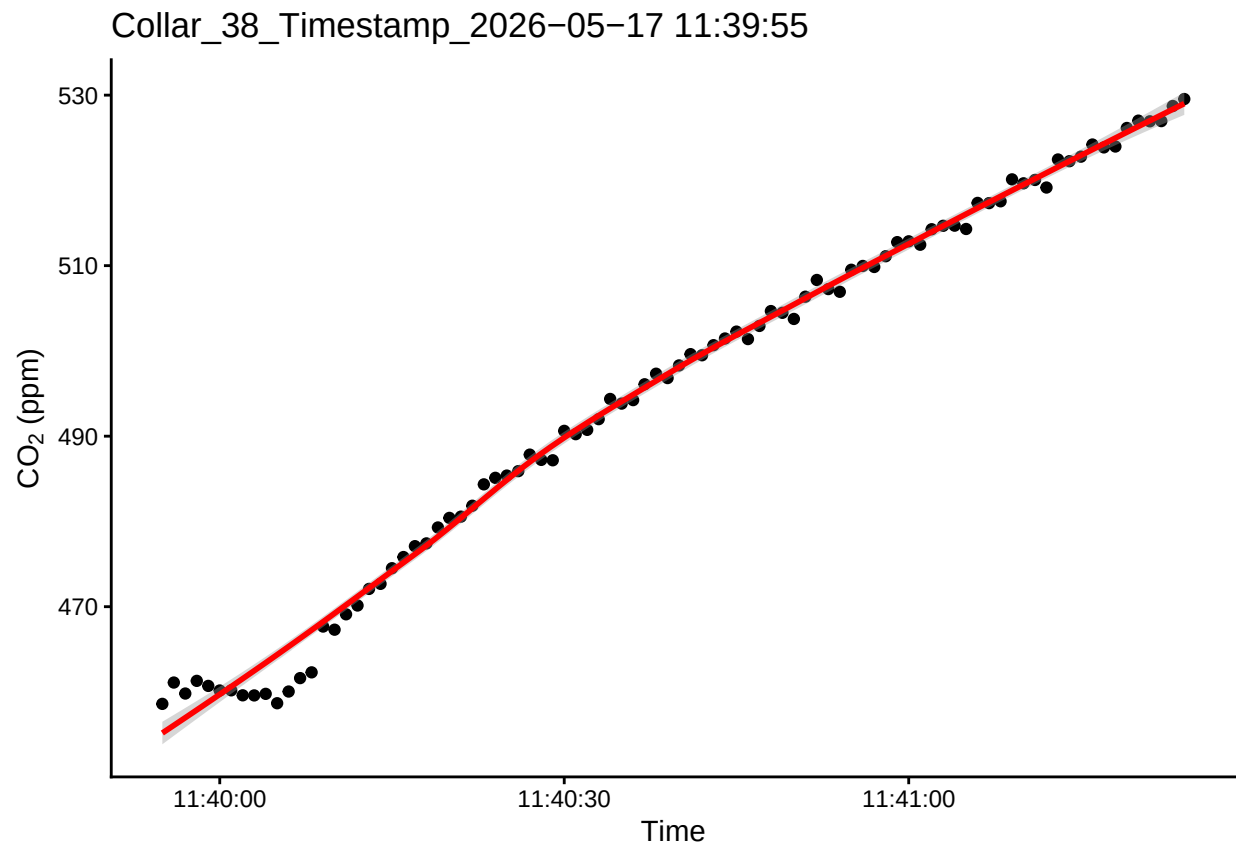
Look at the negative CO₂ fluxes

Plot all fluxes

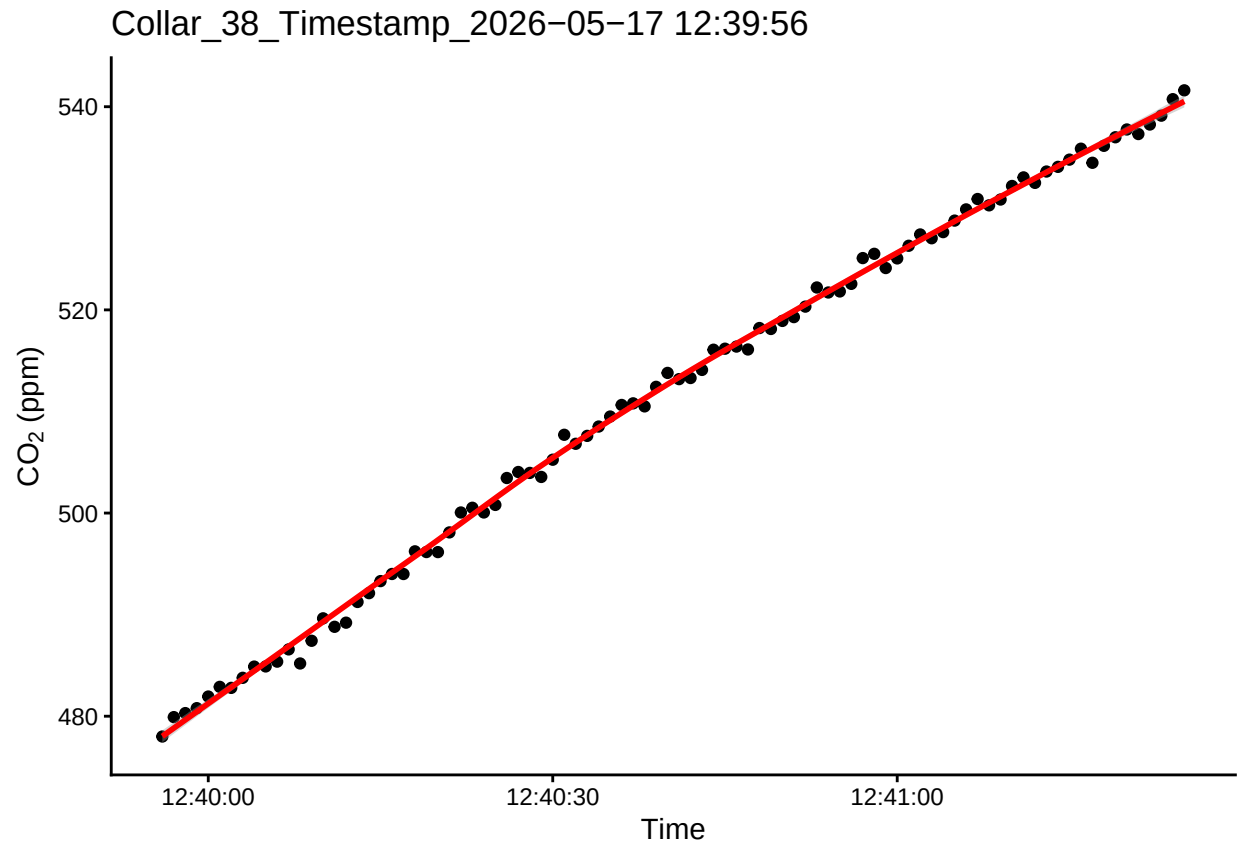
```
## 'geom_smooth()' using method = 'loess' and formula = 'y ~ x'
```



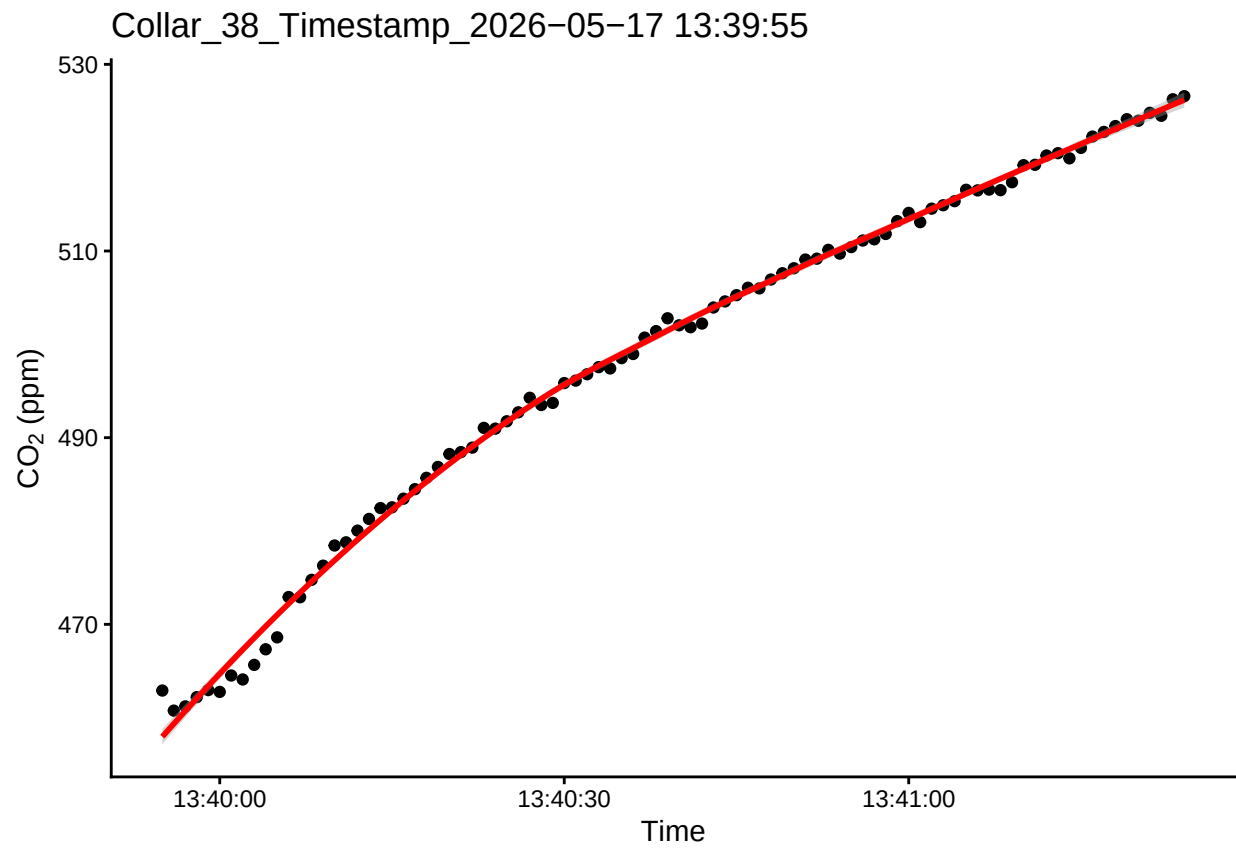
```
## 'geom_smooth()' using method = 'loess' and formula = 'y ~ x'
```

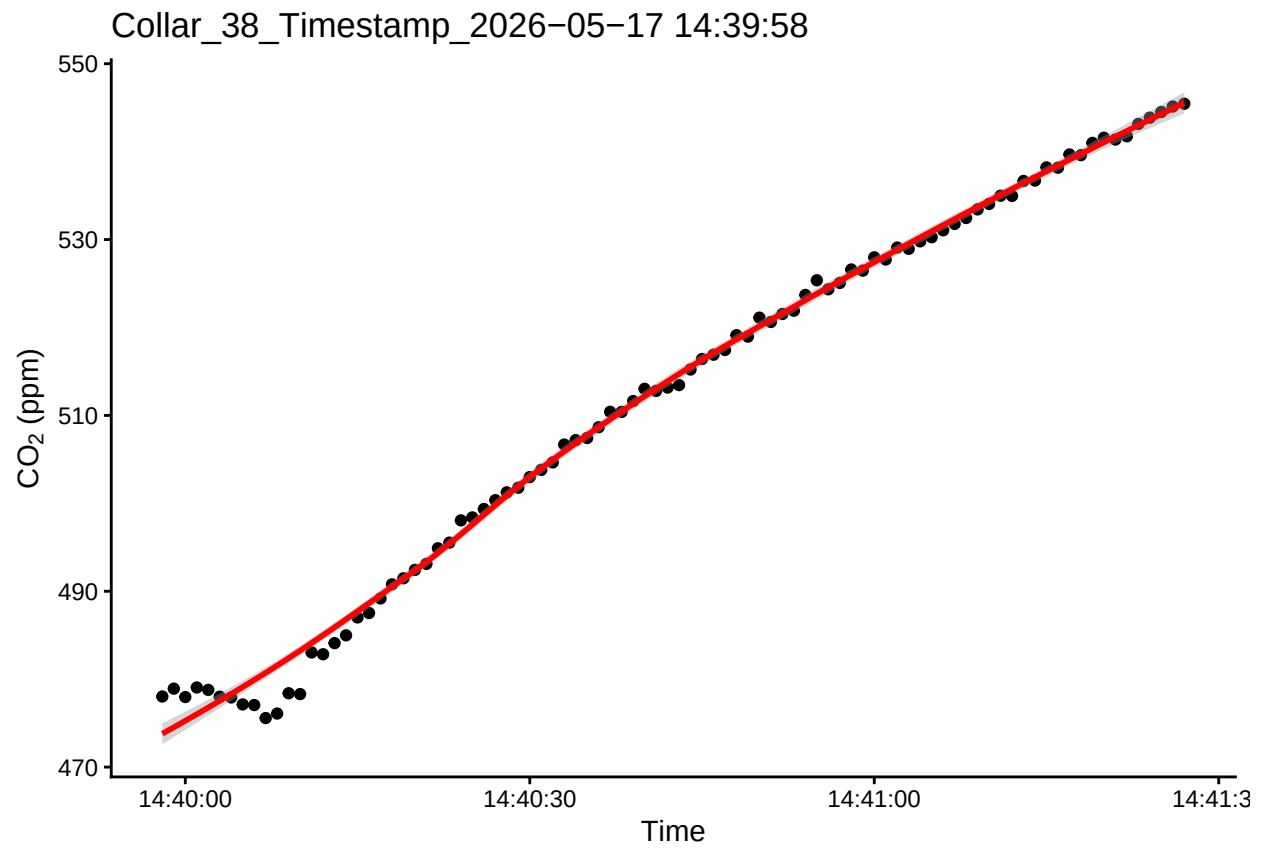
```
## 'geom_smooth()' using method = 'loess' and formula = 'y ~ x'
```



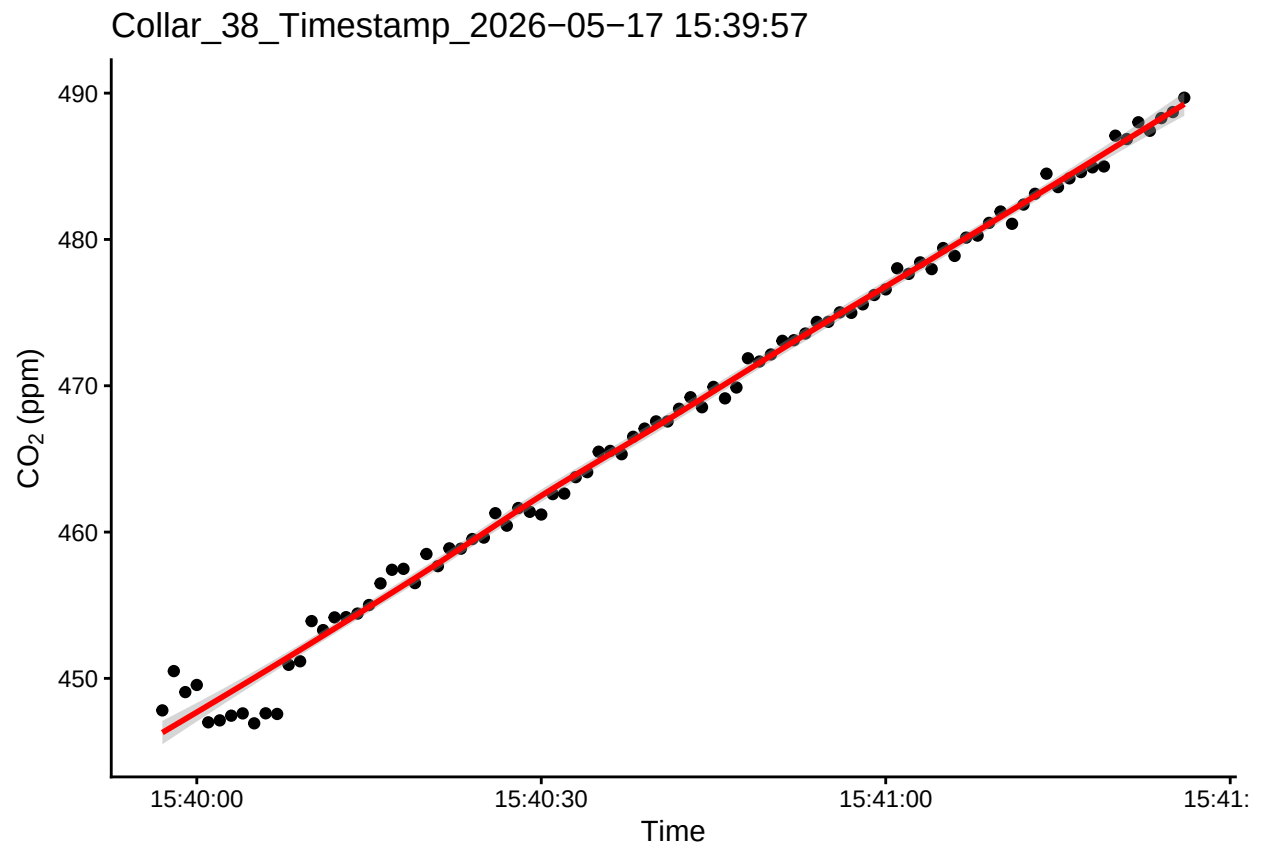
```
## 'geom_smooth()' using method = 'loess' and formula = 'y ~ x'
```



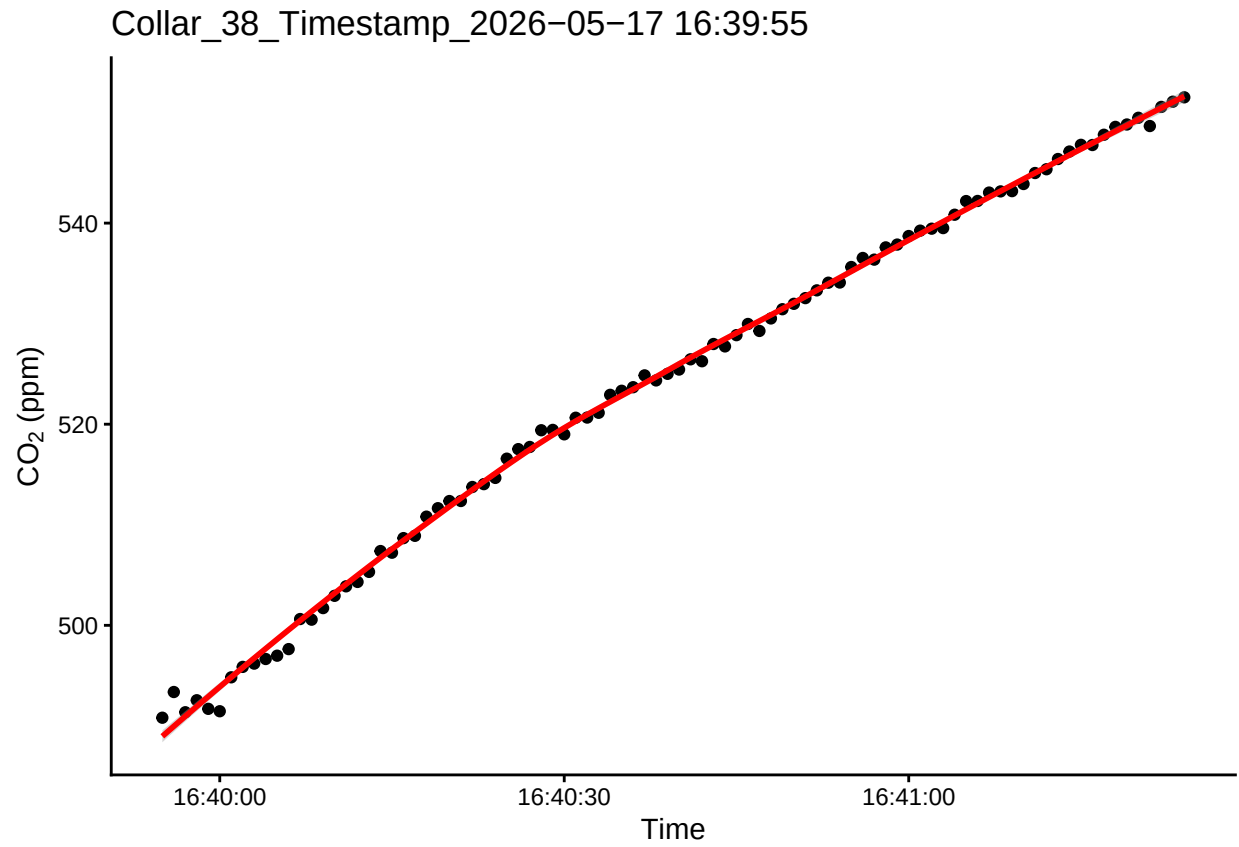
```
## 'geom_smooth()' using method = 'loess' and formula = 'y ~ x'
```



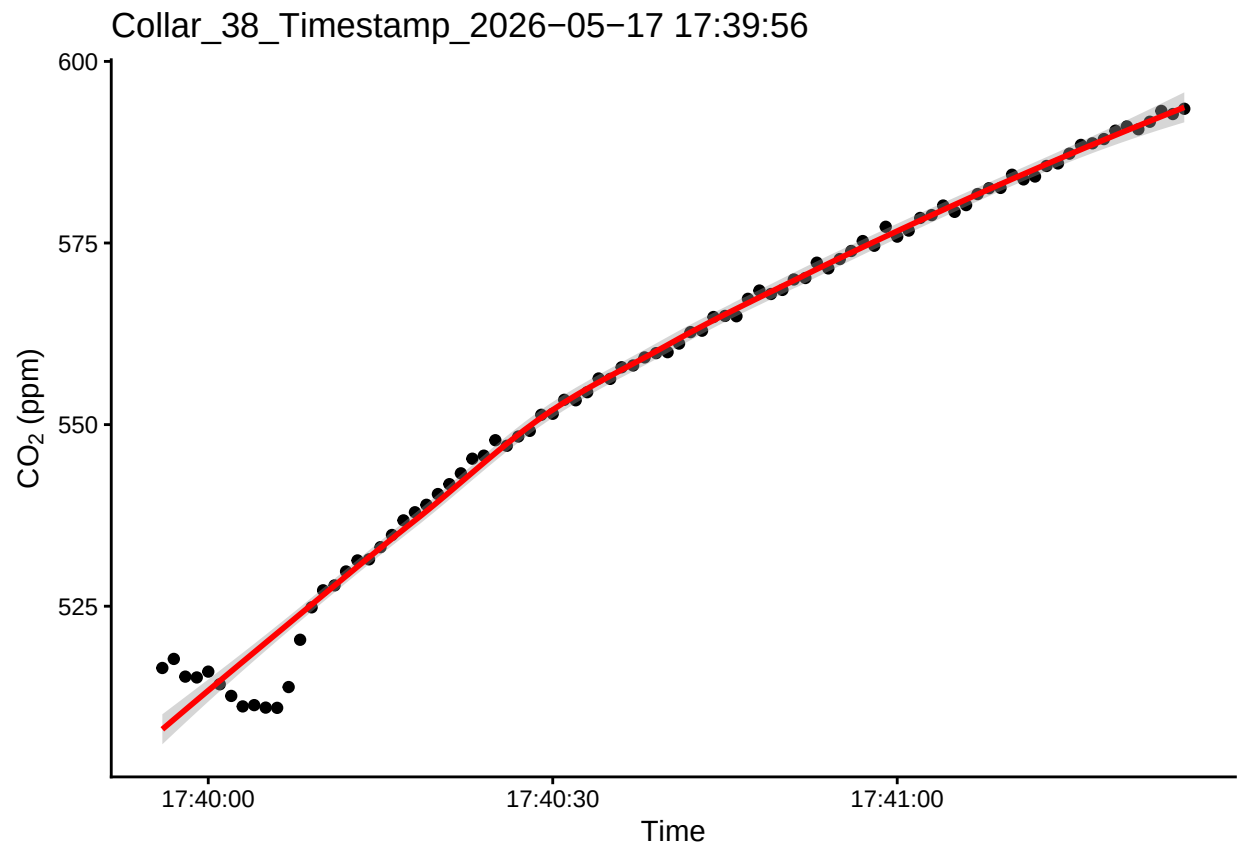
```
## 'geom_smooth()' using method = 'loess' and formula = 'y ~ x'
```



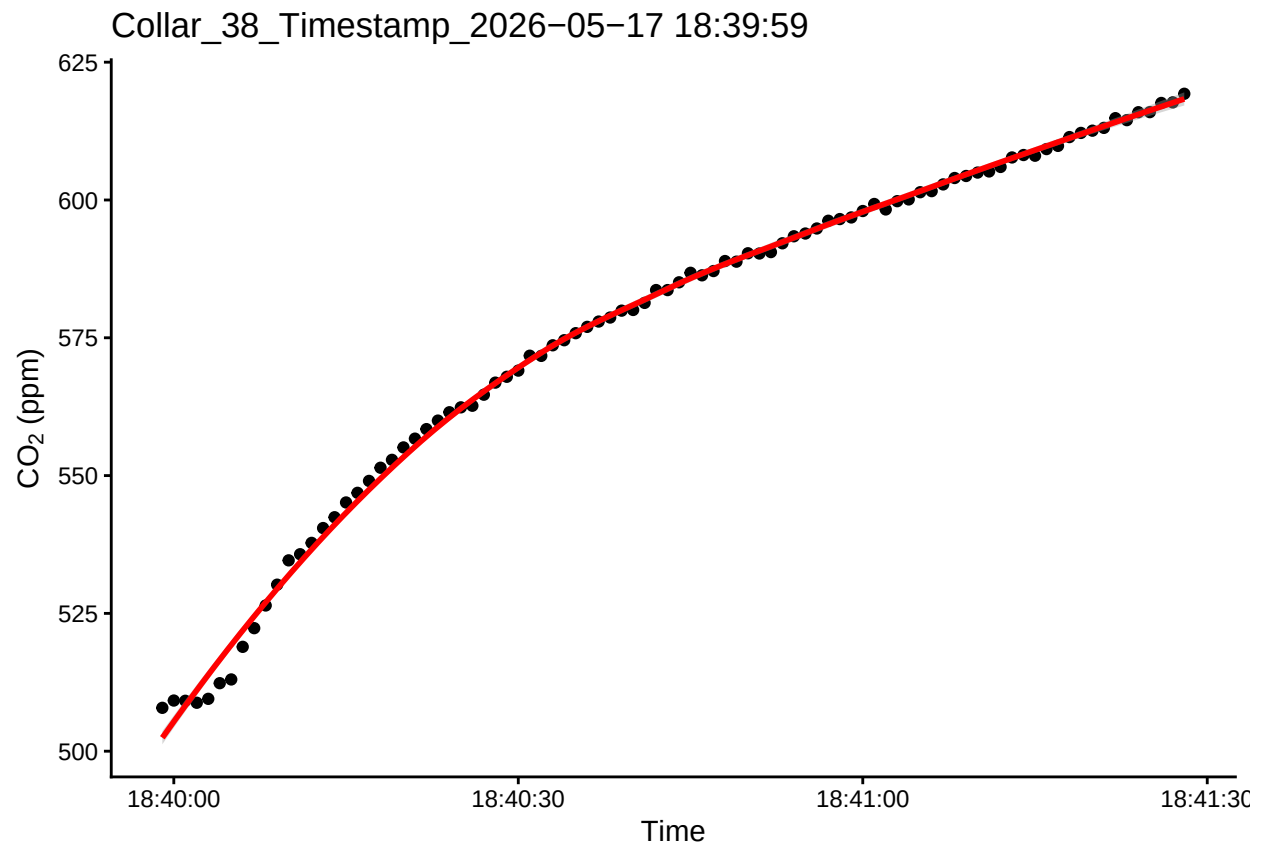
```
## 'geom_smooth()' using method = 'loess' and formula = 'y ~ x'
```



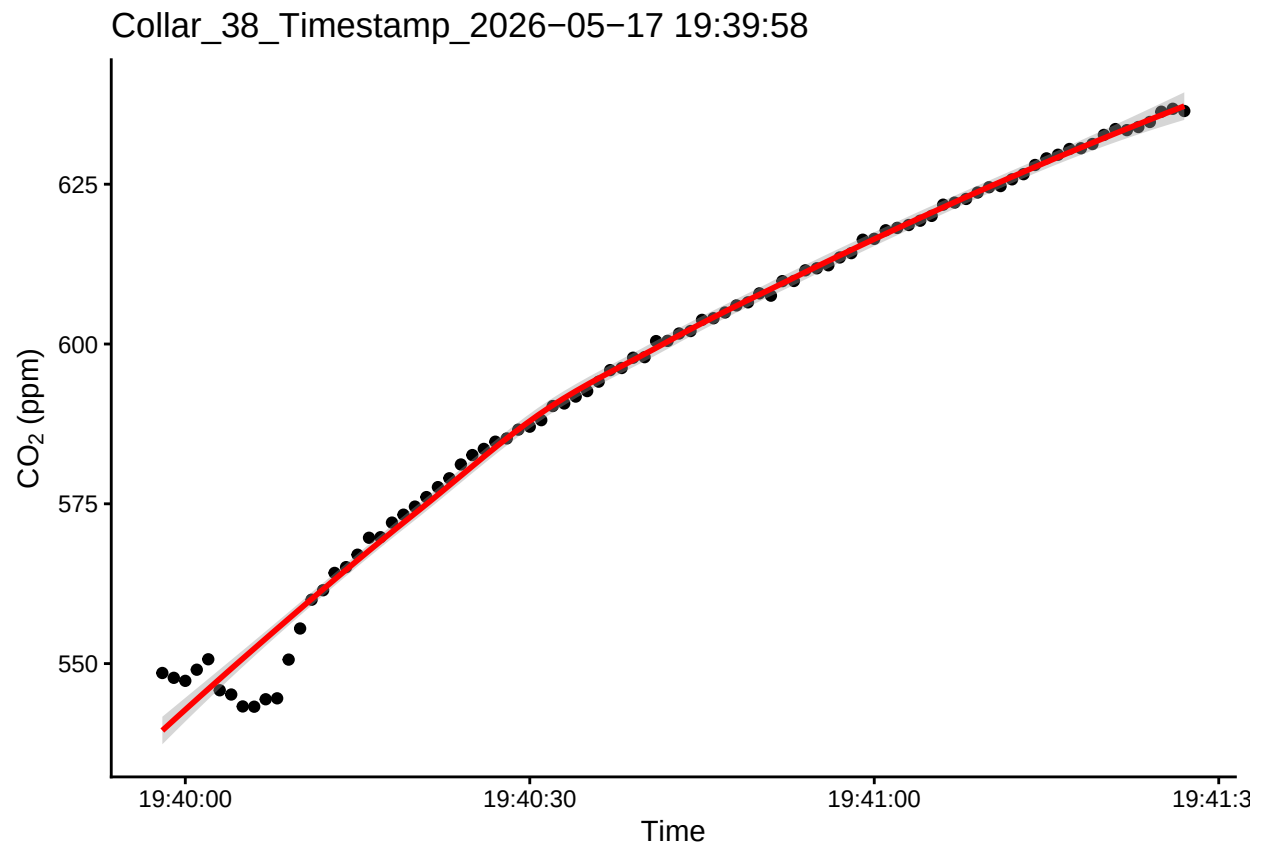
```
## 'geom_smooth()' using method = 'loess' and formula = 'y ~ x'
```



```
## 'geom_smooth()' using method = 'loess' and formula = 'y ~ x'
```

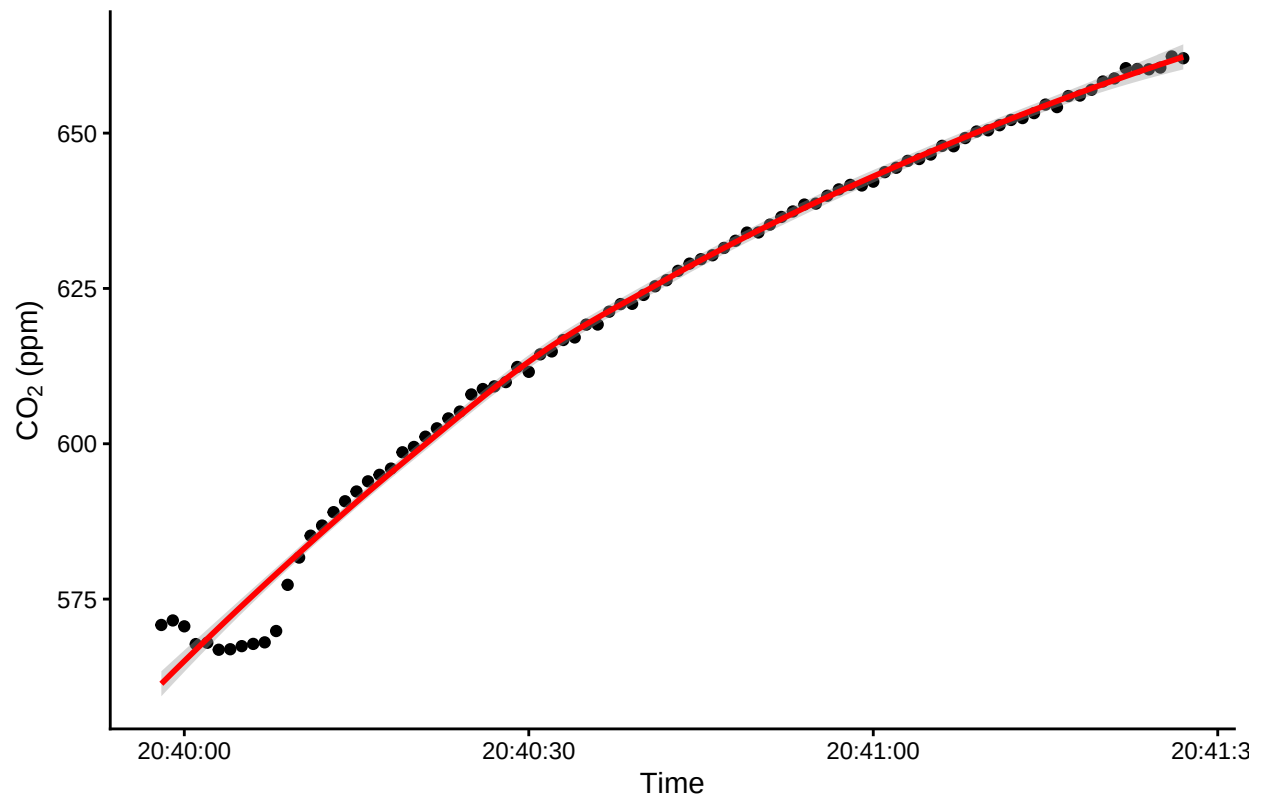


```
## 'geom_smooth()' using method = 'loess' and formula = 'y ~ x'
```

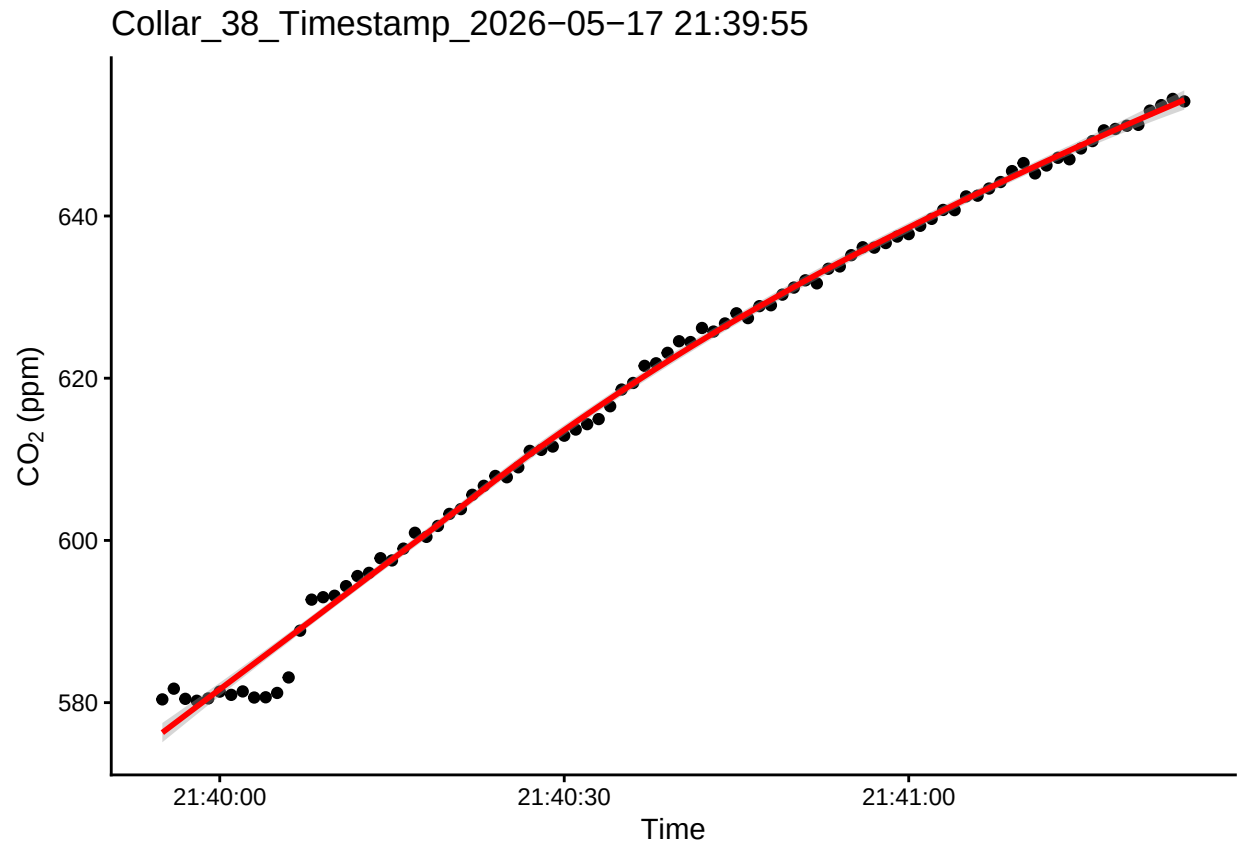



```
## 'geom_smooth()' using method = 'loess' and formula = 'y ~ x'
```

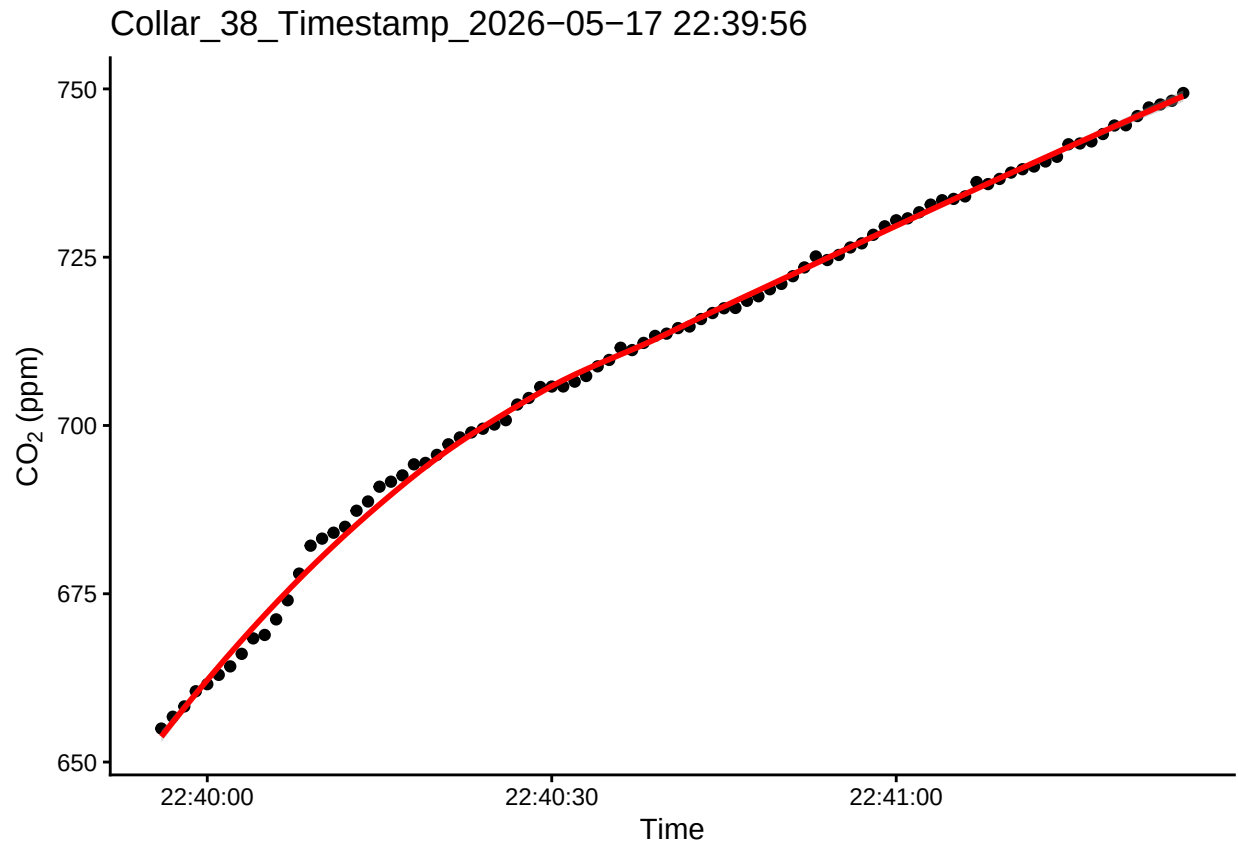
Collar_38_Timestamp_2026-05-17 20:39:58



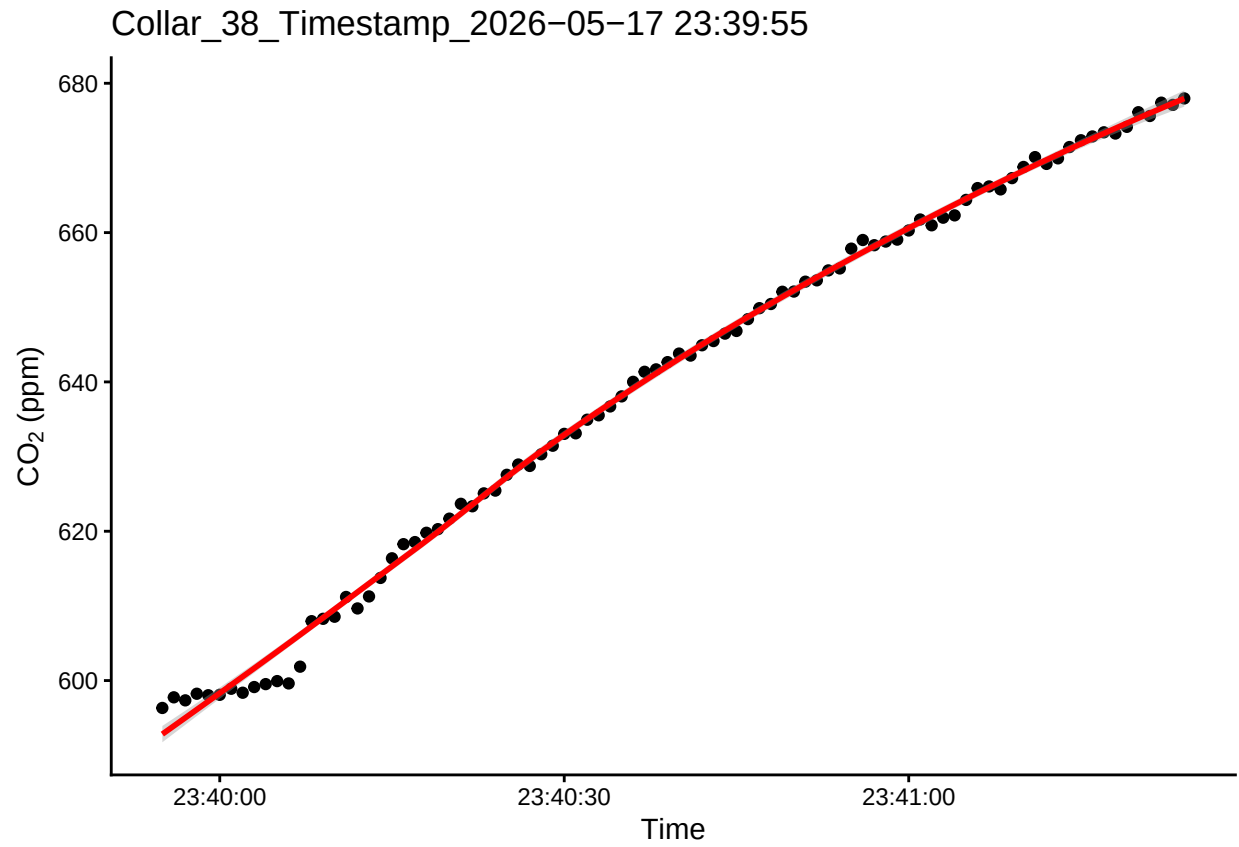
```
## 'geom_smooth()' using method = 'loess' and formula = 'y ~ x'
```



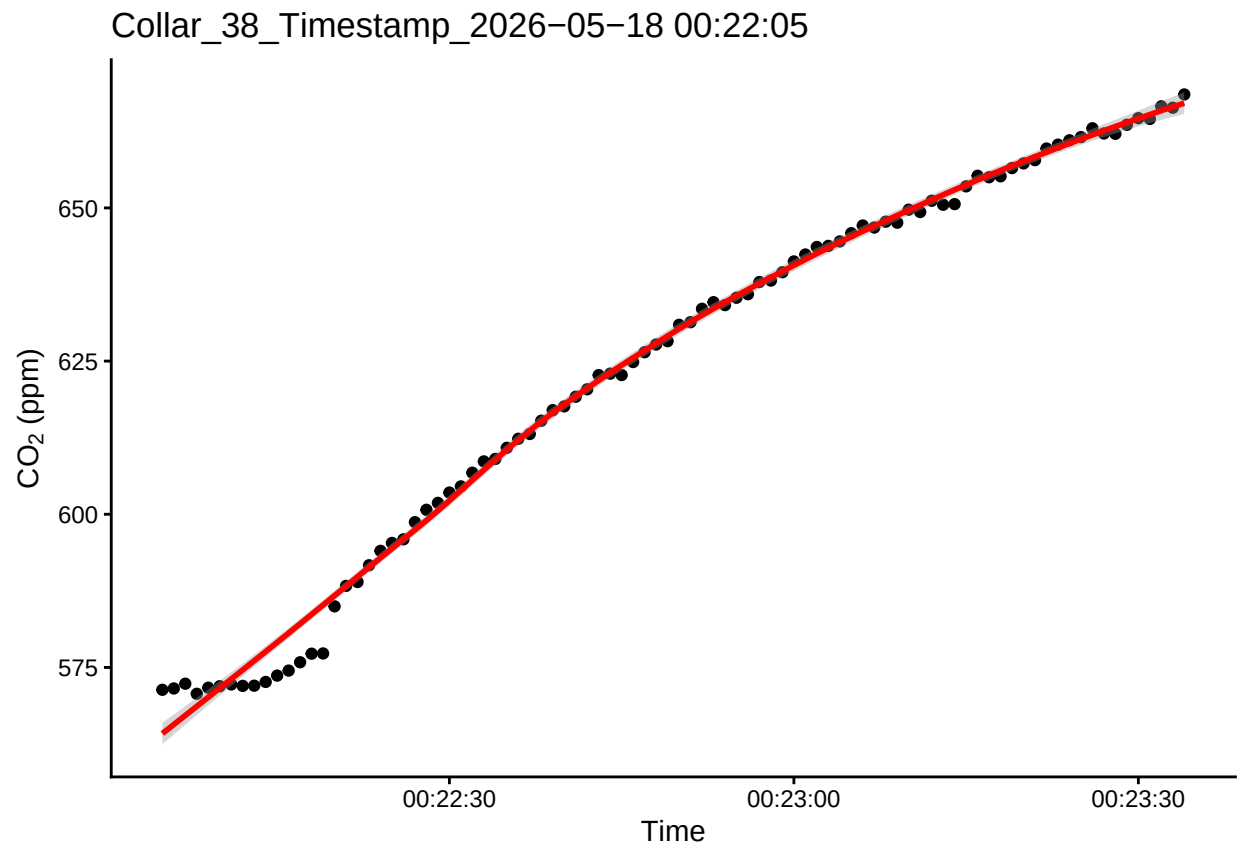
```
## 'geom_smooth()' using method = 'loess' and formula = 'y ~ x'
```



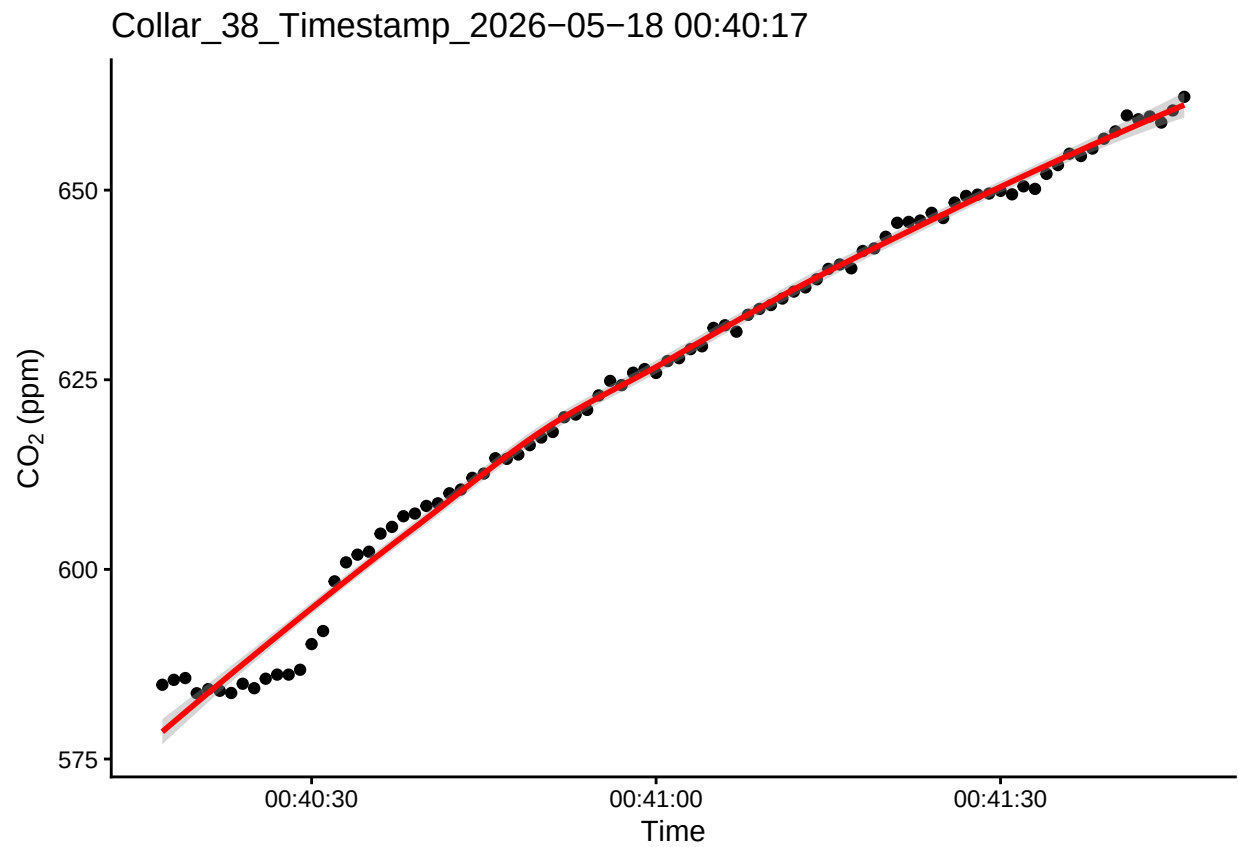
```
## 'geom_smooth()' using method = 'loess' and formula = 'y ~ x'
```



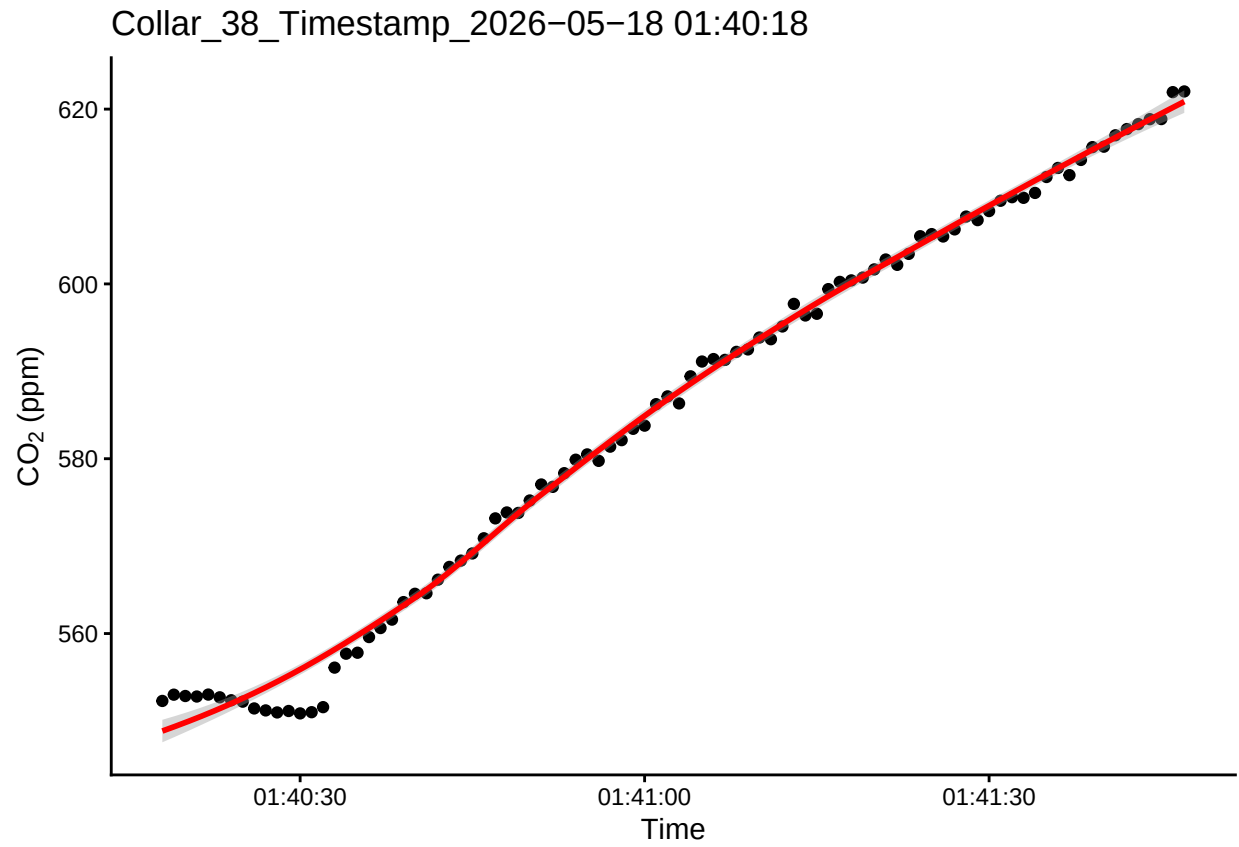
```
## 'geom_smooth()' using method = 'loess' and formula = 'y ~ x'
```



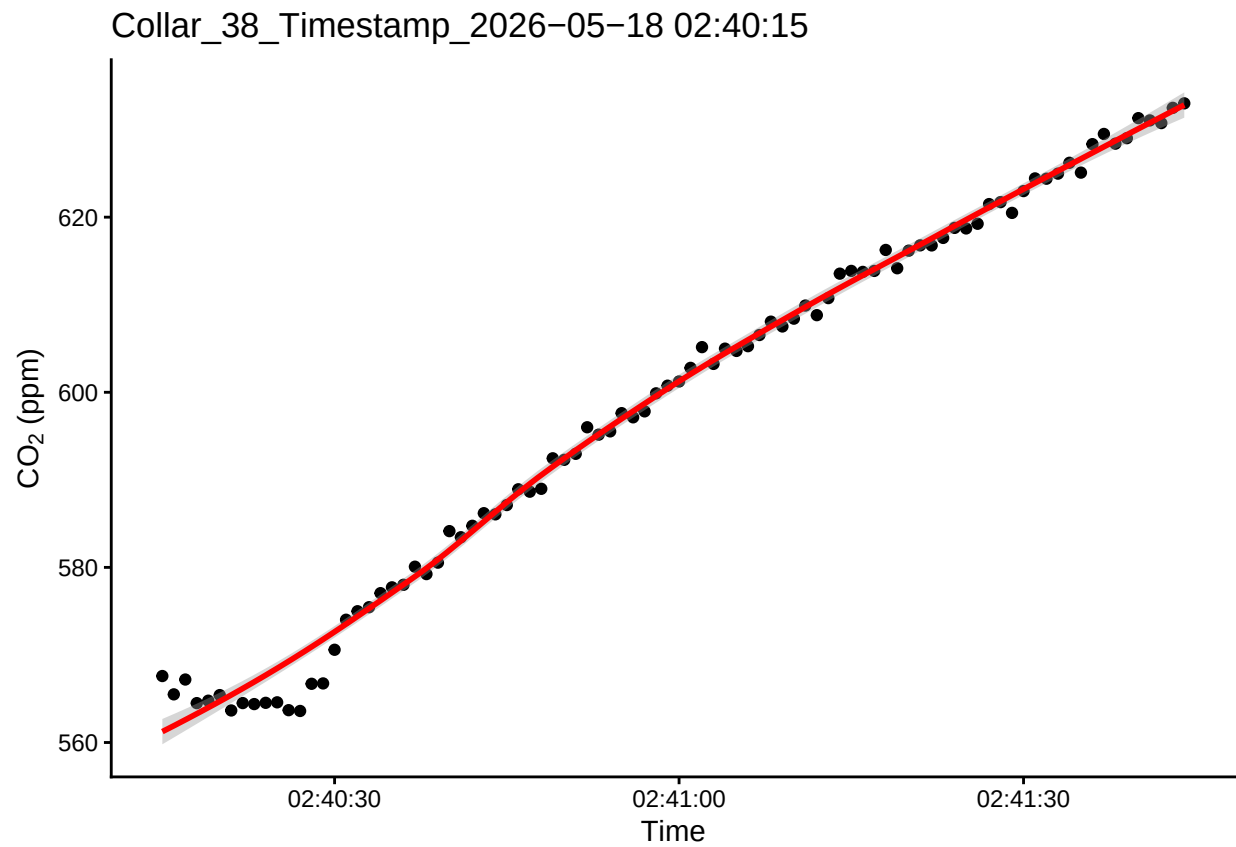
```
## 'geom_smooth()' using method = 'loess' and formula = 'y ~ x'
```



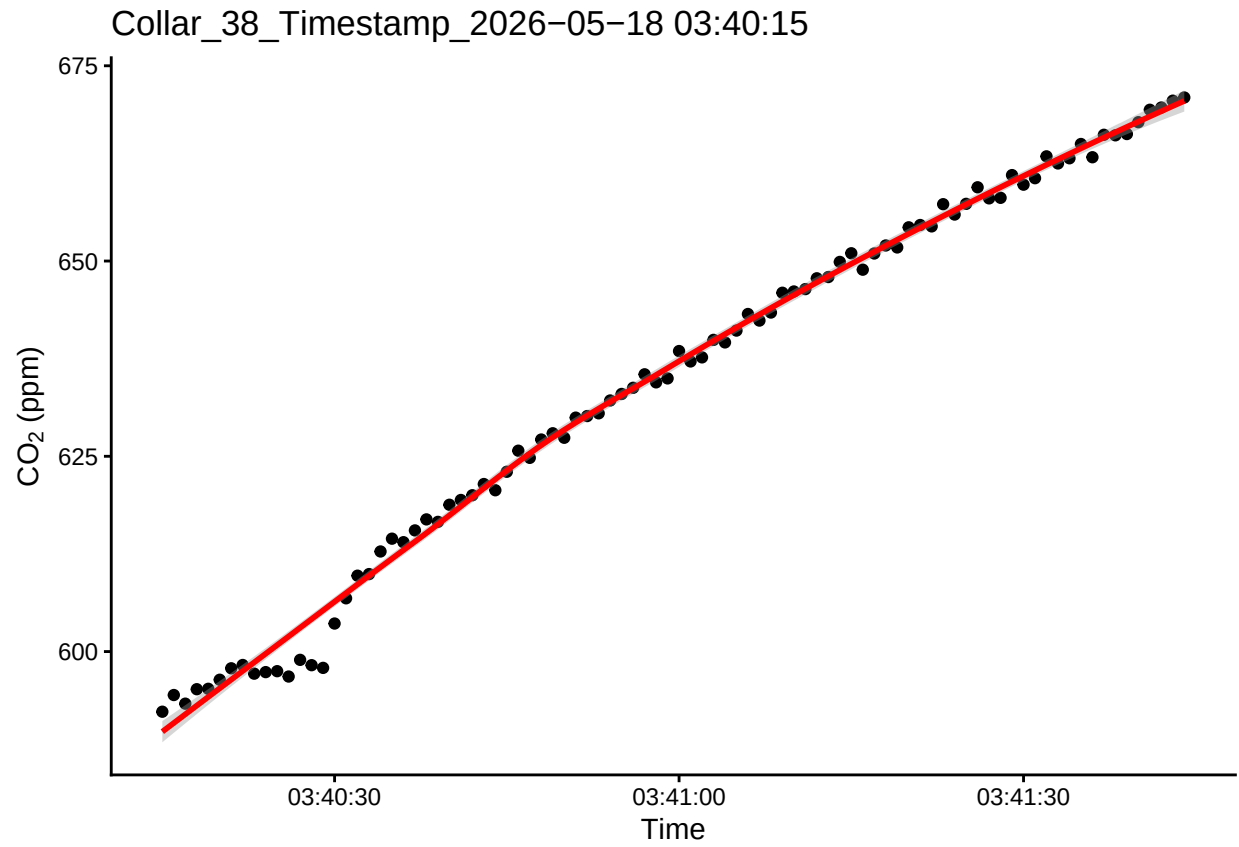
```
## 'geom_smooth()' using method = 'loess' and formula = 'y ~ x'
```



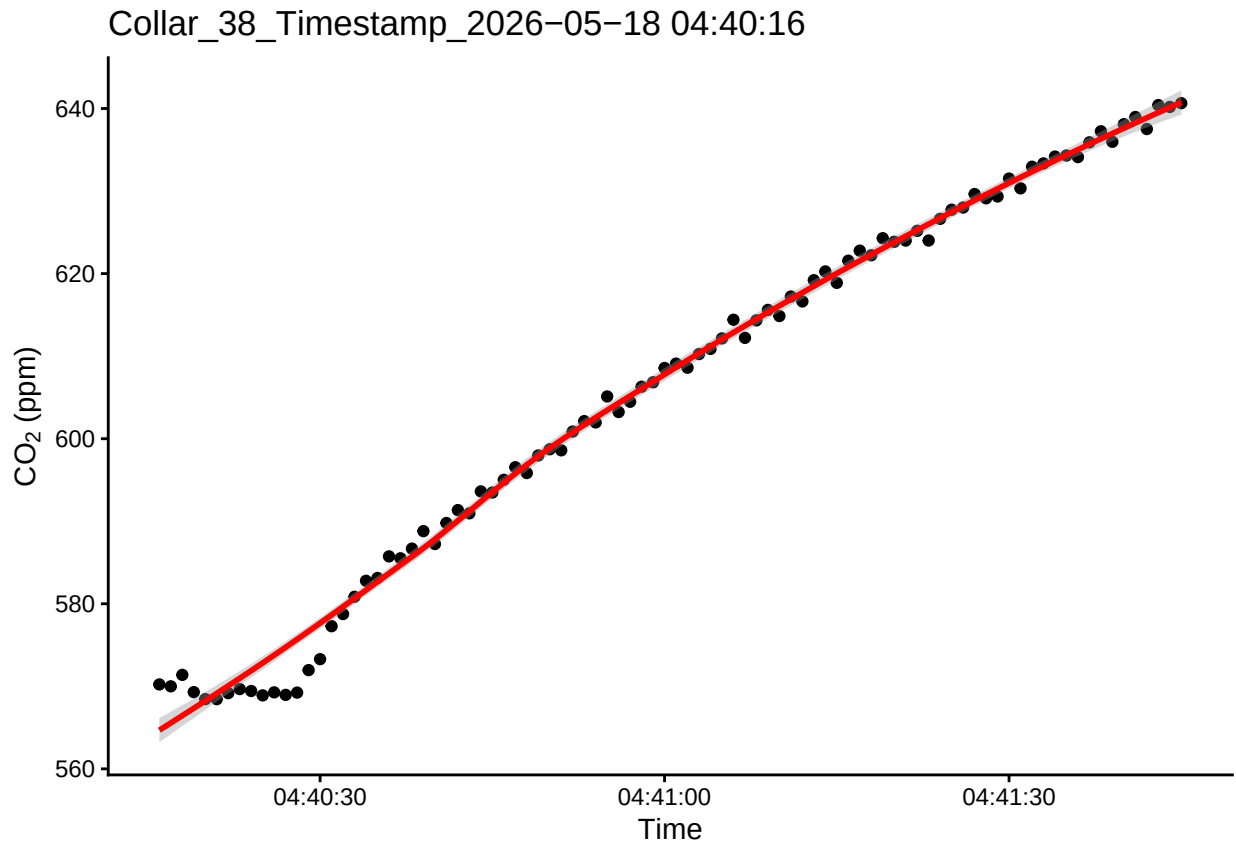
```
## 'geom_smooth()' using method = 'loess' and formula = 'y ~ x'
```

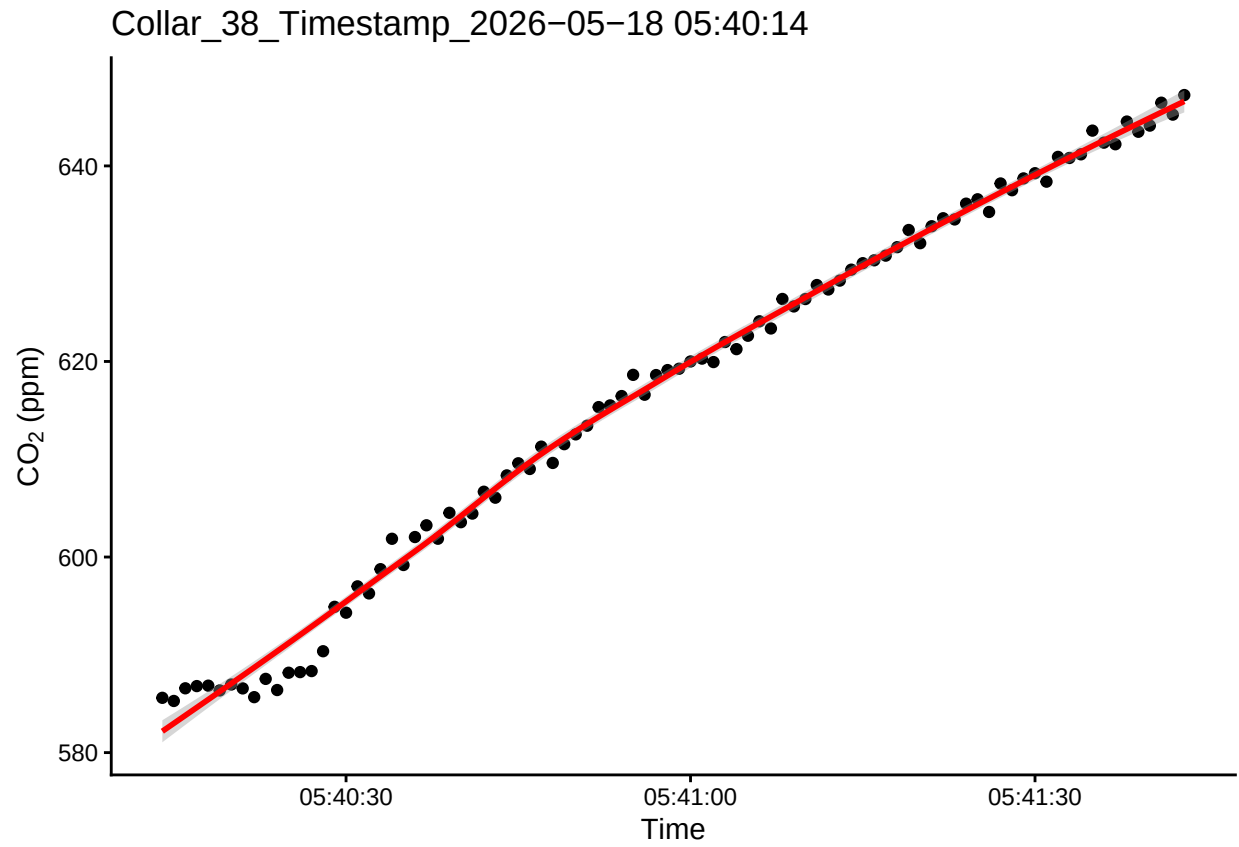
```
## 'geom_smooth()' using method = 'loess' and formula = 'y ~ x'
```



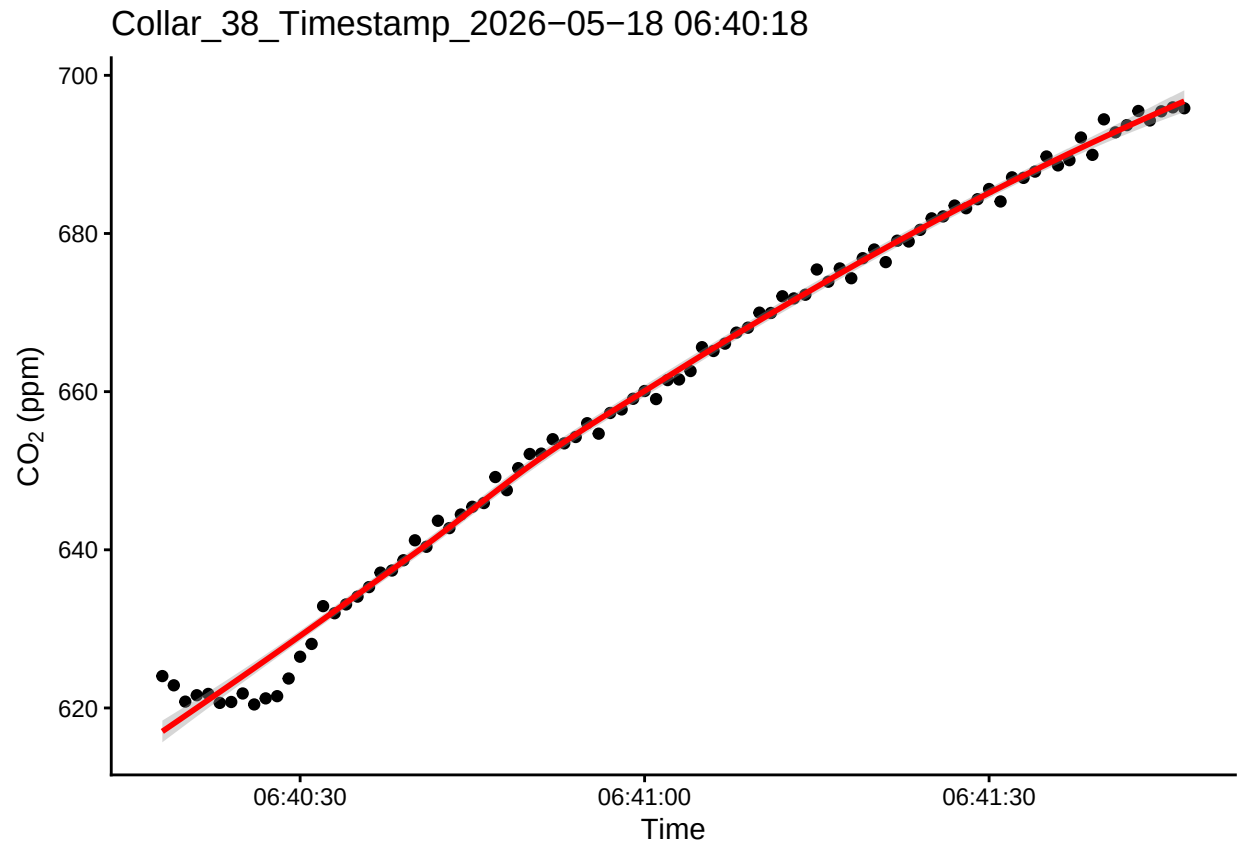
```
## 'geom_smooth()' using method = 'loess' and formula = 'y ~ x'
```



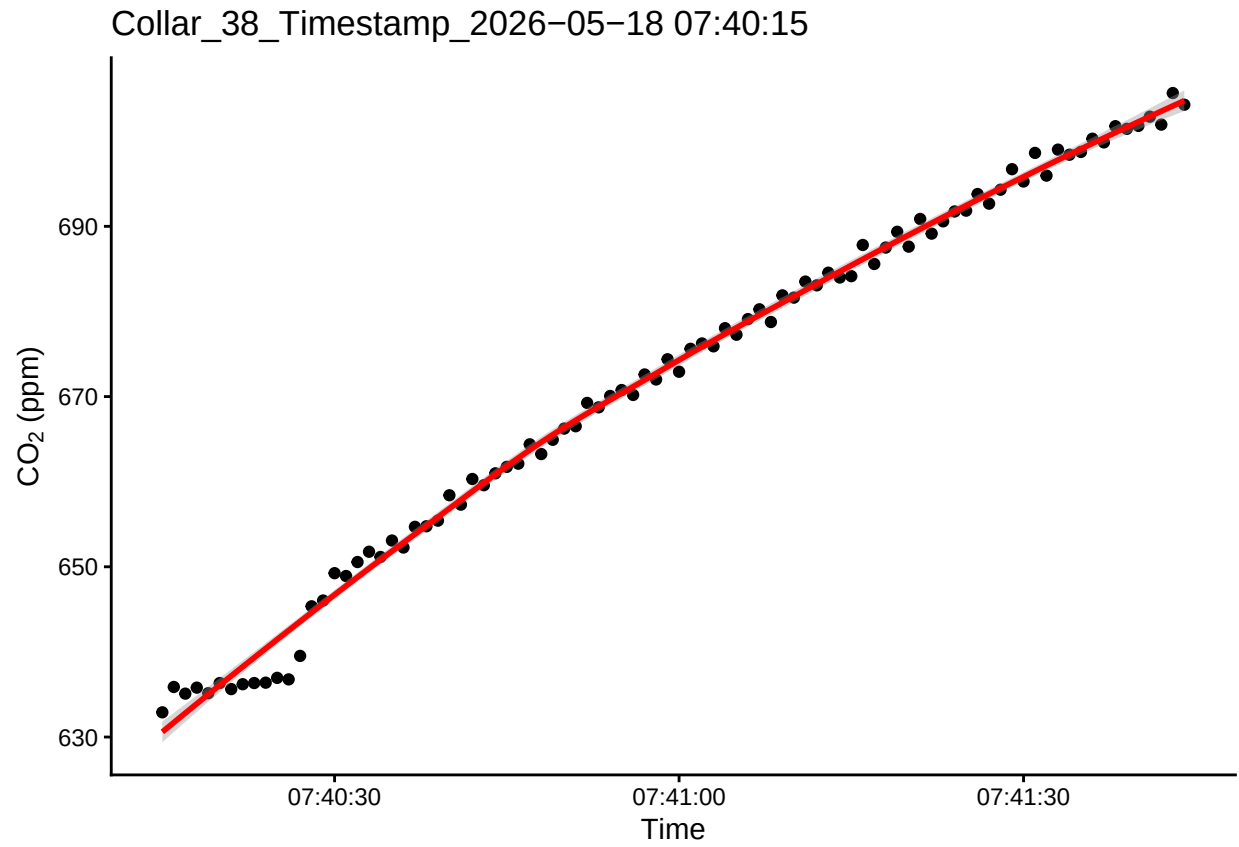
```
## 'geom_smooth()' using method = 'loess' and formula = 'y ~ x'
```



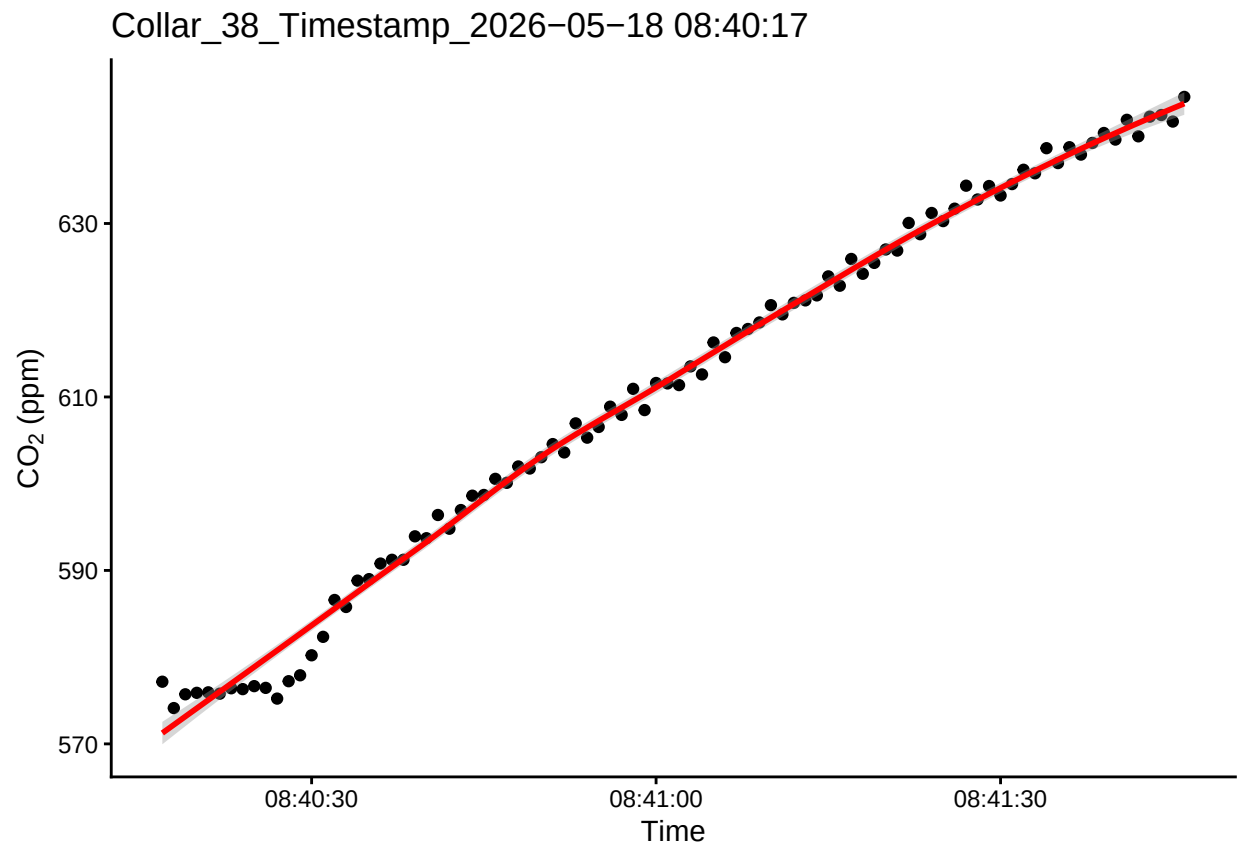
```
## 'geom_smooth()' using method = 'loess' and formula = 'y ~ x'
```



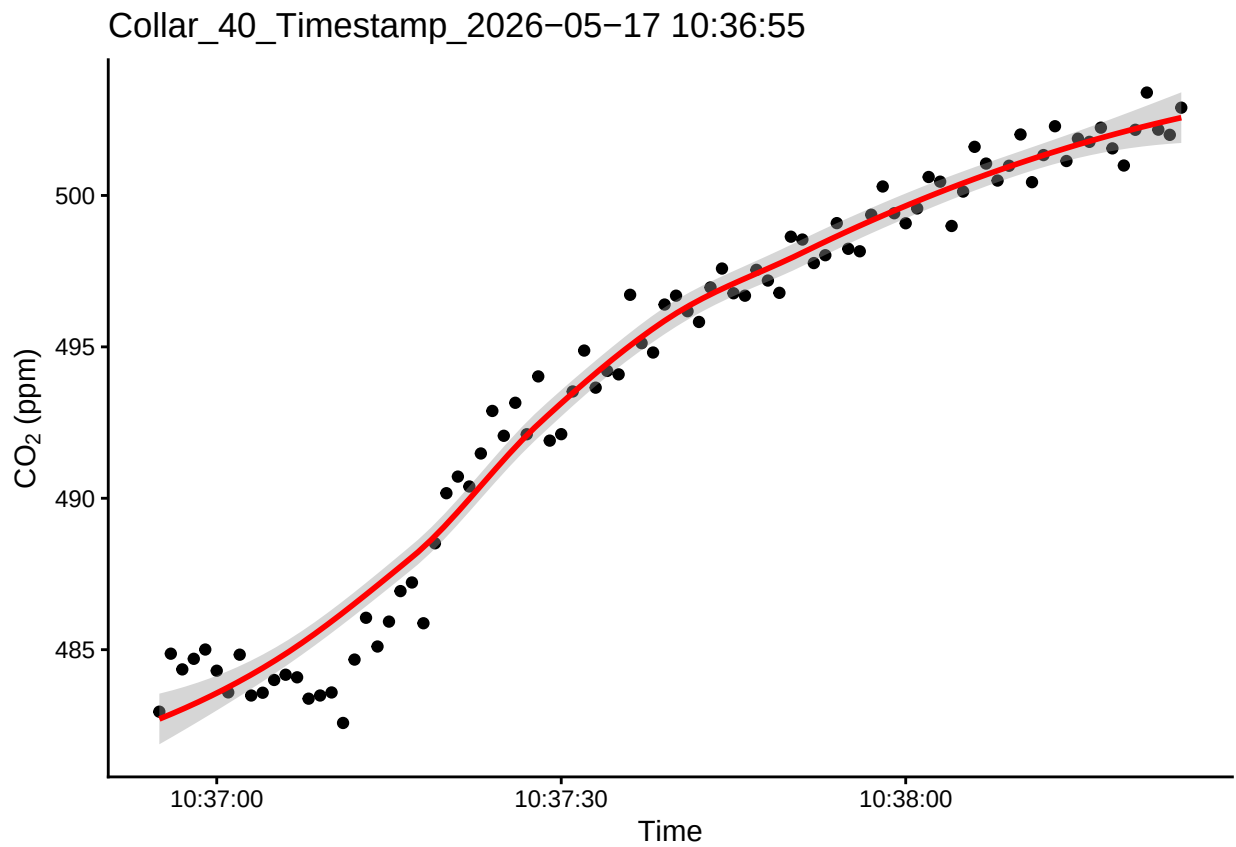
```
## 'geom_smooth()' using method = 'loess' and formula = 'y ~ x'
```



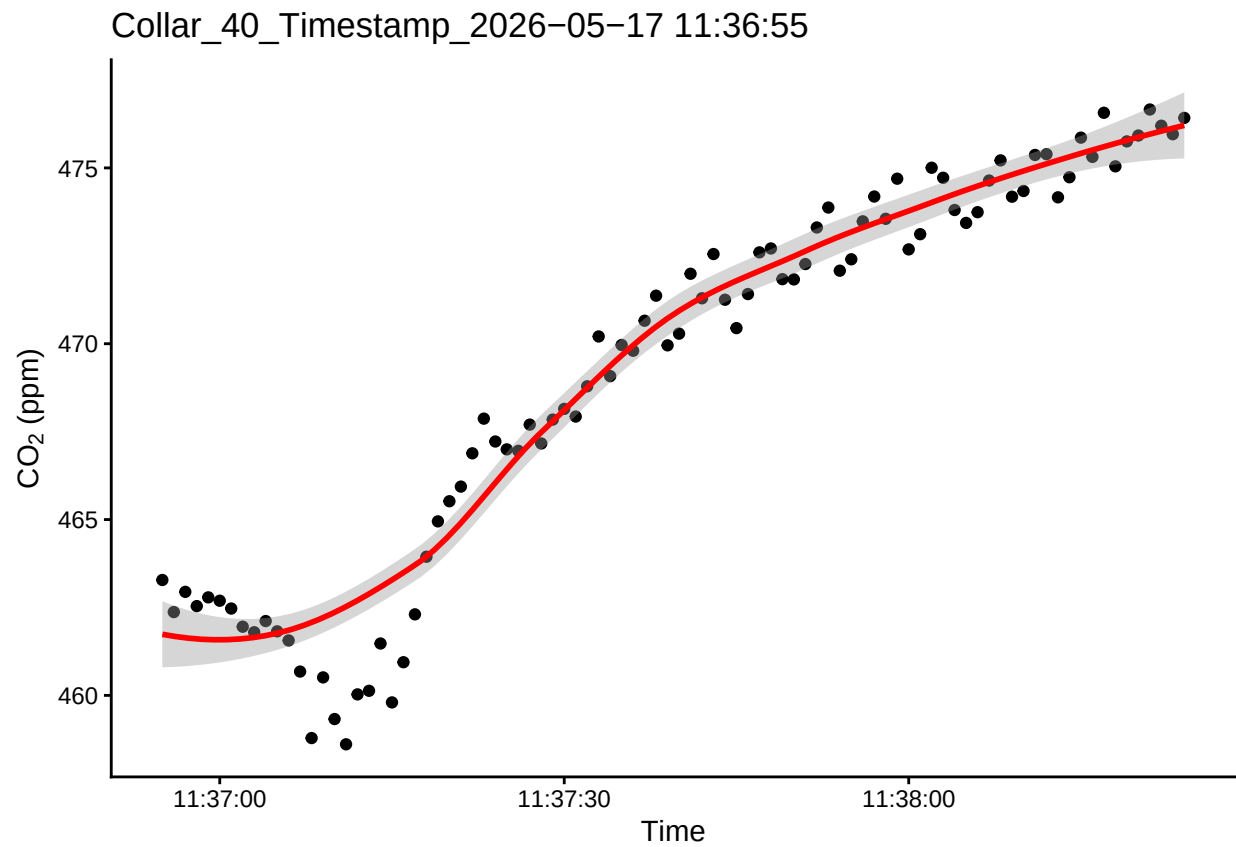
```
## 'geom_smooth()' using method = 'loess' and formula = 'y ~ x'
```



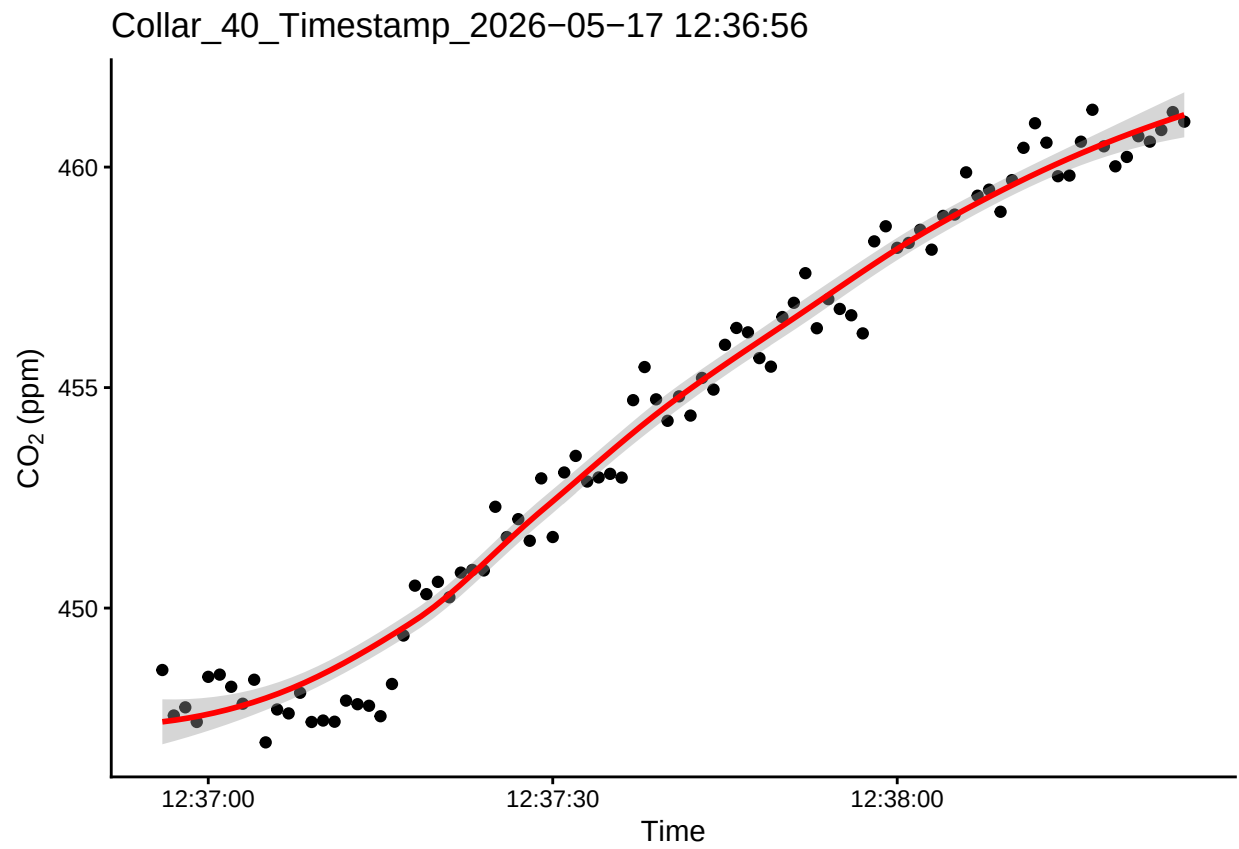
```
## 'geom_smooth()' using method = 'loess' and formula = 'y ~ x'
```



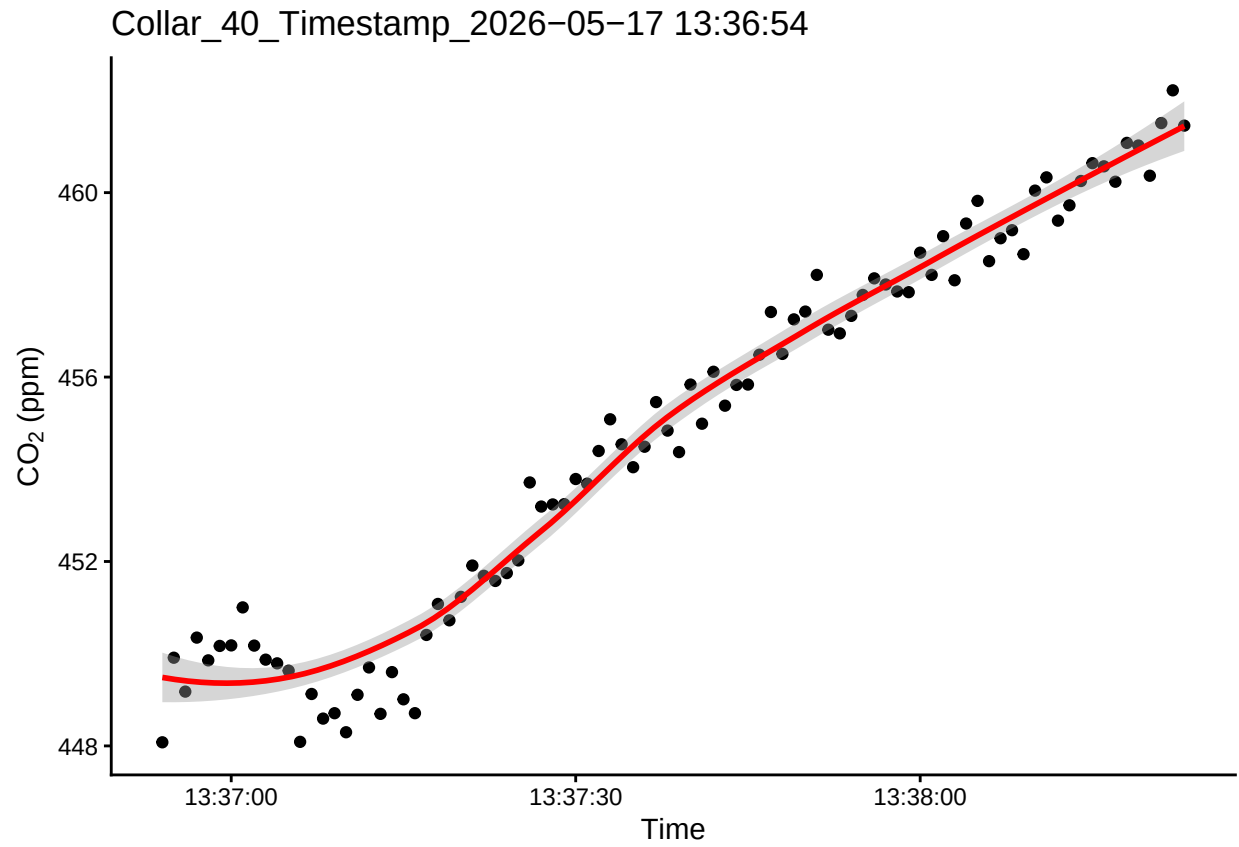
```
## 'geom_smooth()' using method = 'loess' and formula = 'y ~ x'
```

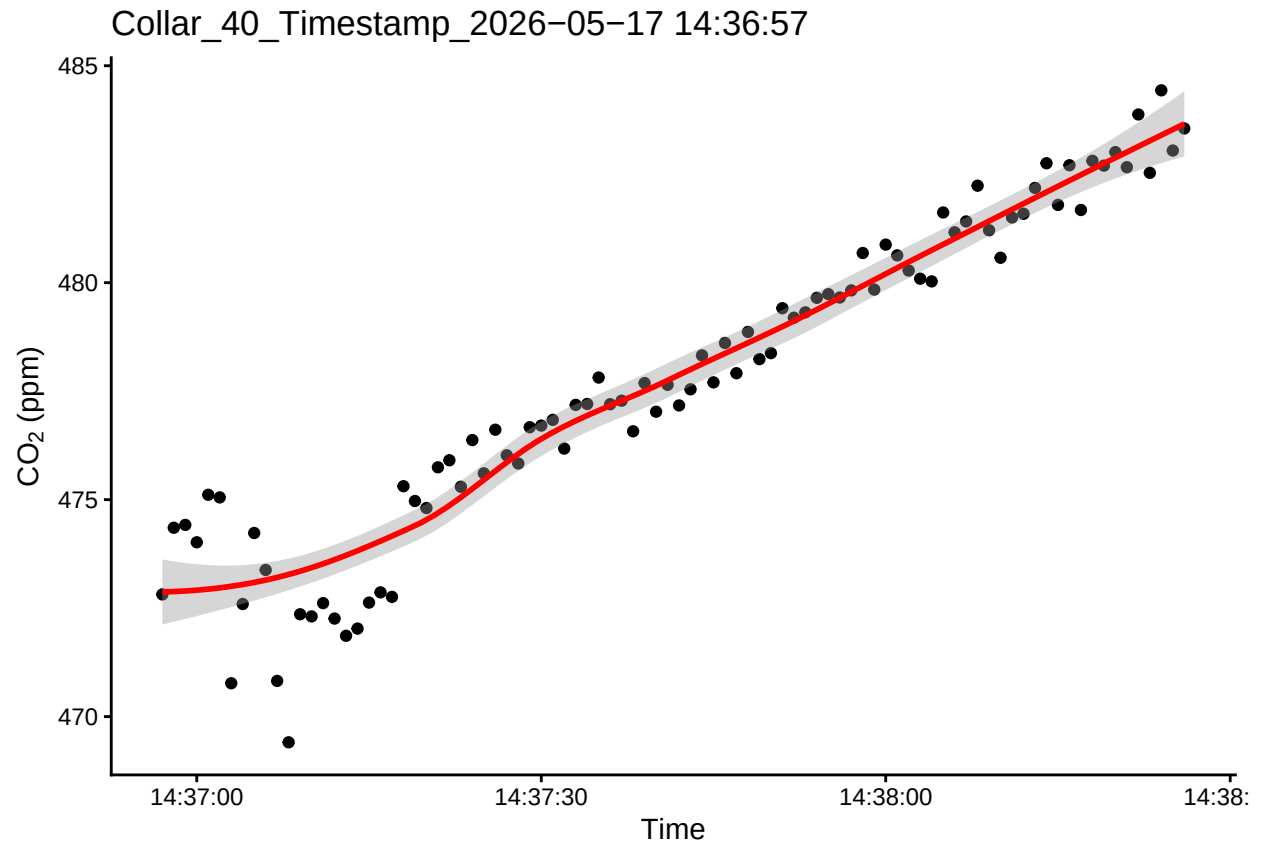
```
## 'geom_smooth()' using method = 'loess' and formula = 'y ~ x'
```



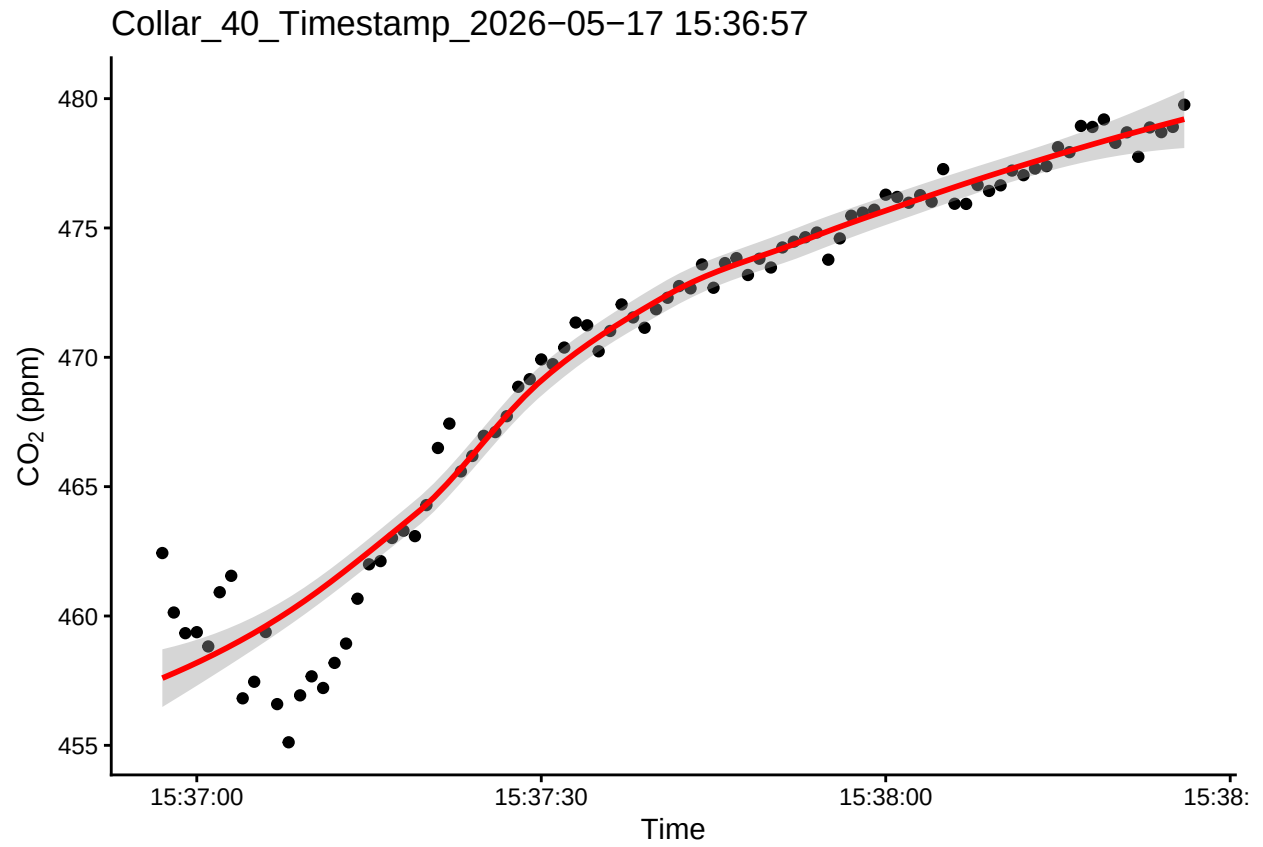
```
## 'geom_smooth()' using method = 'loess' and formula = 'y ~ x'
```



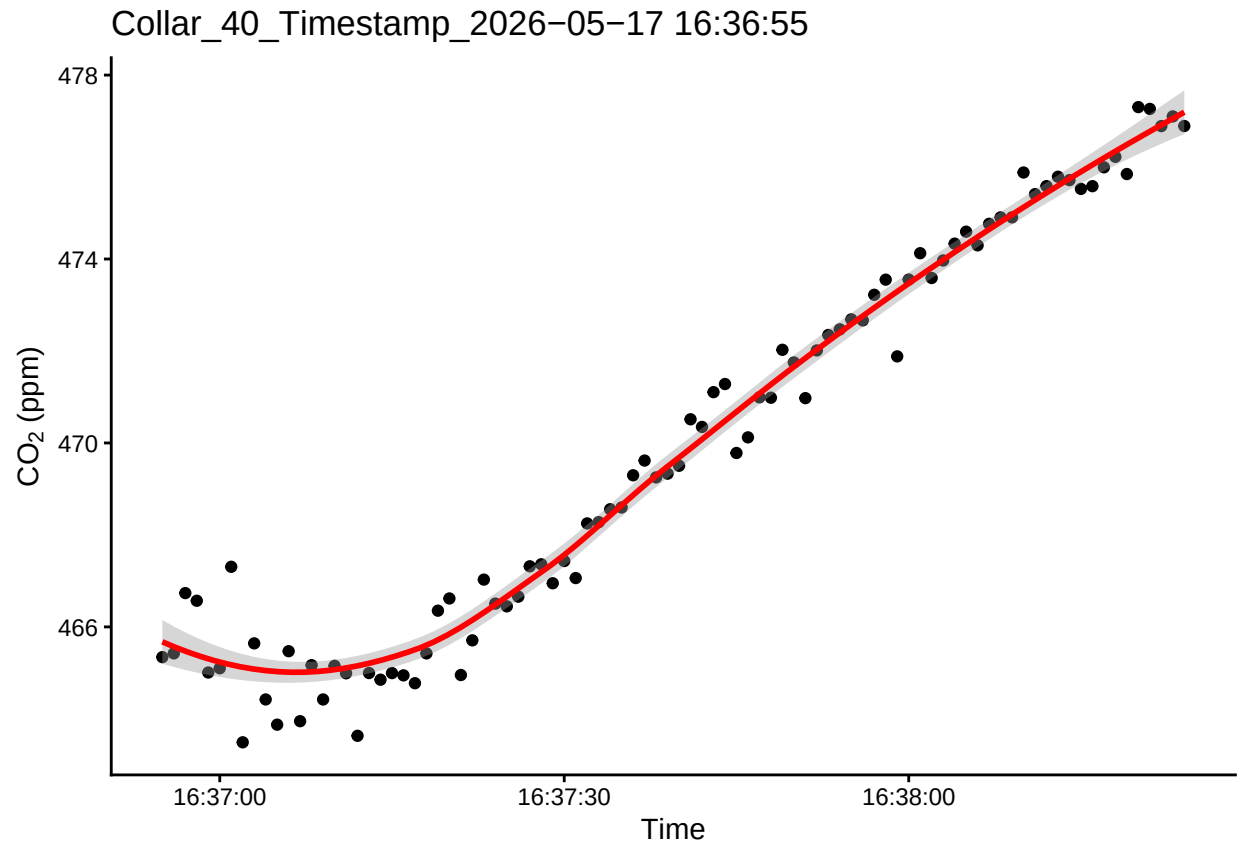
```
## 'geom_smooth()' using method = 'loess' and formula = 'y ~ x'
```



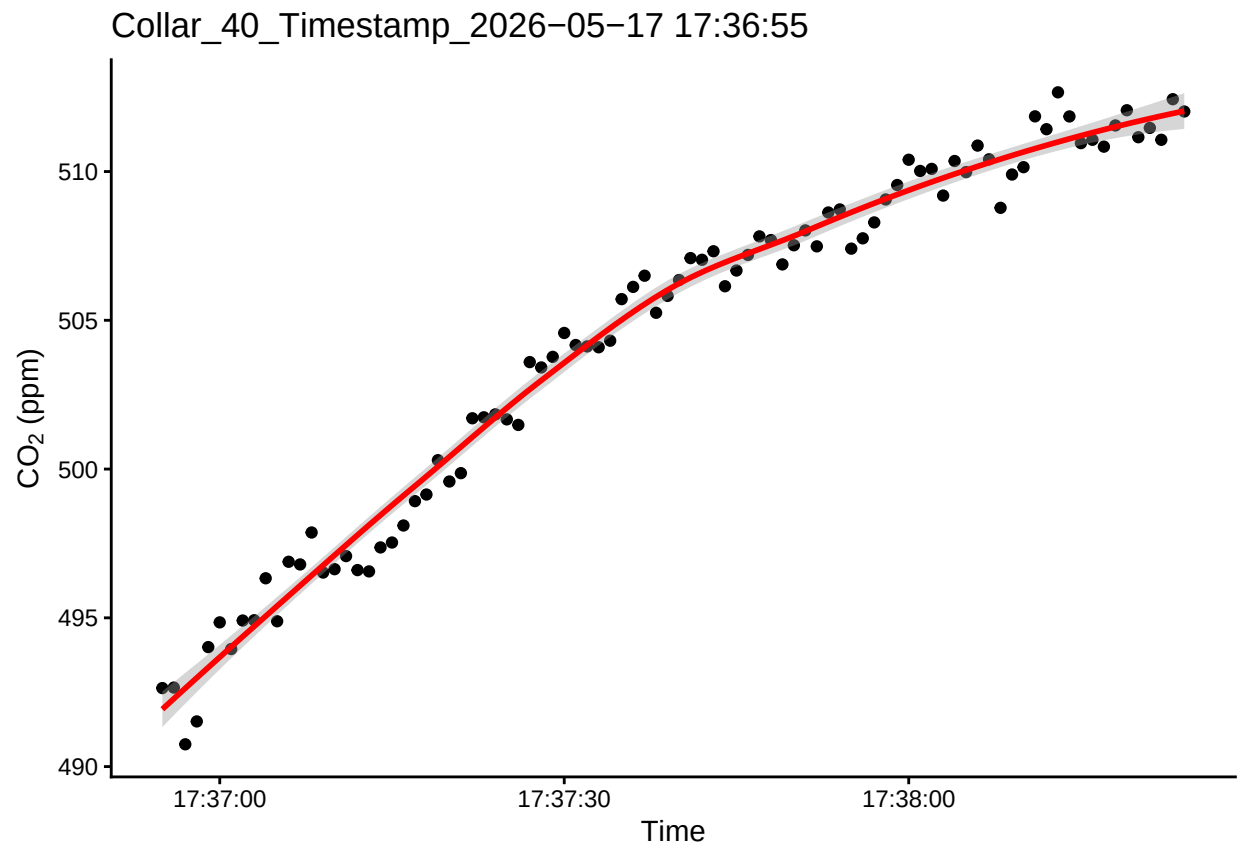
```
## 'geom_smooth()' using method = 'loess' and formula = 'y ~ x'
```



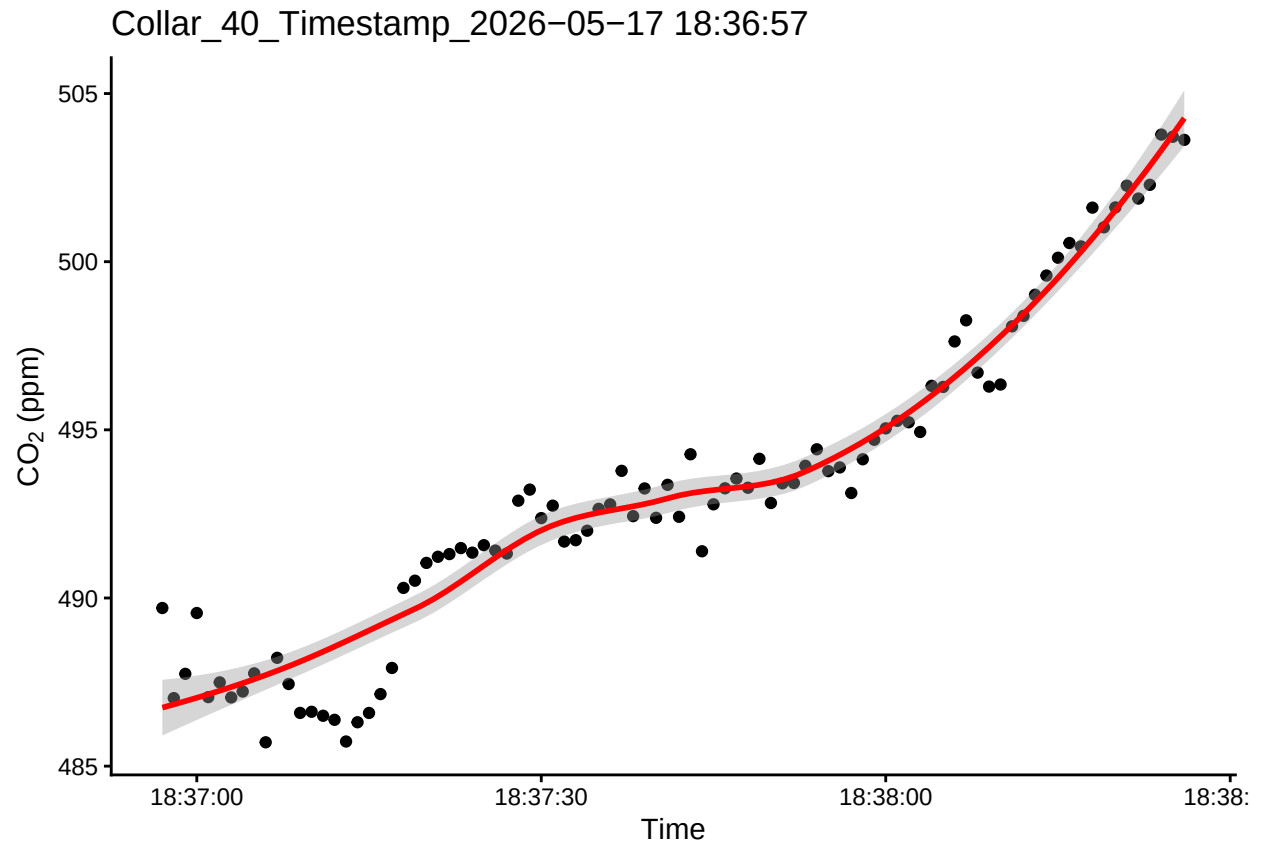
```
## 'geom_smooth()' using method = 'loess' and formula = 'y ~ x'
```



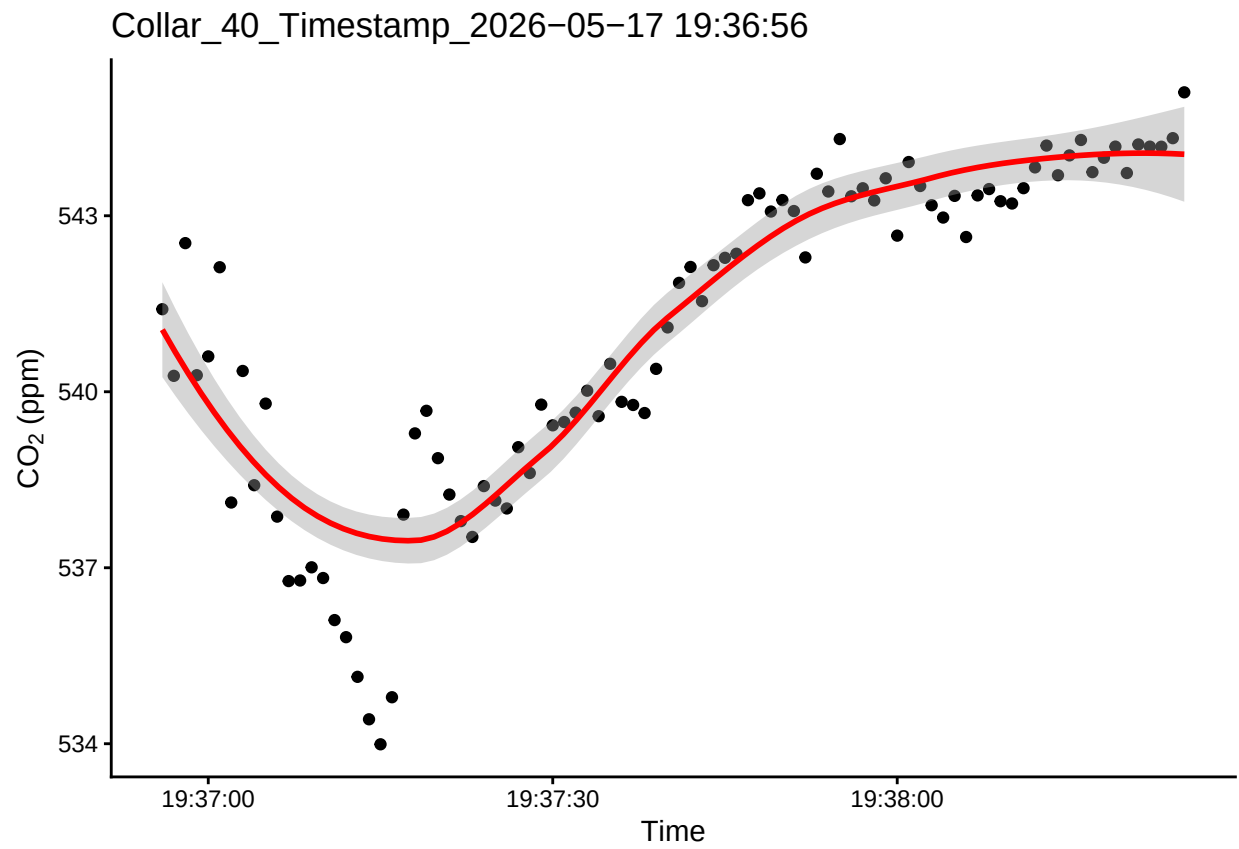
```
## 'geom_smooth()' using method = 'loess' and formula = 'y ~ x'
```



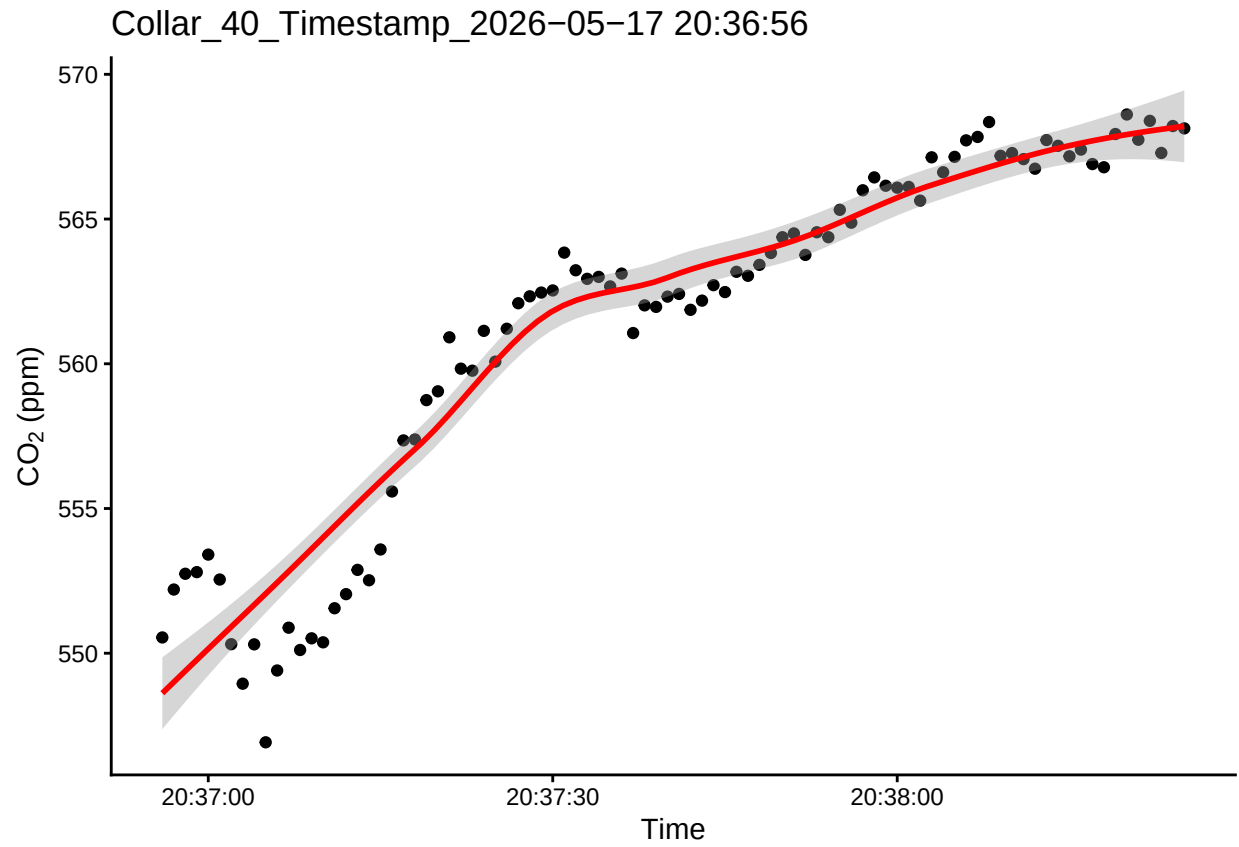
```
## 'geom_smooth()' using method = 'loess' and formula = 'y ~ x'
```



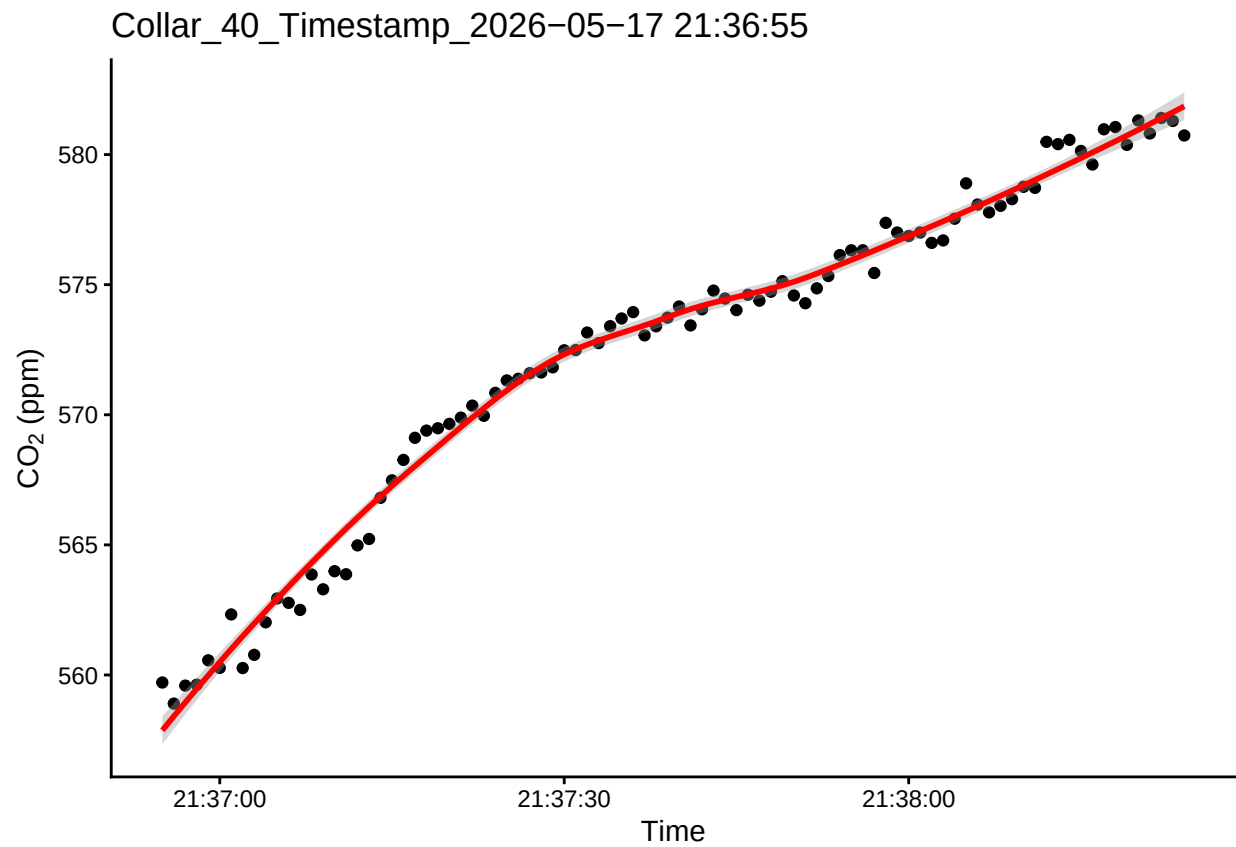
```
## 'geom_smooth()' using method = 'loess' and formula = 'y ~ x'
```

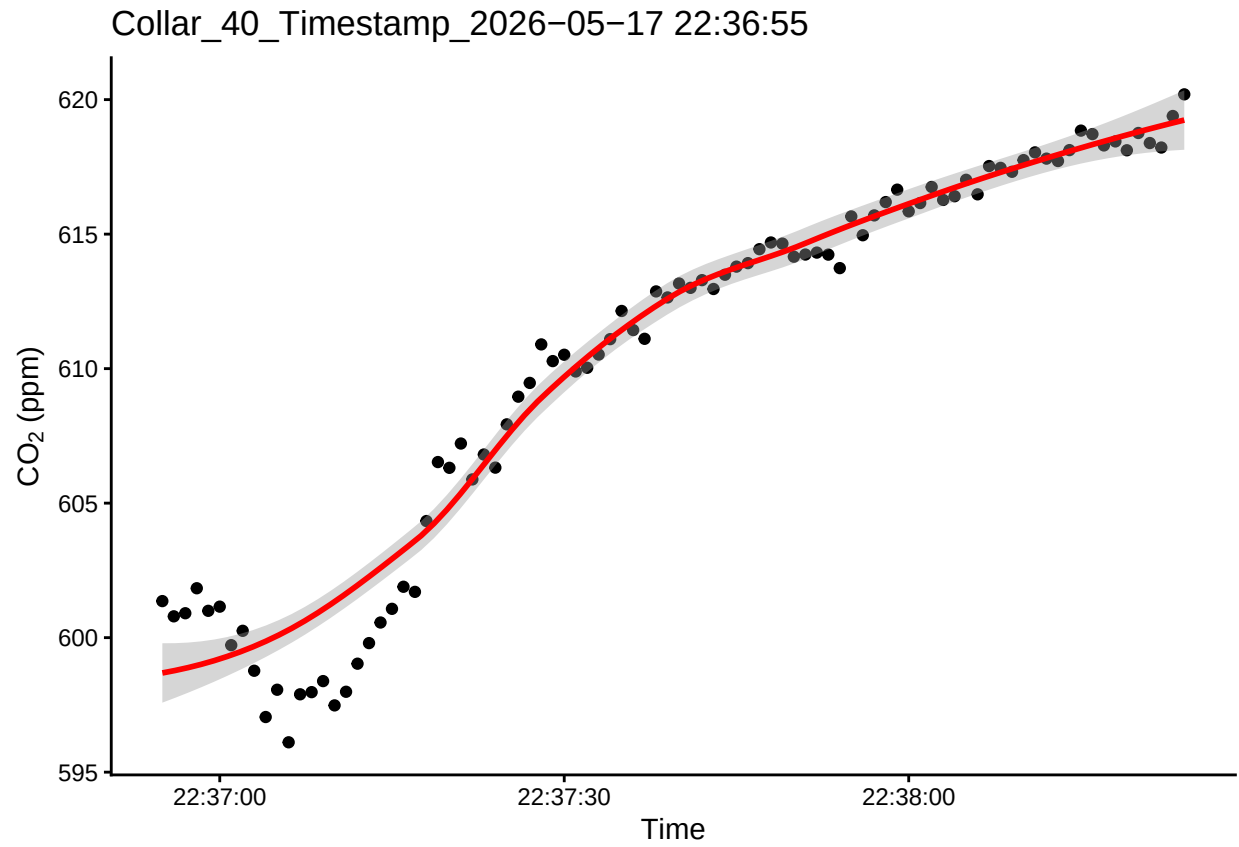
```
## 'geom_smooth()' using method = 'loess' and formula = 'y ~ x'
```



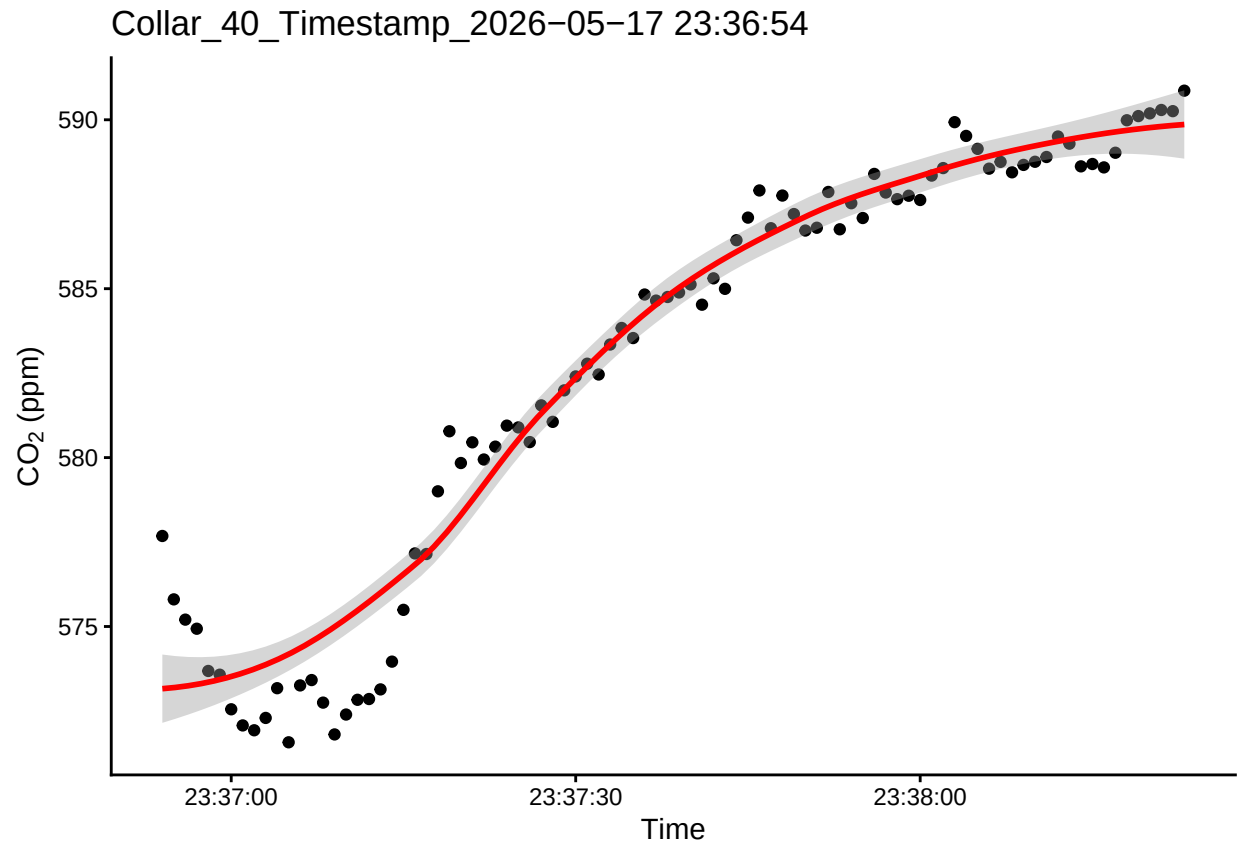
```
## 'geom_smooth()' using method = 'loess' and formula = 'y ~ x'
```



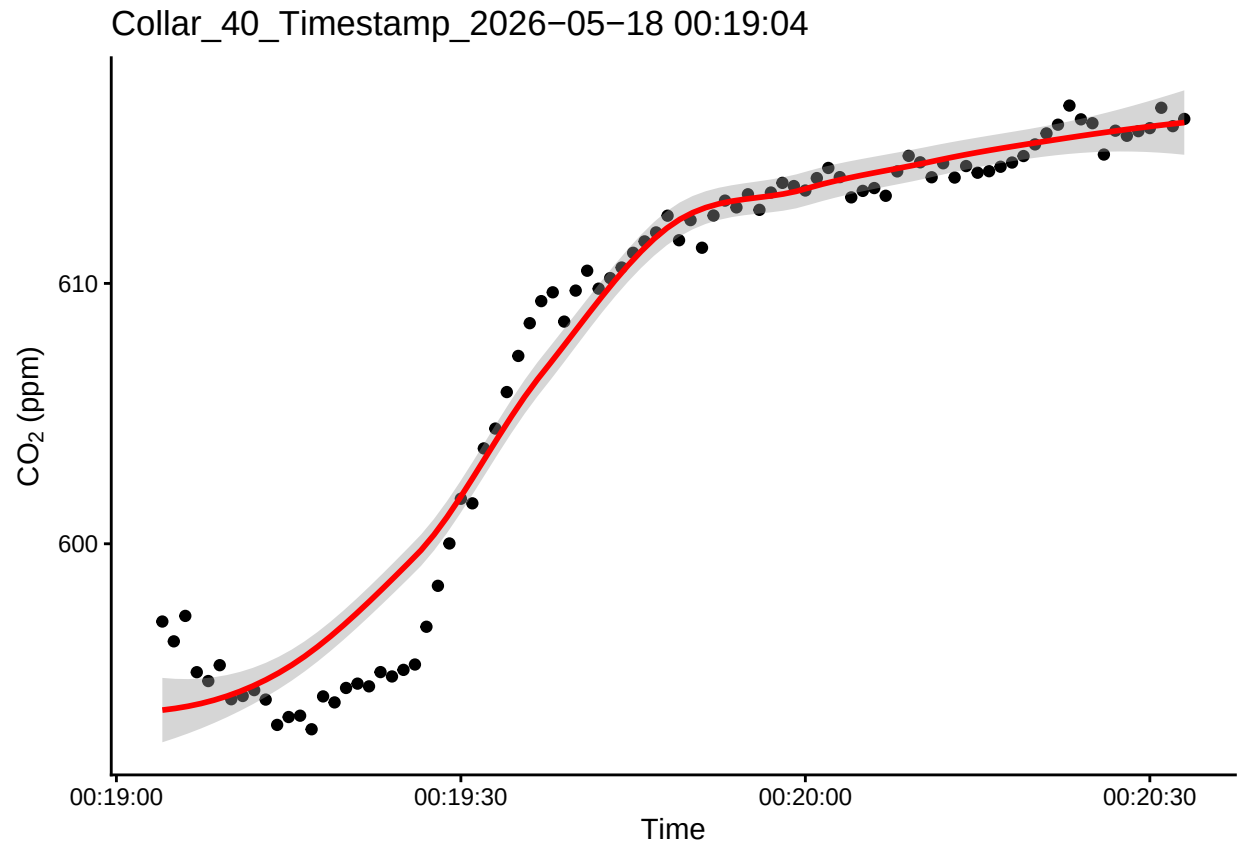
```
## 'geom_smooth()' using method = 'loess' and formula = 'y ~ x'
```



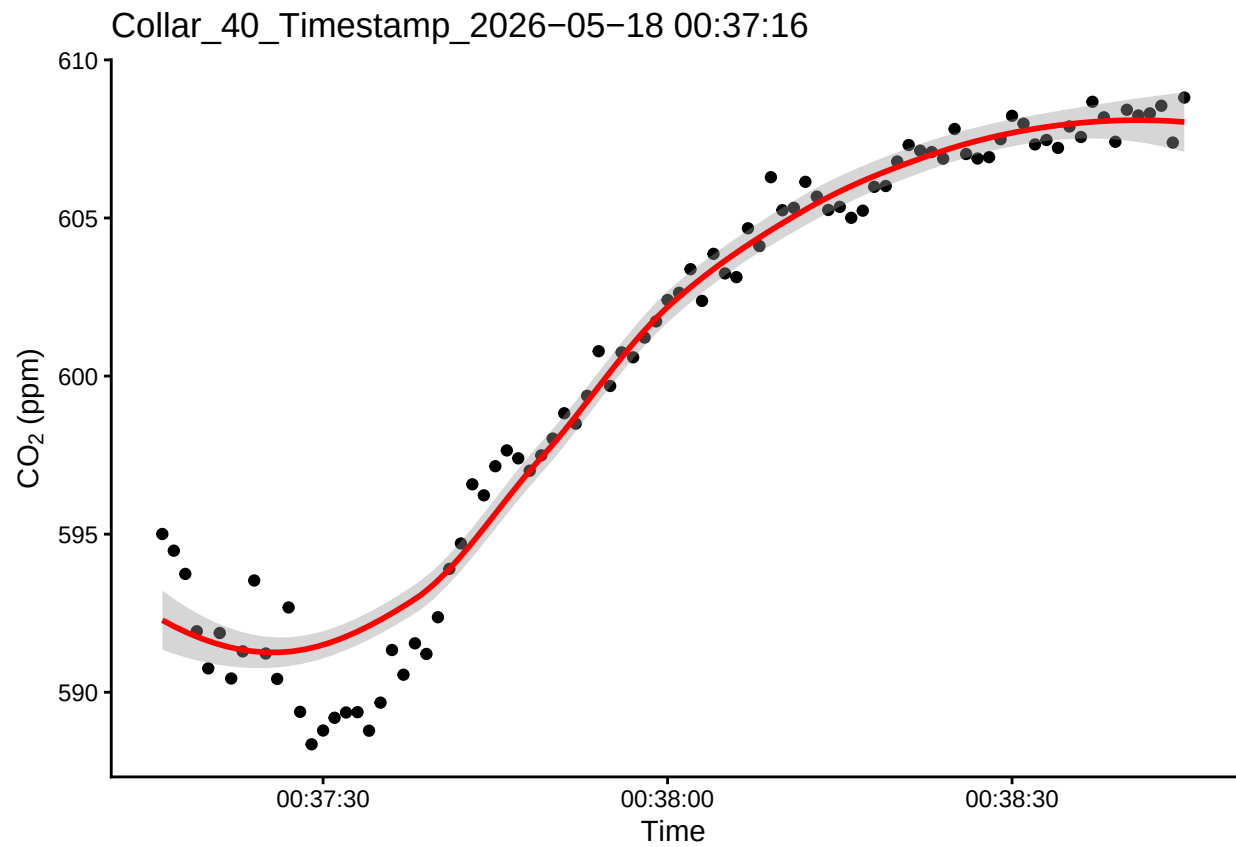
```
## 'geom_smooth()' using method = 'loess' and formula = 'y ~ x'
```



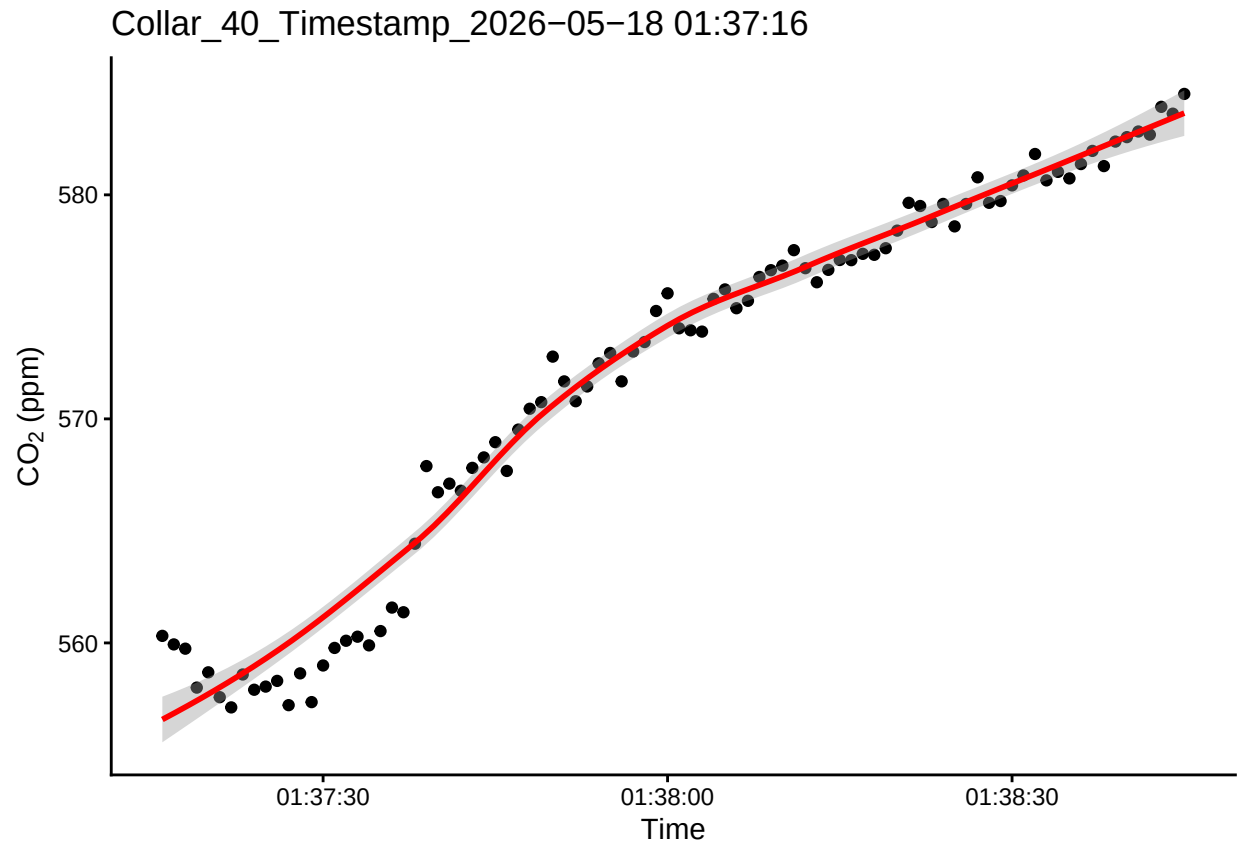
```
## 'geom_smooth()' using method = 'loess' and formula = 'y ~ x'
```



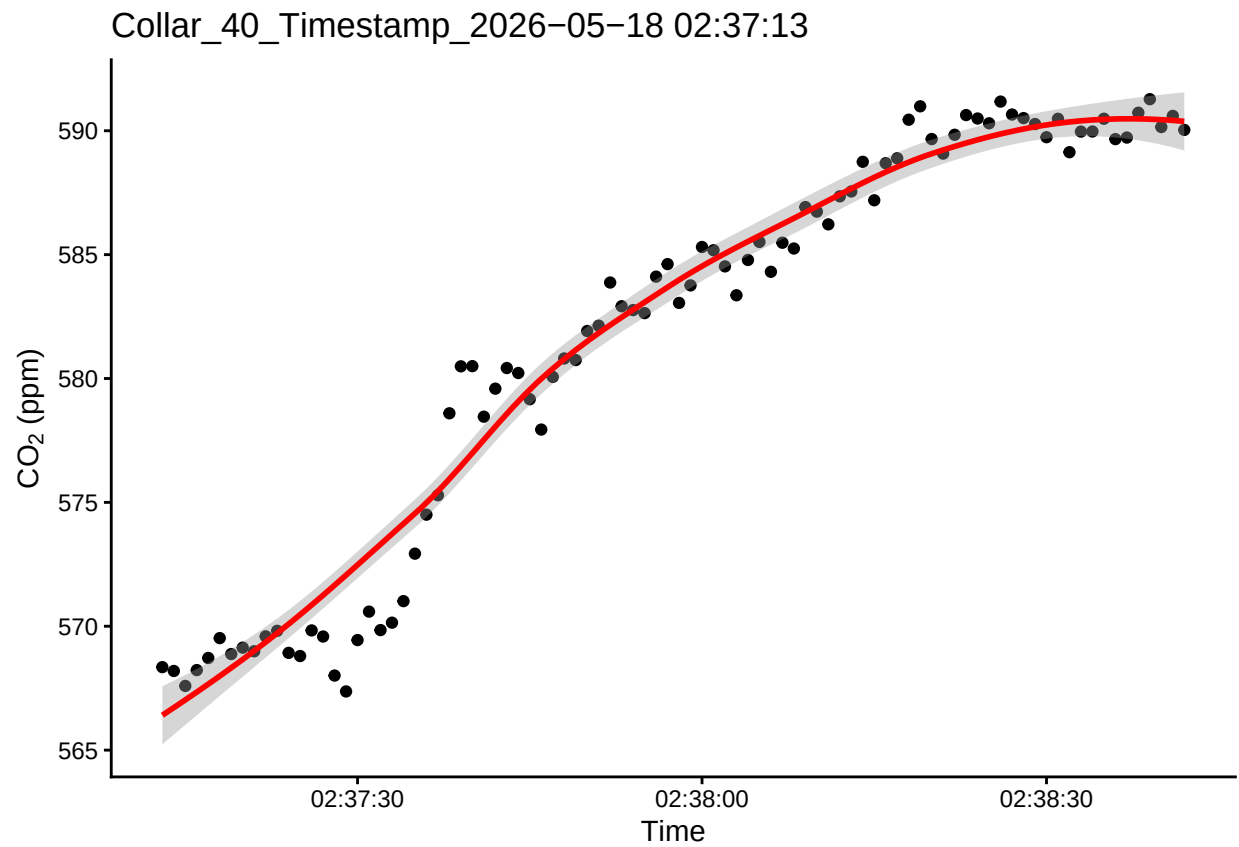
```
## 'geom_smooth()' using method = 'loess' and formula = 'y ~ x'
```



```
## 'geom_smooth()' using method = 'loess' and formula = 'y ~ x'
```

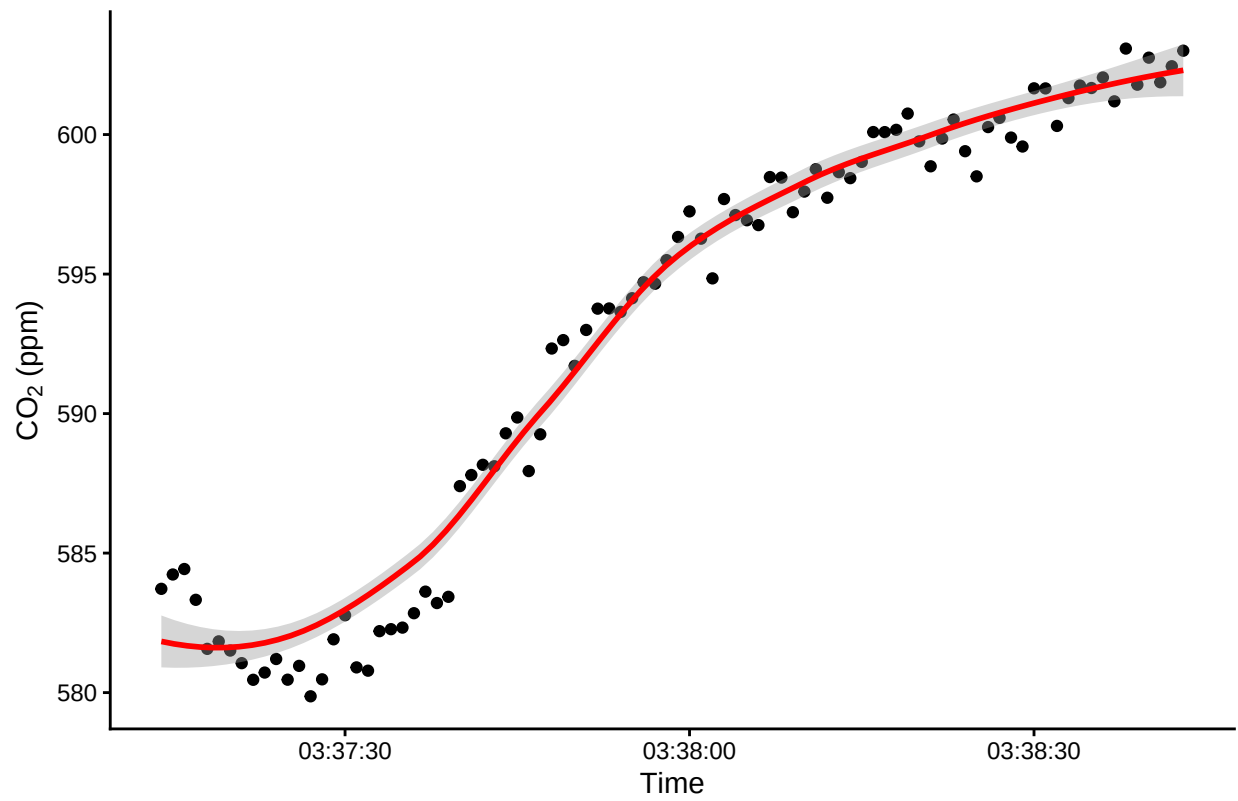


```
## 'geom_smooth()' using method = 'loess' and formula = 'y ~ x'
```

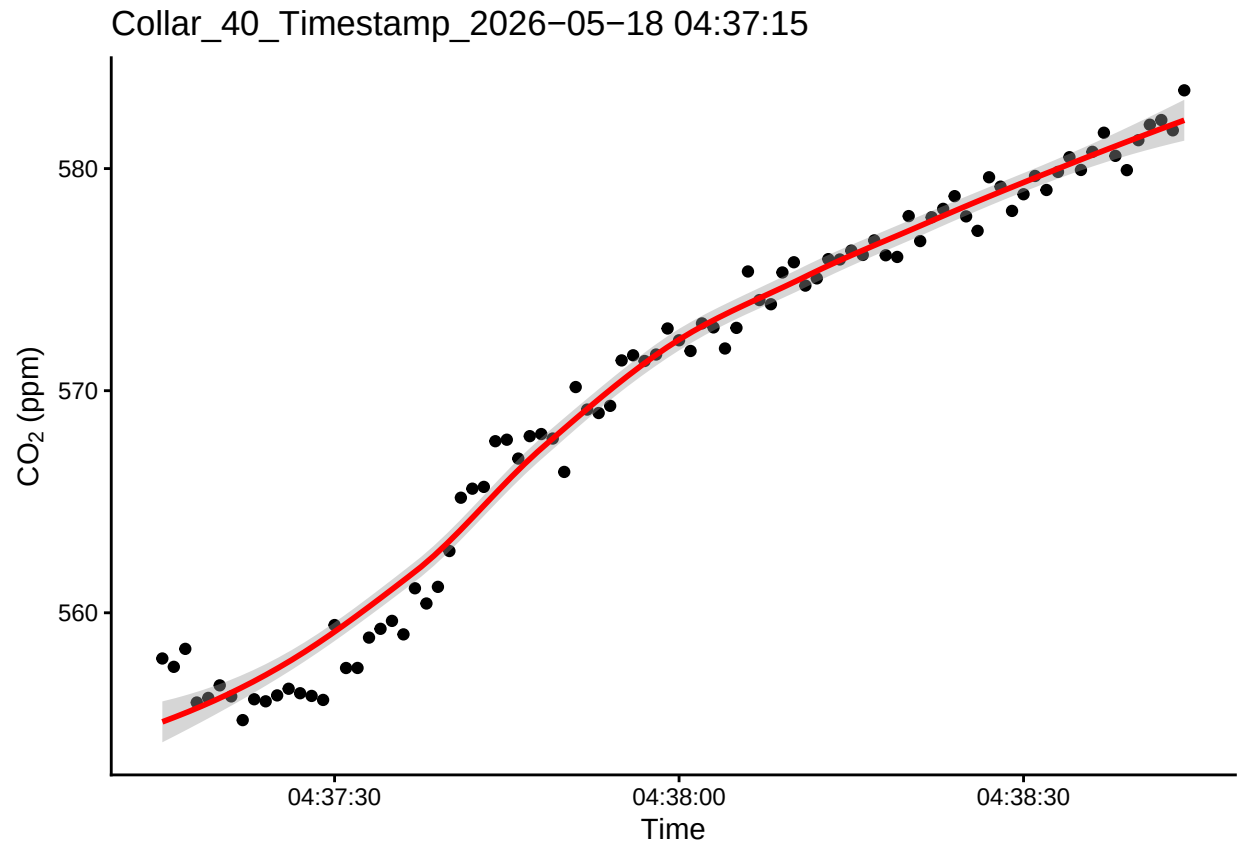



```
## 'geom_smooth()' using method = 'loess' and formula = 'y ~ x'
```

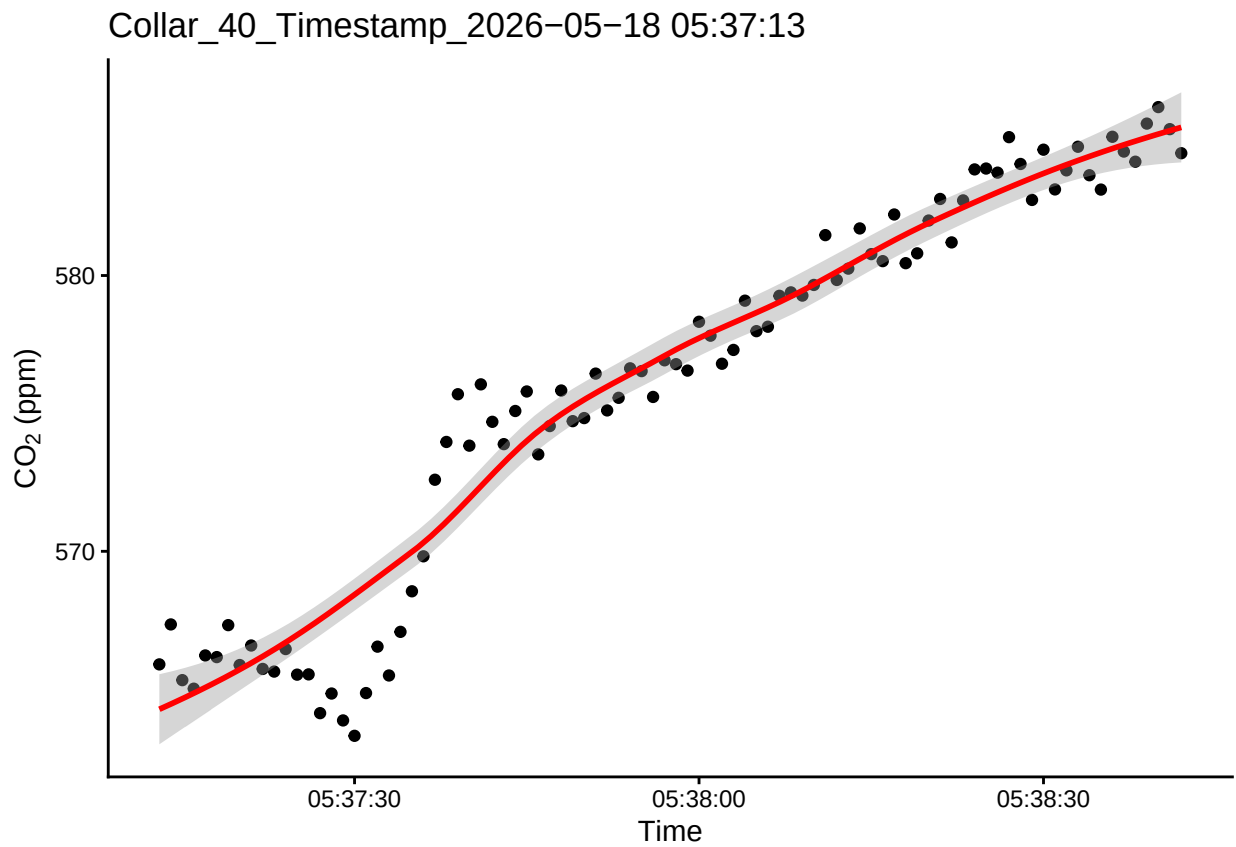
Collar_40_Timestamp_2026-05-18 03:37:14



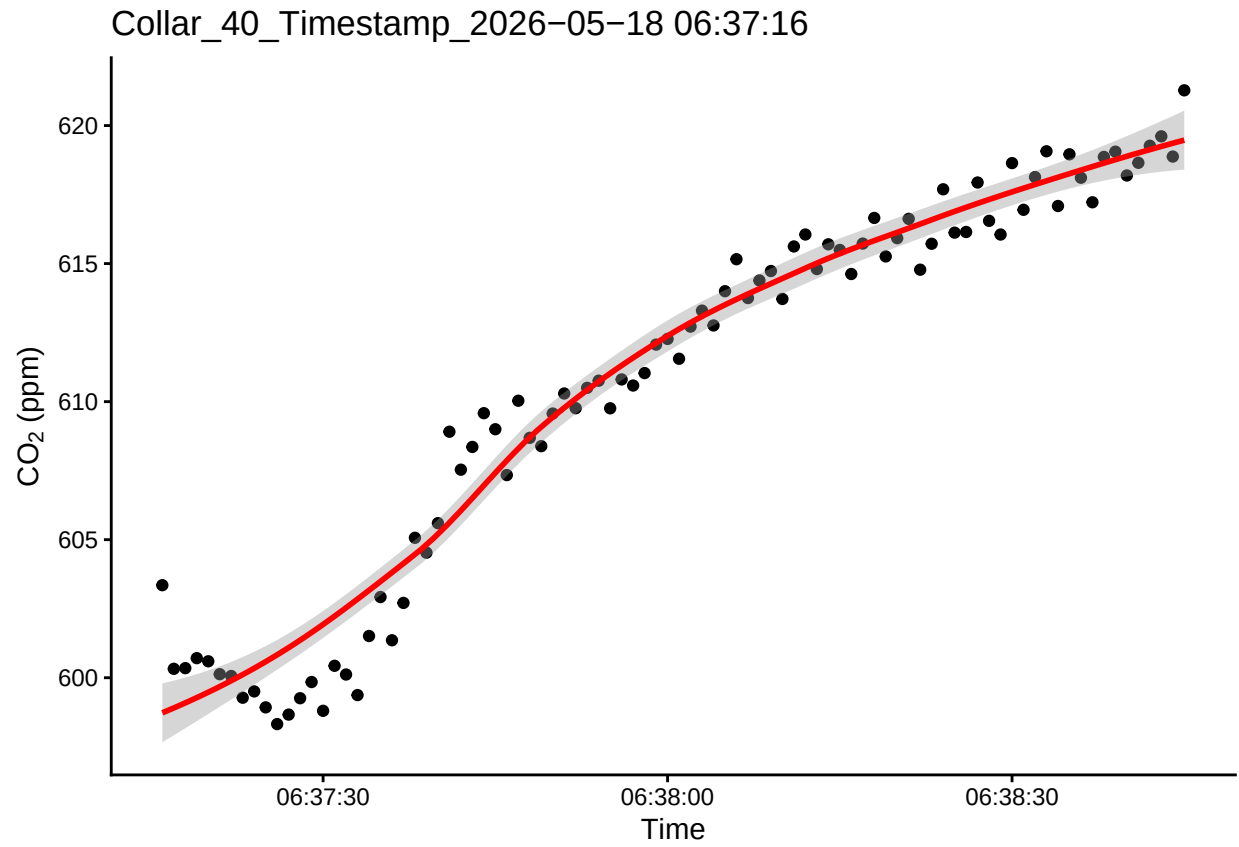
```
## 'geom_smooth()' using method = 'loess' and formula = 'y ~ x'
```



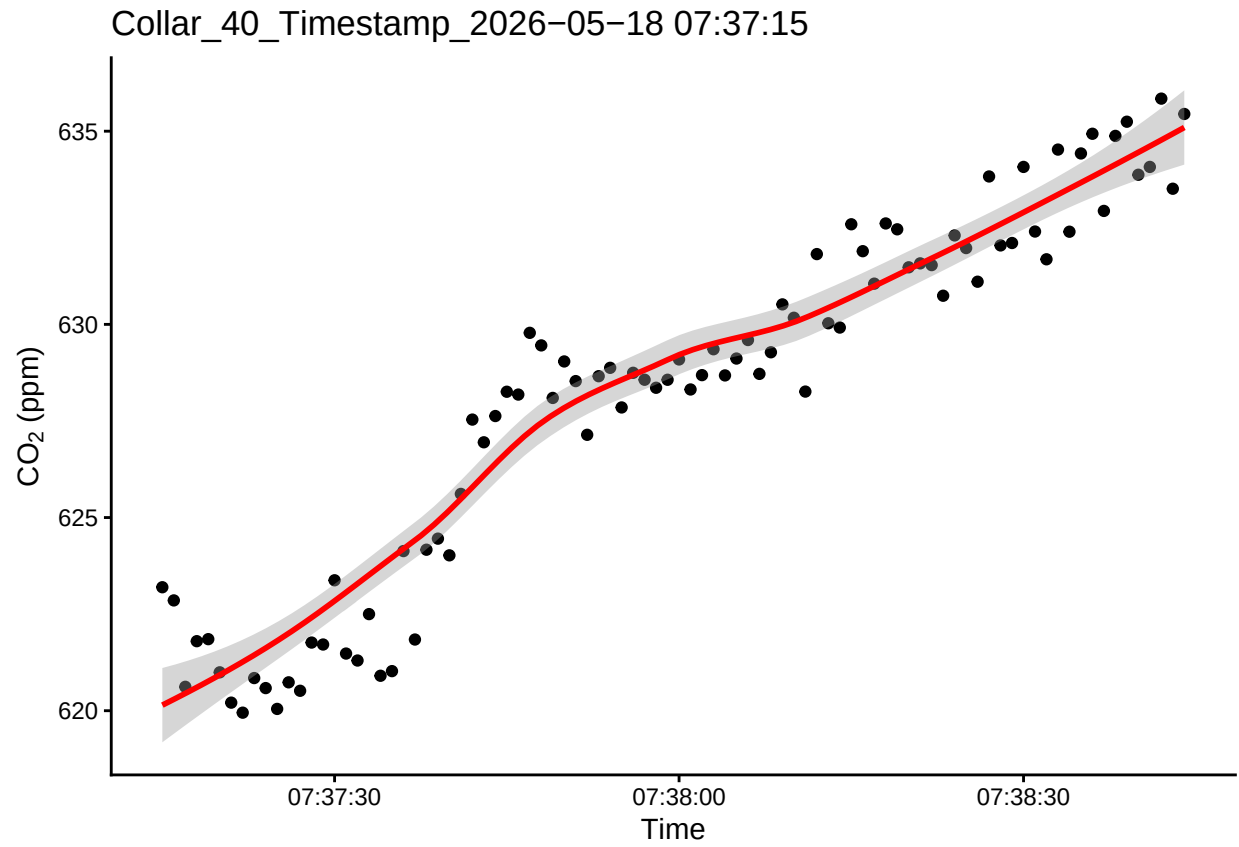
```
## 'geom_smooth()' using method = 'loess' and formula = 'y ~ x'
```



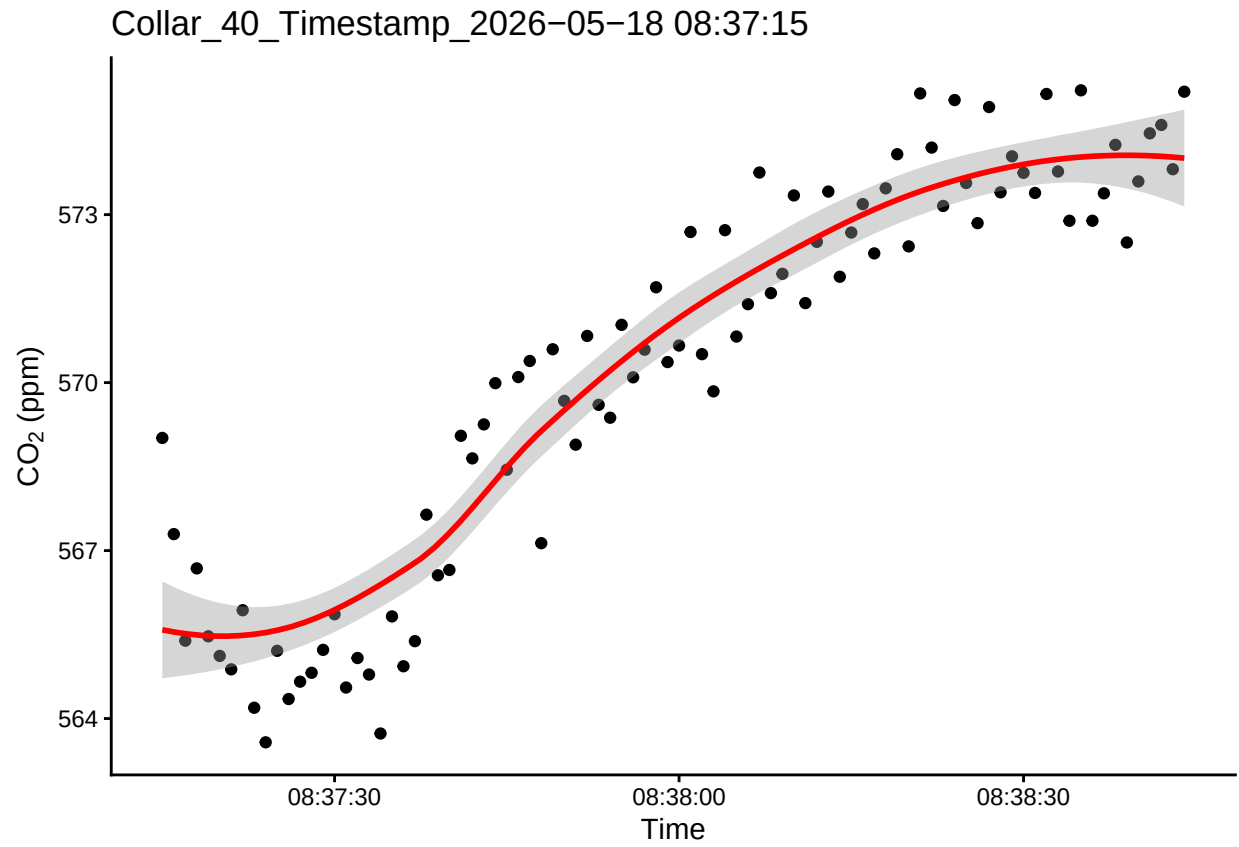
```
## 'geom_smooth()' using method = 'loess' and formula = 'y ~ x'
```



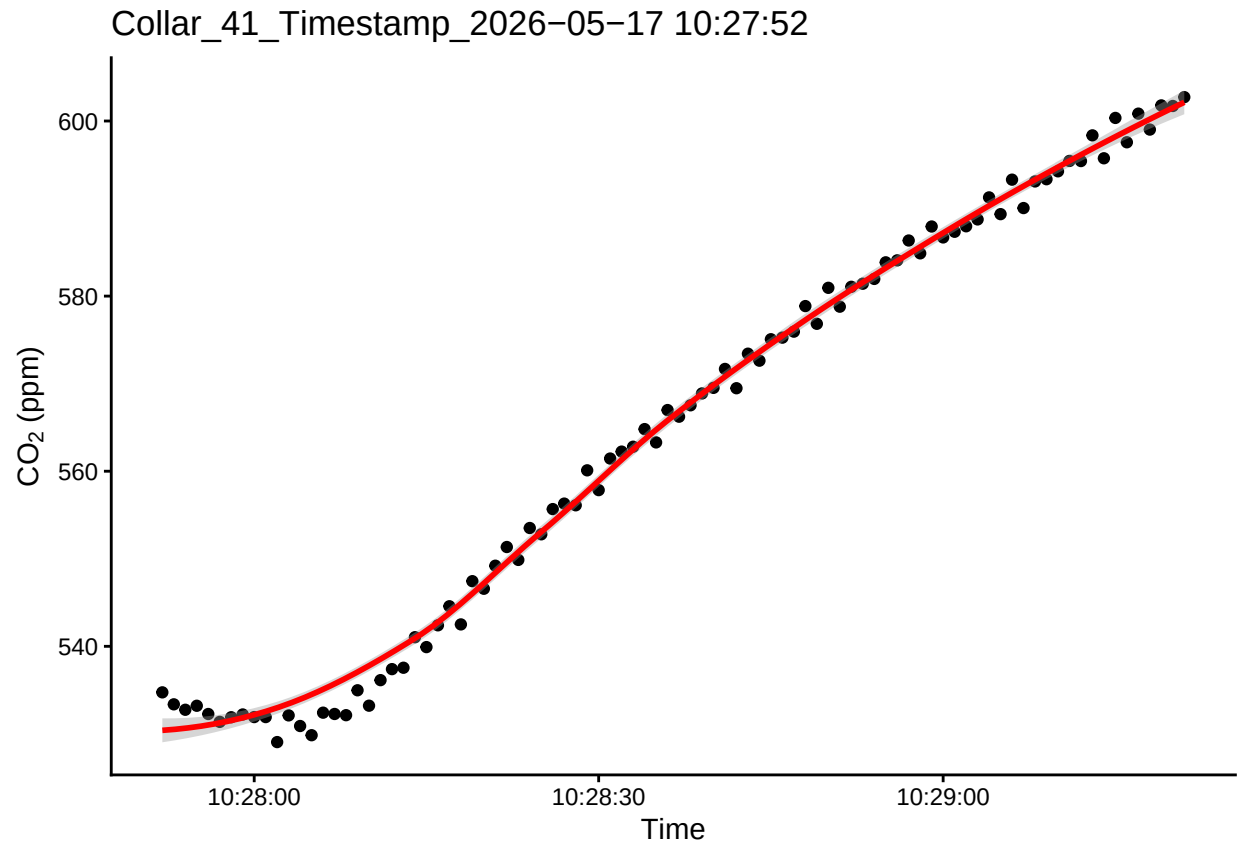
```
## 'geom_smooth()' using method = 'loess' and formula = 'y ~ x'
```



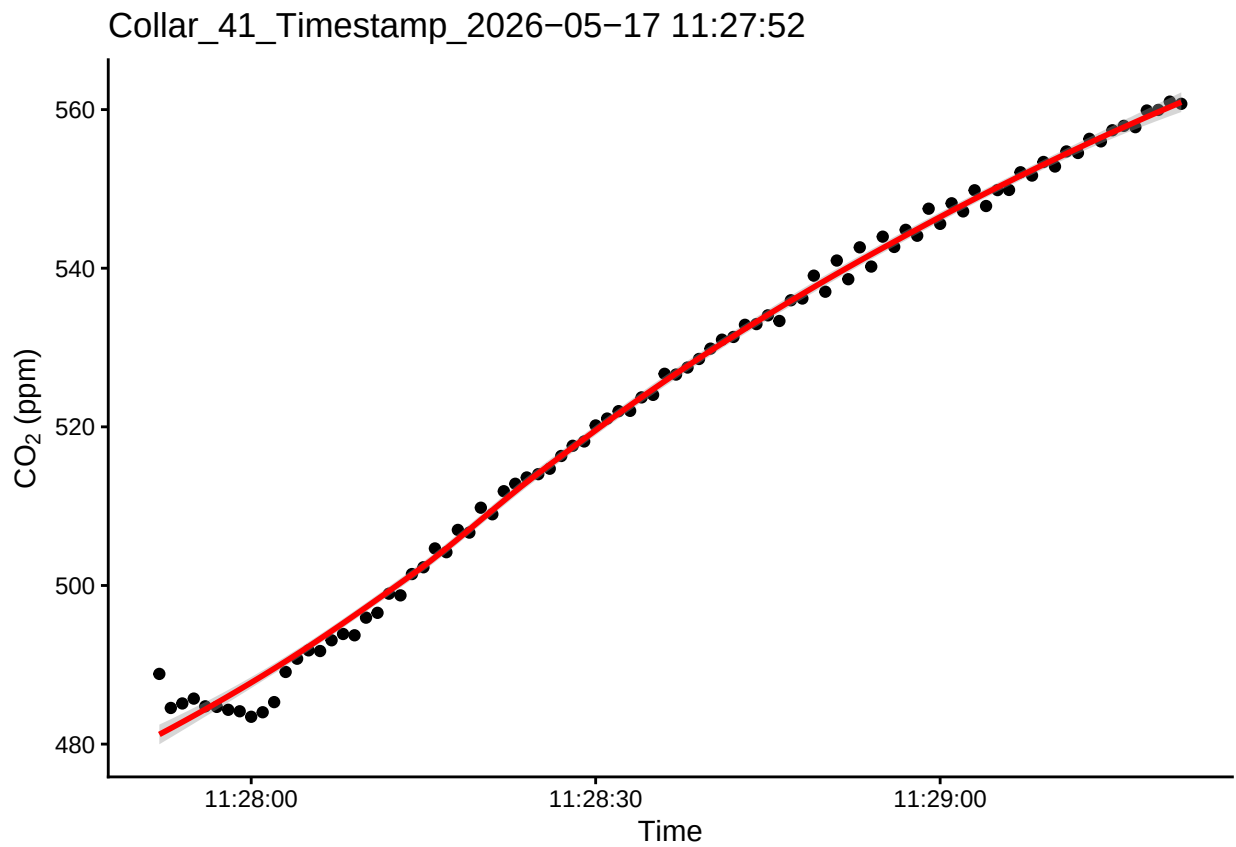
```
## 'geom_smooth()' using method = 'loess' and formula = 'y ~ x'
```



```
## 'geom_smooth()' using method = 'loess' and formula = 'y ~ x'
```

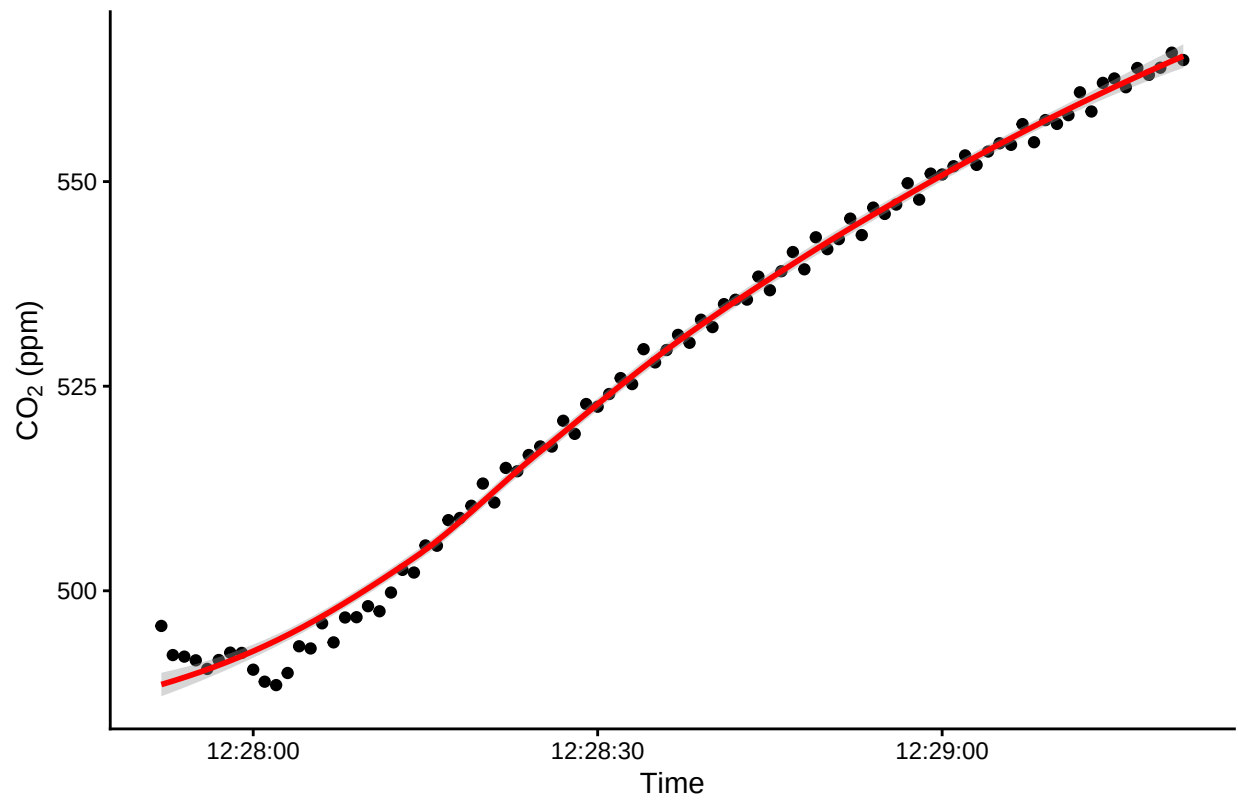


```
## 'geom_smooth()' using method = 'loess' and formula = 'y ~ x'
```

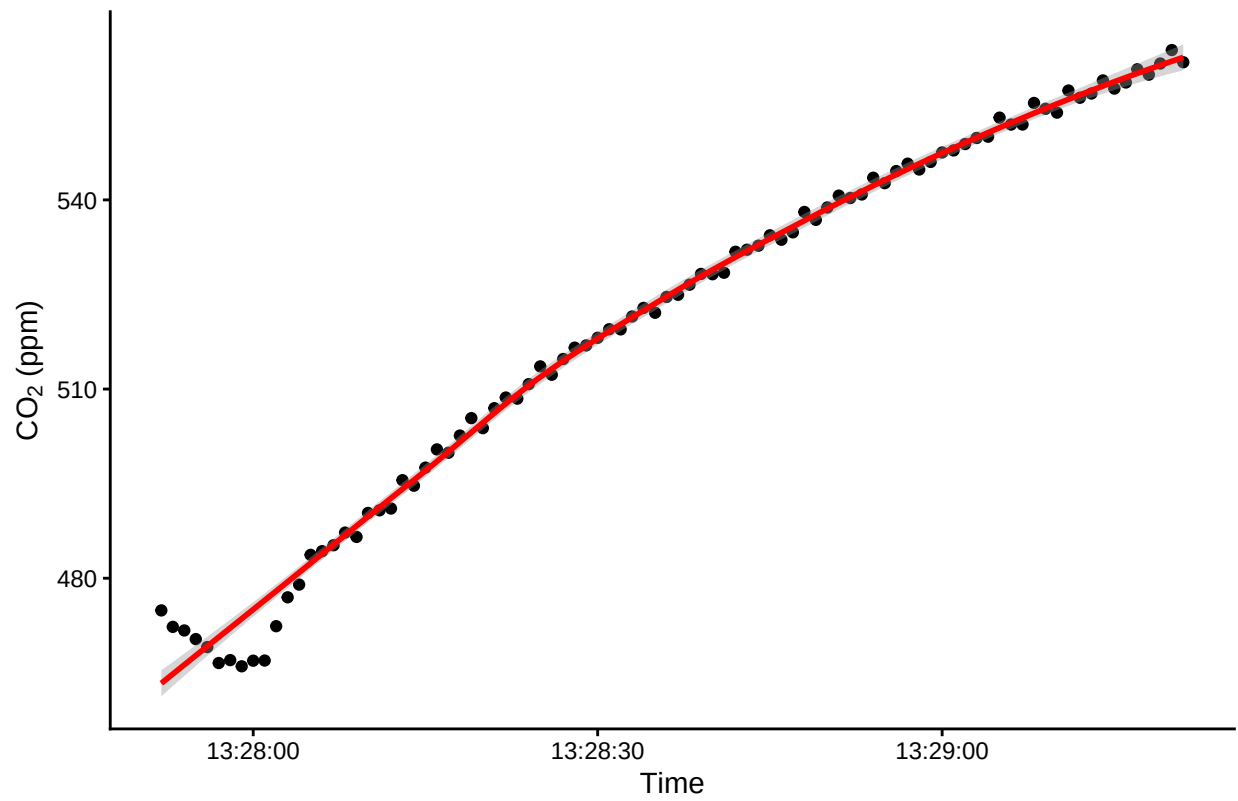
```
## 'geom_smooth()' using method = 'loess' and formula = 'y ~ x'
```

Collar_41_Timestamp_2026-05-17 12:27:52



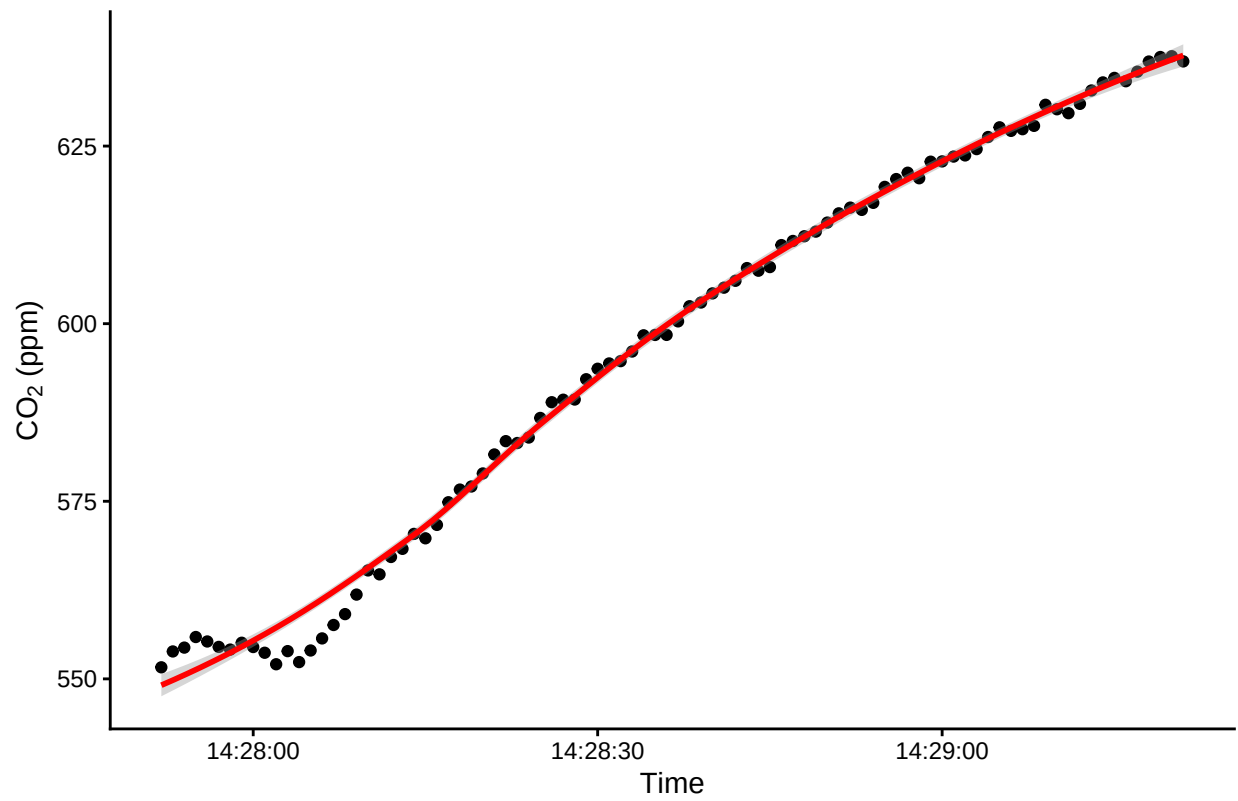
```
## 'geom_smooth()' using method = 'loess' and formula = 'y ~ x'
```

Collar_41_Timestamp_2026-05-17 13:27:52



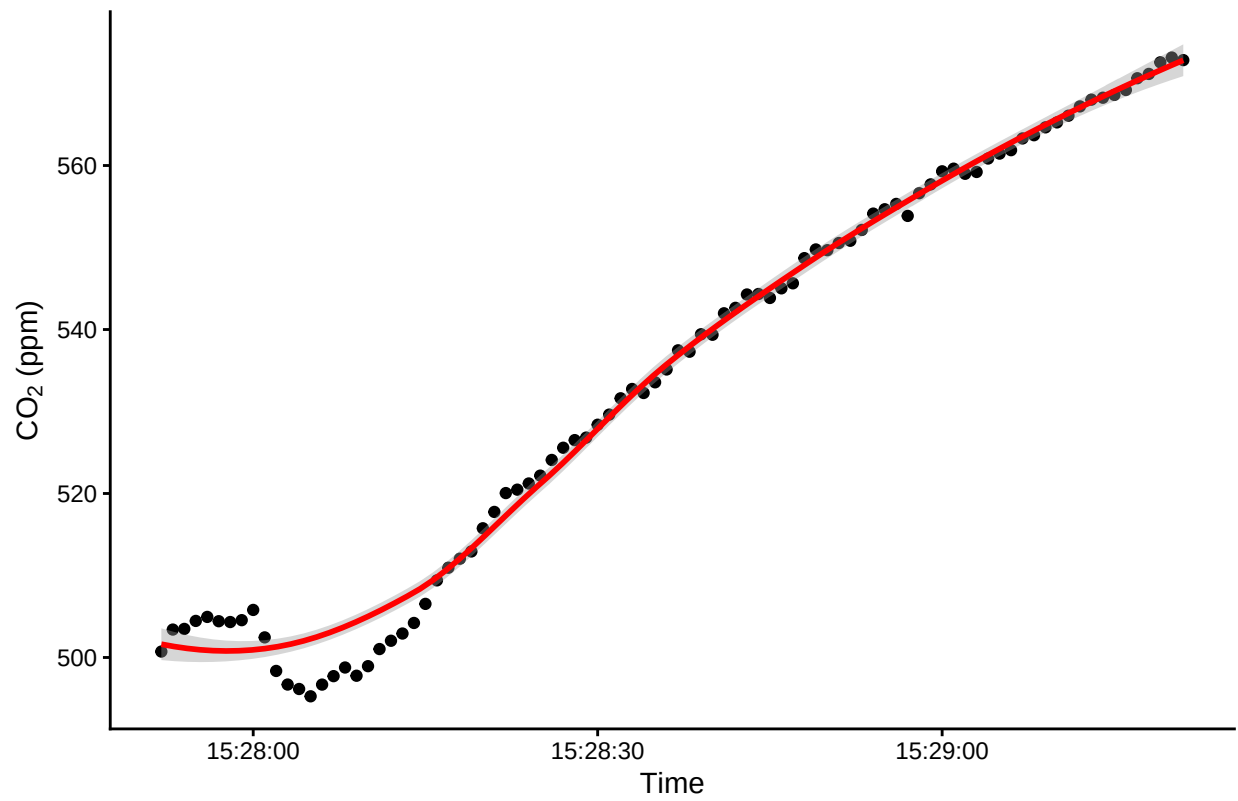
```
## 'geom_smooth()' using method = 'loess' and formula = 'y ~ x'
```

Collar_41_Timestamp_2026-05-17 14:27:52



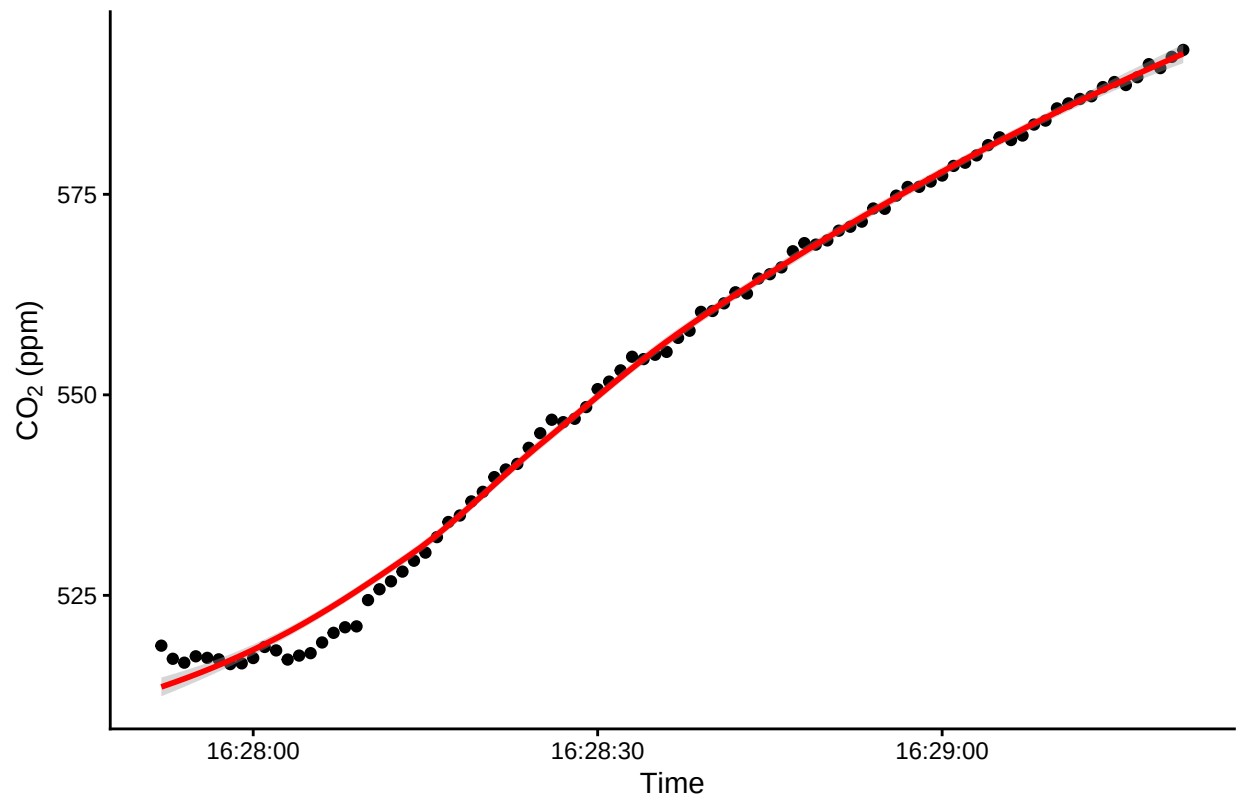
```
## 'geom_smooth()' using method = 'loess' and formula = 'y ~ x'
```

Collar_41_Timestamp_2026-05-17 15:27:52



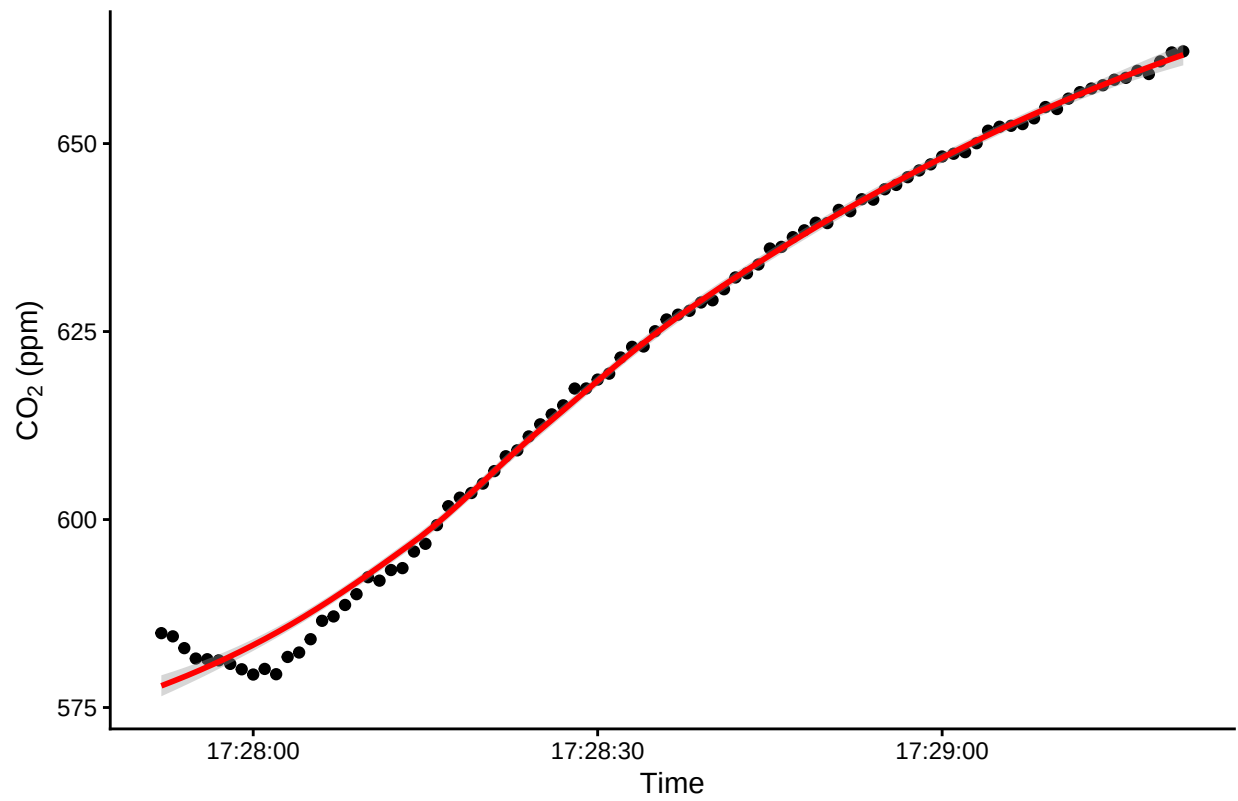
```
## 'geom_smooth()' using method = 'loess' and formula = 'y ~ x'
```

Collar_41_Timestamp_2026-05-17 16:27:52

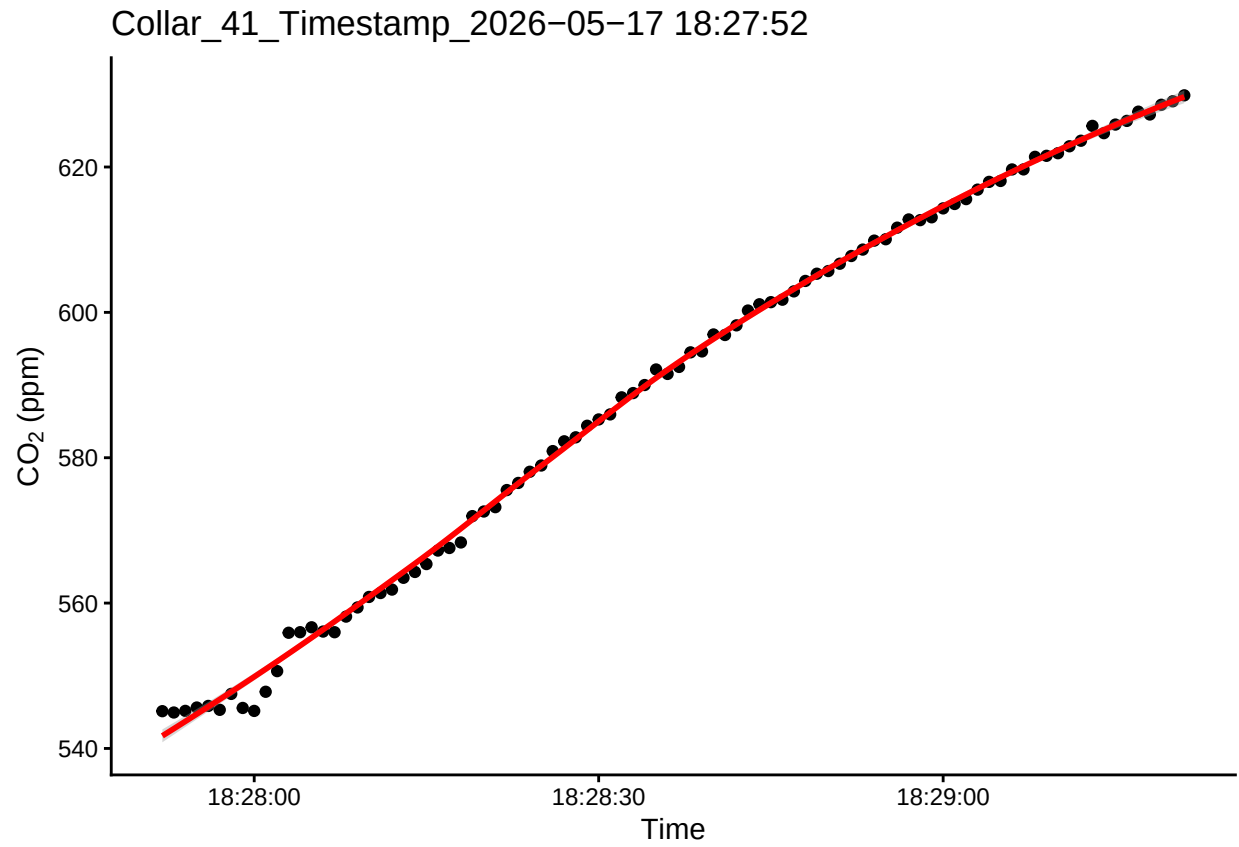


```
## 'geom_smooth()' using method = 'loess' and formula = 'y ~ x'
```

Collar_41_Timestamp_2026-05-17 17:27:52

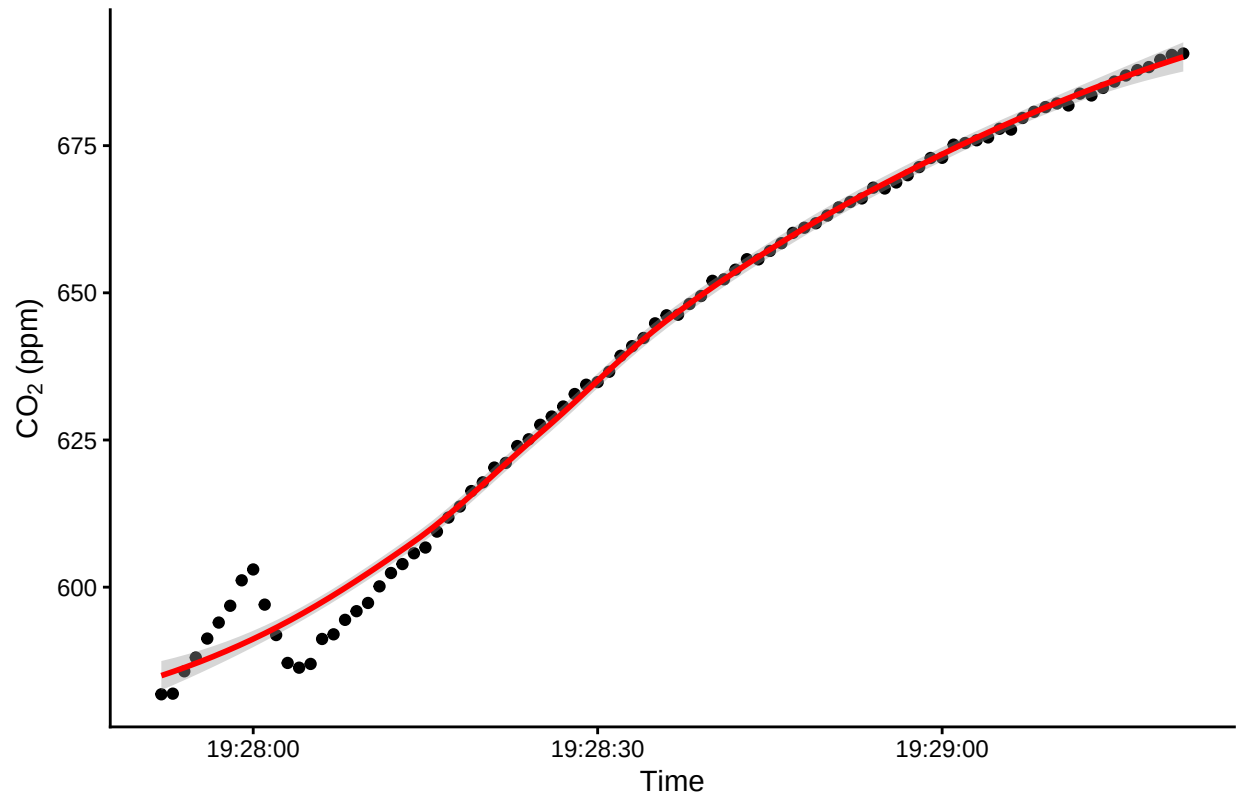


```
## 'geom_smooth()' using method = 'loess' and formula = 'y ~ x'
```

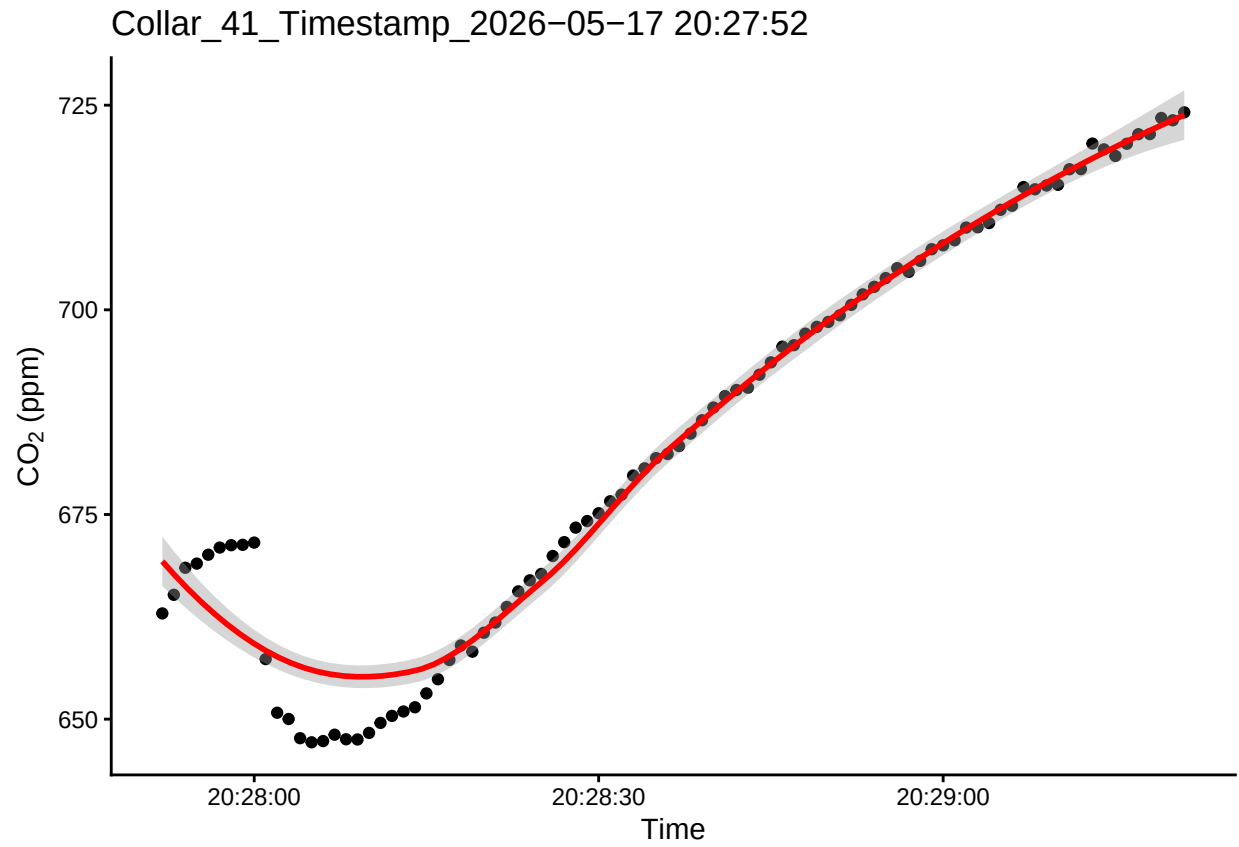


```
## 'geom_smooth()' using method = 'loess' and formula = 'y ~ x'
```

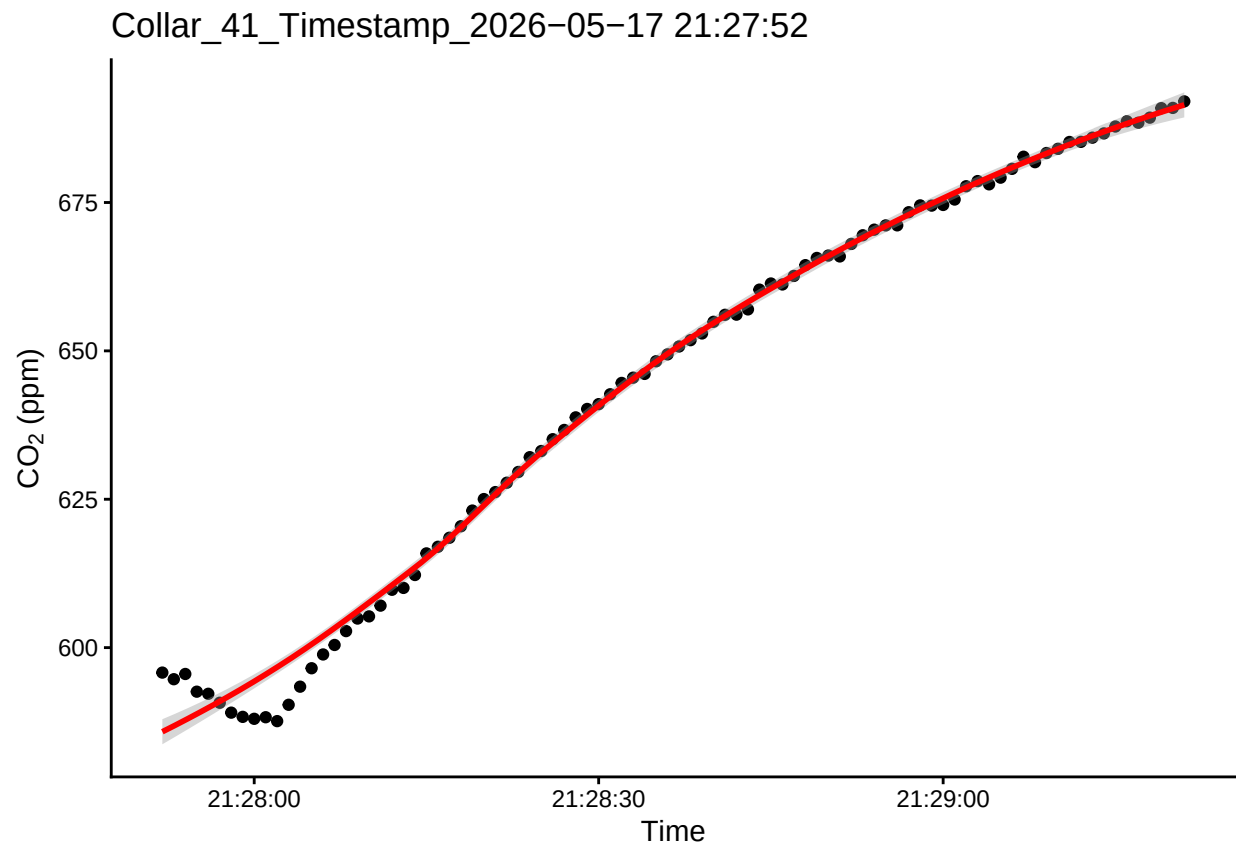

Collar_41_Timestamp_2026-05-17 19:27:52



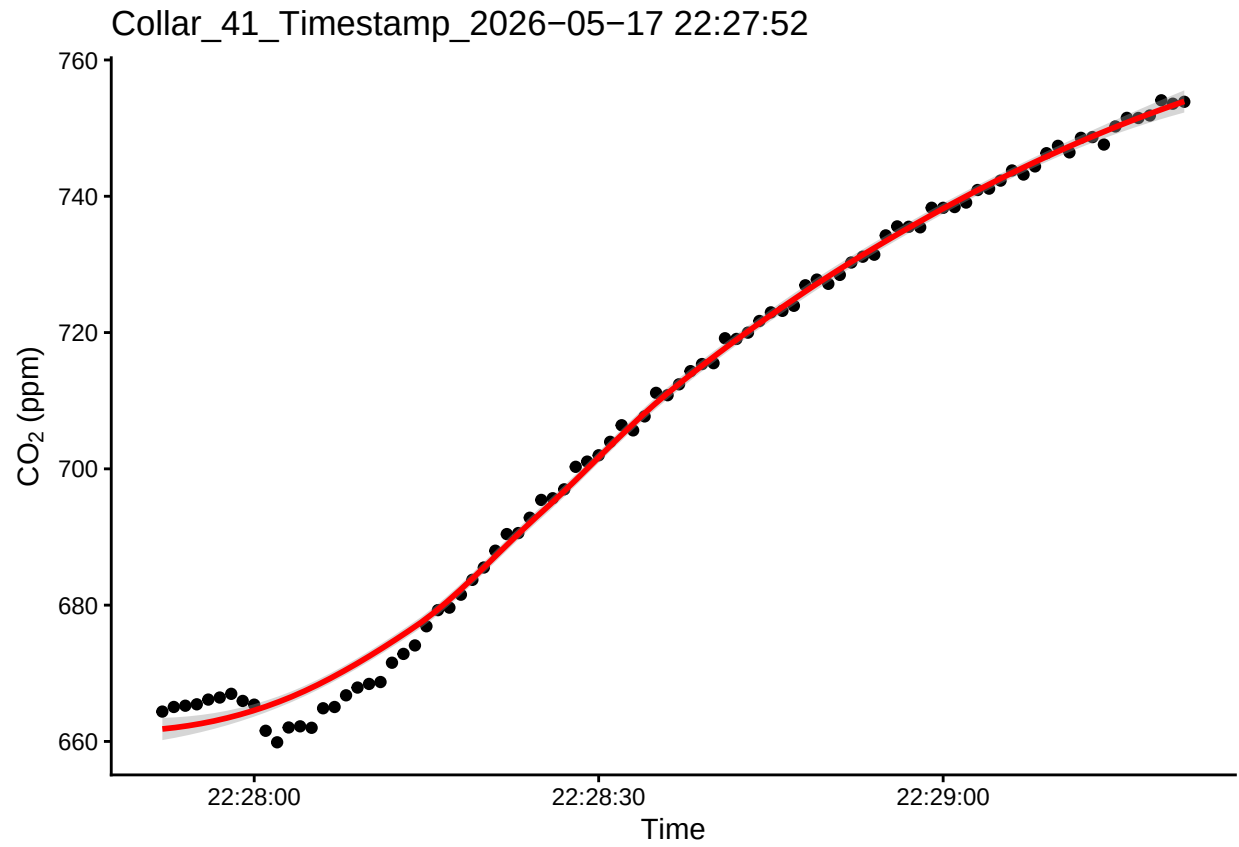
```
## 'geom_smooth()' using method = 'loess' and formula = 'y ~ x'
```



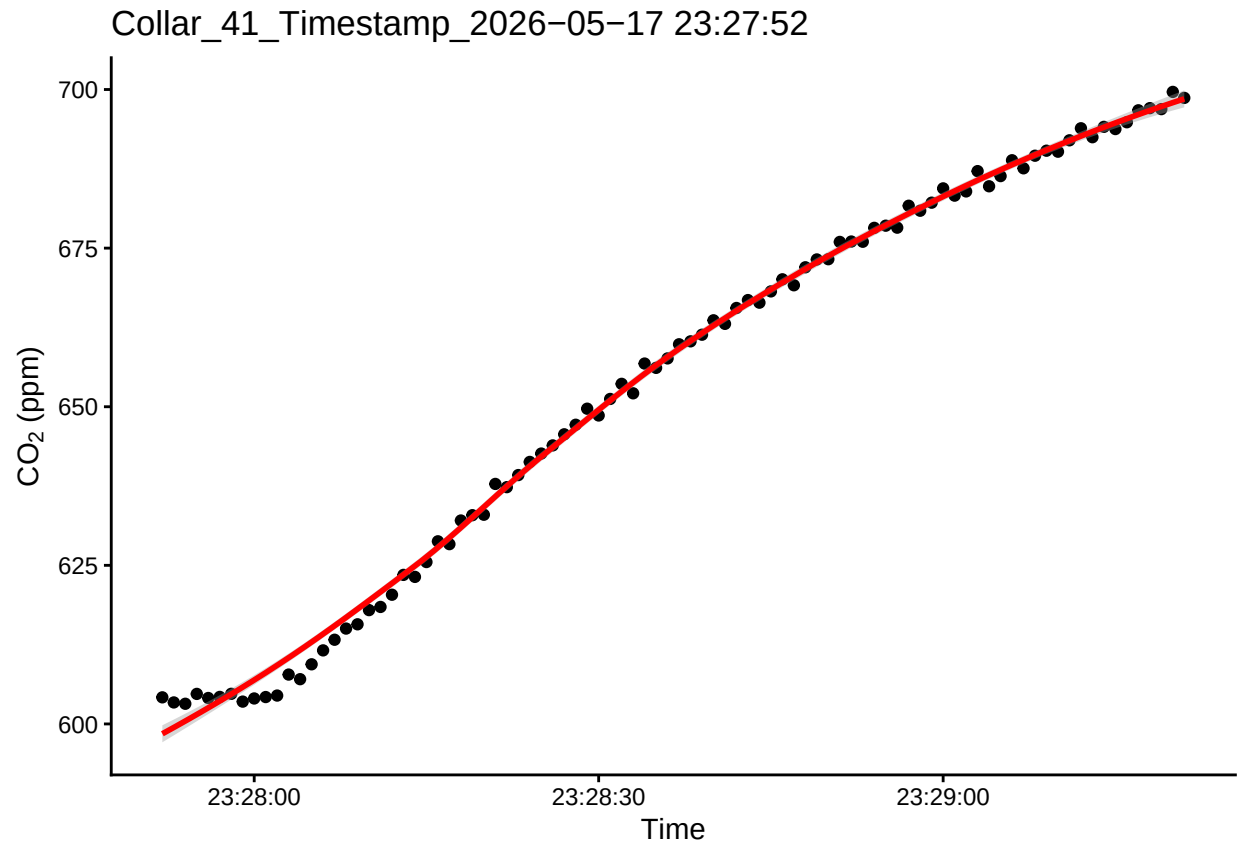
```
## 'geom_smooth()' using method = 'loess' and formula = 'y ~ x'
```



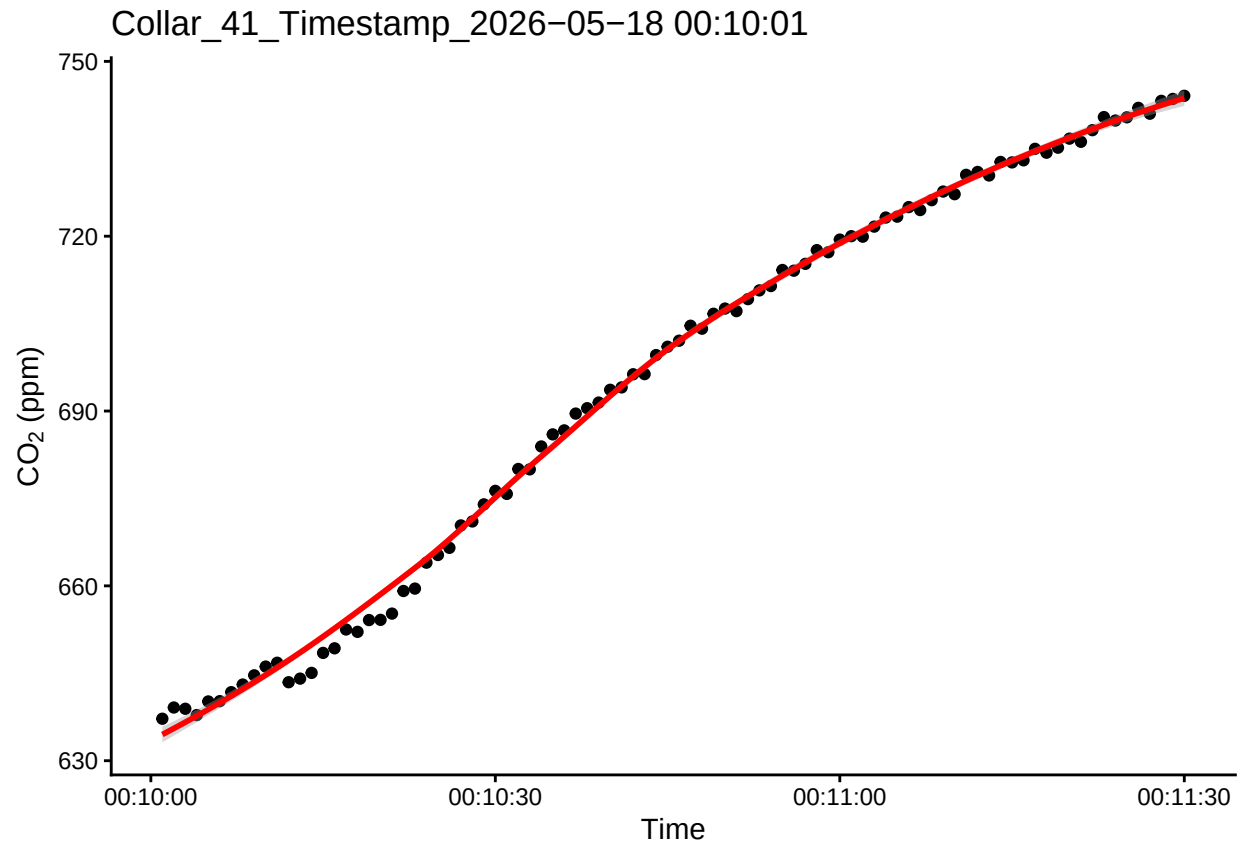
```
## 'geom_smooth()' using method = 'loess' and formula = 'y ~ x'
```



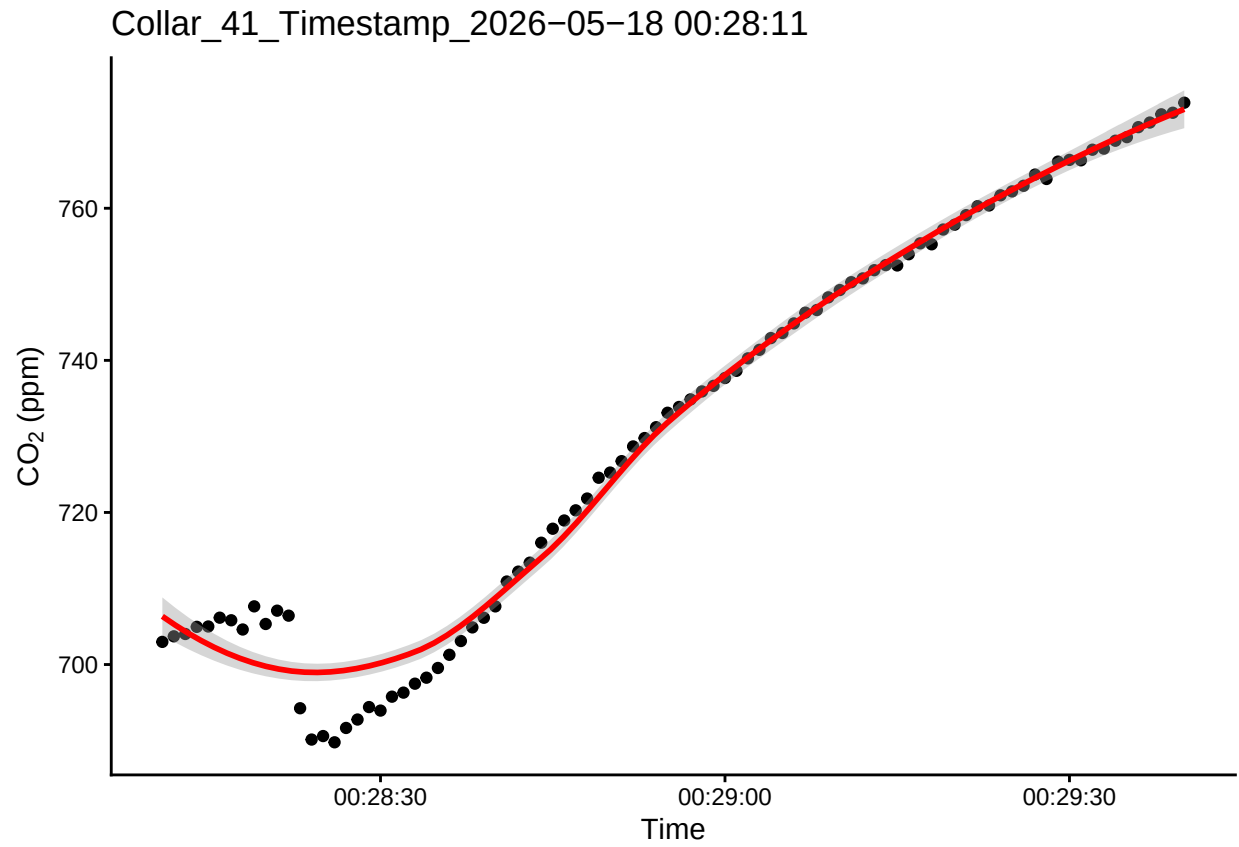
```
## 'geom_smooth()' using method = 'loess' and formula = 'y ~ x'
```



```
## 'geom_smooth()' using method = 'loess' and formula = 'y ~ x'
```

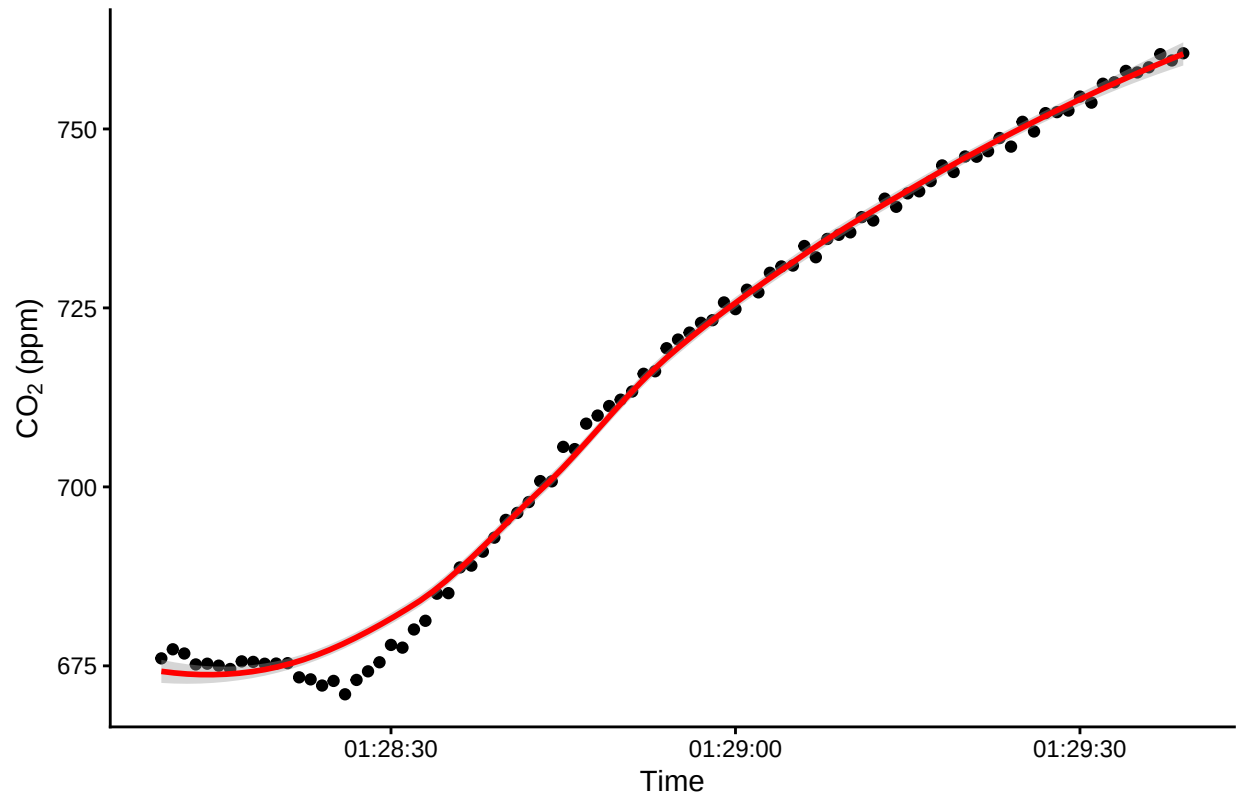


```
## 'geom_smooth()' using method = 'loess' and formula = 'y ~ x'
```

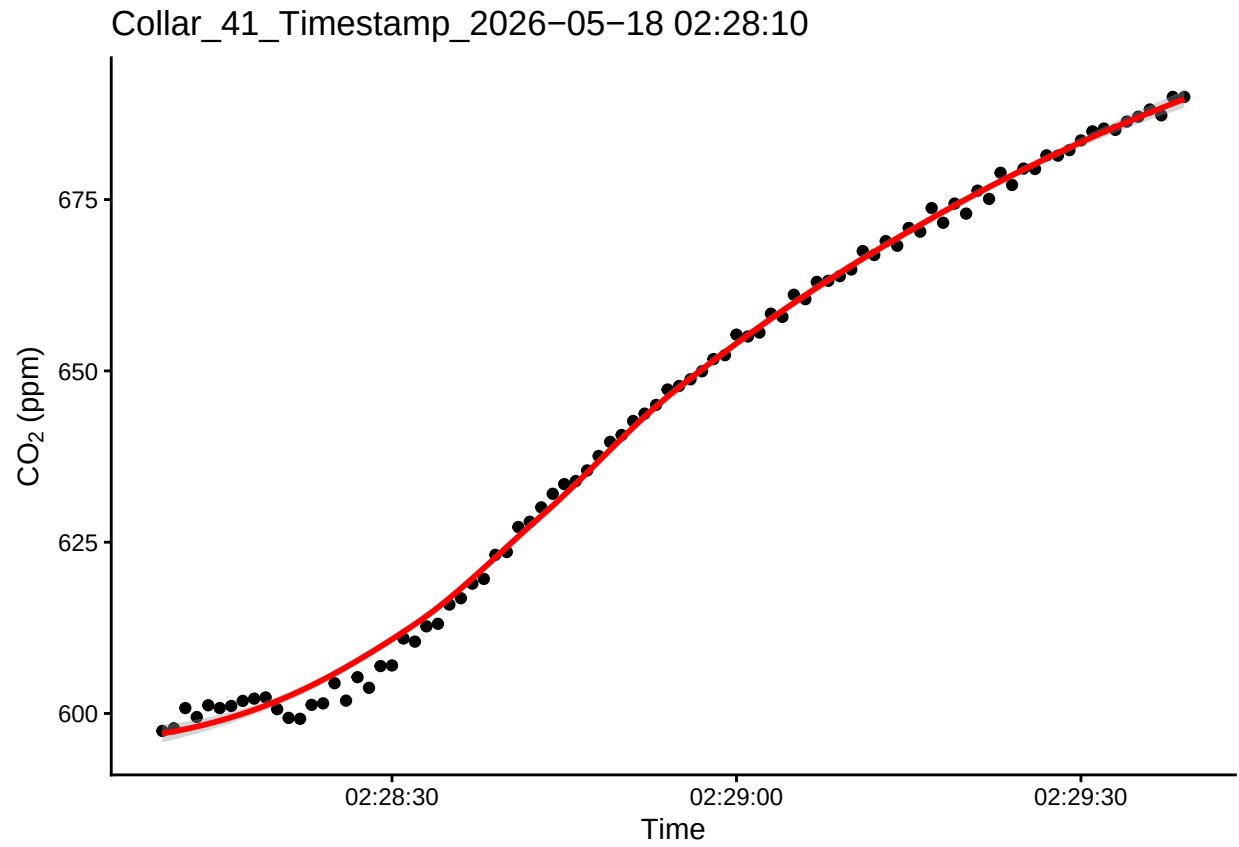


```
## 'geom_smooth()' using method = 'loess' and formula = 'y ~ x'
```

Collar_41_Timestamp_2026-05-18 01:28:10

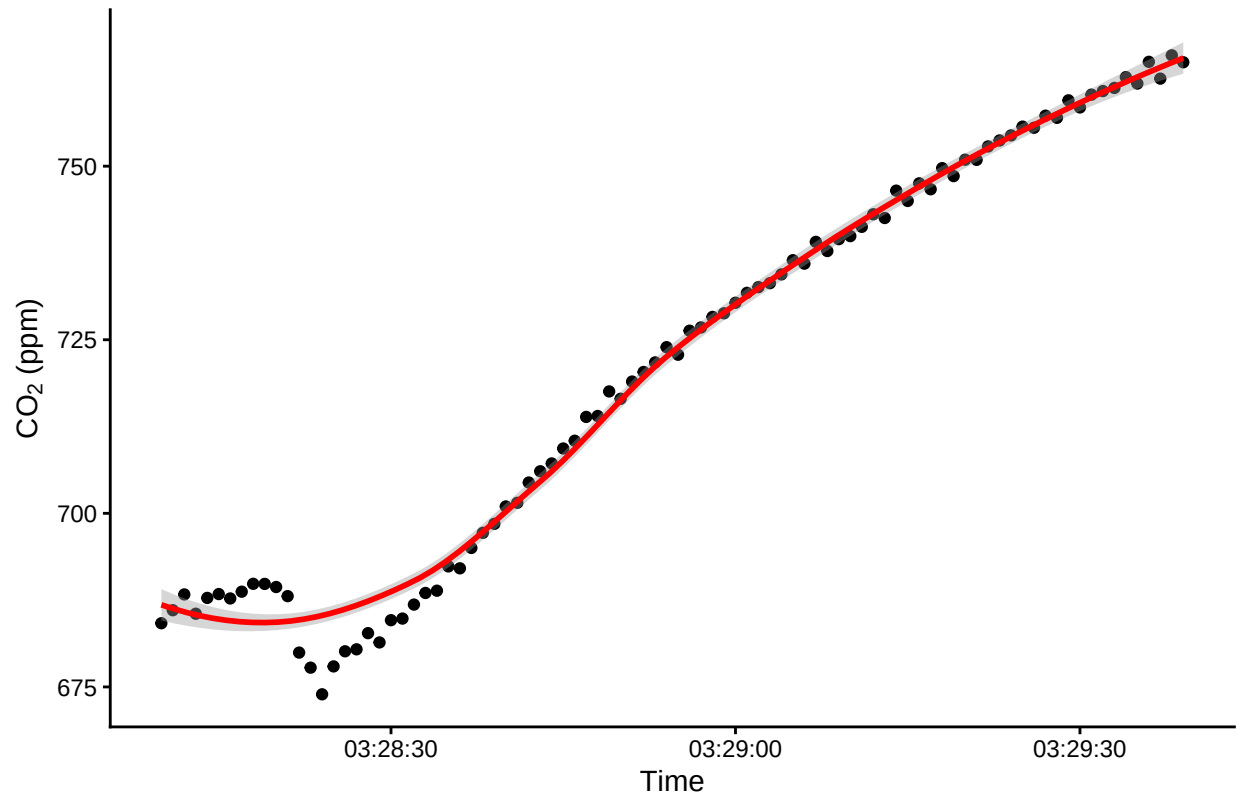


```
## 'geom_smooth()' using method = 'loess' and formula = 'y ~ x'
```

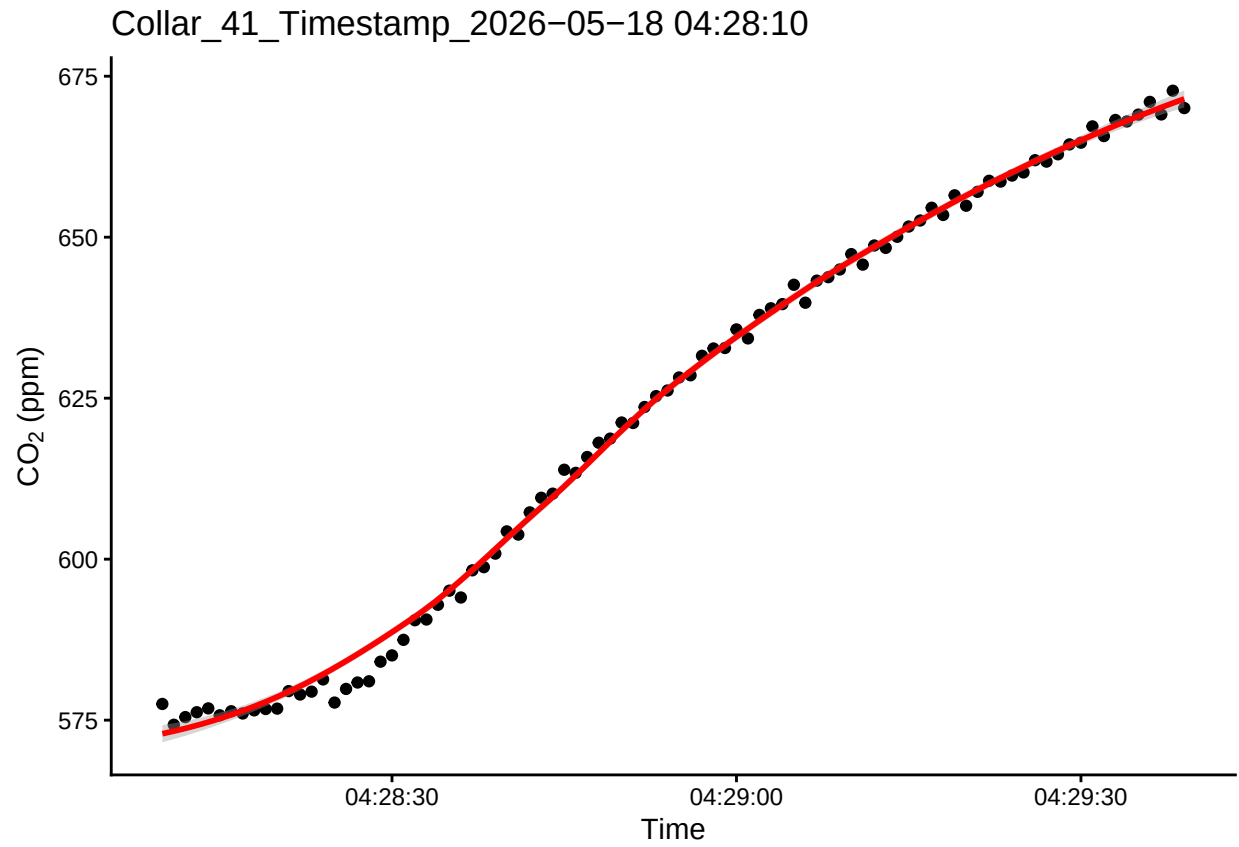



```
## 'geom_smooth()' using method = 'loess' and formula = 'y ~ x'
```

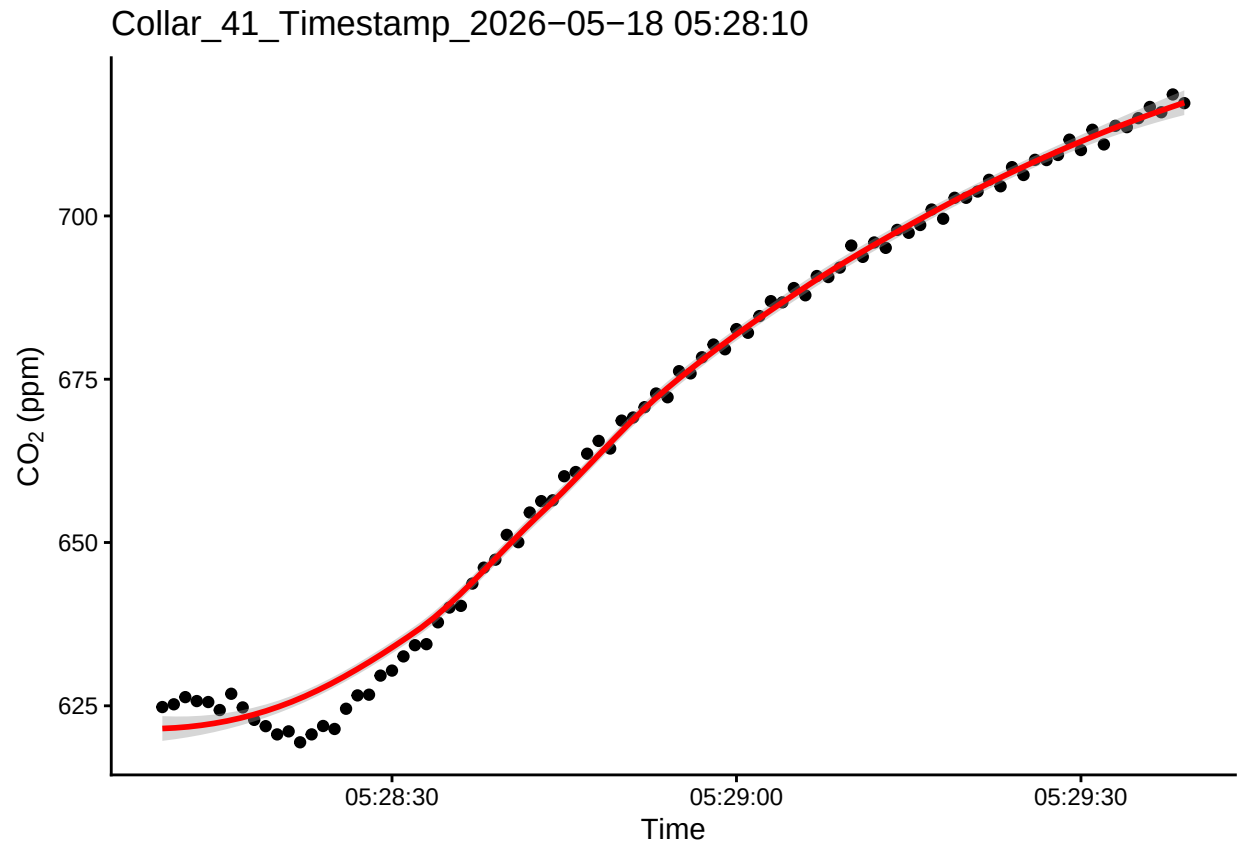
Collar_41_Timestamp_2026-05-18 03:28:10



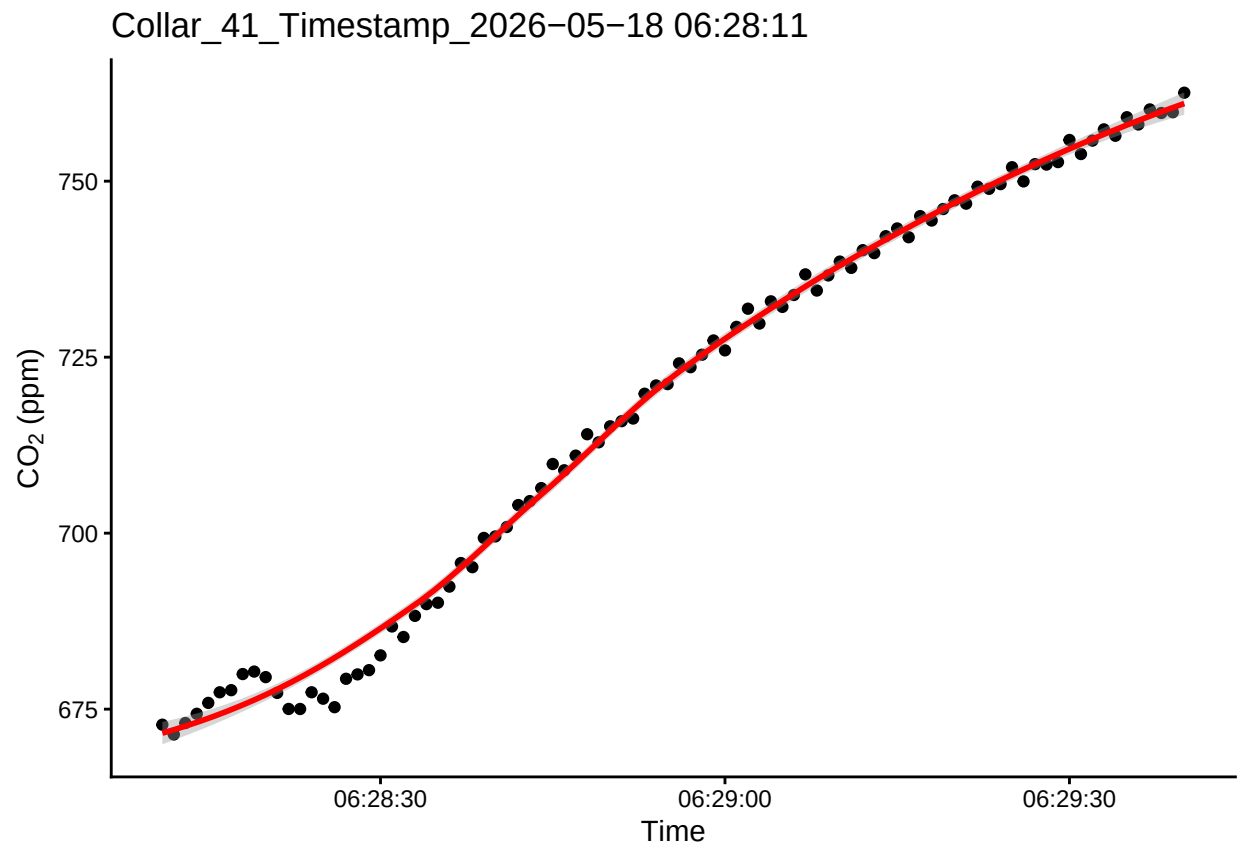
```
## 'geom_smooth()' using method = 'loess' and formula = 'y ~ x'
```



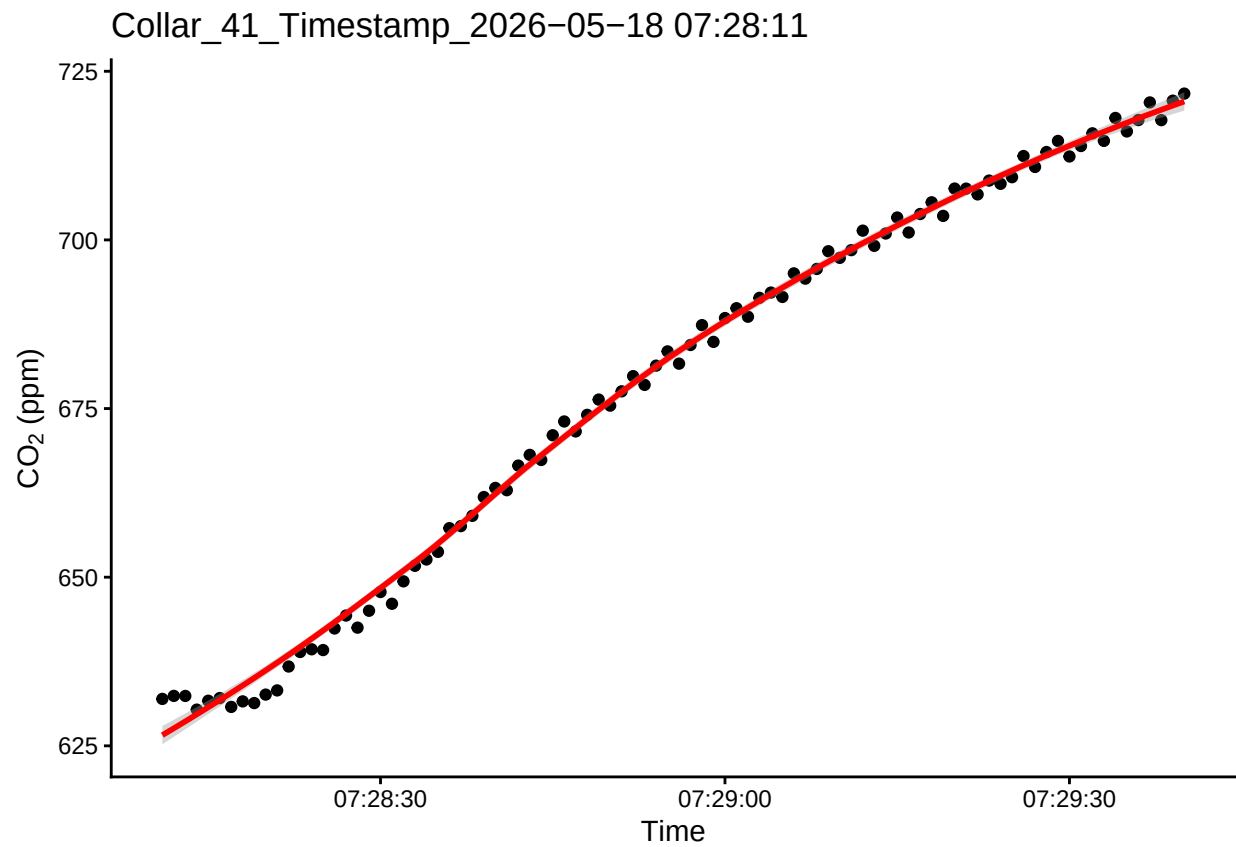
```
## 'geom_smooth()' using method = 'loess' and formula = 'y ~ x'
```



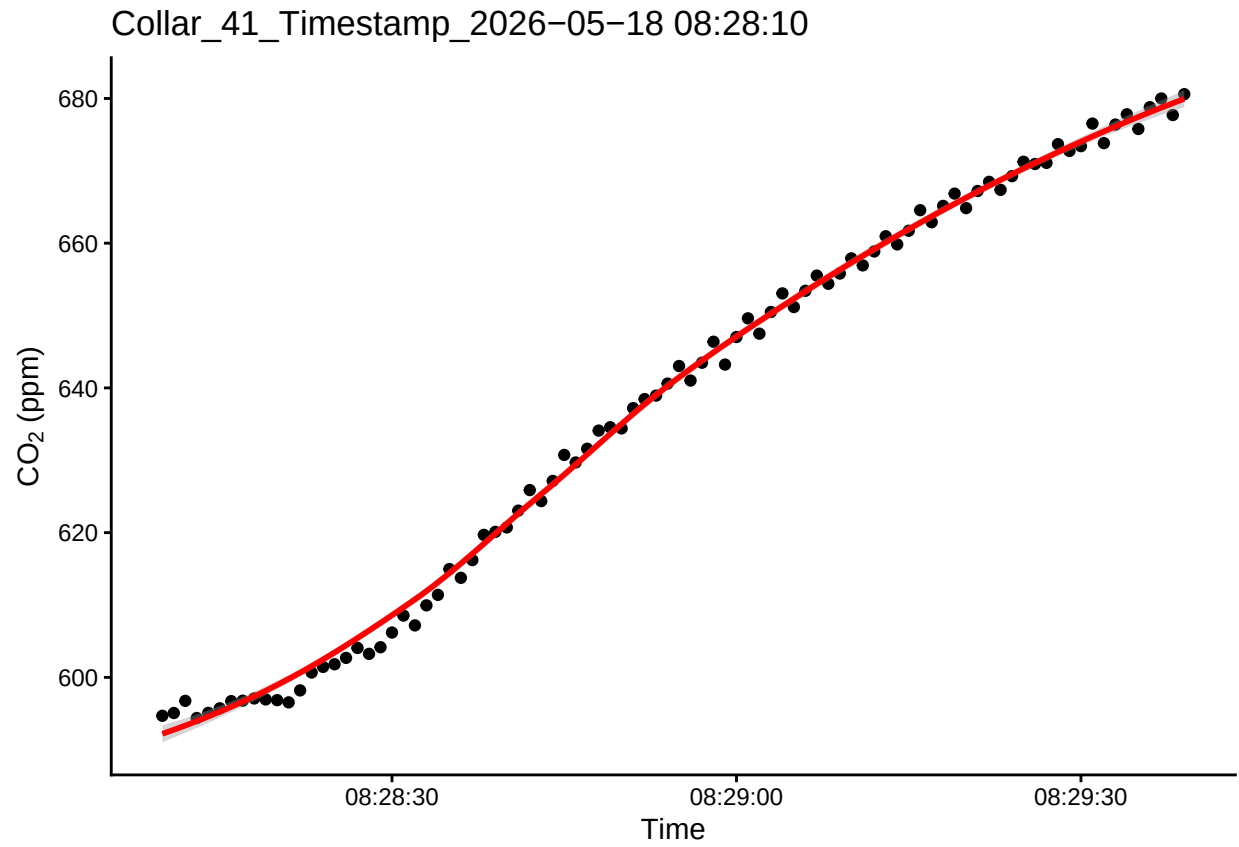
```
## 'geom_smooth()' using method = 'loess' and formula = 'y ~ x'
```



```
## 'geom_smooth()' using method = 'loess' and formula = 'y ~ x'
```

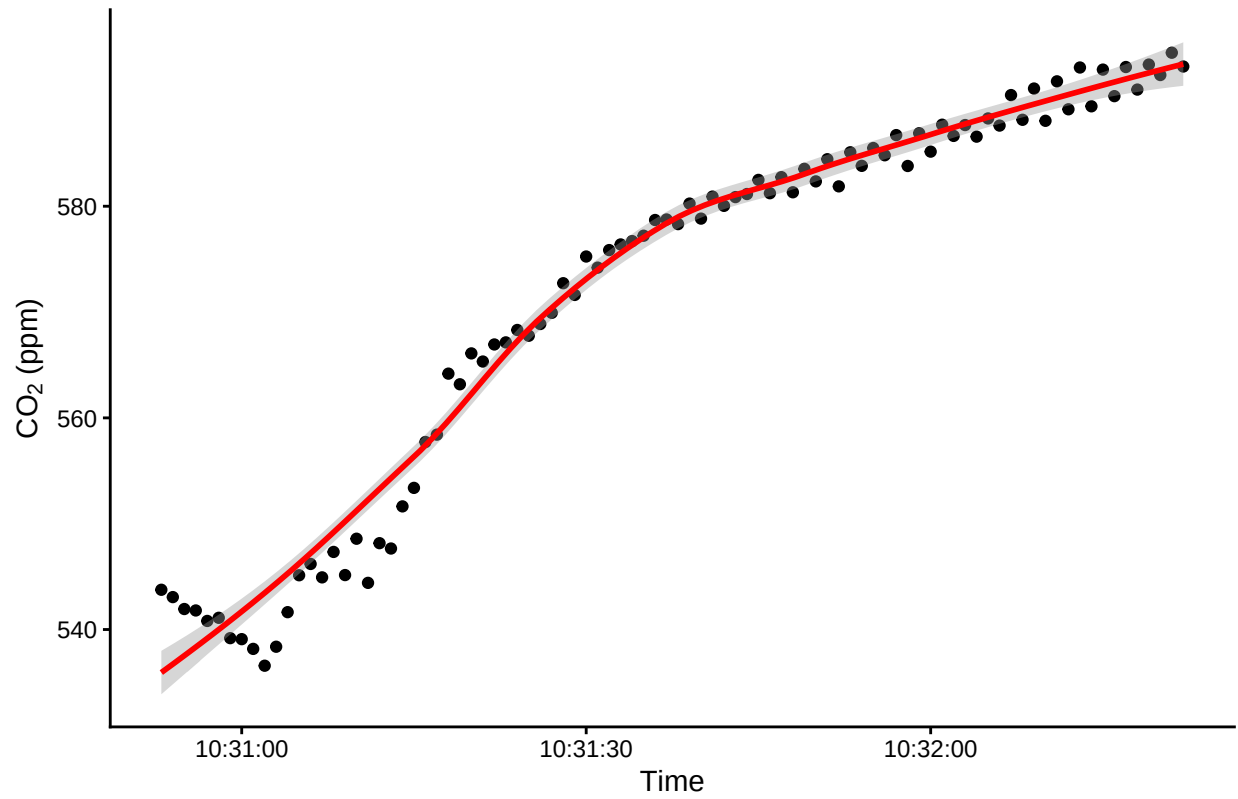


```
## 'geom_smooth()' using method = 'loess' and formula = 'y ~ x'
```

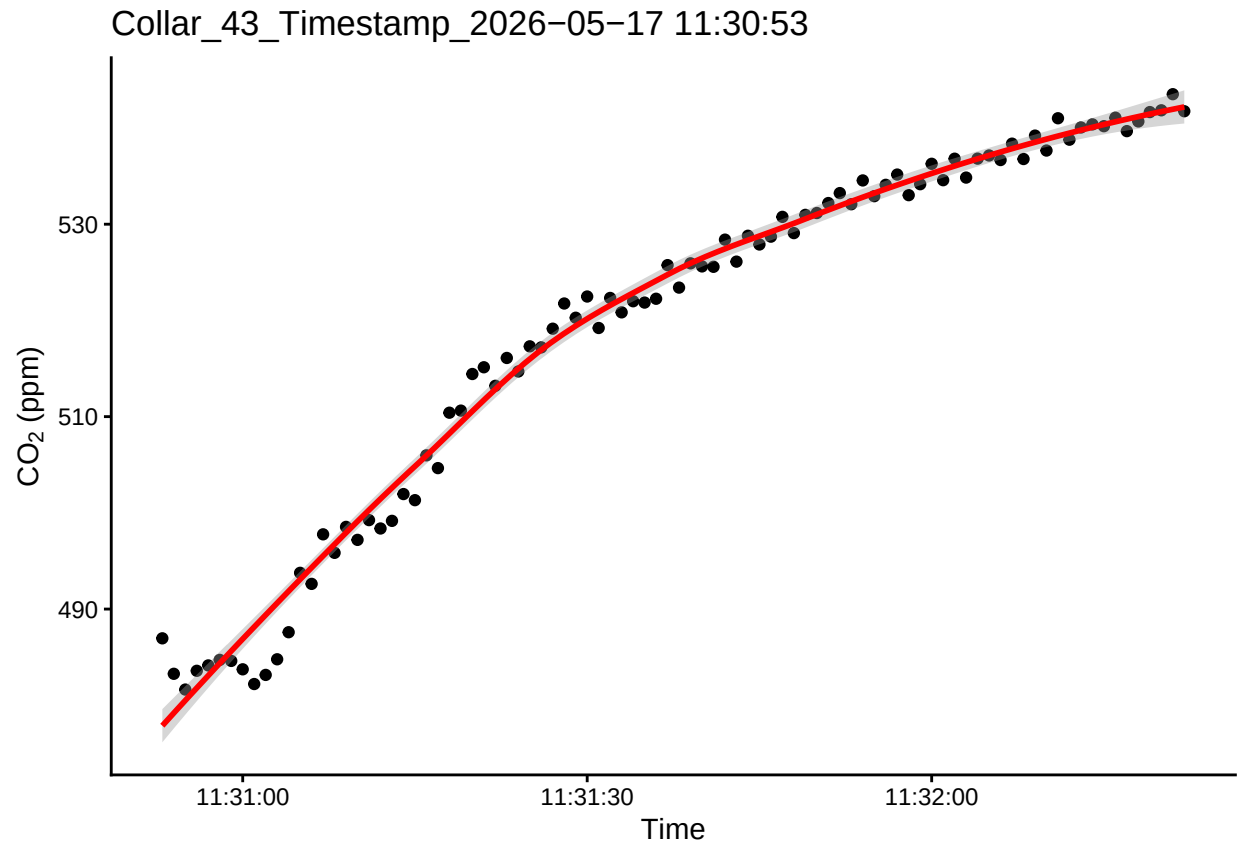


```
## 'geom_smooth()' using method = 'loess' and formula = 'y ~ x'
```

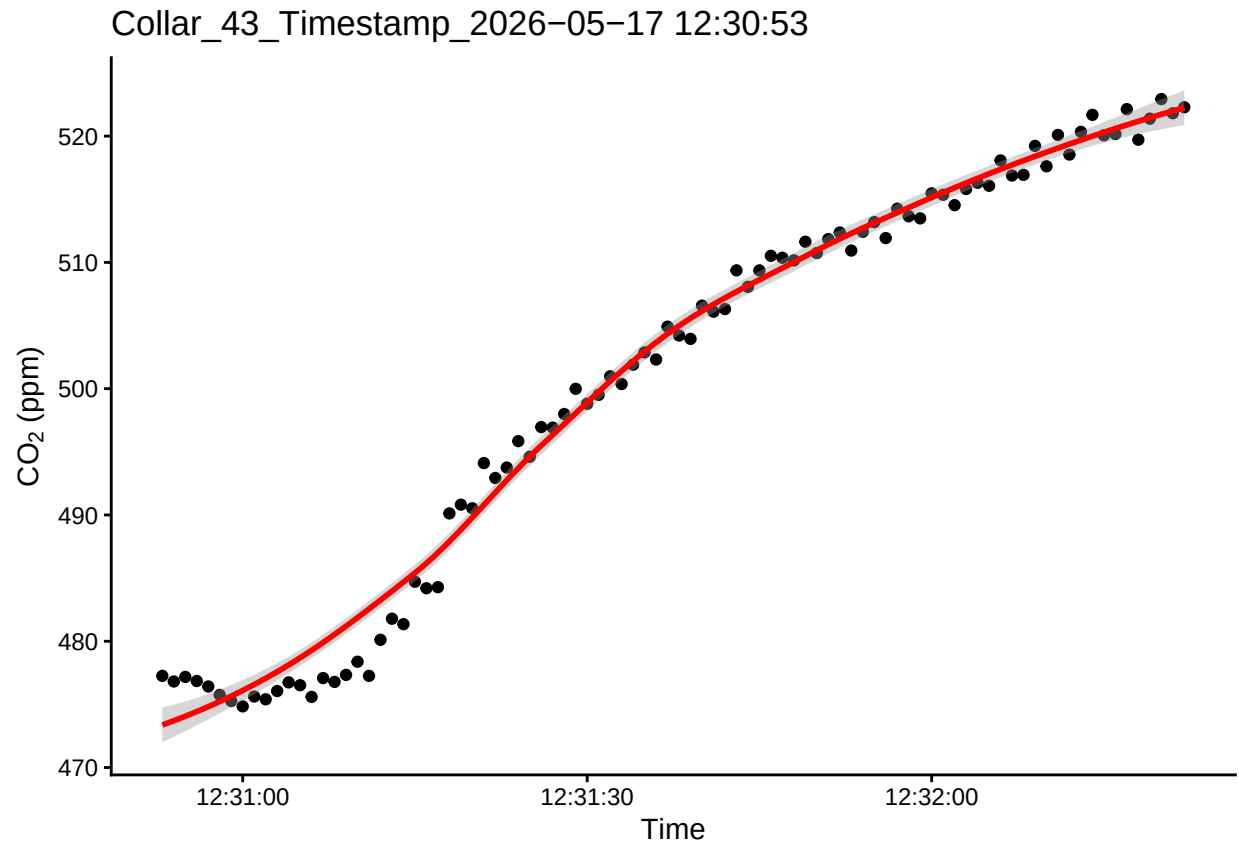
Collar_43_Timestamp_2026-05-17 10:30:53



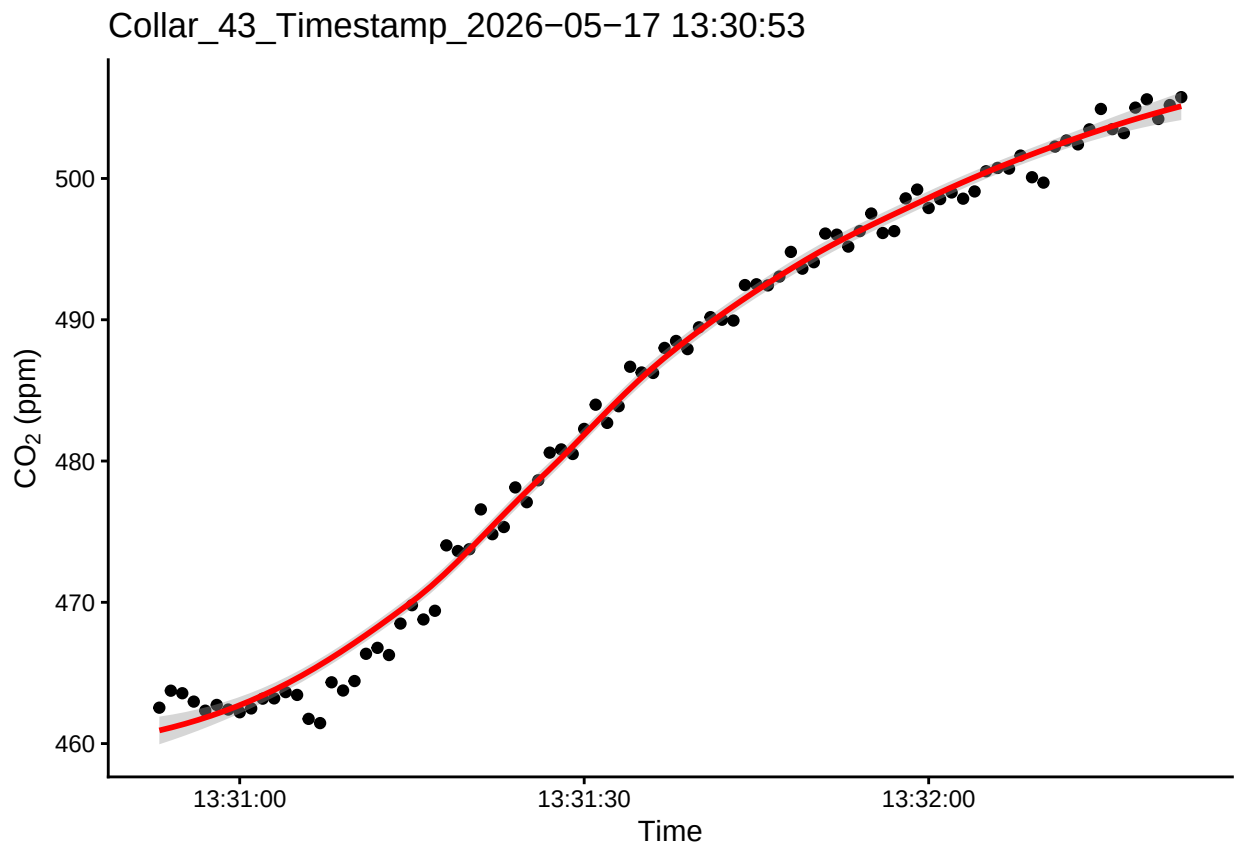
```
## 'geom_smooth()' using method = 'loess' and formula = 'y ~ x'
```

```
## 'geom_smooth()' using method = 'loess' and formula = 'y ~ x'
```

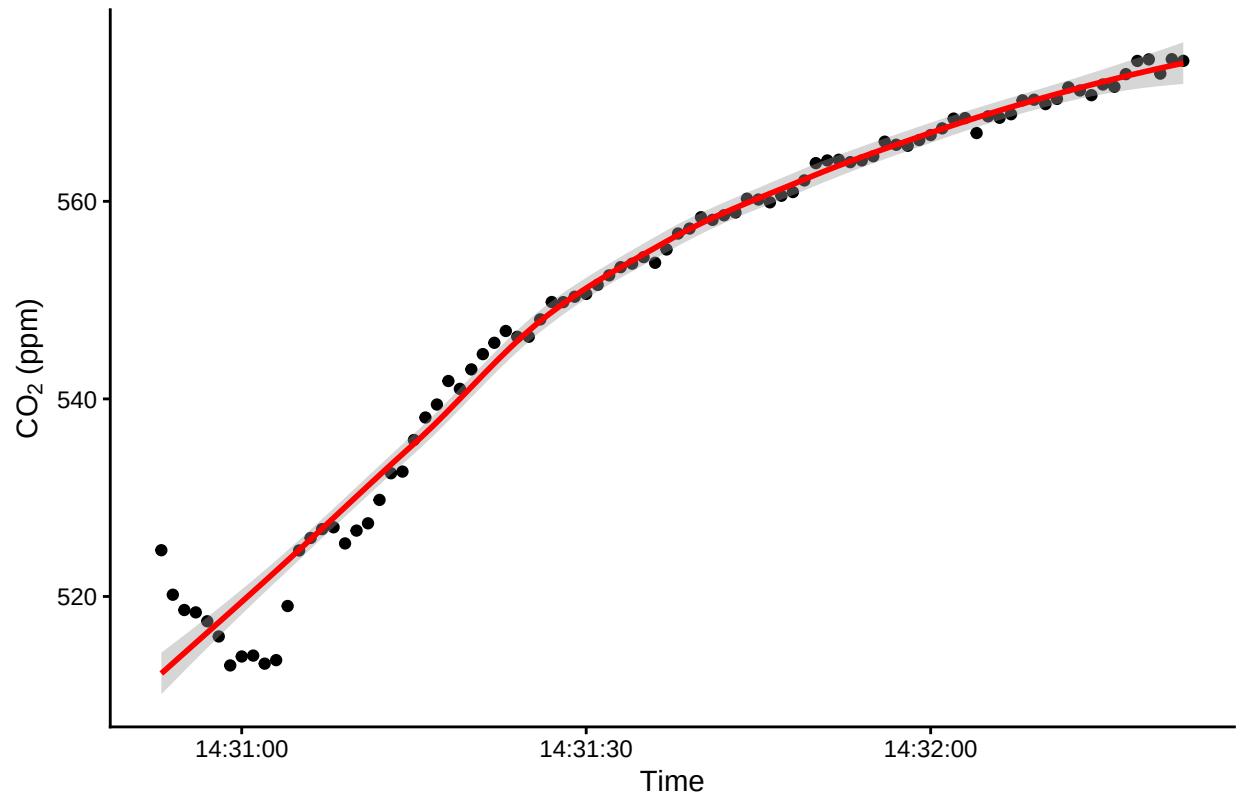


```
## 'geom_smooth()' using method = 'loess' and formula = 'y ~ x'
```



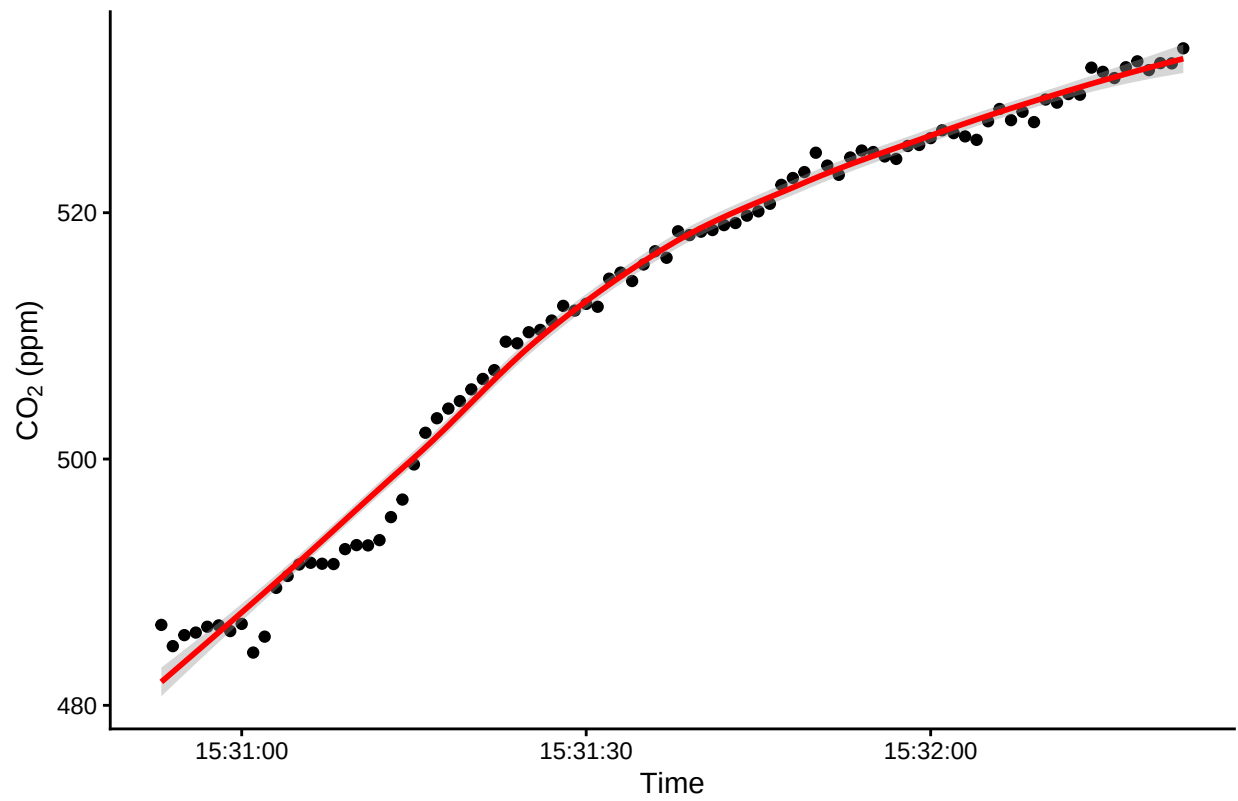
```
## 'geom_smooth()' using method = 'loess' and formula = 'y ~ x'
```

Collar_43_Timestamp_2026-05-17 14:30:53

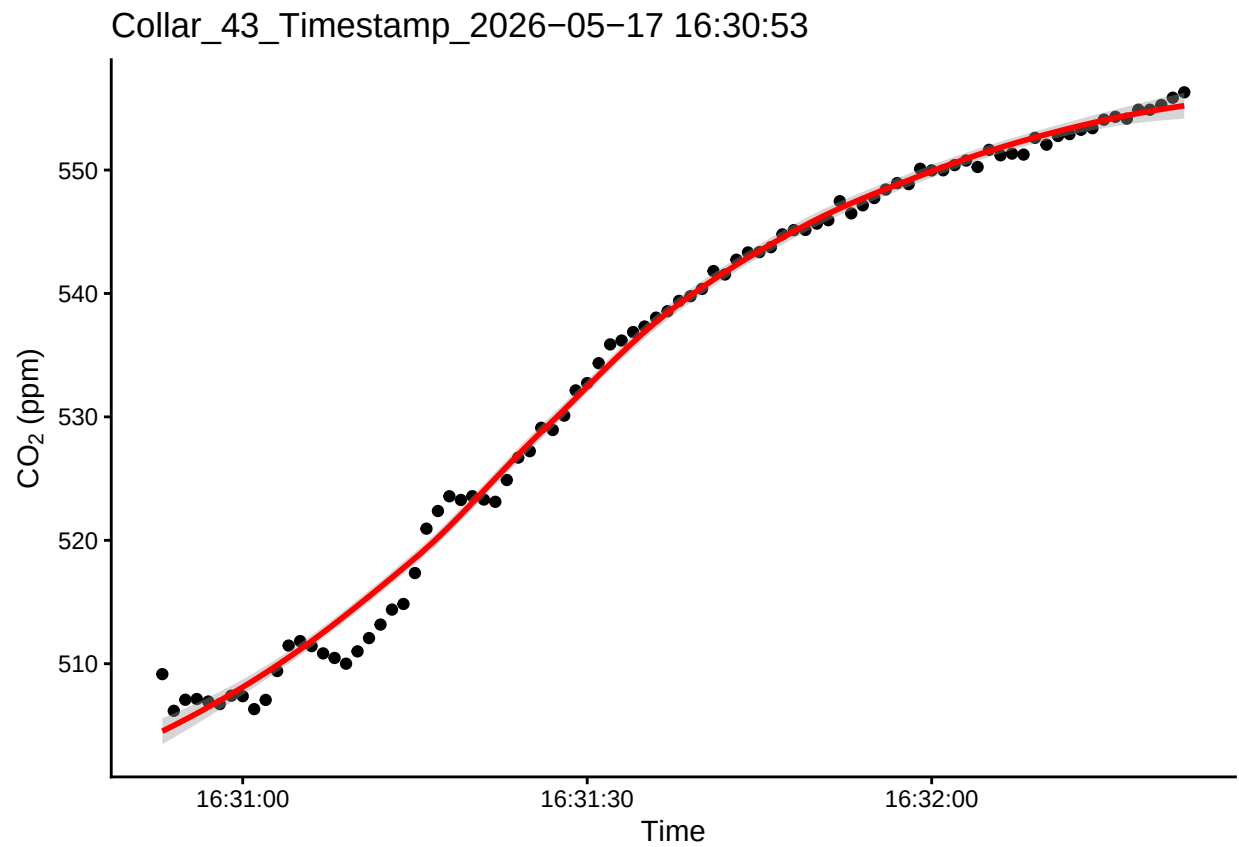


```
## 'geom_smooth()' using method = 'loess' and formula = 'y ~ x'
```

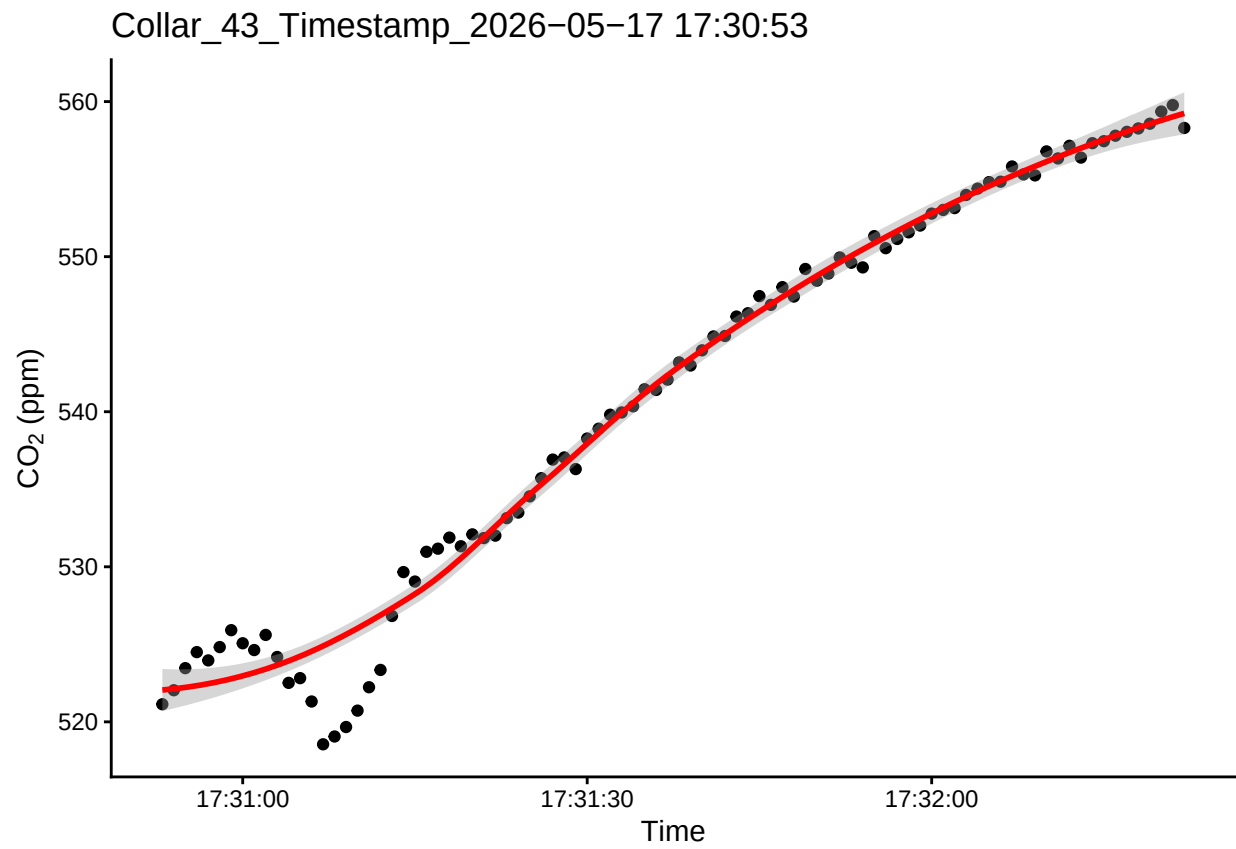
Collar_43_Timestamp_2026-05-17 15:30:53



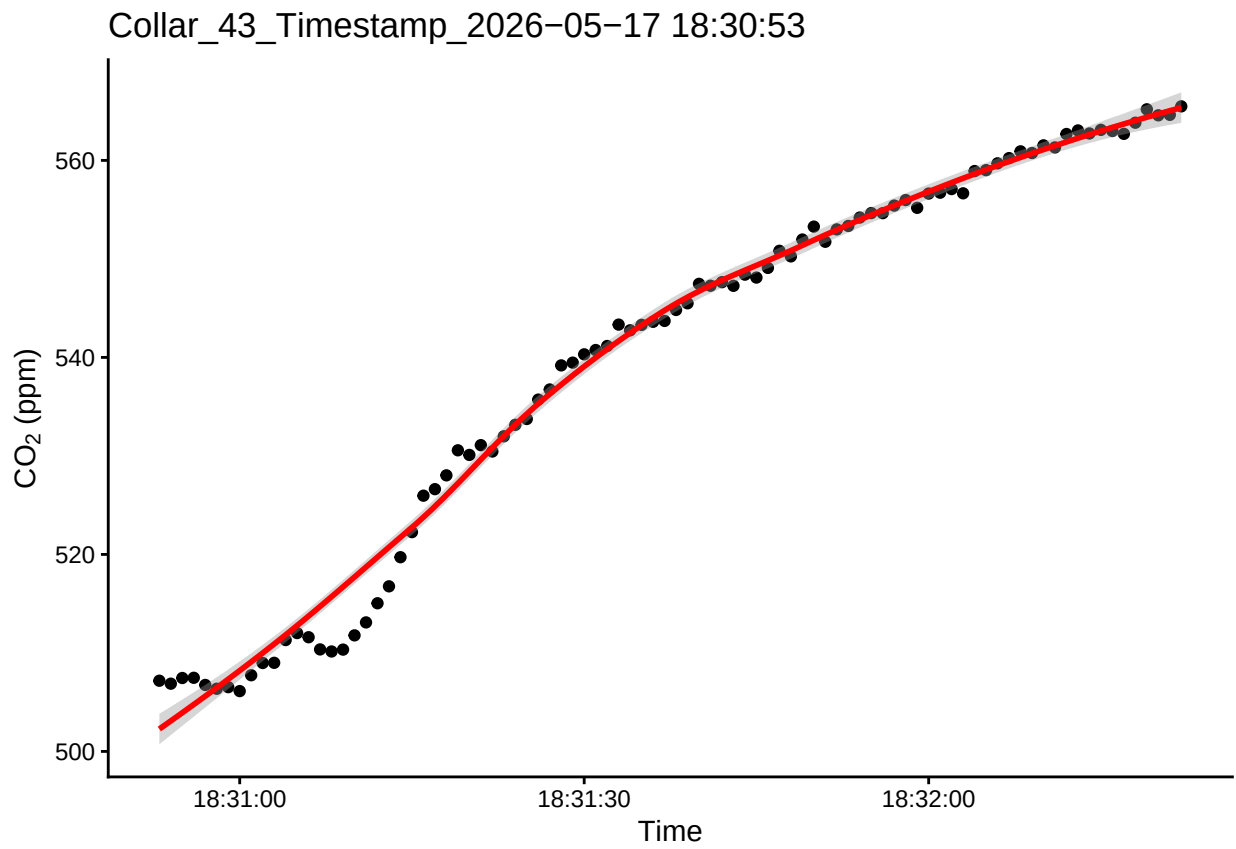
```
## 'geom_smooth()' using method = 'loess' and formula = 'y ~ x'
```



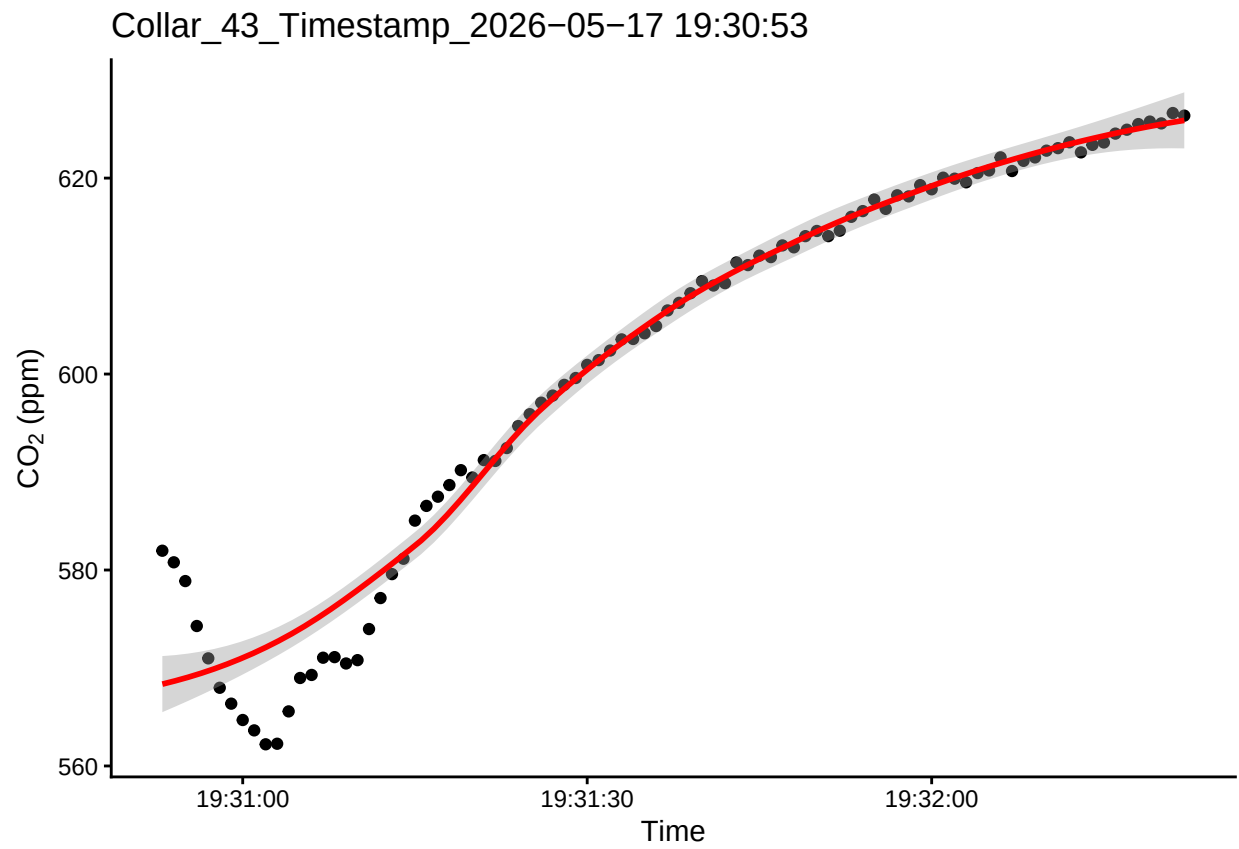
```
## 'geom_smooth()' using method = 'loess' and formula = 'y ~ x'
```



```
## 'geom_smooth()' using method = 'loess' and formula = 'y ~ x'
```

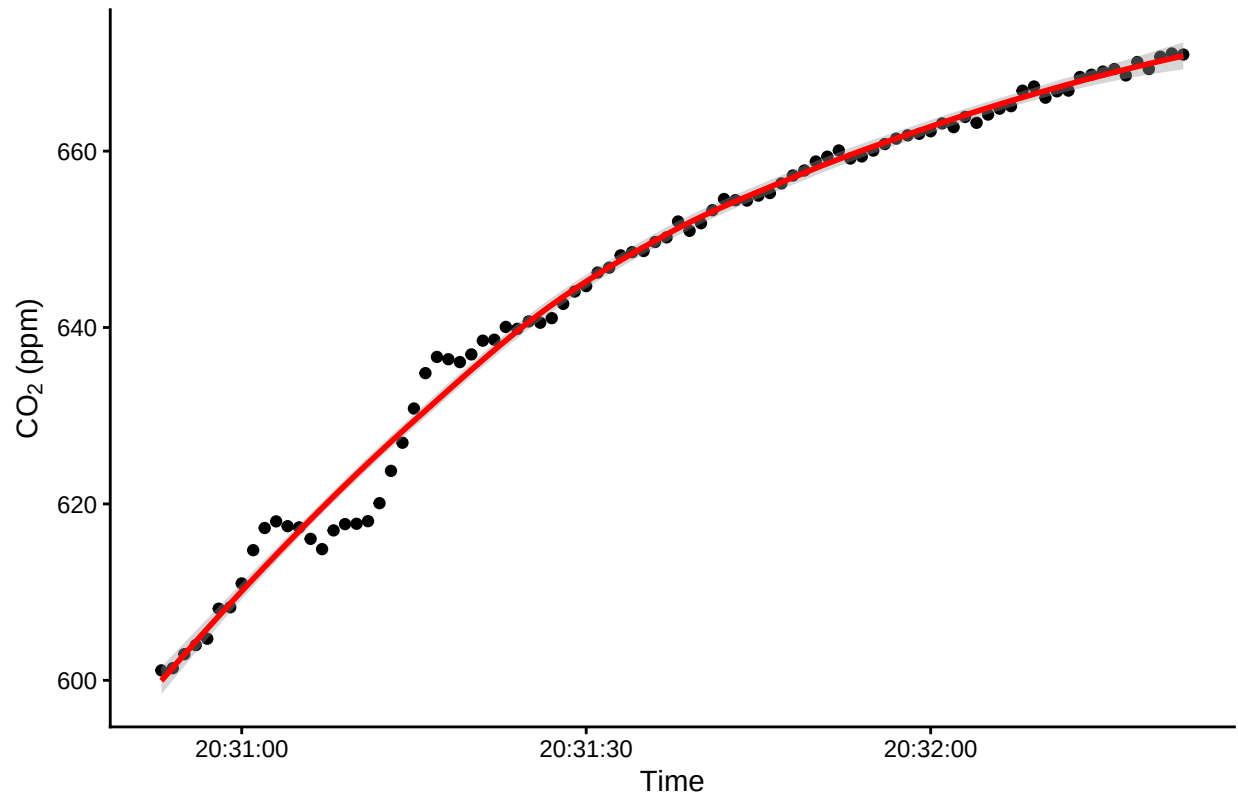


```
## 'geom_smooth()' using method = 'loess' and formula = 'y ~ x'
```

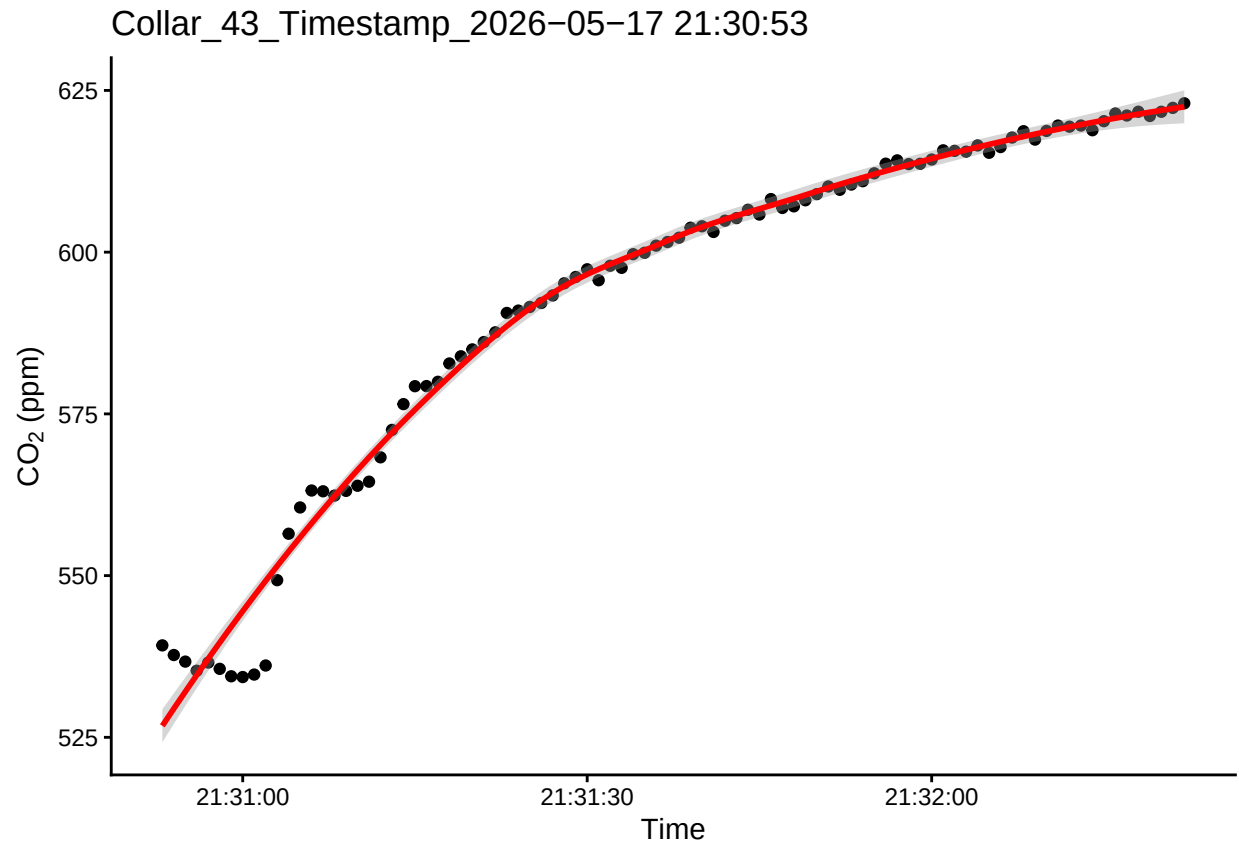



```
## 'geom_smooth()' using method = 'loess' and formula = 'y ~ x'
```

Collar_43_Timestamp_2026-05-17 20:30:53

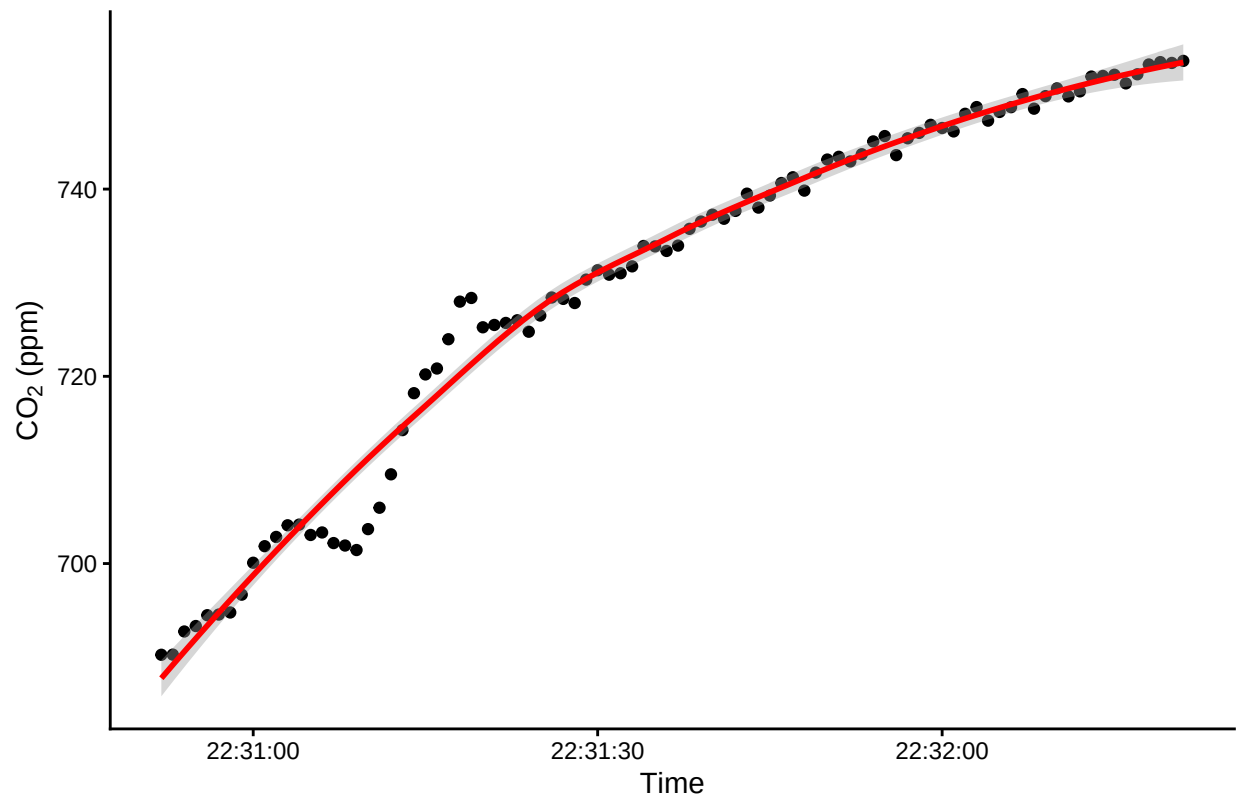


```
## 'geom_smooth()' using method = 'loess' and formula = 'y ~ x'
```

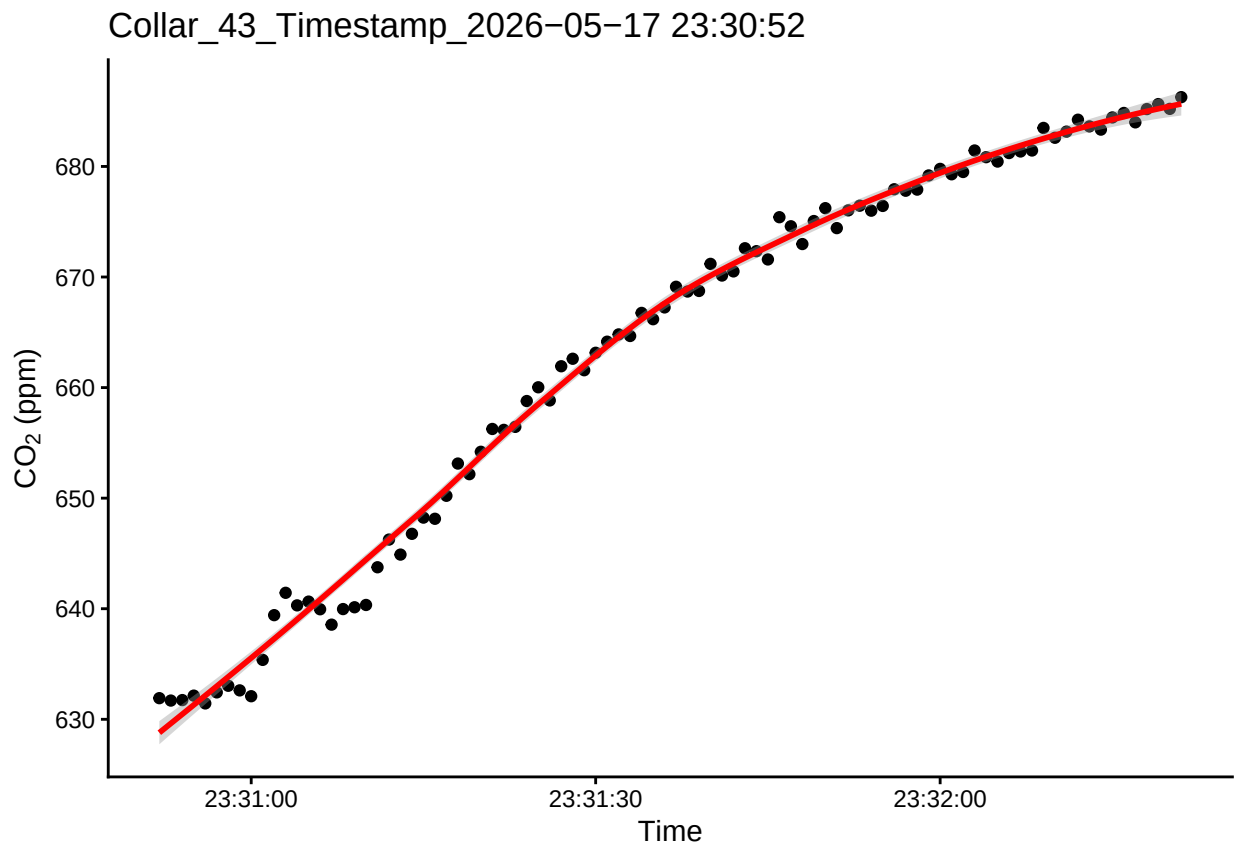


```
## 'geom_smooth()' using method = 'loess' and formula = 'y ~ x'
```

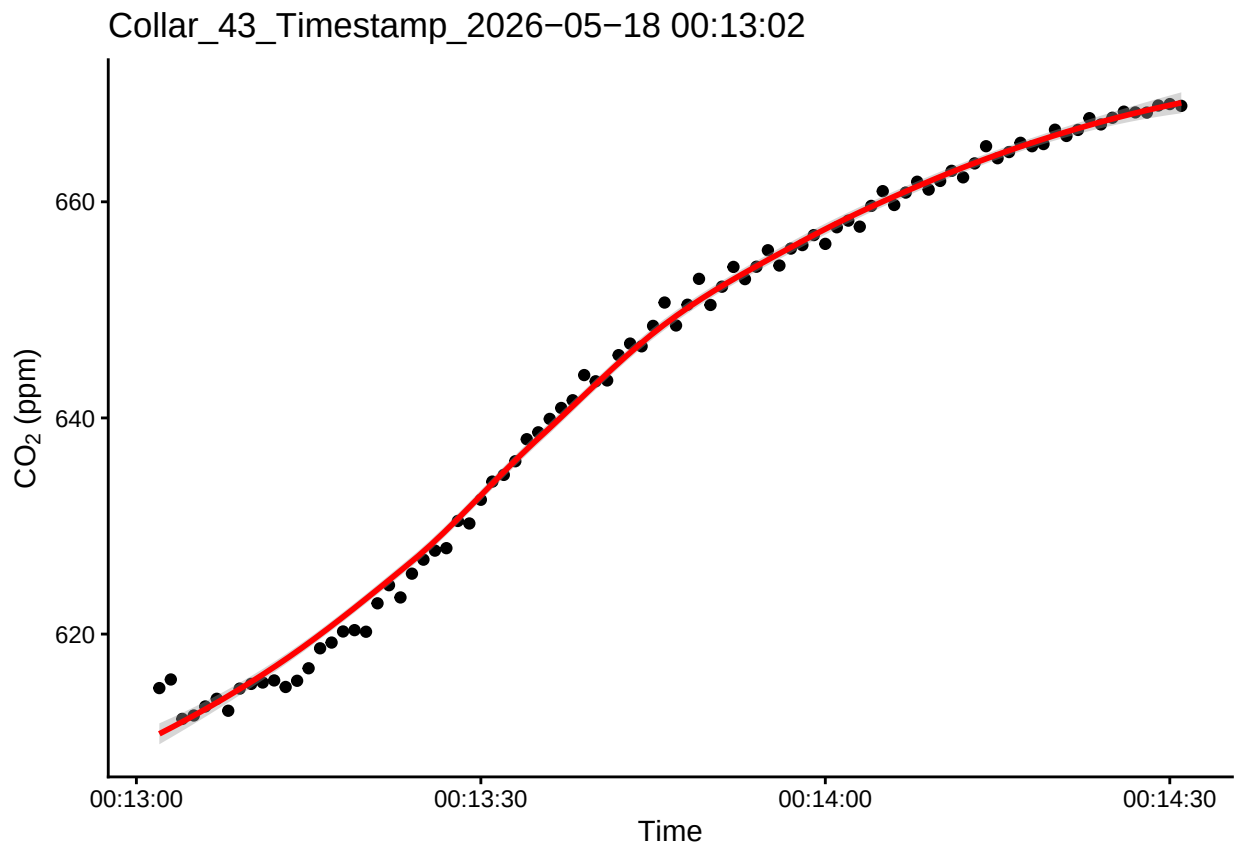
Collar_43_Timestamp_2026-05-17 22:30:52



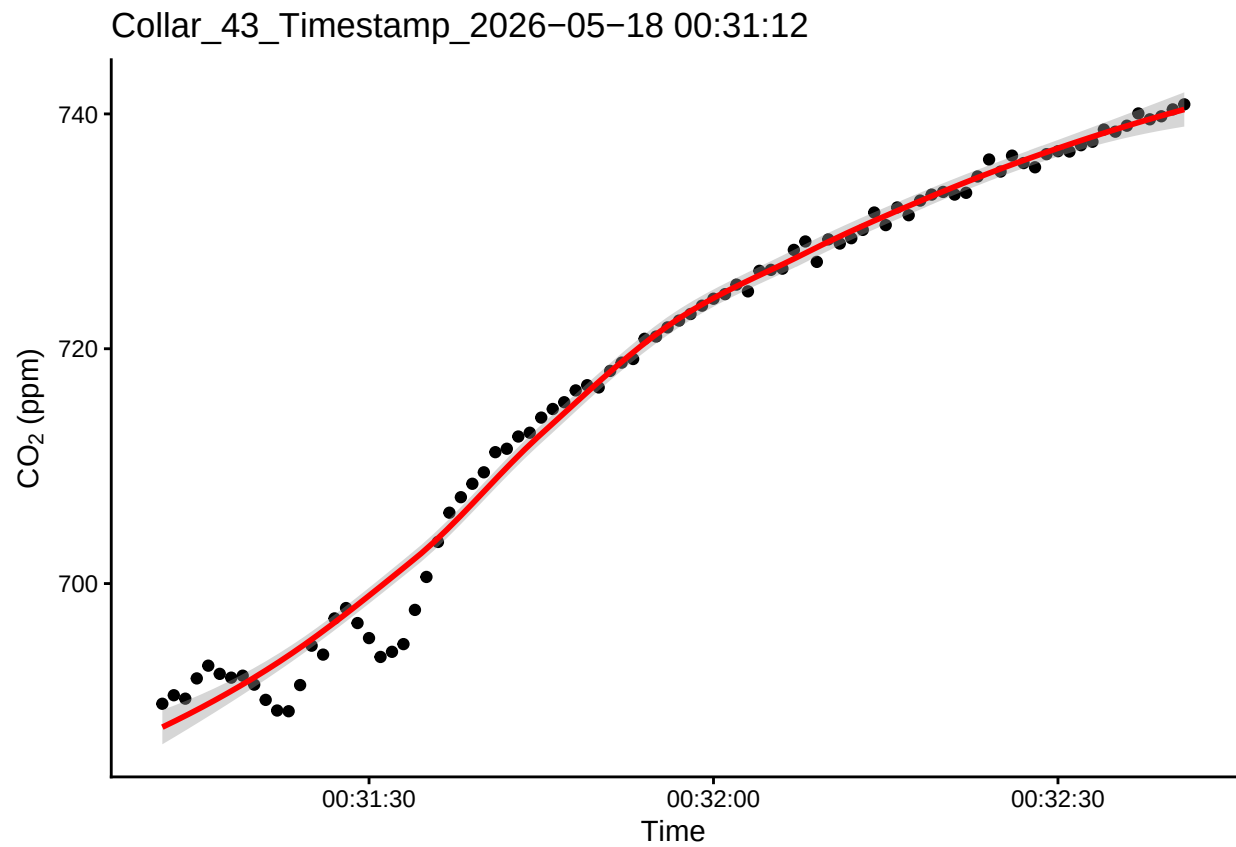
```
## 'geom_smooth()' using method = 'loess' and formula = 'y ~ x'
```



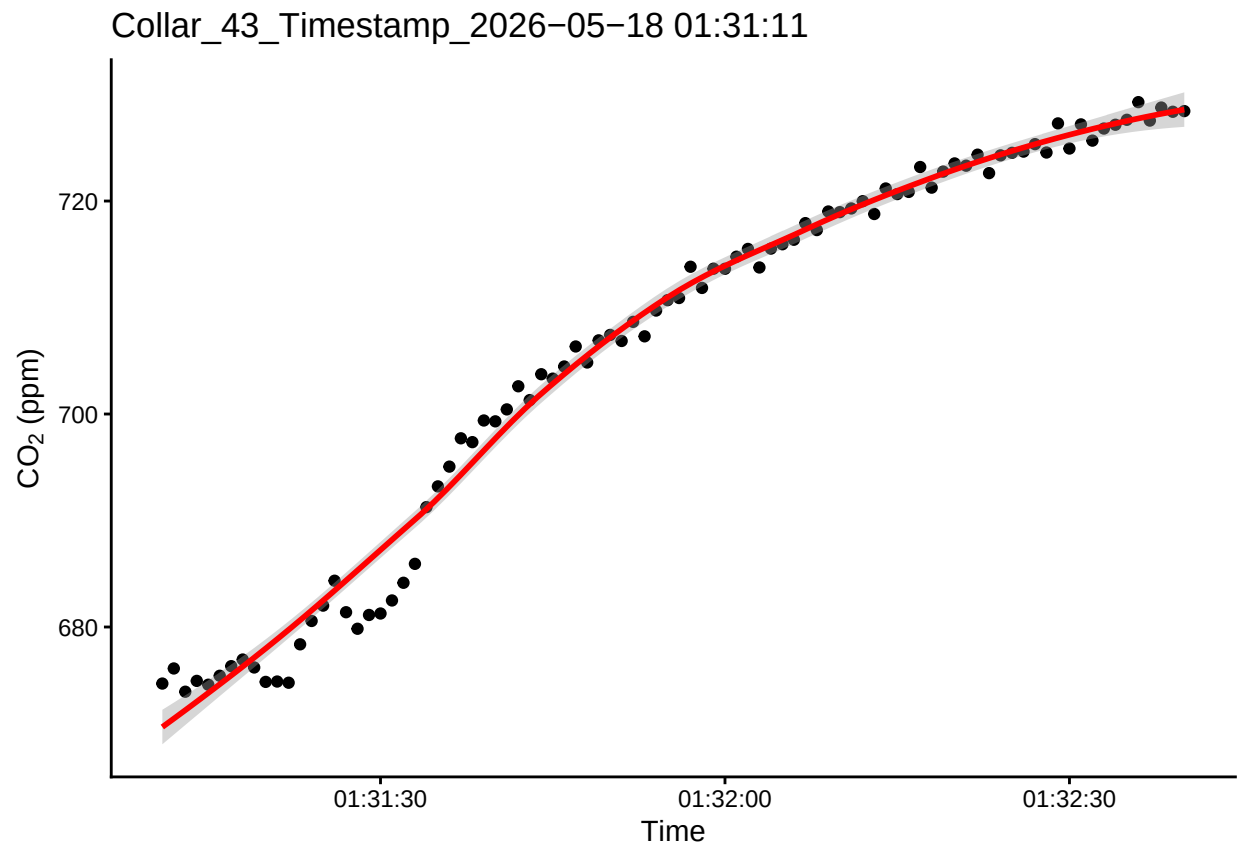
```
## 'geom_smooth()' using method = 'loess' and formula = 'y ~ x'
```



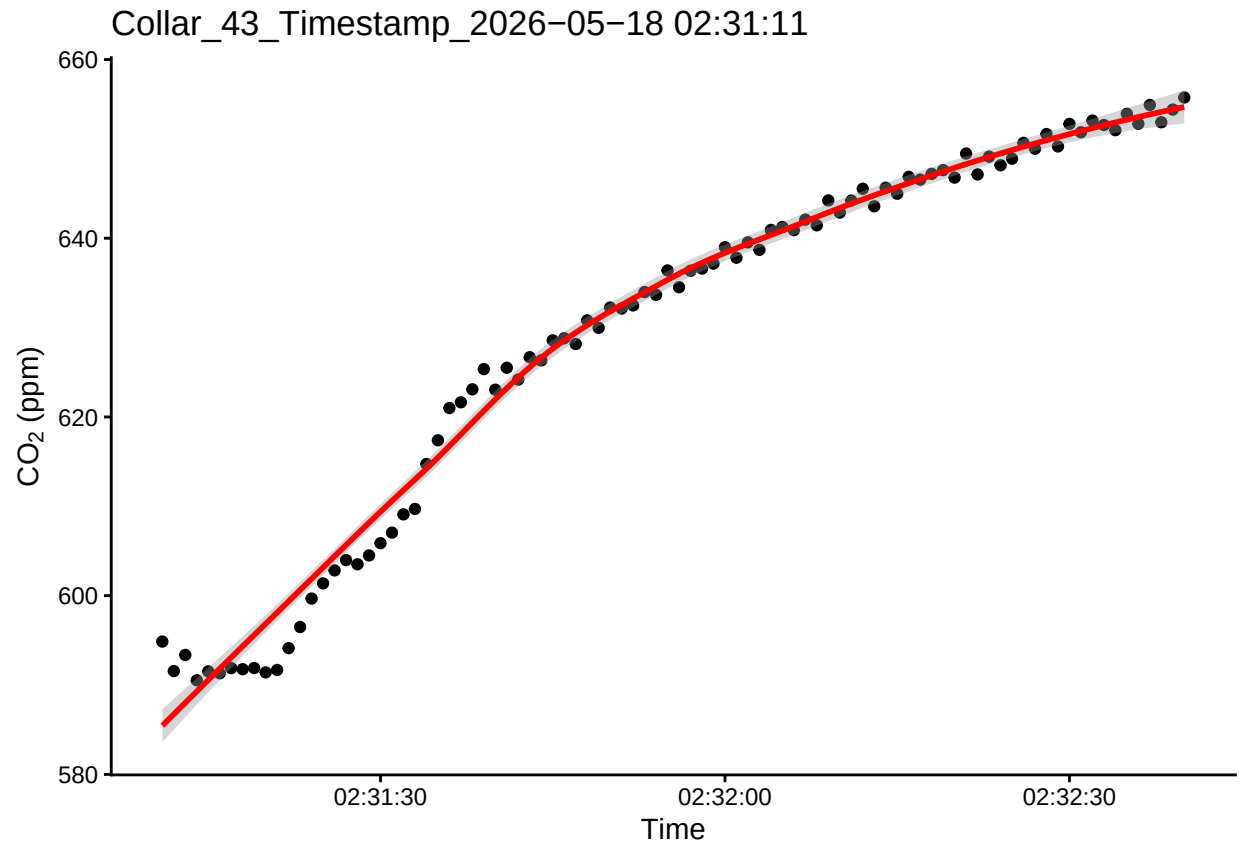
```
## 'geom_smooth()' using method = 'loess' and formula = 'y ~ x'
```



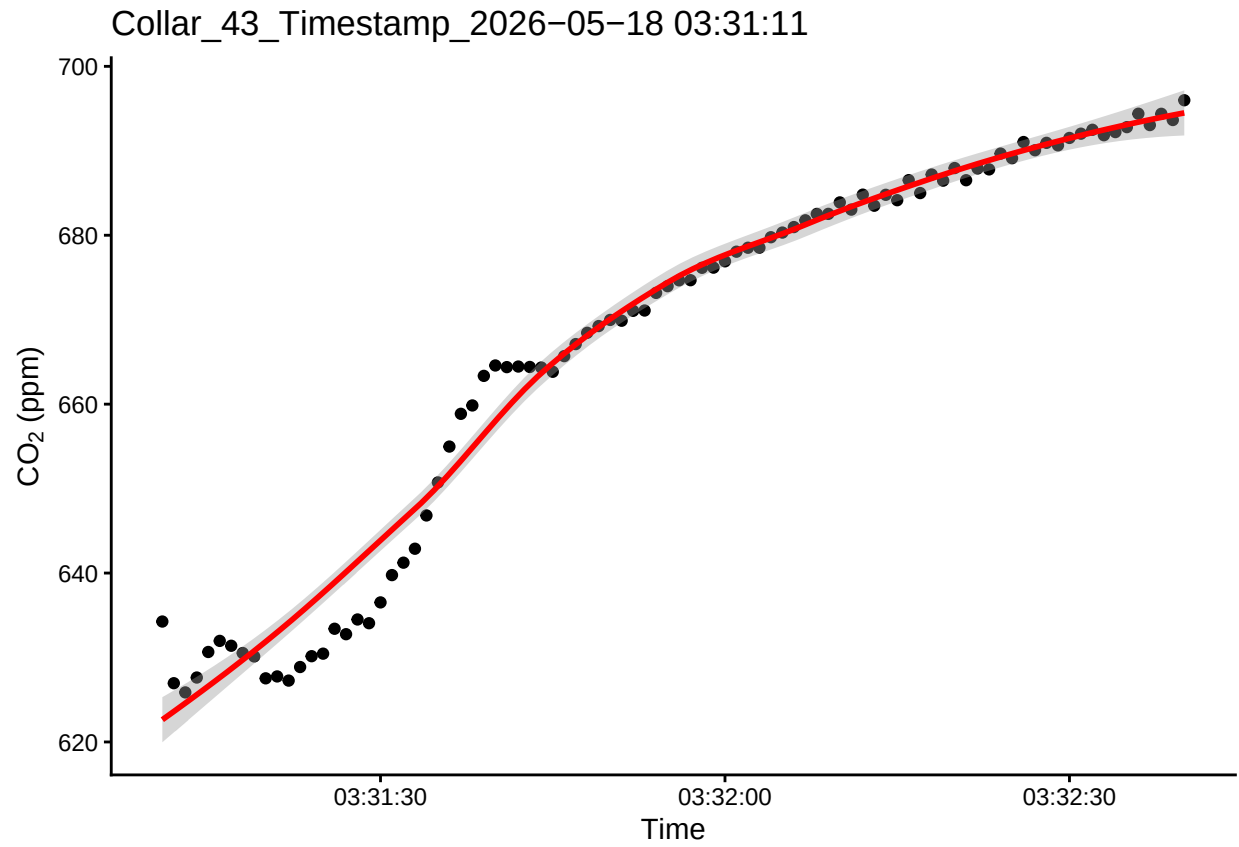
```
## 'geom_smooth()' using method = 'loess' and formula = 'y ~ x'
```



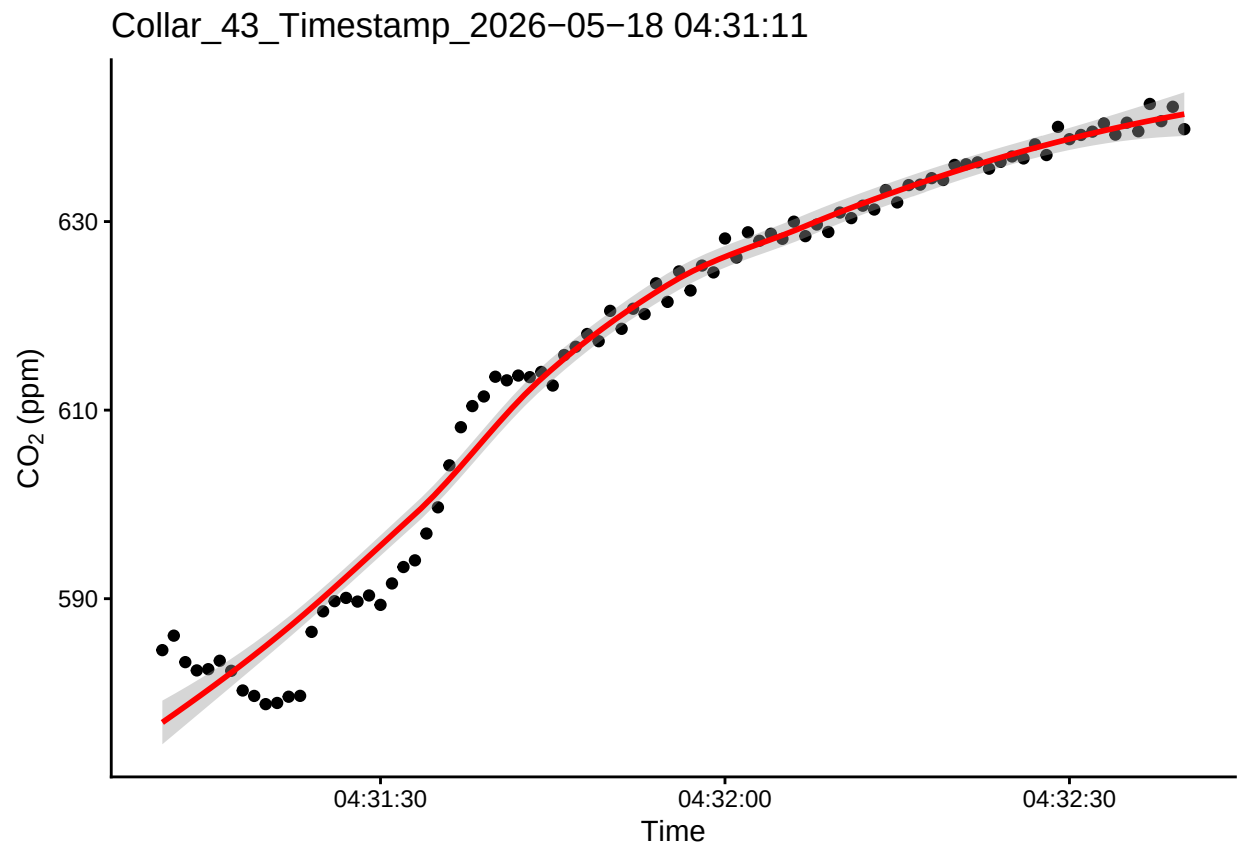
```
## 'geom_smooth()' using method = 'loess' and formula = 'y ~ x'
```

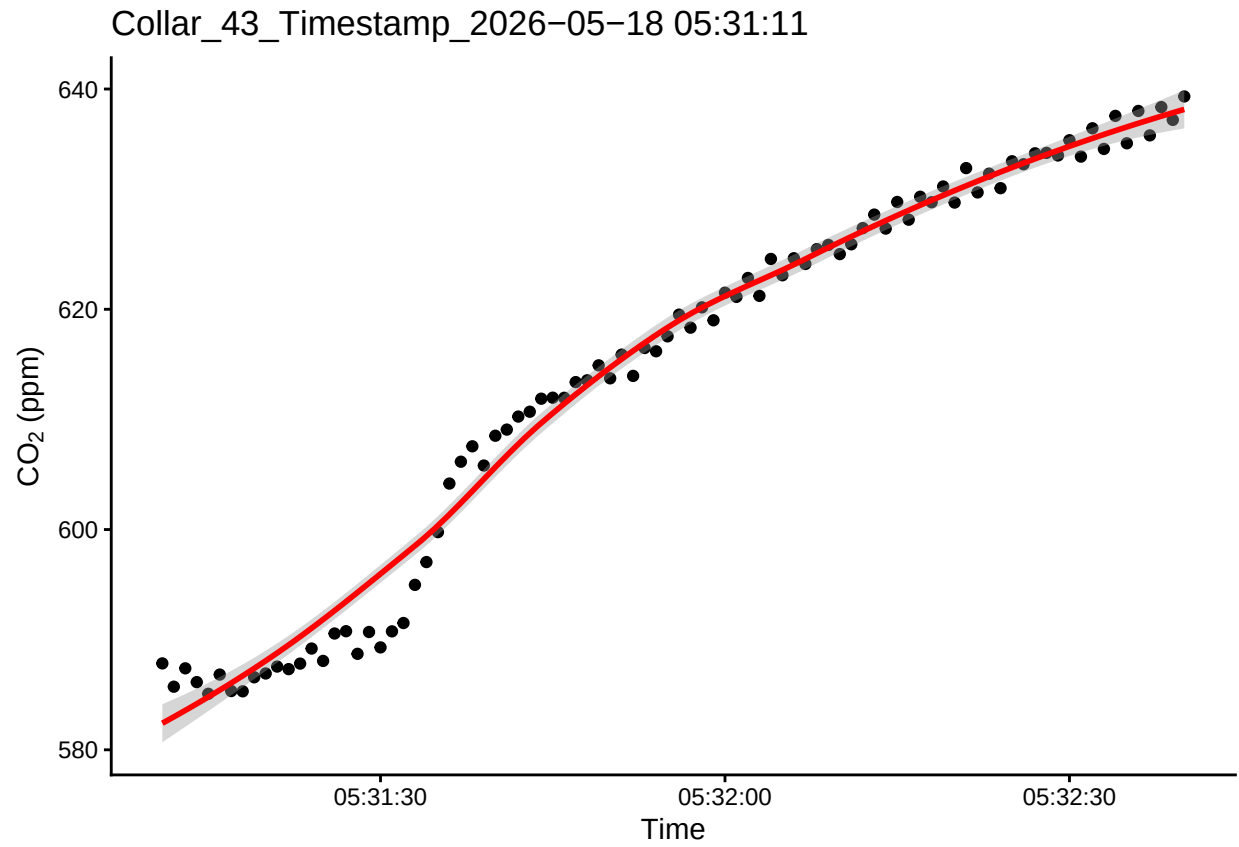
```
## 'geom_smooth()' using method = 'loess' and formula = 'y ~ x'
```



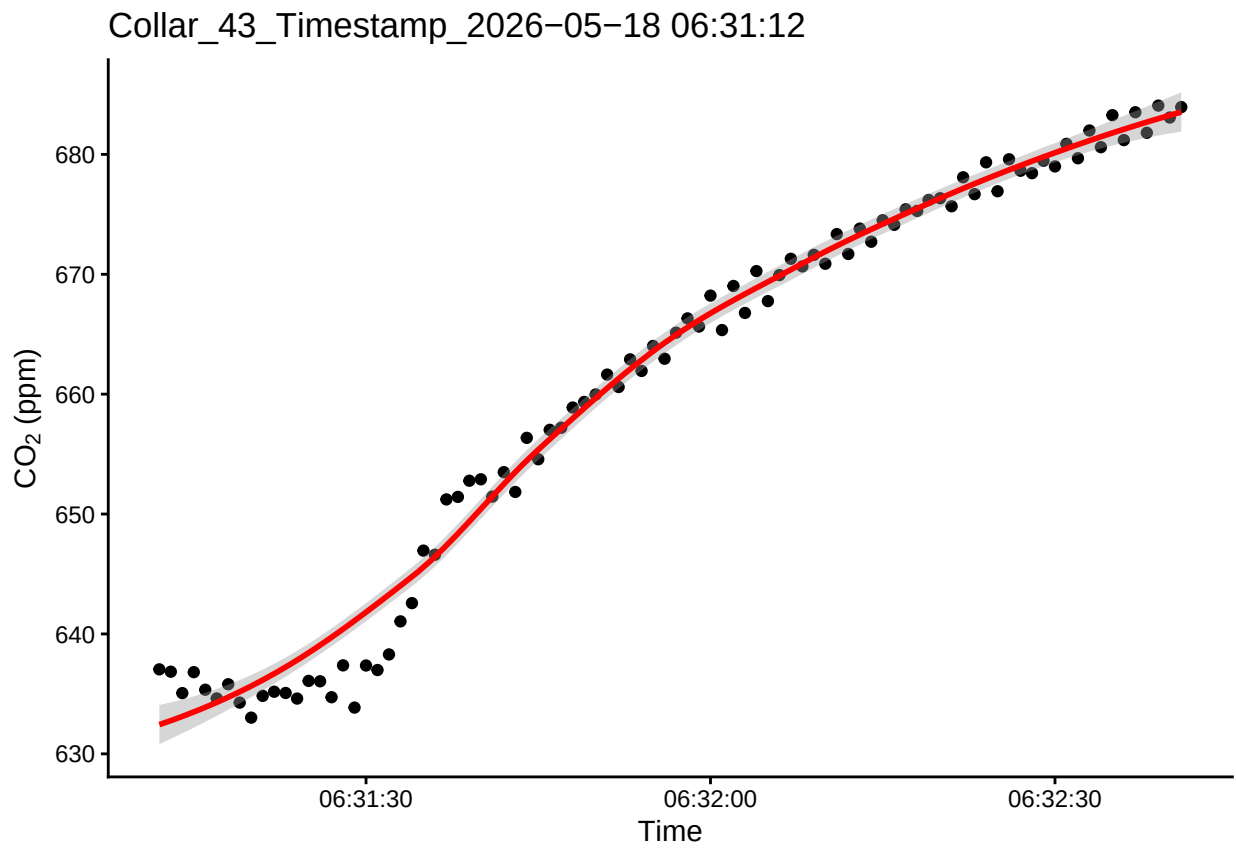
```
## 'geom_smooth()' using method = 'loess' and formula = 'y ~ x'
```



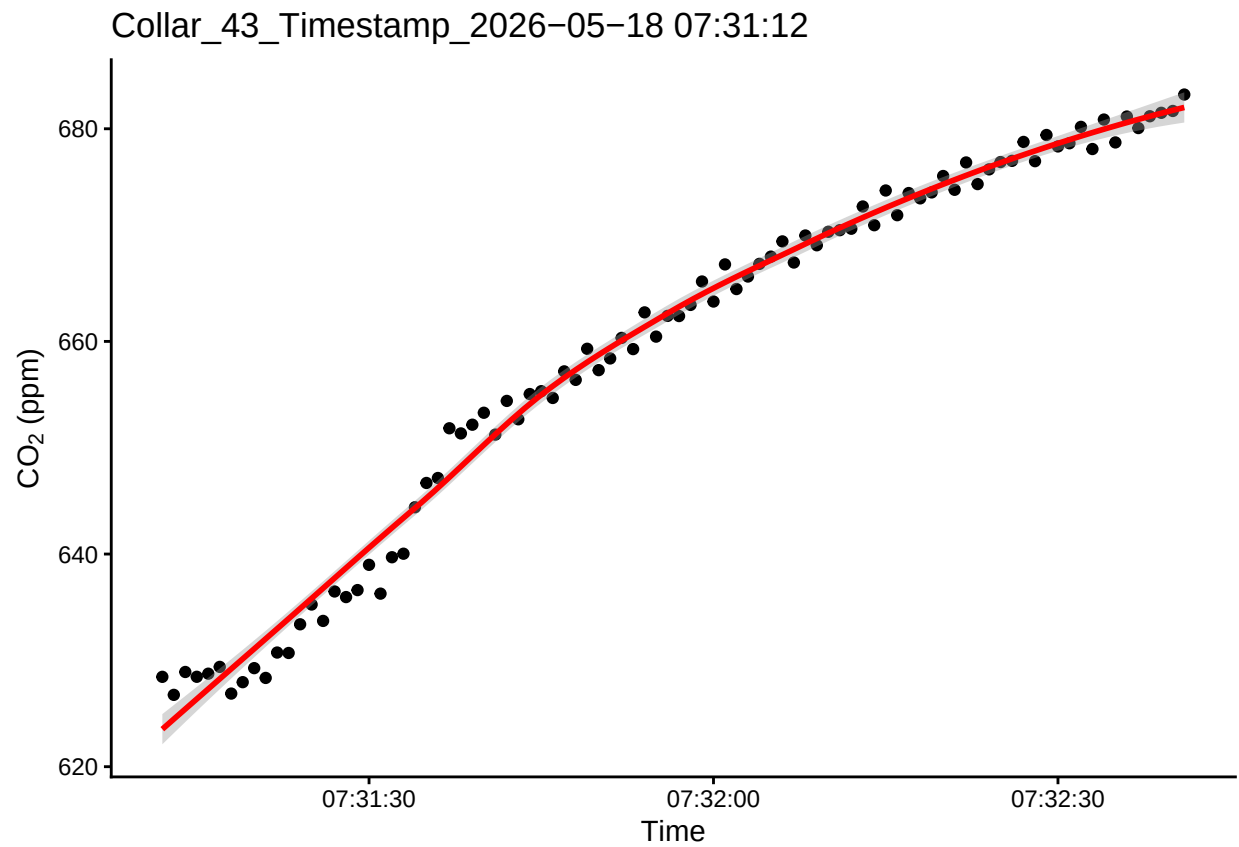
```
## 'geom_smooth()' using method = 'loess' and formula = 'y ~ x'
```



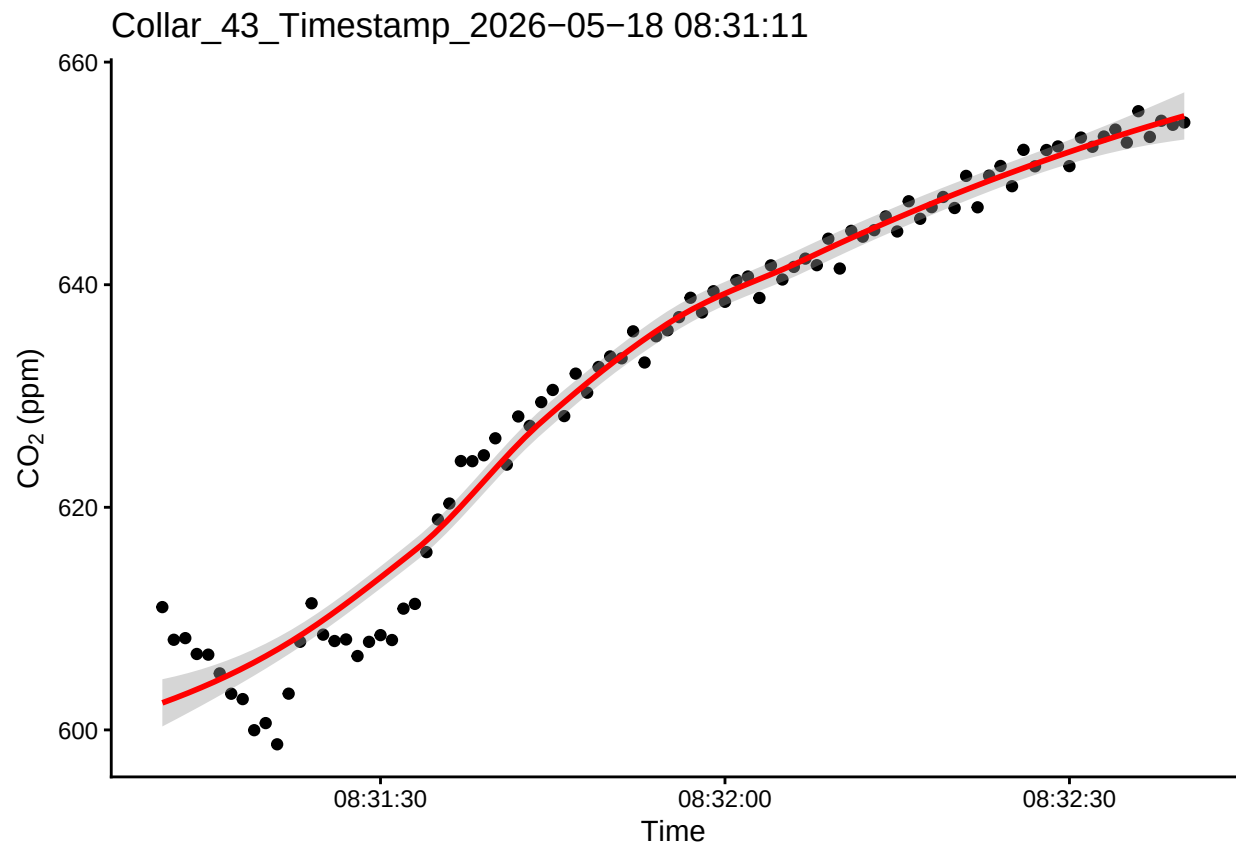
```
## 'geom_smooth()' using method = 'loess' and formula = 'y ~ x'
```



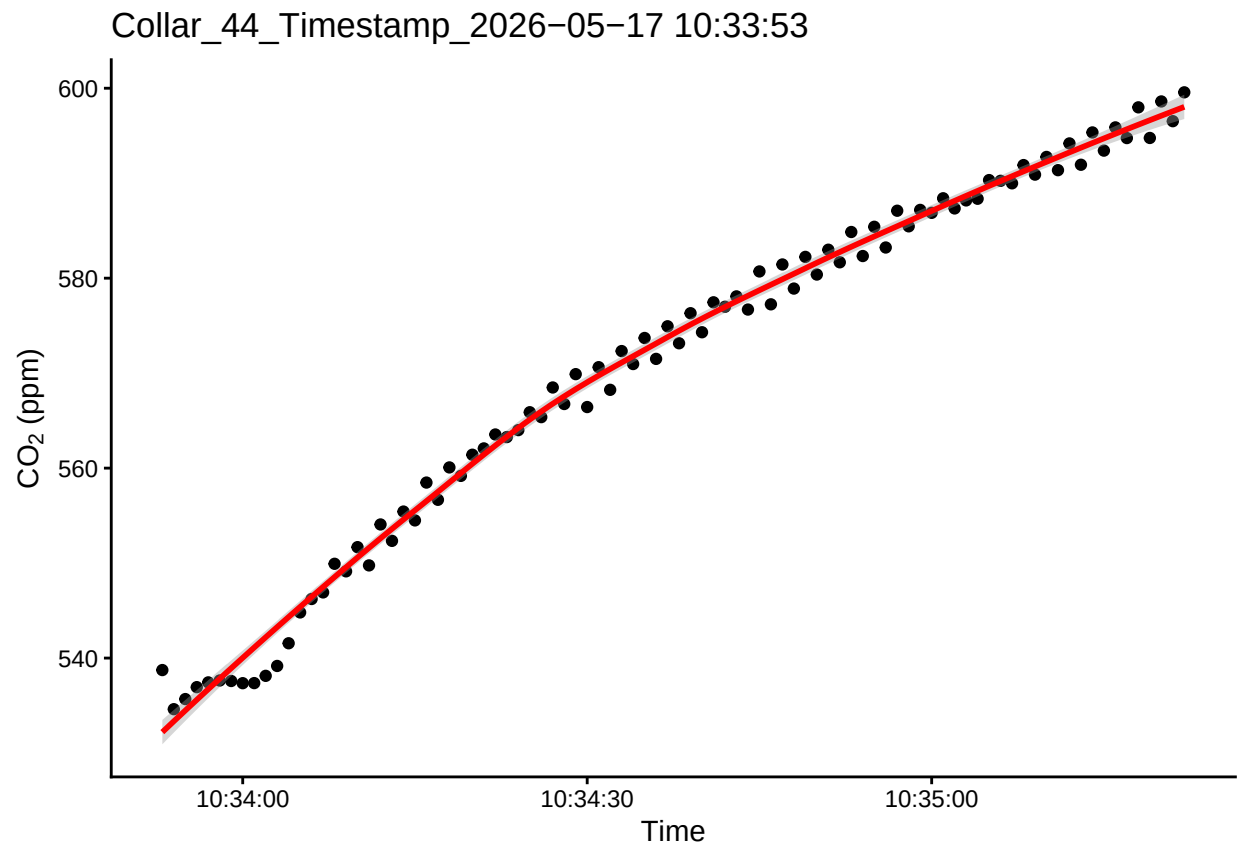
```
## 'geom_smooth()' using method = 'loess' and formula = 'y ~ x'
```



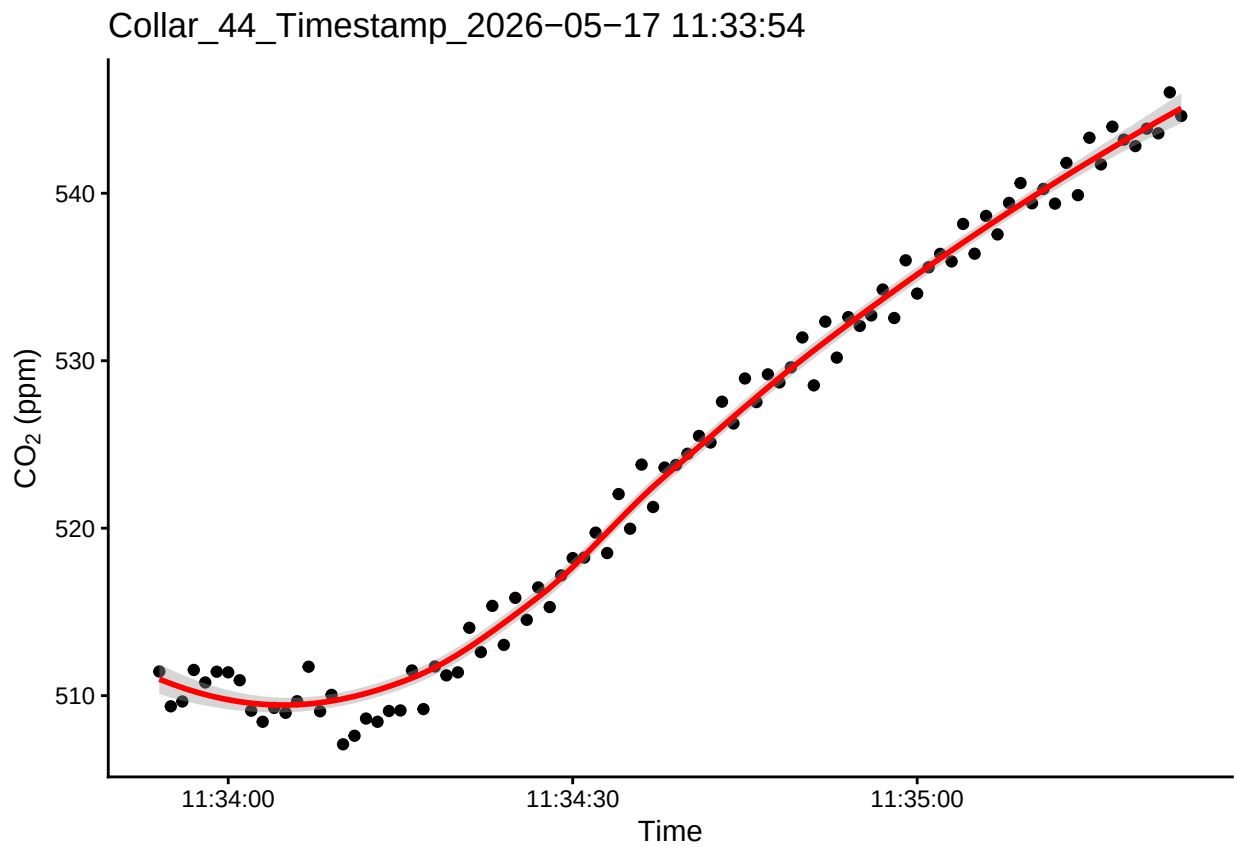
```
## 'geom_smooth()' using method = 'loess' and formula = 'y ~ x'
```



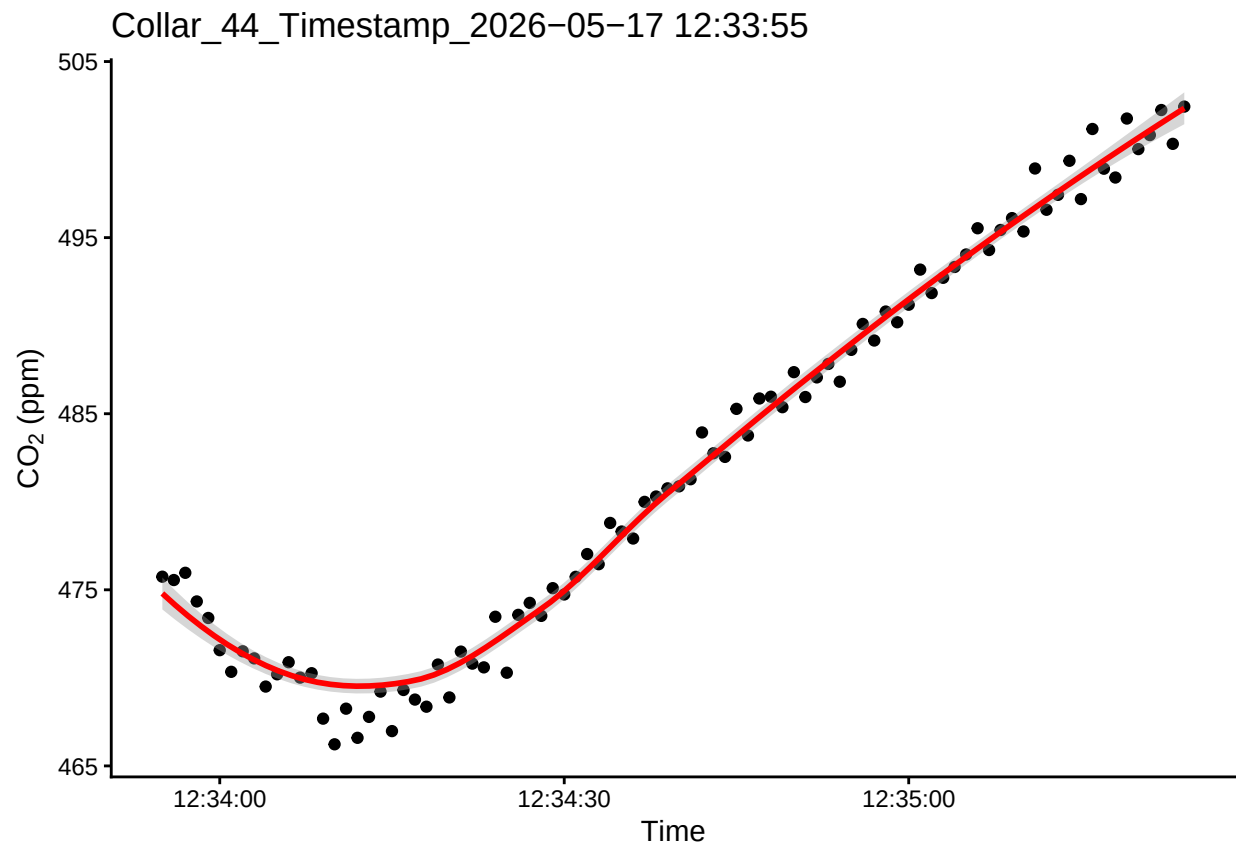
```
## 'geom_smooth()' using method = 'loess' and formula = 'y ~ x'
```



```
## 'geom_smooth()' using method = 'loess' and formula = 'y ~ x'
```

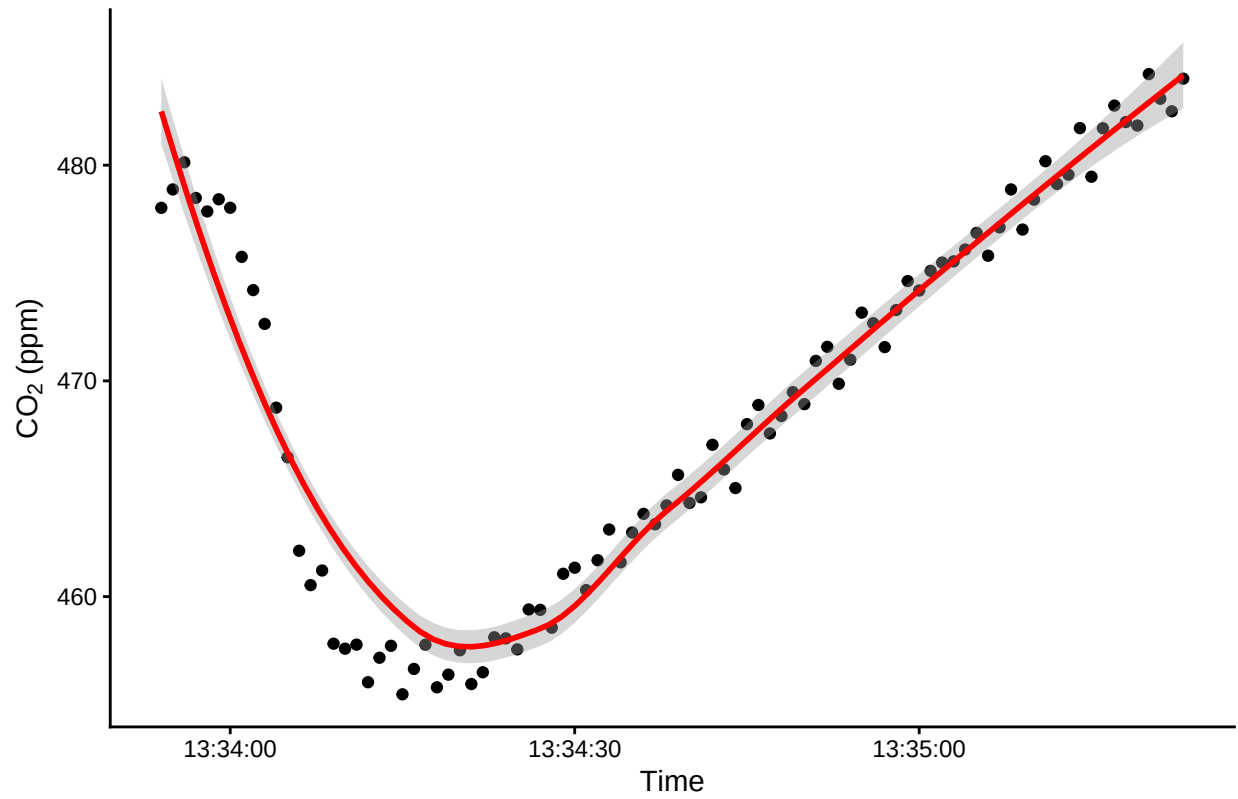



```
## 'geom_smooth()' using method = 'loess' and formula = 'y ~ x'
```

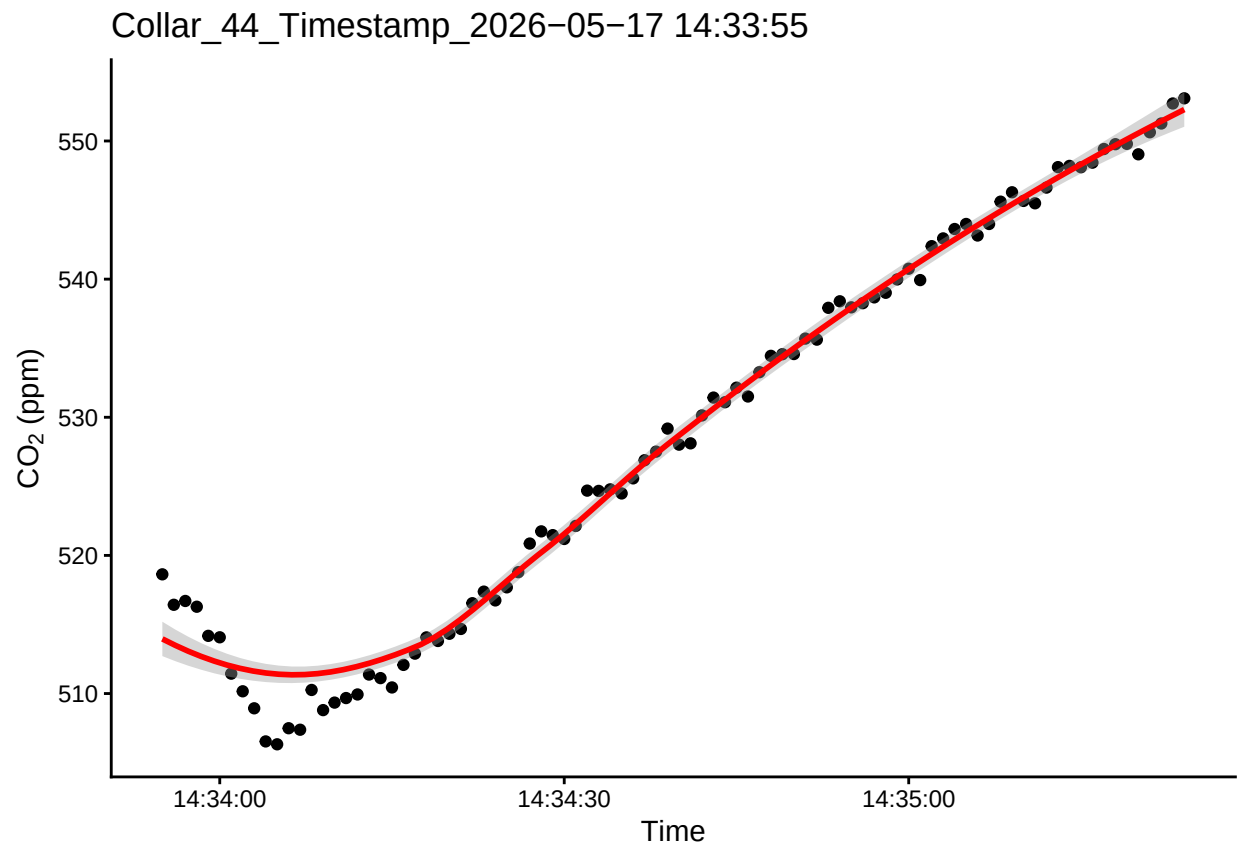


```
## 'geom_smooth()' using method = 'loess' and formula = 'y ~ x'
```

Collar_44_Timestamp_2026-05-17 13:33:54

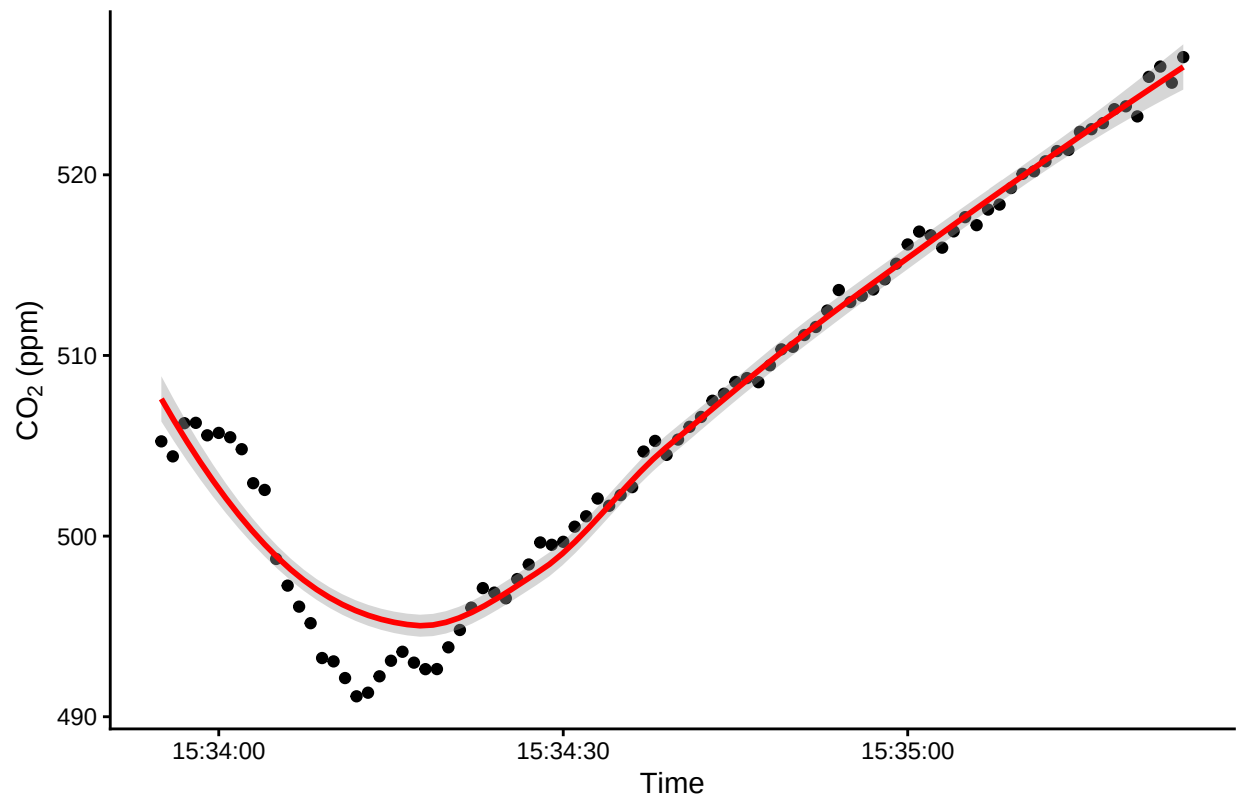


```
## 'geom_smooth()' using method = 'loess' and formula = 'y ~ x'
```

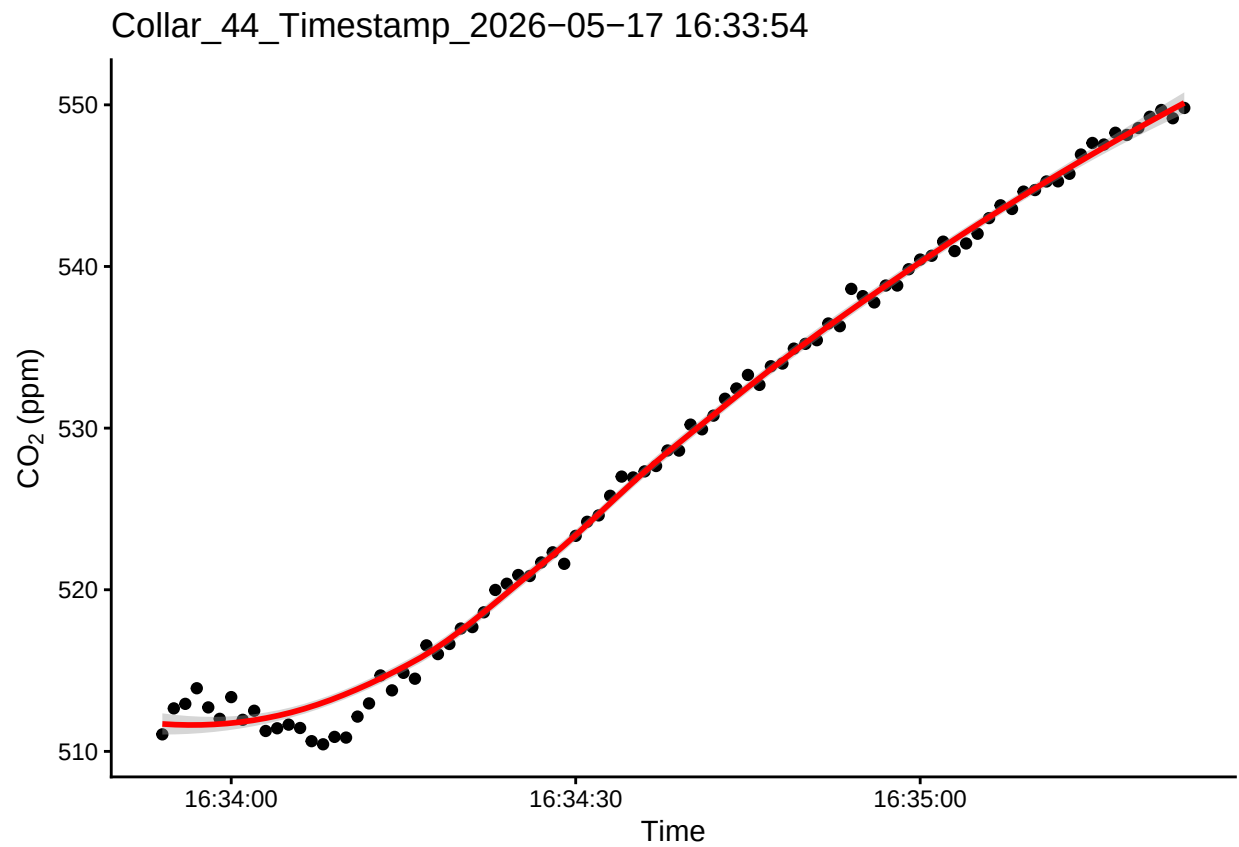


```
## 'geom_smooth()' using method = 'loess' and formula = 'y ~ x'
```

Collar_44_Timestamp_2026-05-17 15:33:55

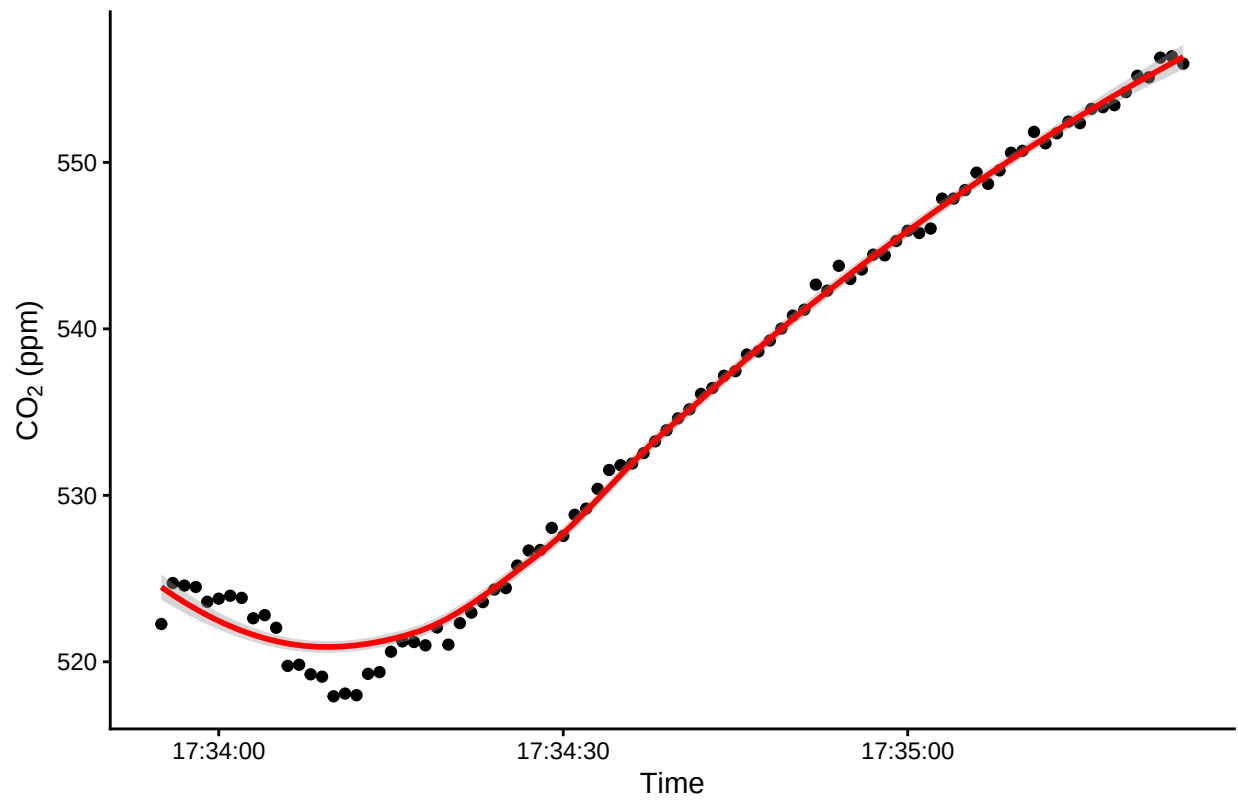


```
## 'geom_smooth()' using method = 'loess' and formula = 'y ~ x'
```



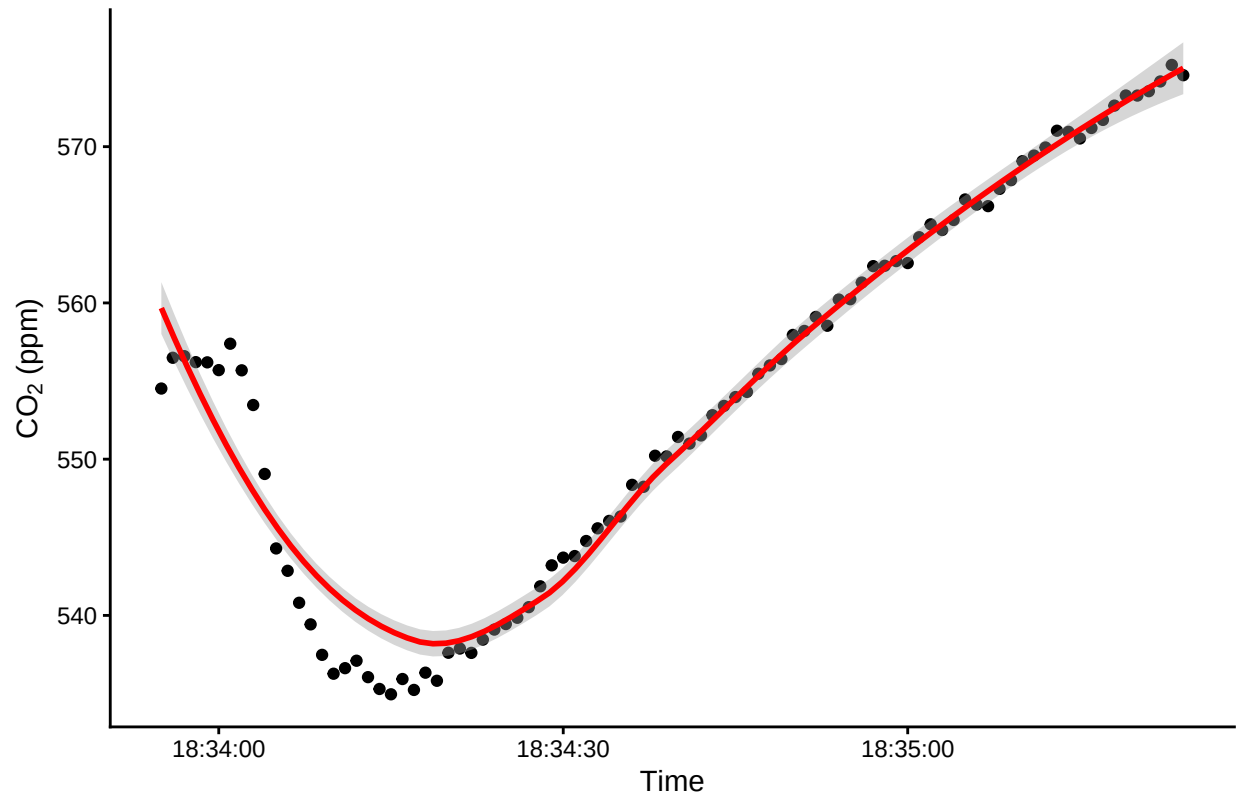
```
## 'geom_smooth()' using method = 'loess' and formula = 'y ~ x'
```

Collar_44_Timestamp_2026-05-17 17:33:55

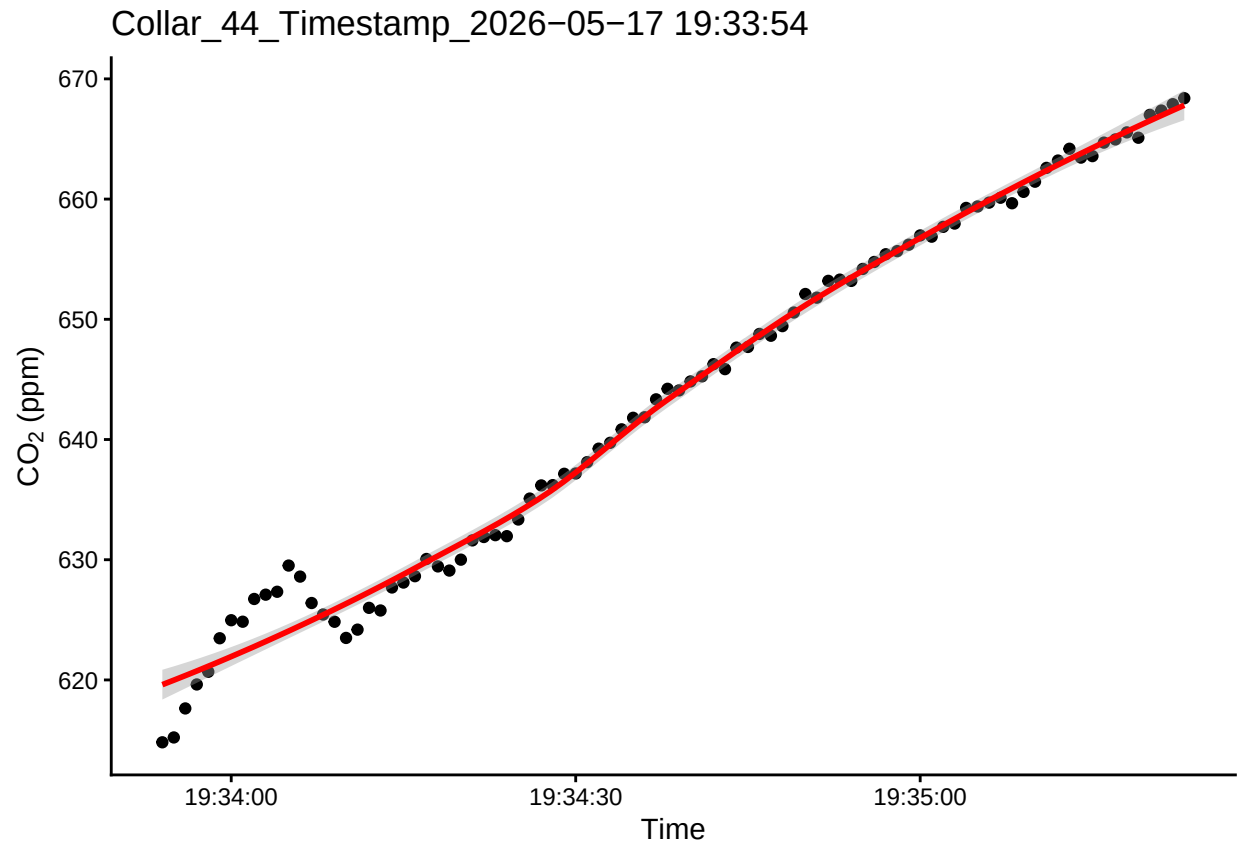


```
## 'geom_smooth()' using method = 'loess' and formula = 'y ~ x'
```

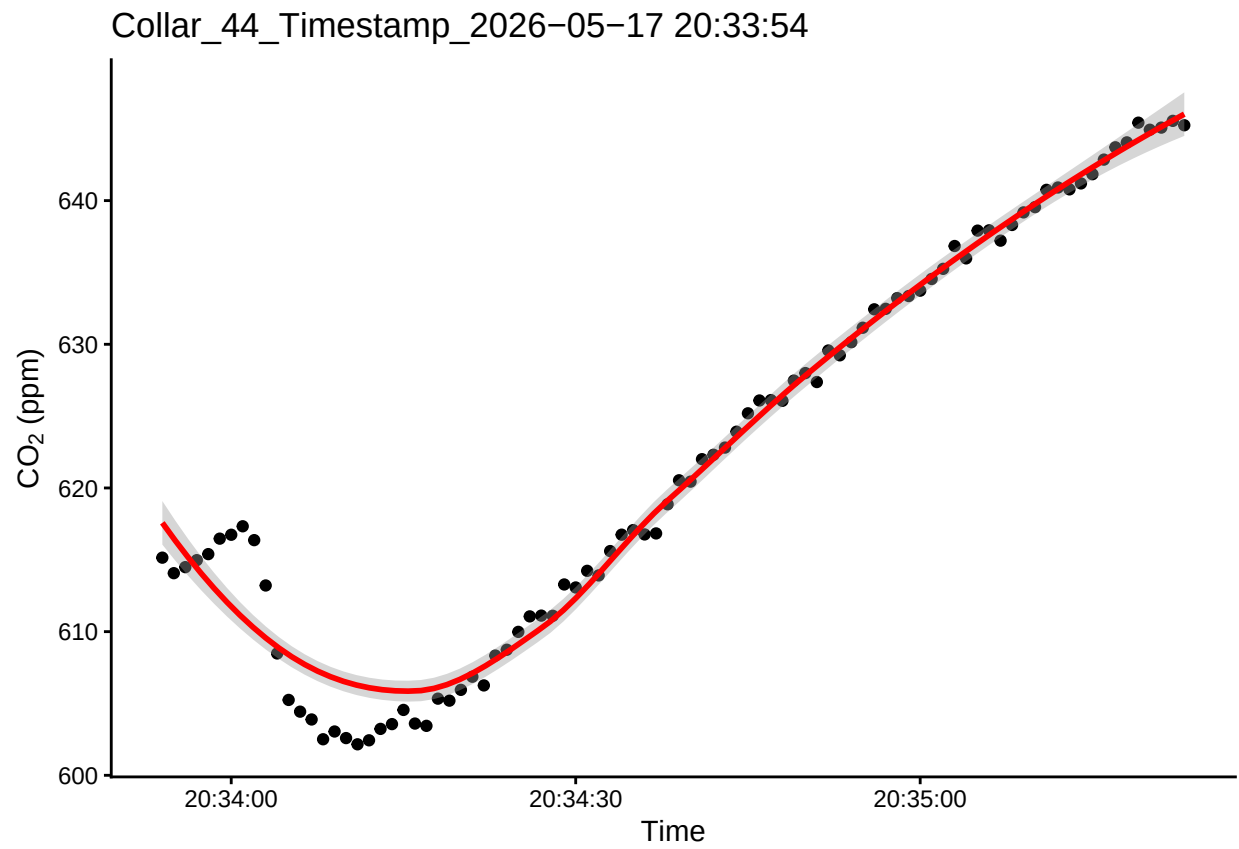
Collar_44_Timestamp_2026-05-17 18:33:55



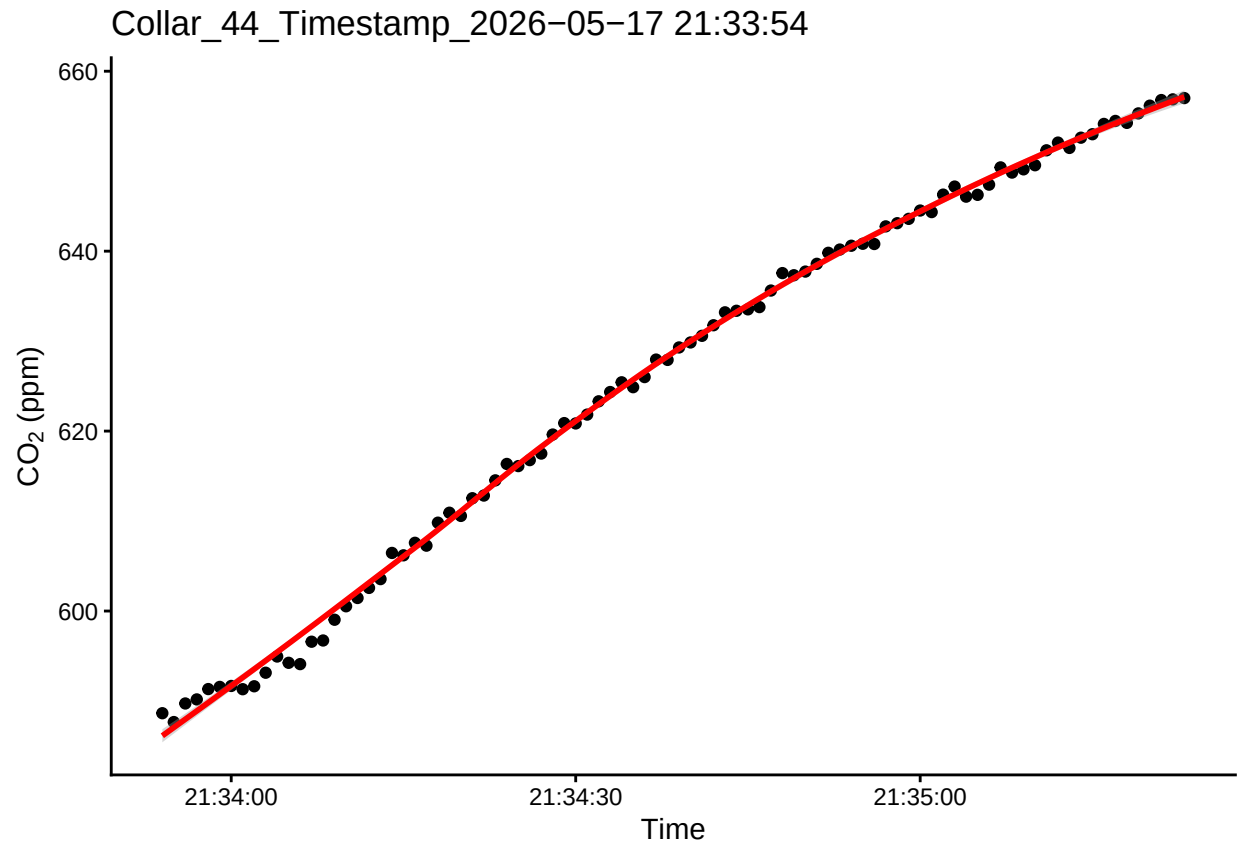
```
## 'geom_smooth()' using method = 'loess' and formula = 'y ~ x'
```

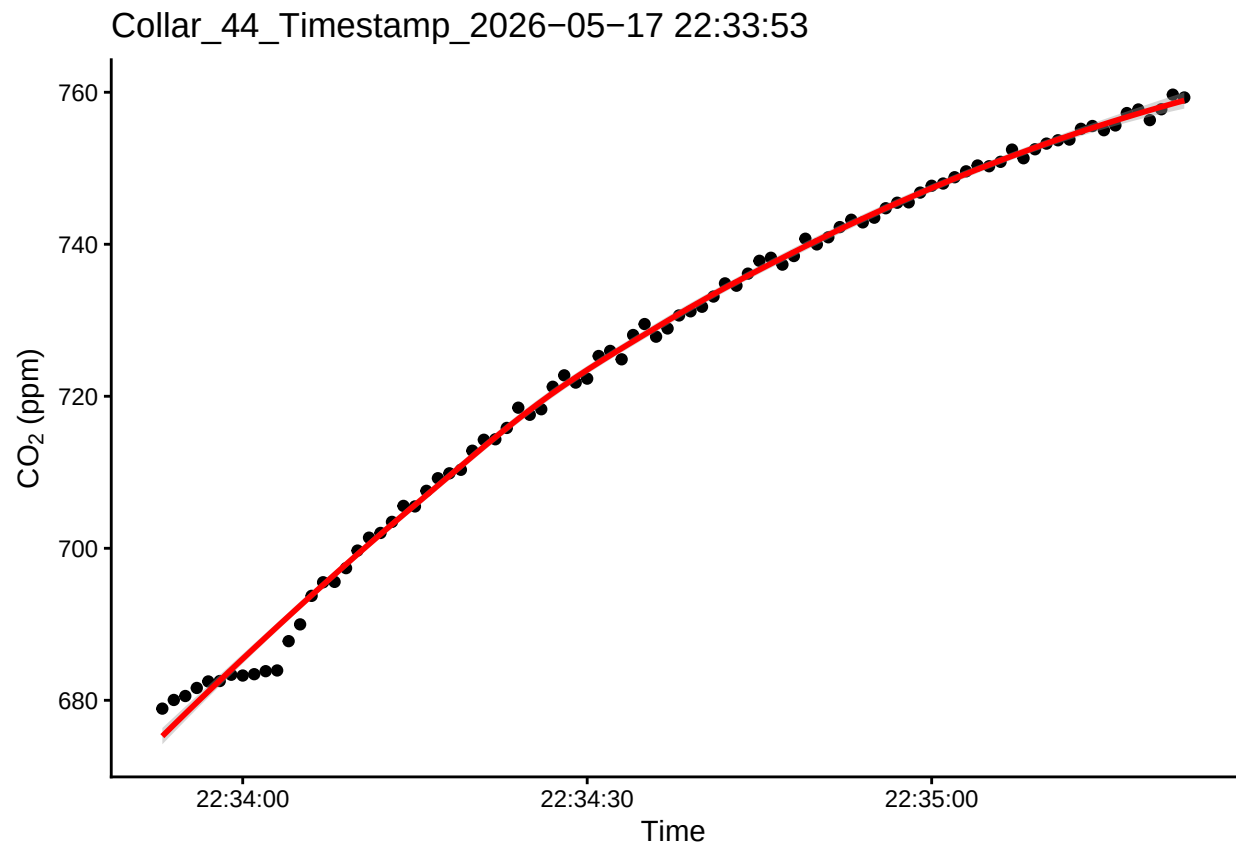
```
## 'geom_smooth()' using method = 'loess' and formula = 'y ~ x'
```



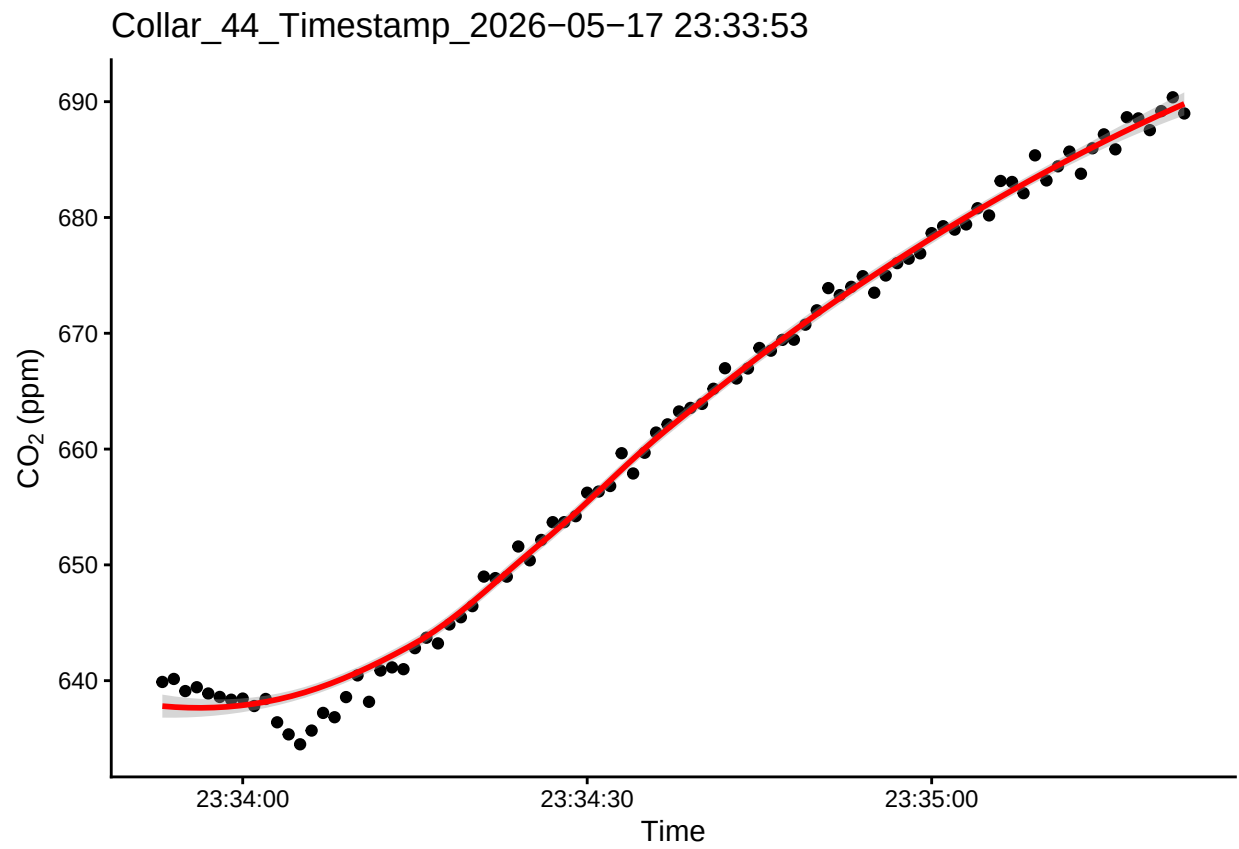
```
## 'geom_smooth()' using method = 'loess' and formula = 'y ~ x'
```



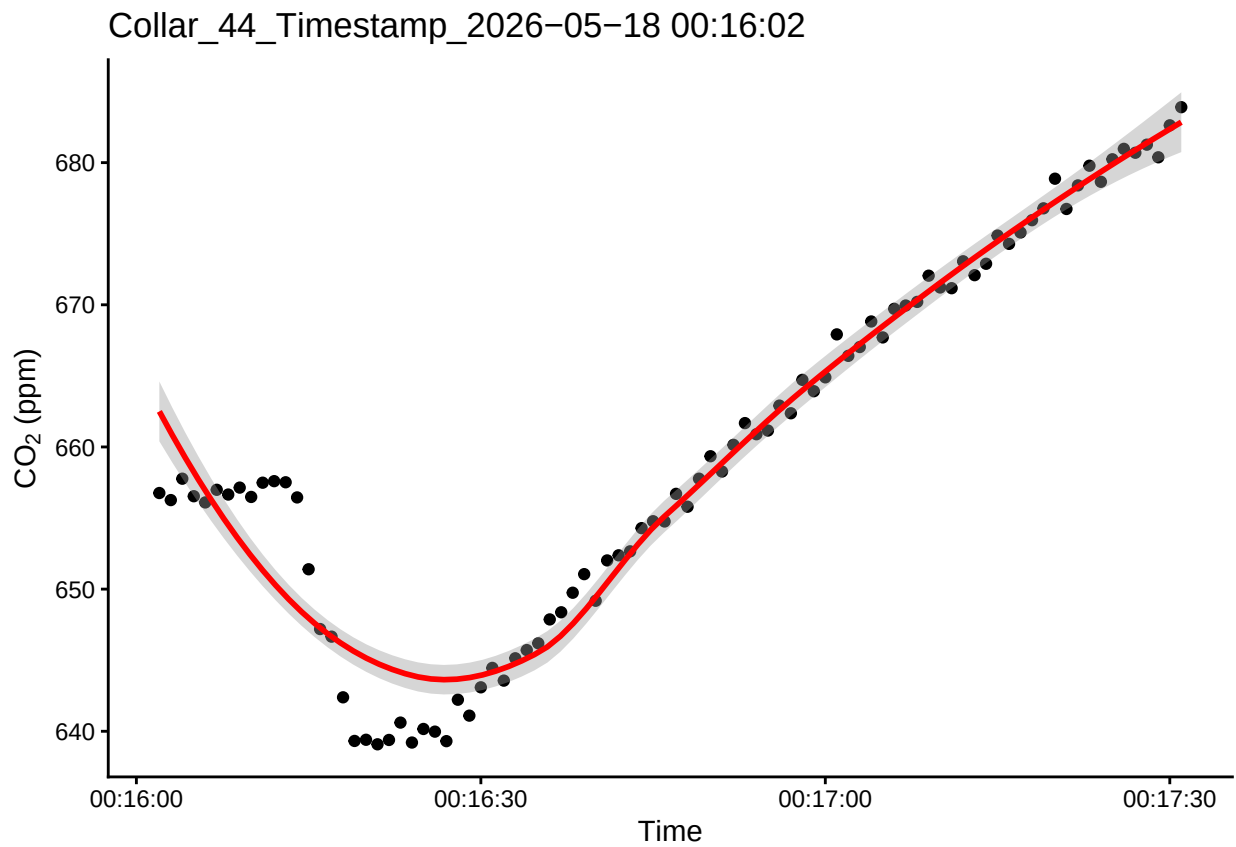
```
## 'geom_smooth()' using method = 'loess' and formula = 'y ~ x'
```



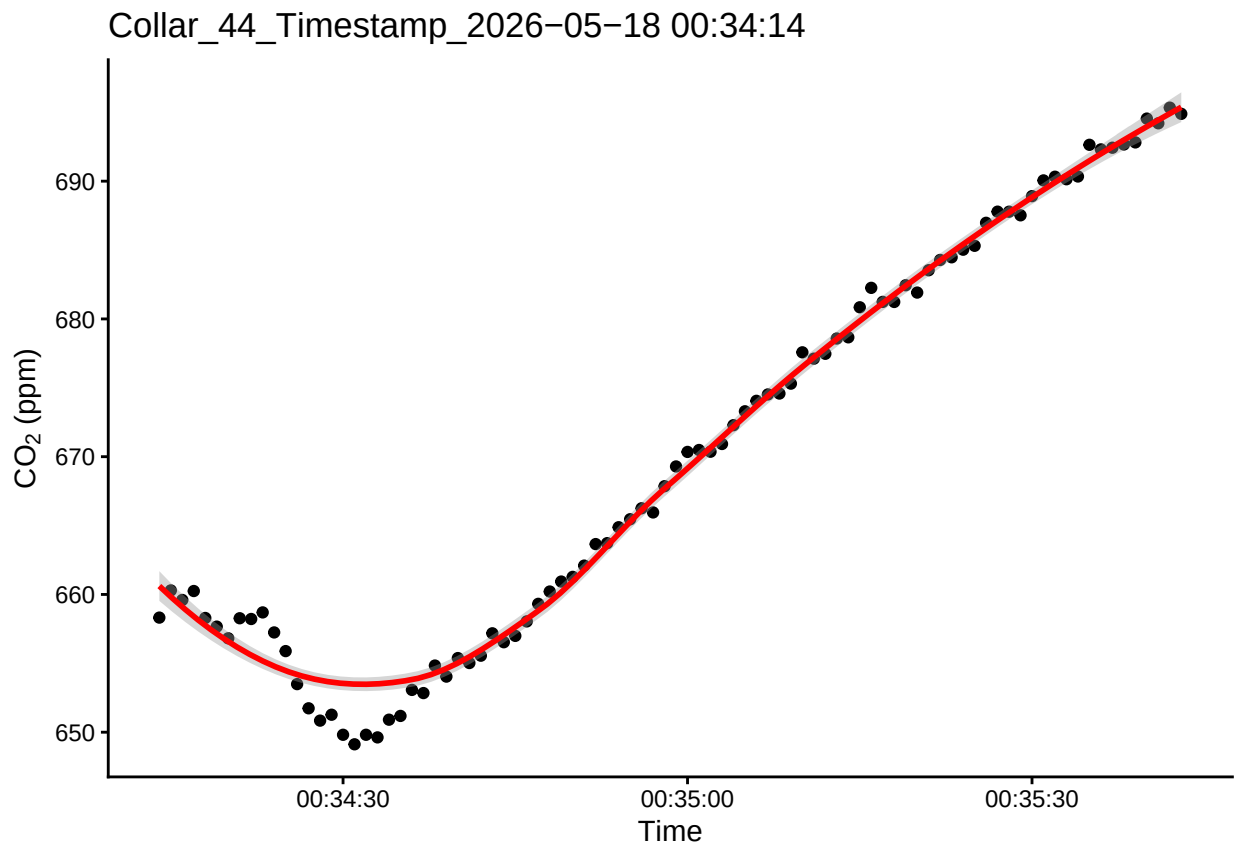
```
## 'geom_smooth()' using method = 'loess' and formula = 'y ~ x'
```



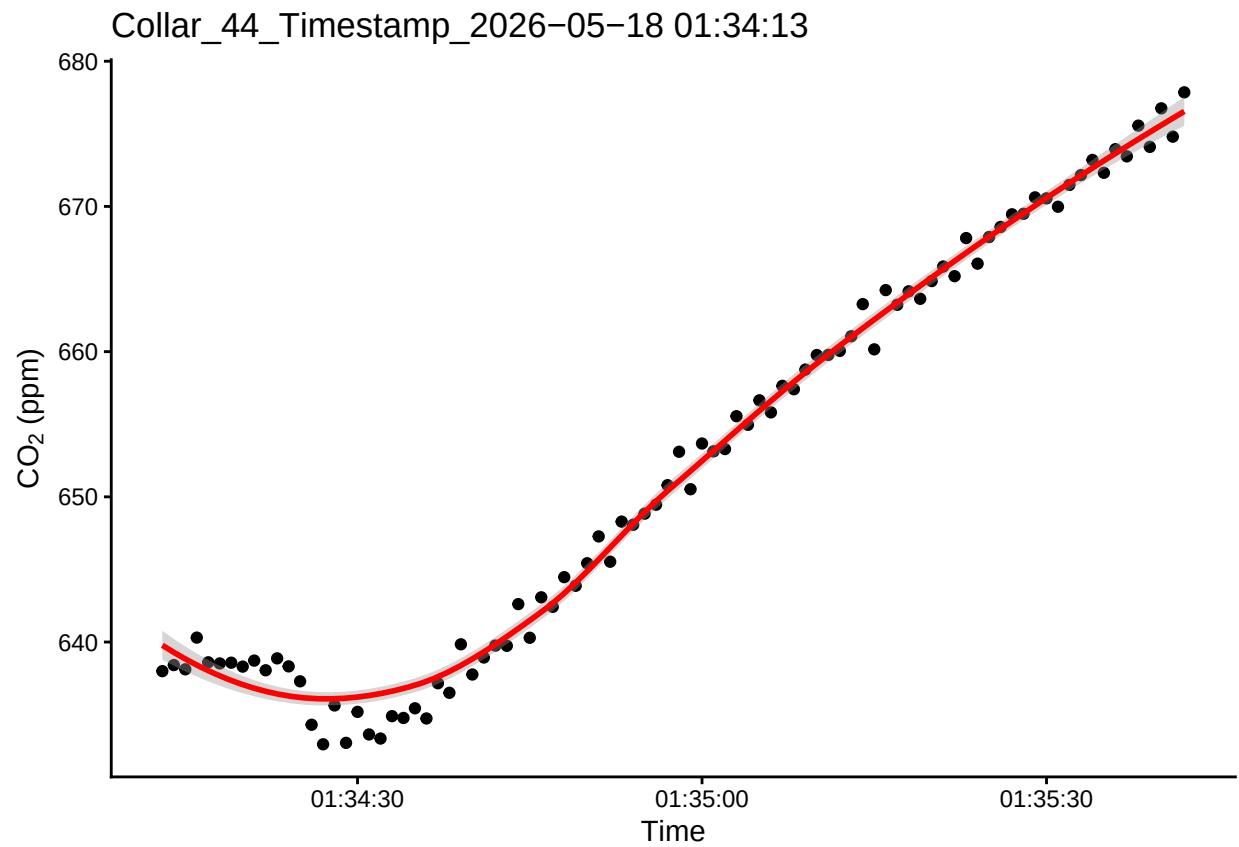
```
## 'geom_smooth()' using method = 'loess' and formula = 'y ~ x'
```



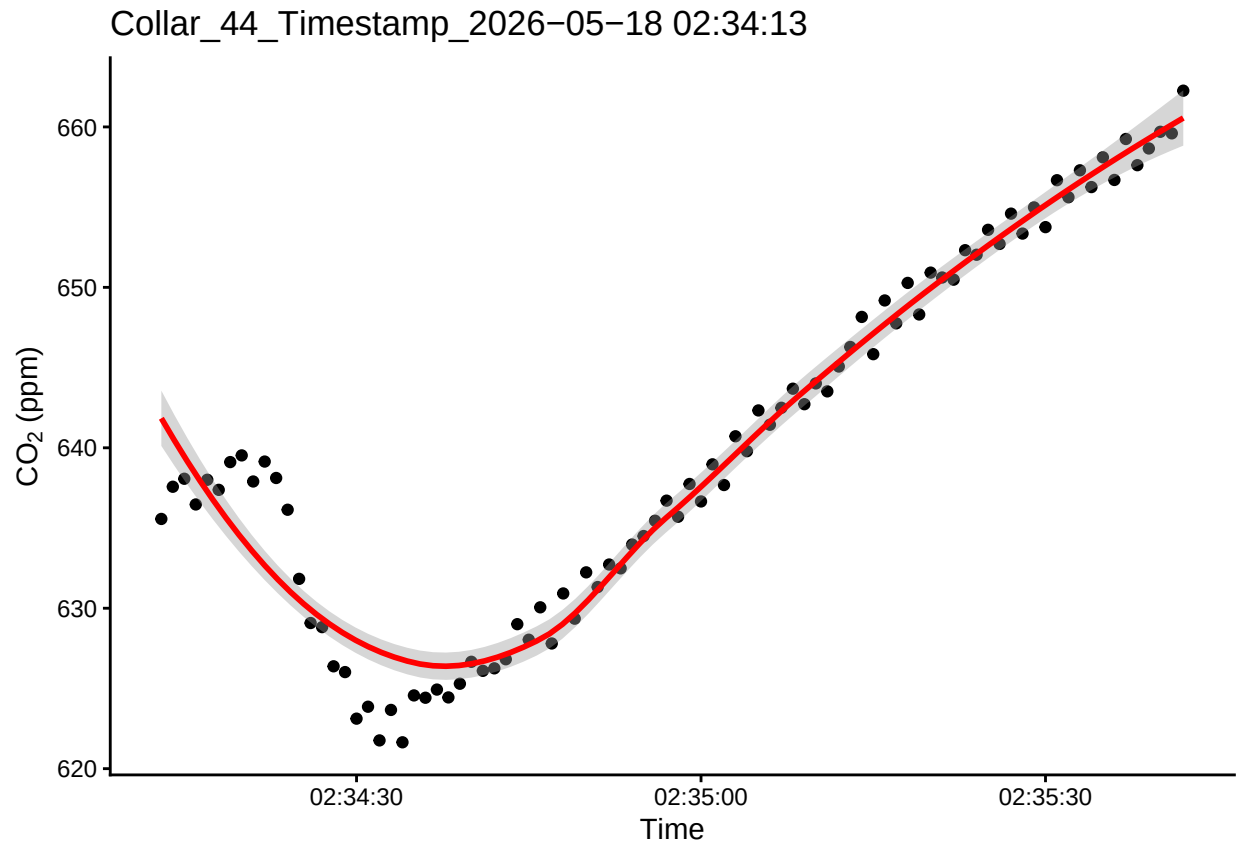
```
## 'geom_smooth()' using method = 'loess' and formula = 'y ~ x'
```



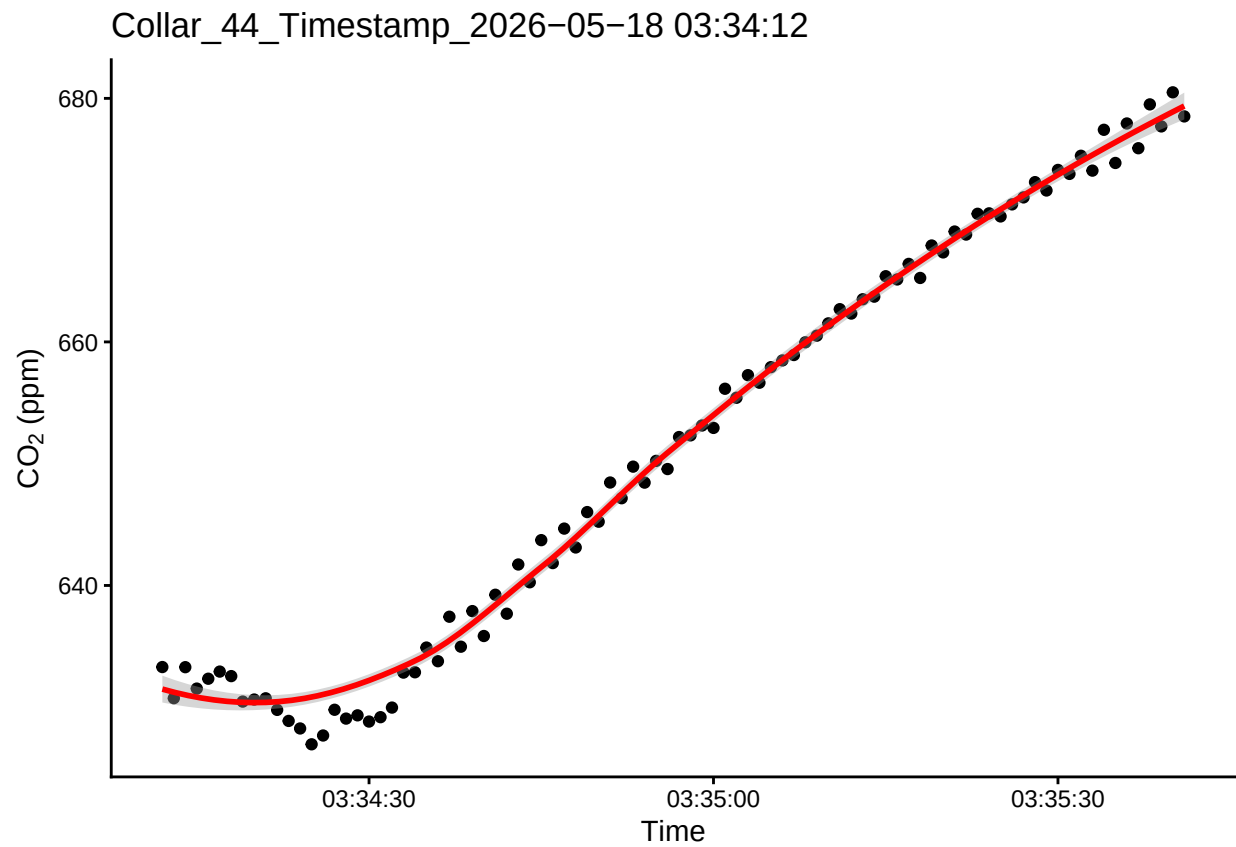
```
## 'geom_smooth()' using method = 'loess' and formula = 'y ~ x'
```



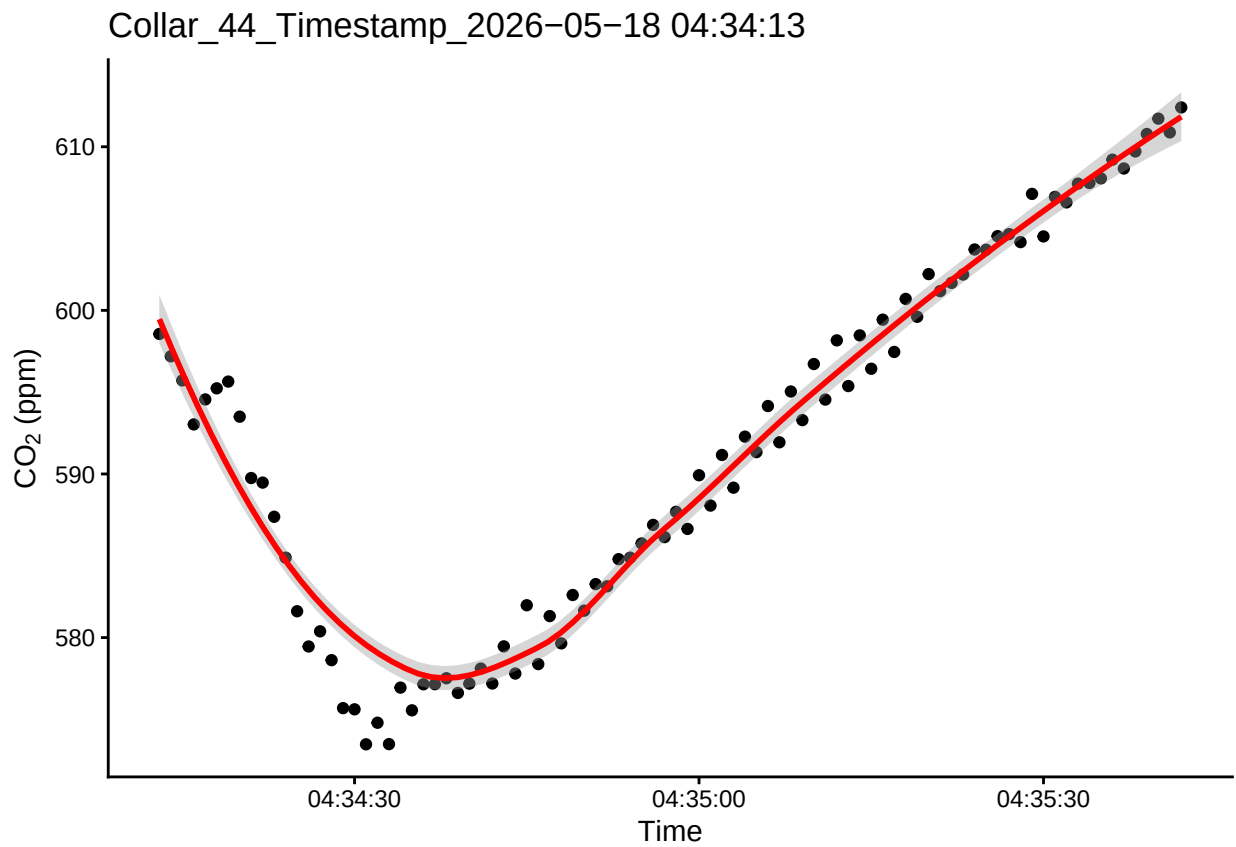
```
## 'geom_smooth()' using method = 'loess' and formula = 'y ~ x'
```

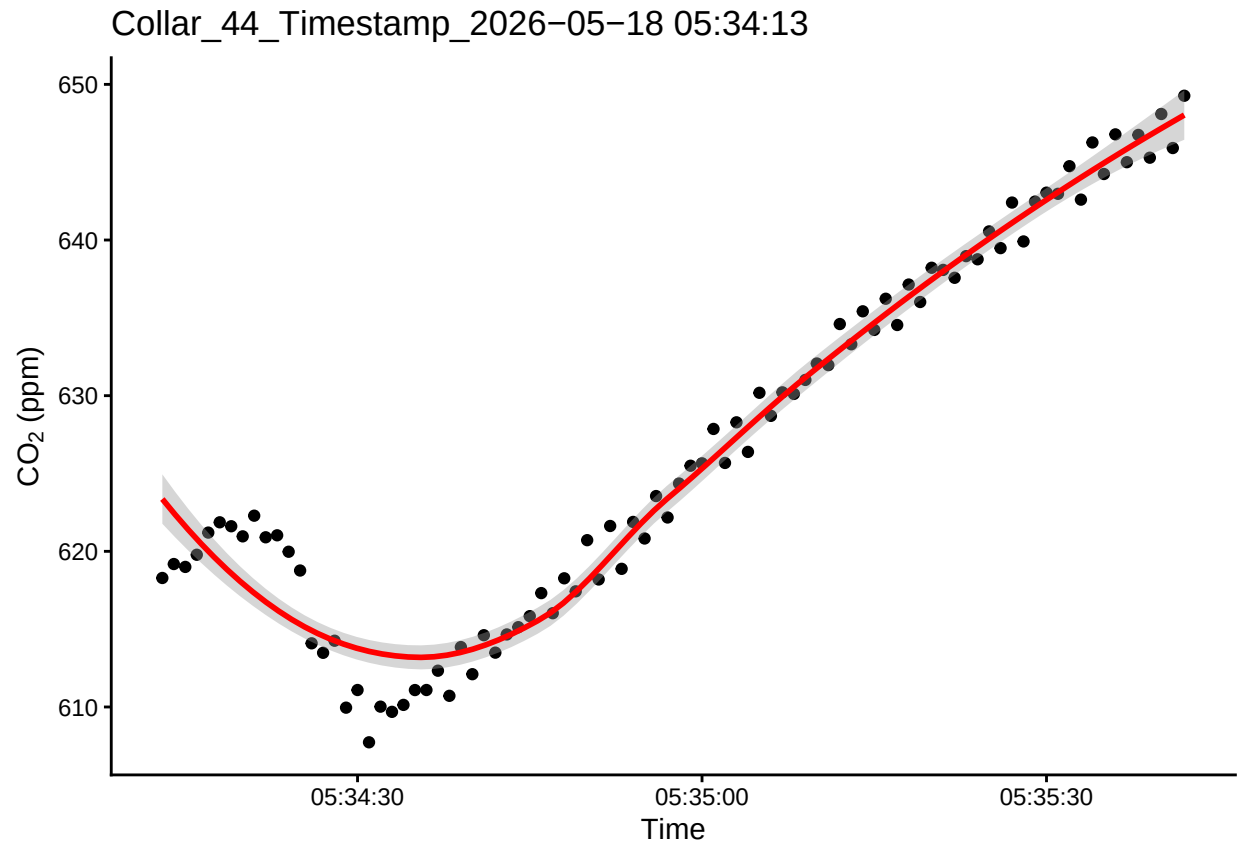
```
## 'geom_smooth()' using method = 'loess' and formula = 'y ~ x'
```



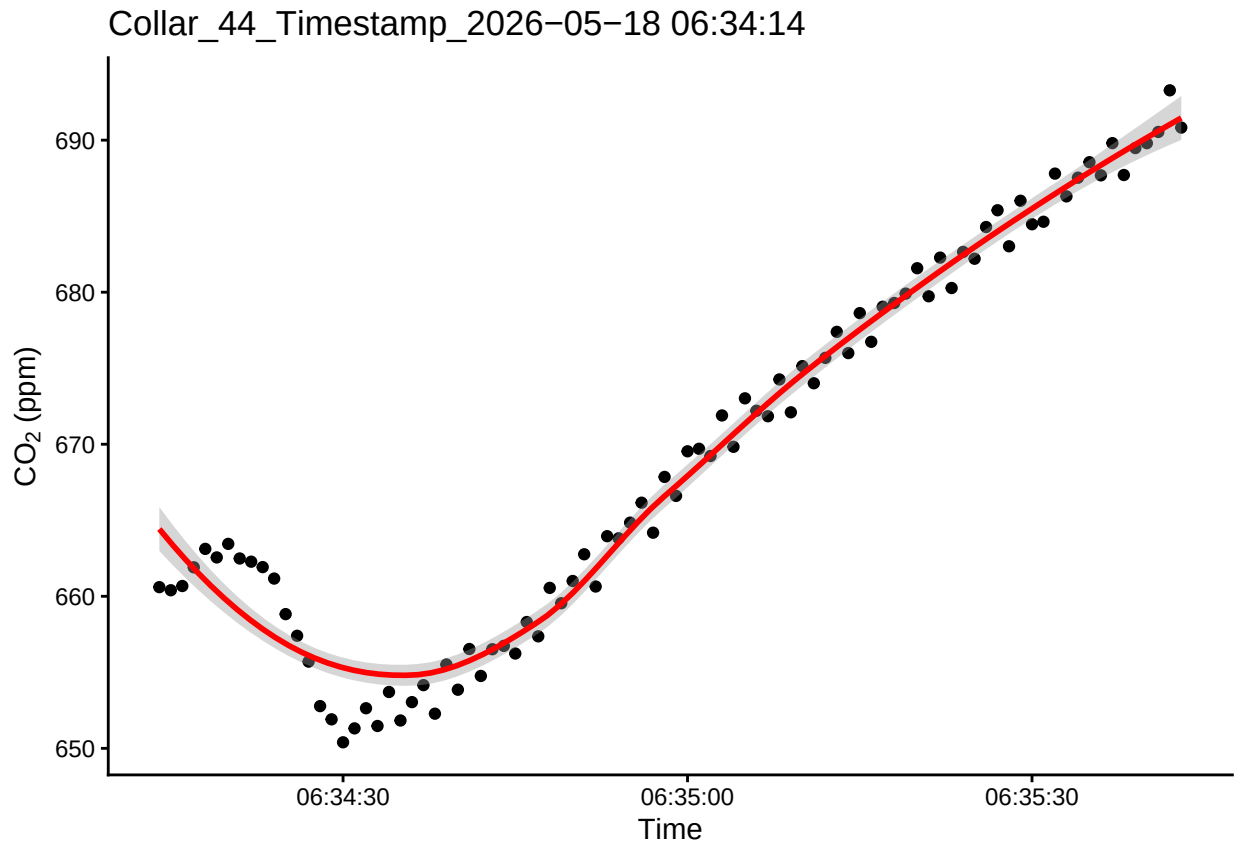
```
## 'geom_smooth()' using method = 'loess' and formula = 'y ~ x'
```



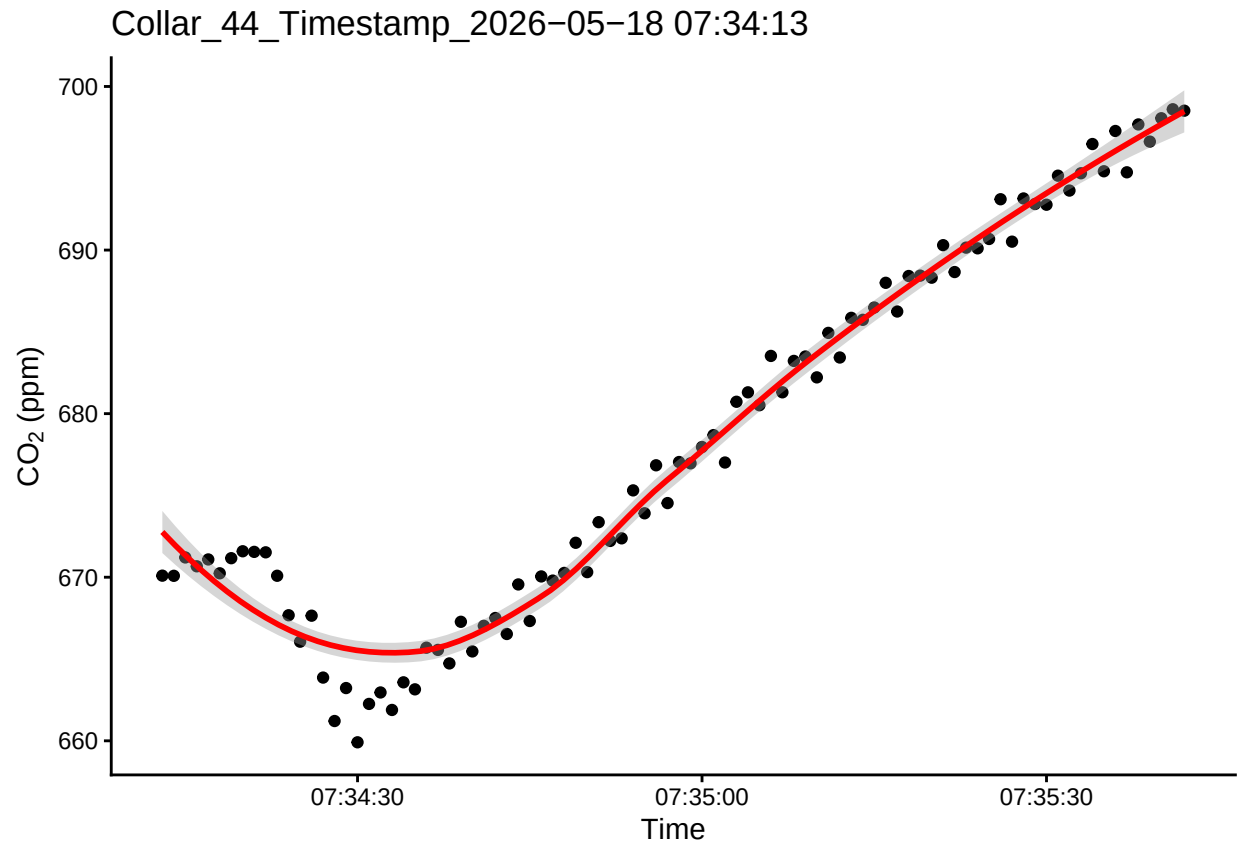
```
## 'geom_smooth()' using method = 'loess' and formula = 'y ~ x'
```



```
## 'geom_smooth()' using method = 'loess' and formula = 'y ~ x'
```

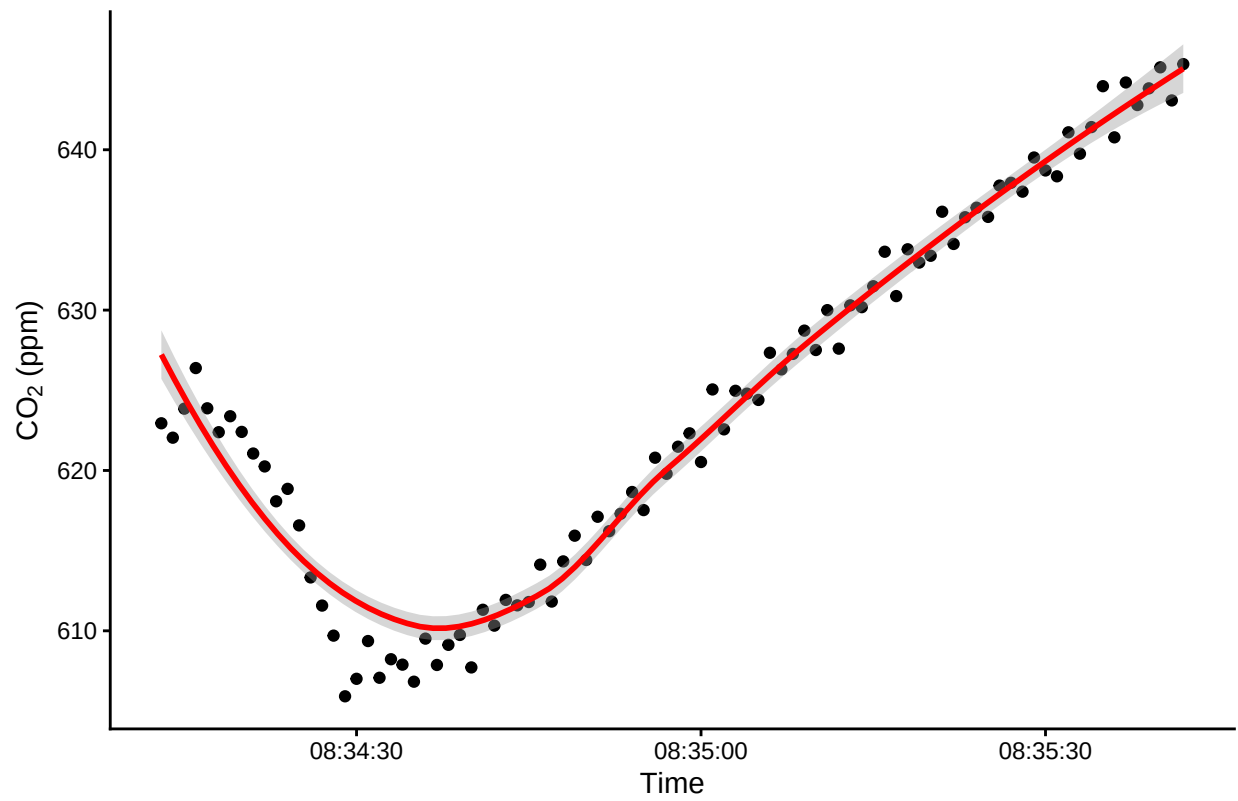


```
## 'geom_smooth()' using method = 'loess' and formula = 'y ~ x'
```



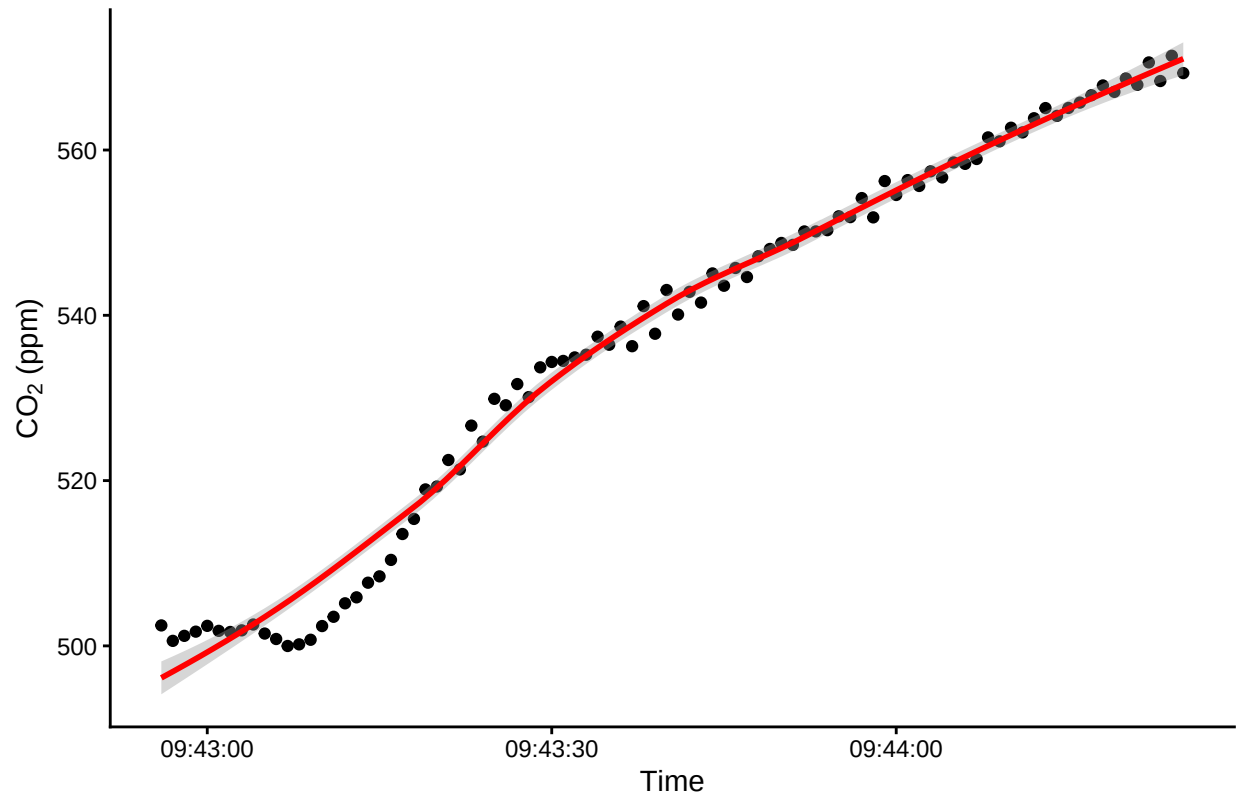
```
## 'geom_smooth()' using method = 'loess' and formula = 'y ~ x'
```

Collar_44_Timestamp_2026-05-18 08:34:13

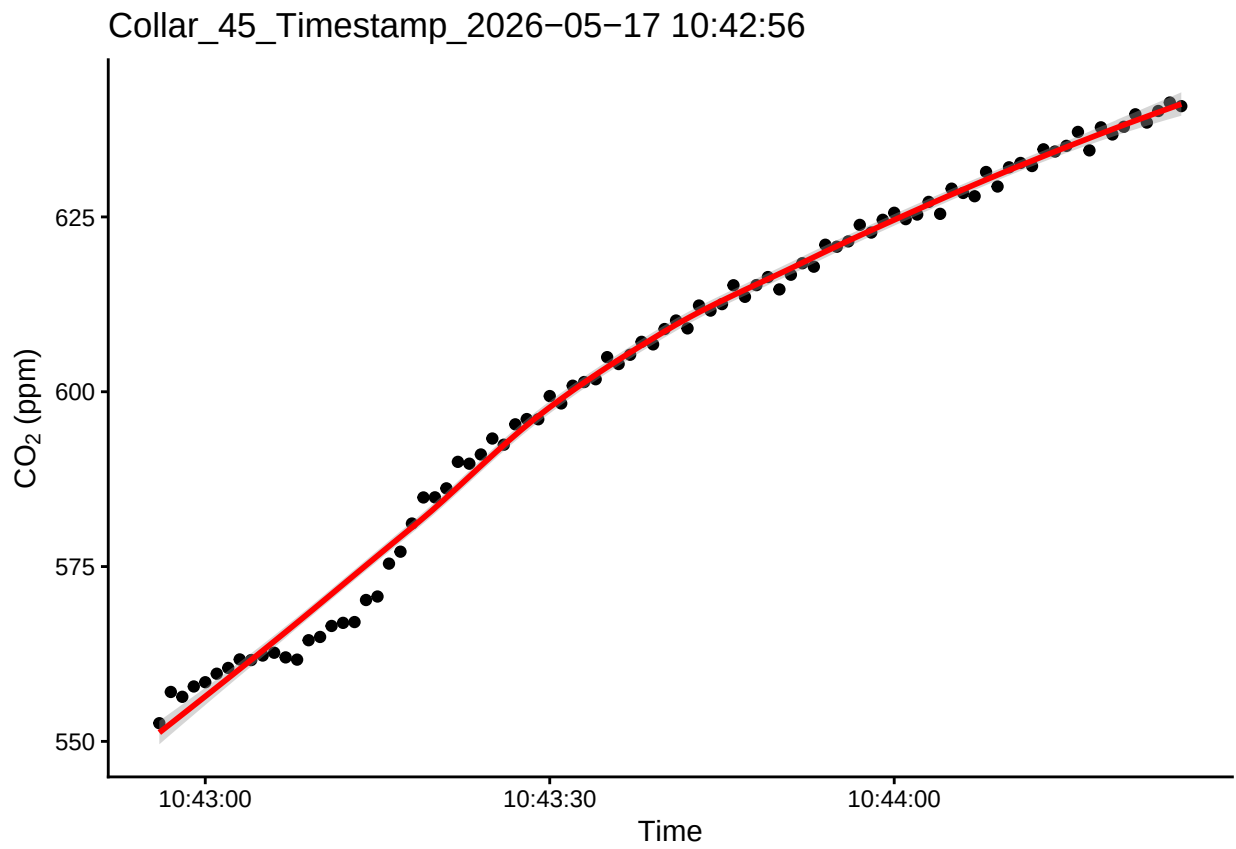


```
## 'geom_smooth()' using method = 'loess' and formula = 'y ~ x'
```

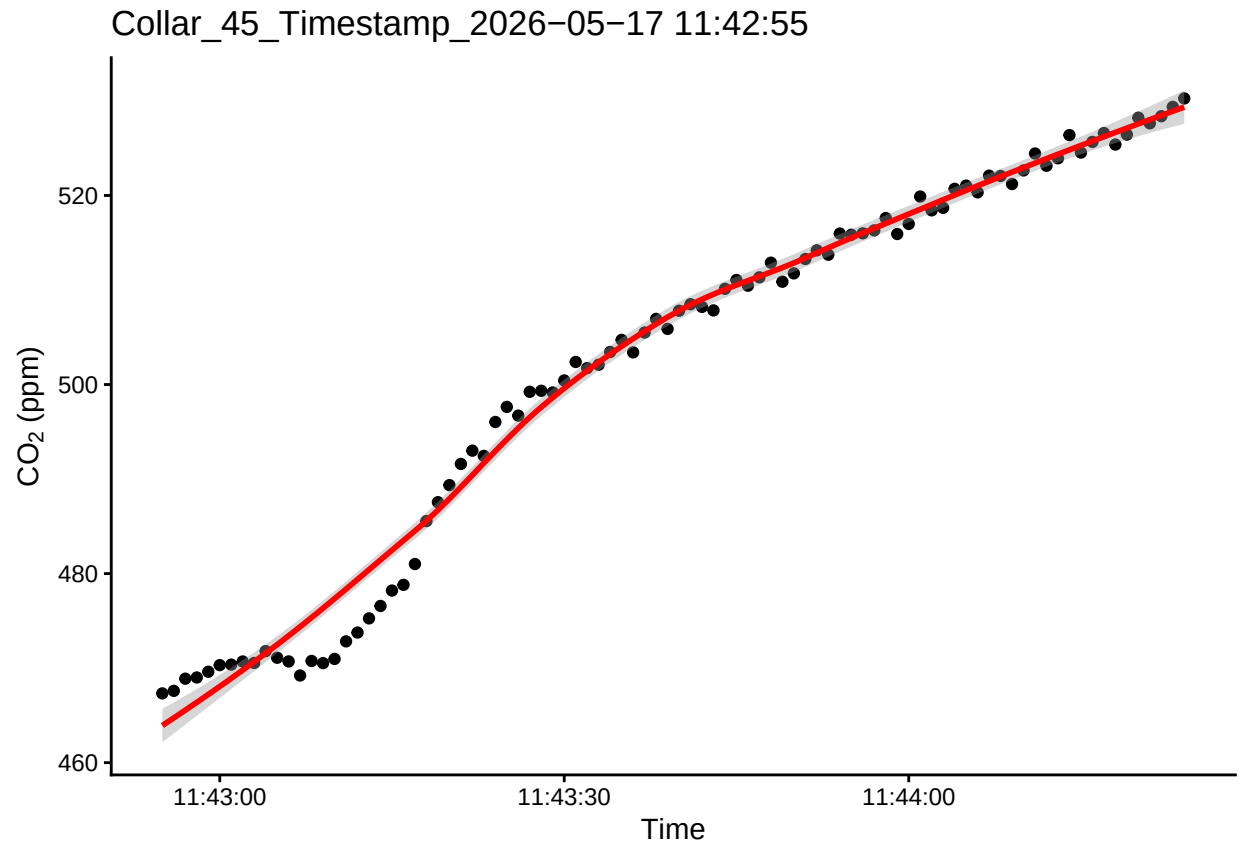
Collar_45_Timestamp_2026-05-17 09:42:56



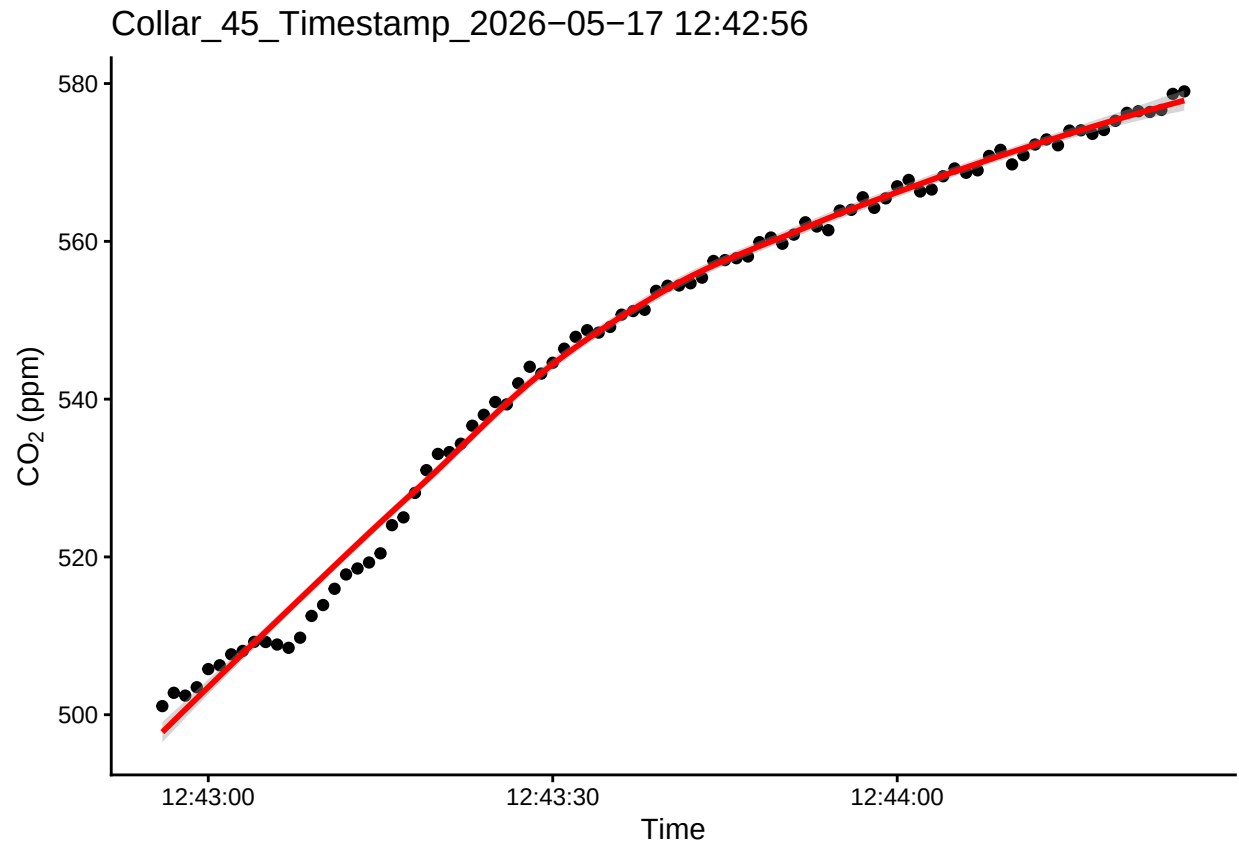
```
## 'geom_smooth()' using method = 'loess' and formula = 'y ~ x'
```

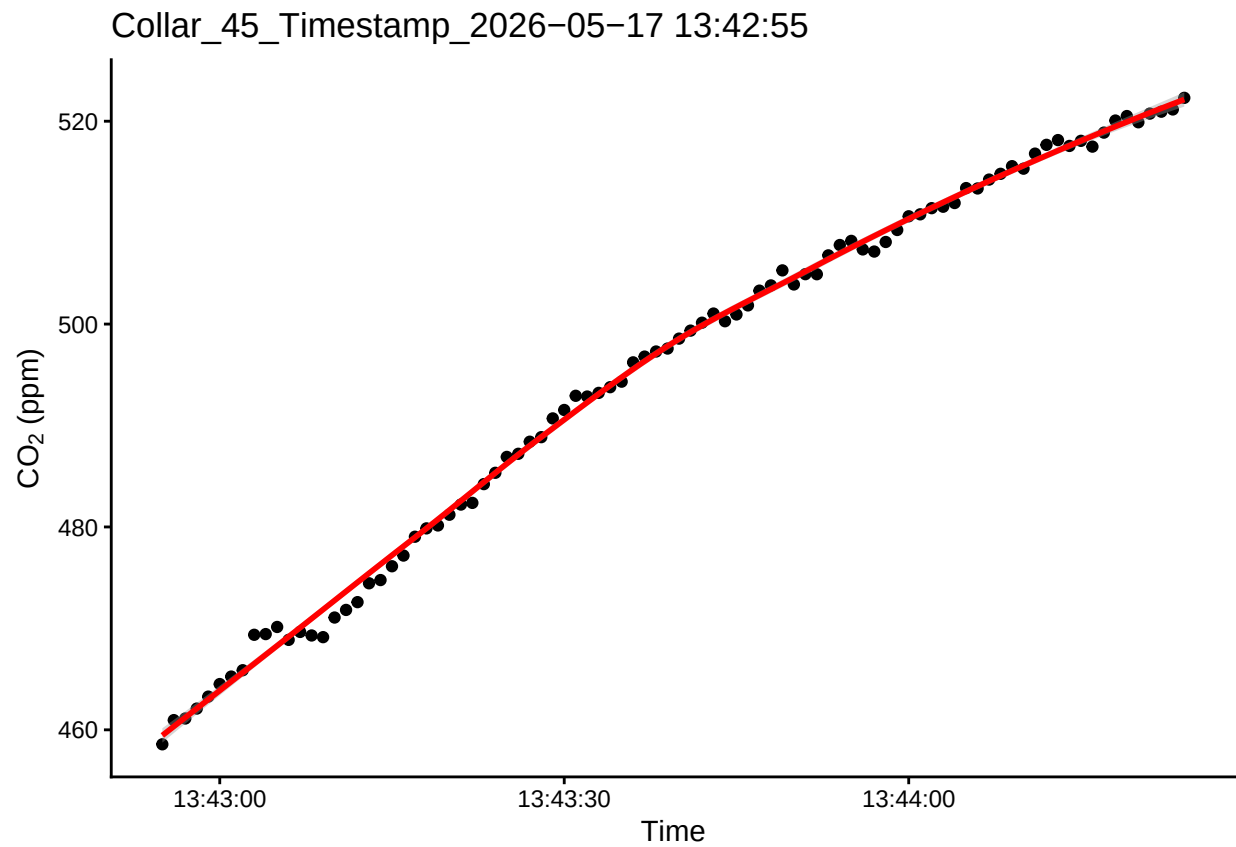
```
## 'geom_smooth()' using method = 'loess' and formula = 'y ~ x'
```



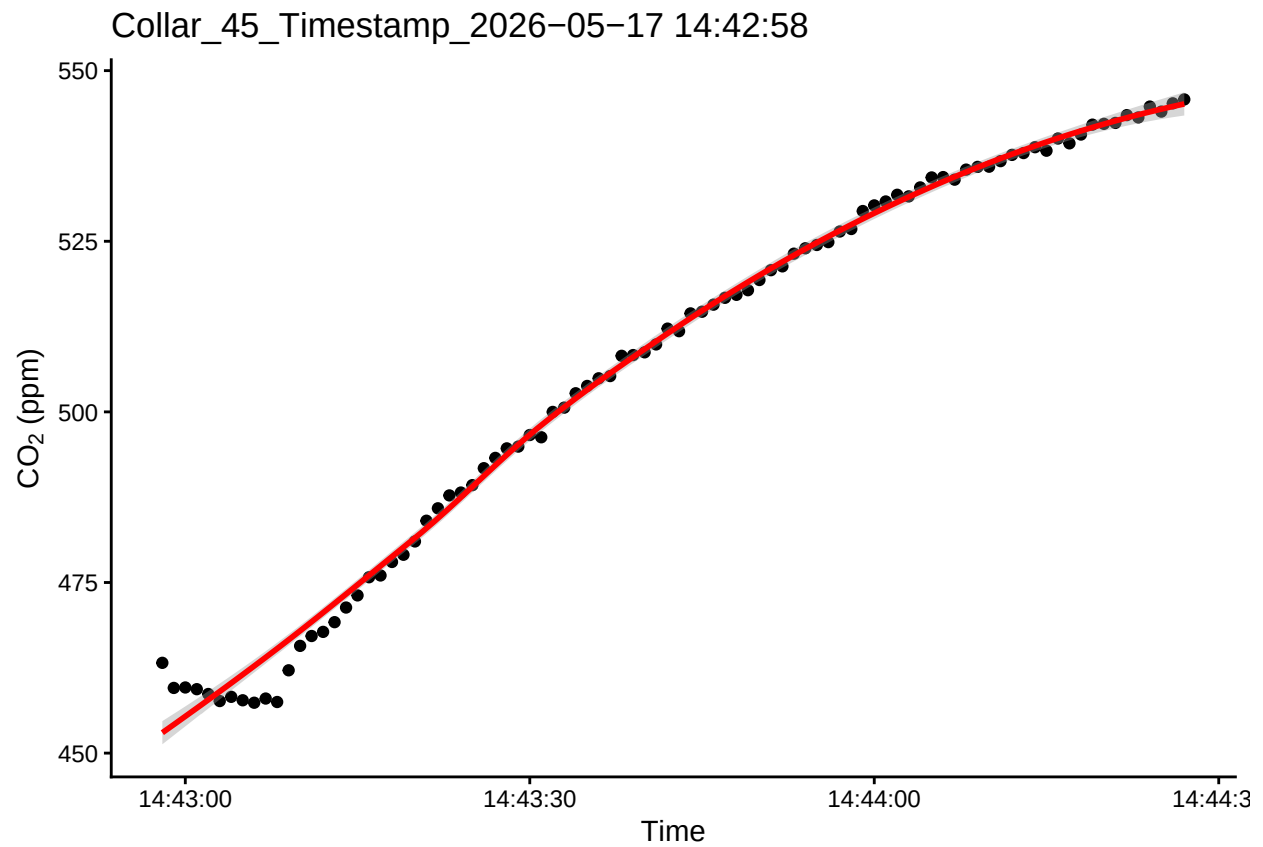
```
## 'geom_smooth()' using method = 'loess' and formula = 'y ~ x'
```



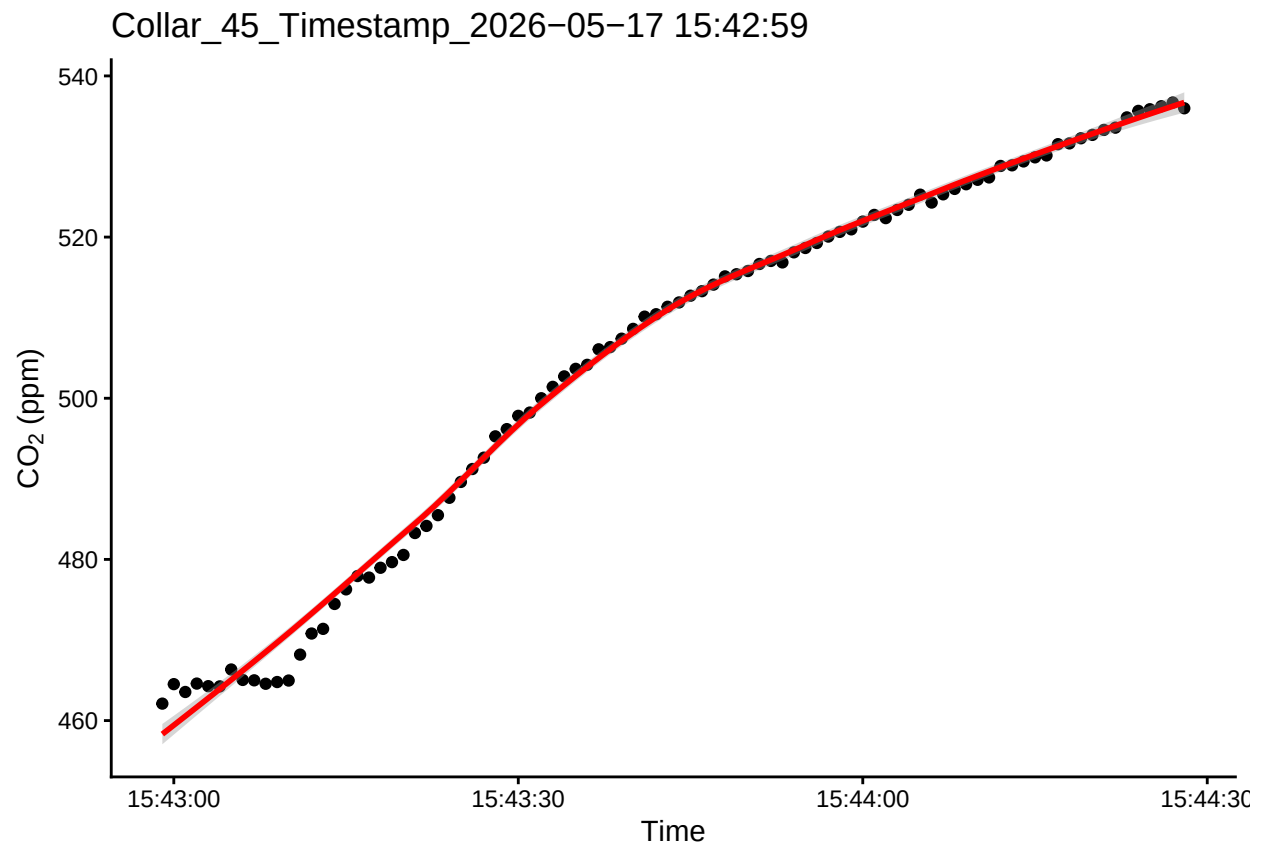
```
## 'geom_smooth()' using method = 'loess' and formula = 'y ~ x'
```



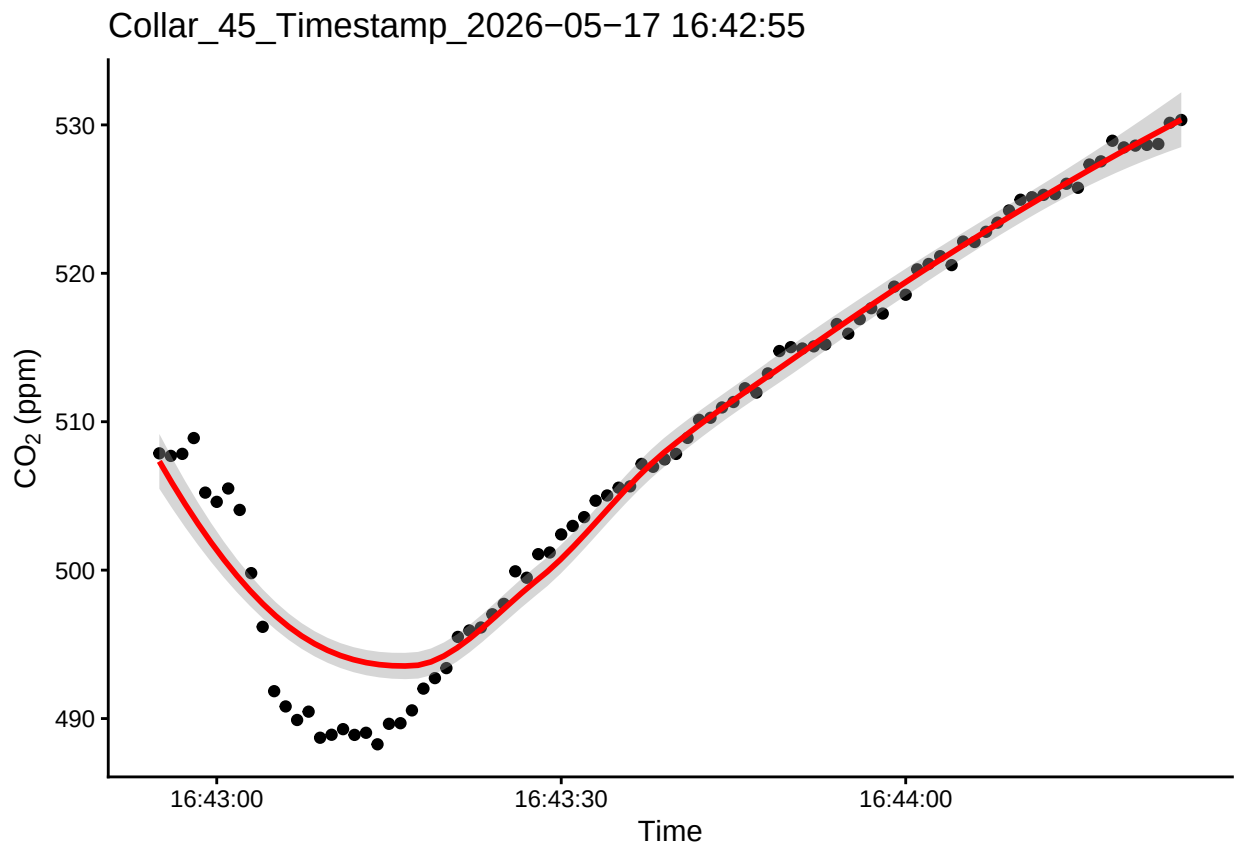
```
## 'geom_smooth()' using method = 'loess' and formula = 'y ~ x'
```



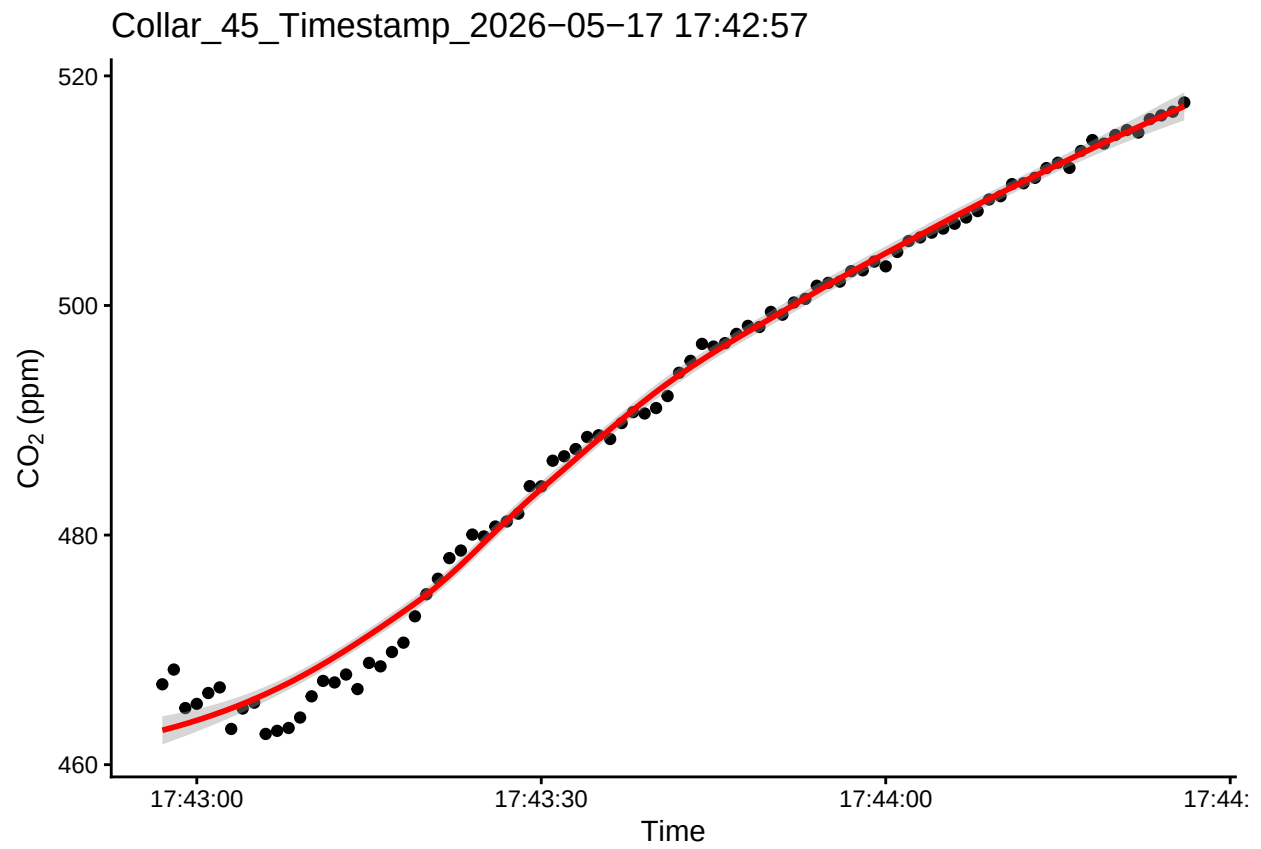
```
## 'geom_smooth()' using method = 'loess' and formula = 'y ~ x'
```



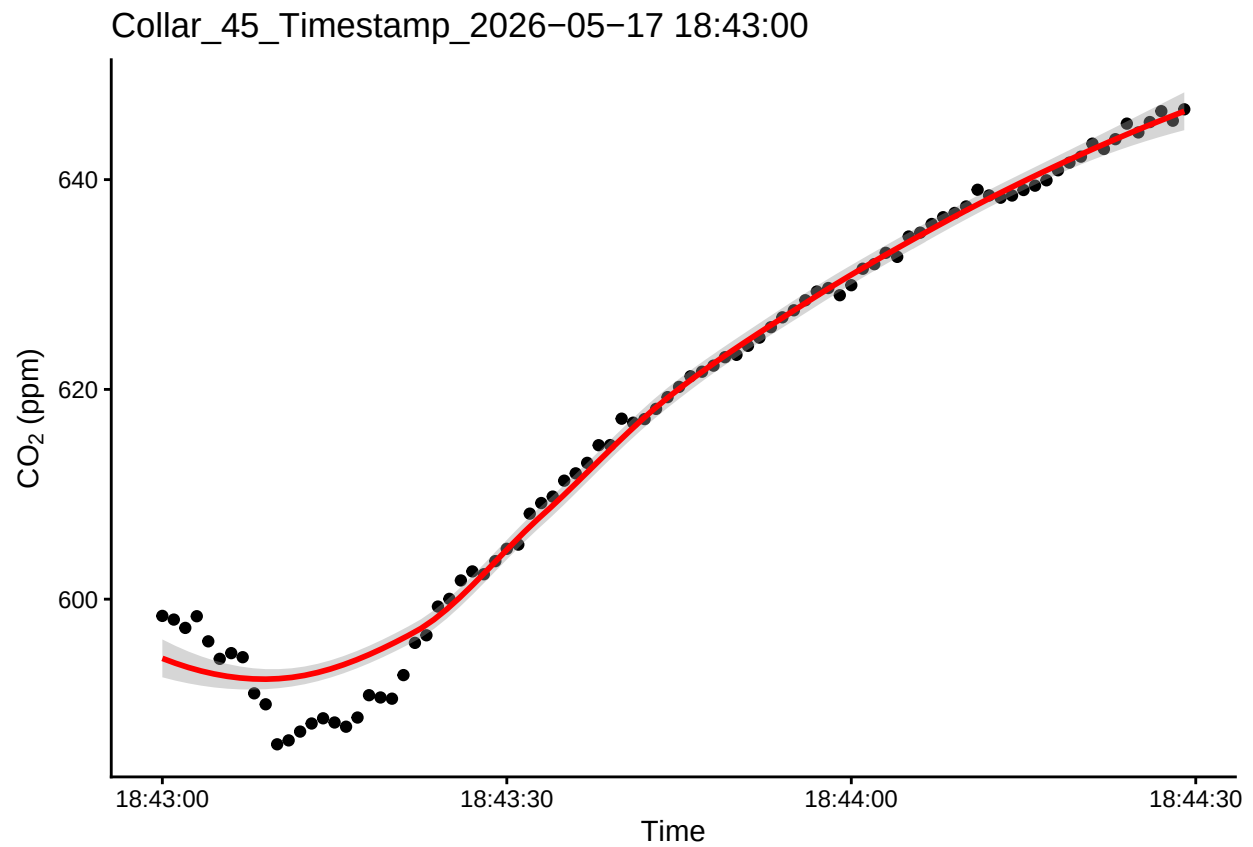
```
## 'geom_smooth()' using method = 'loess' and formula = 'y ~ x'
```



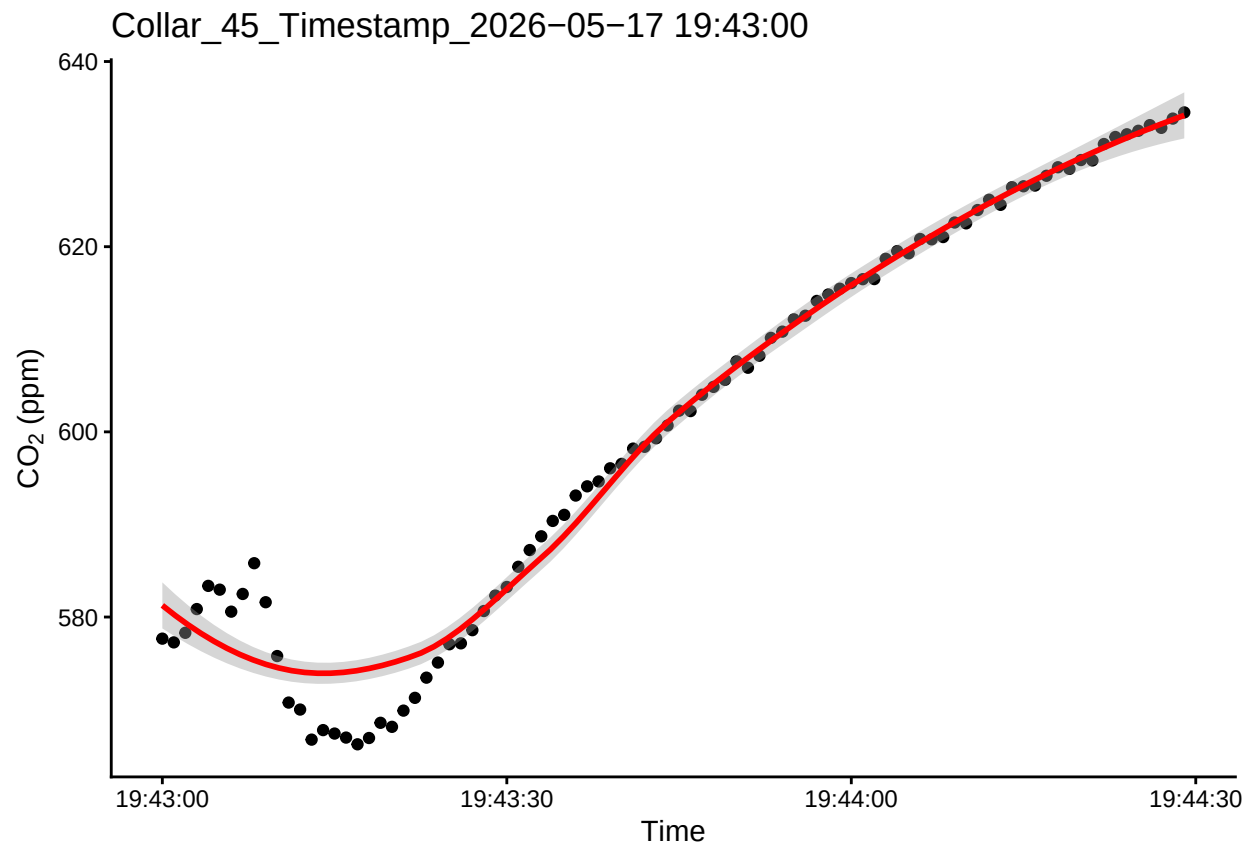
```
## 'geom_smooth()' using method = 'loess' and formula = 'y ~ x'
```



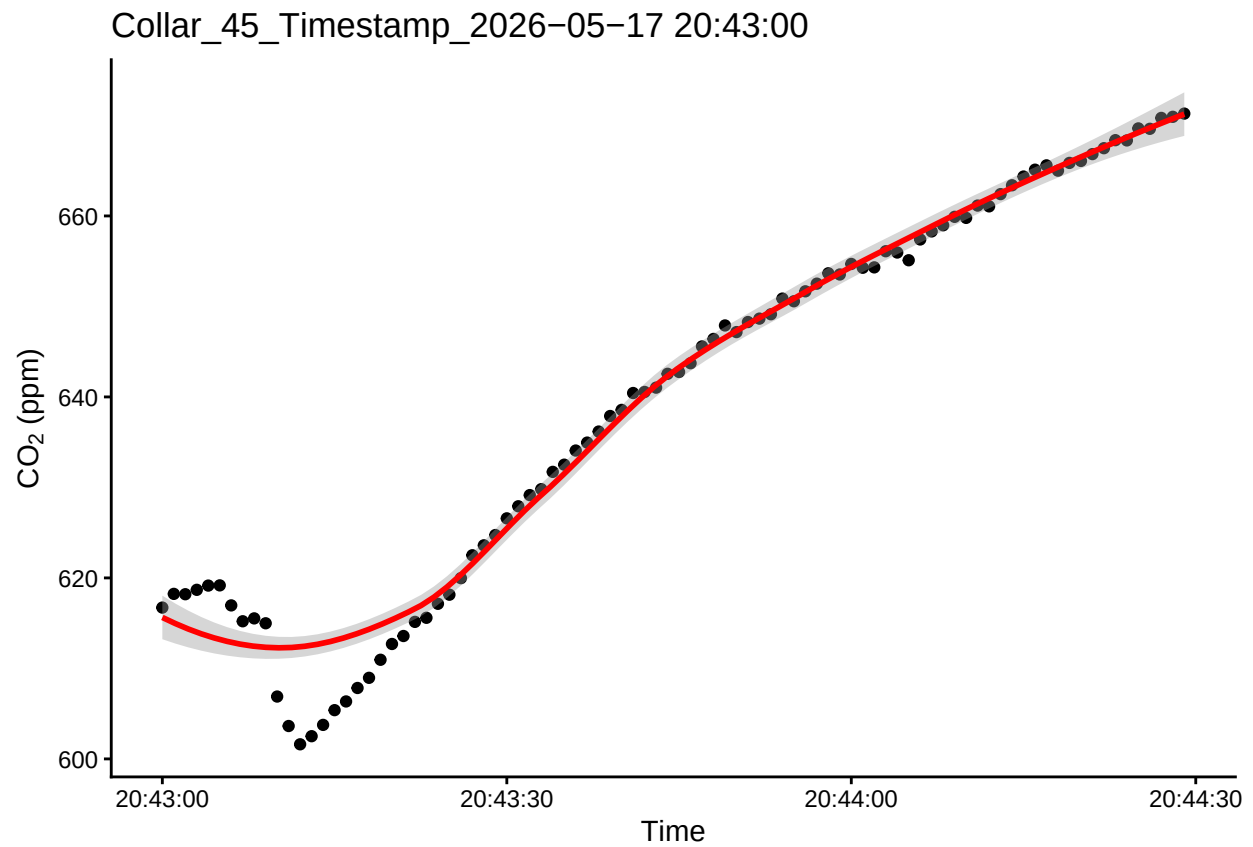
```
## 'geom_smooth()' using method = 'loess' and formula = 'y ~ x'
```

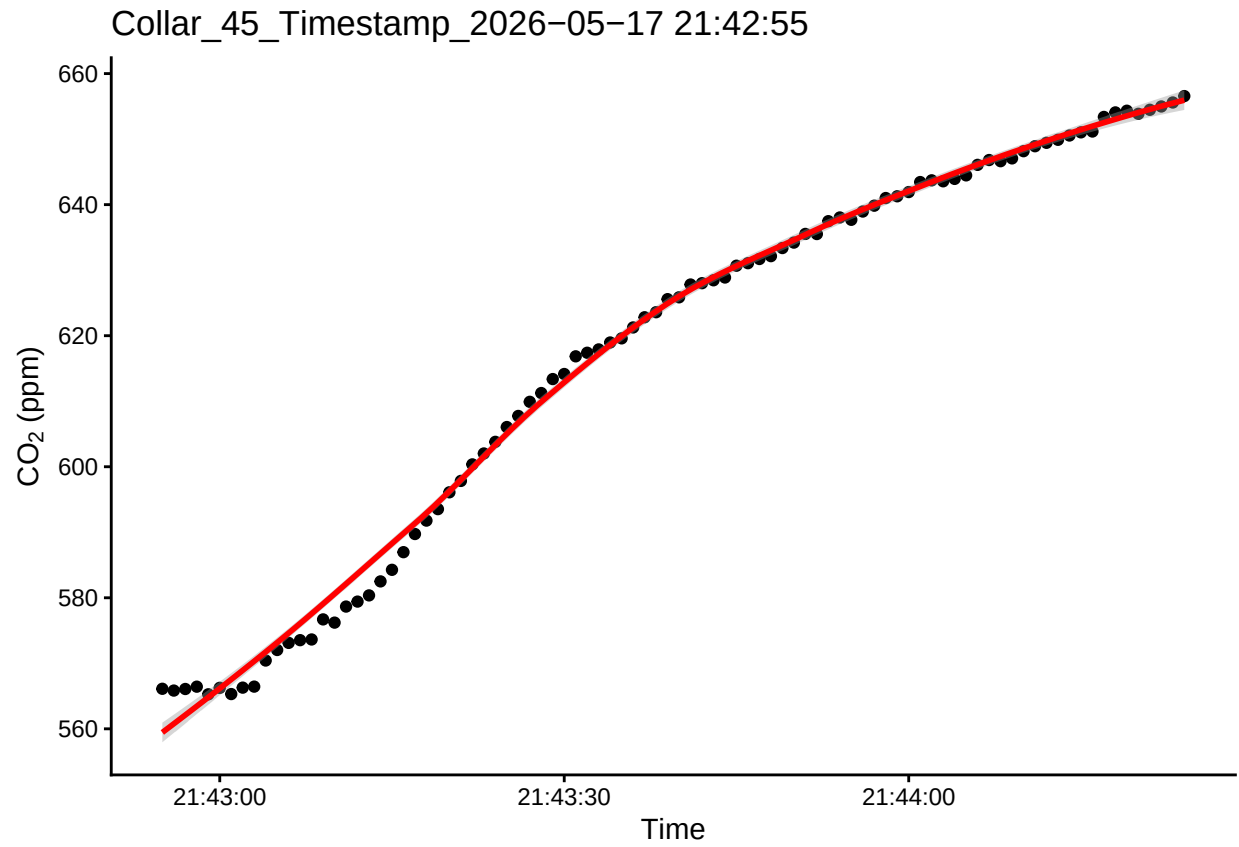
```
## 'geom_smooth()' using method = 'loess' and formula = 'y ~ x'
```



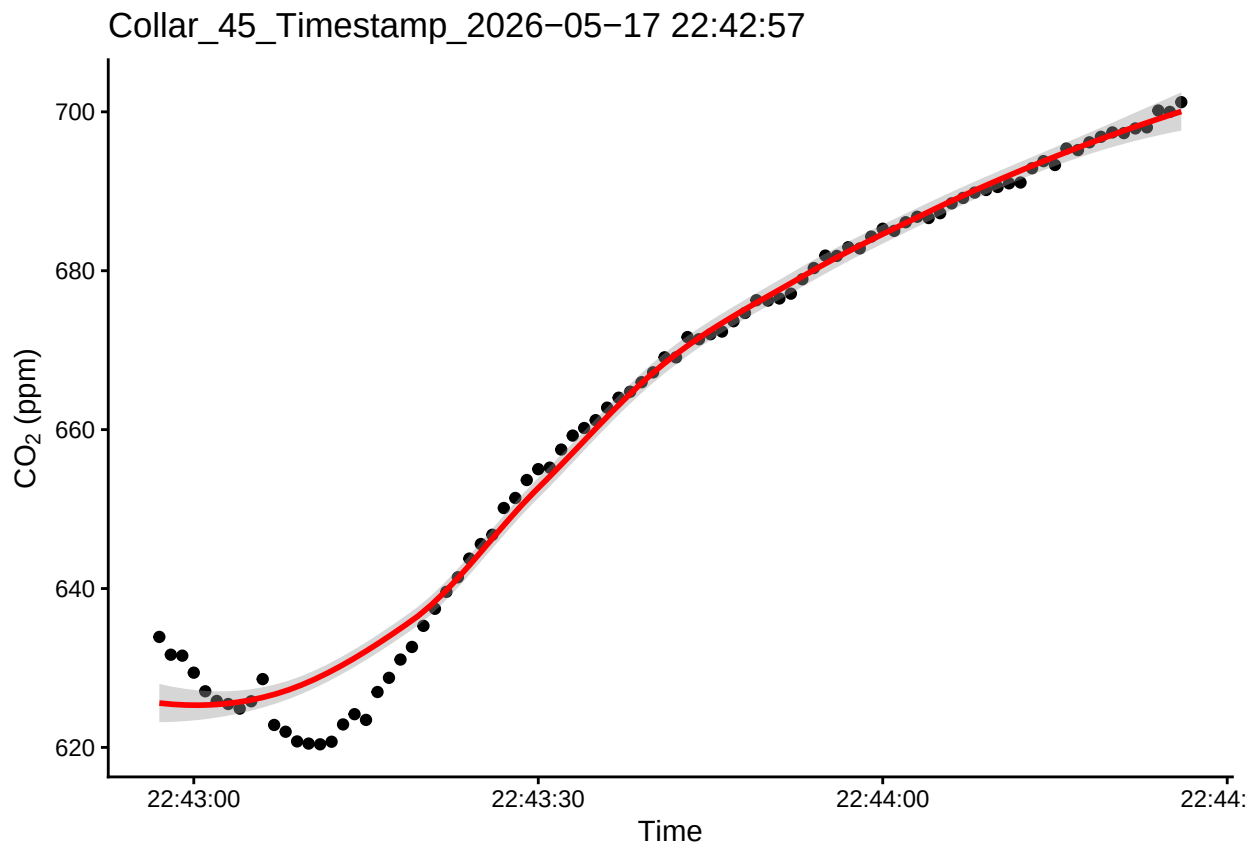
```
## 'geom_smooth()' using method = 'loess' and formula = 'y ~ x'
```



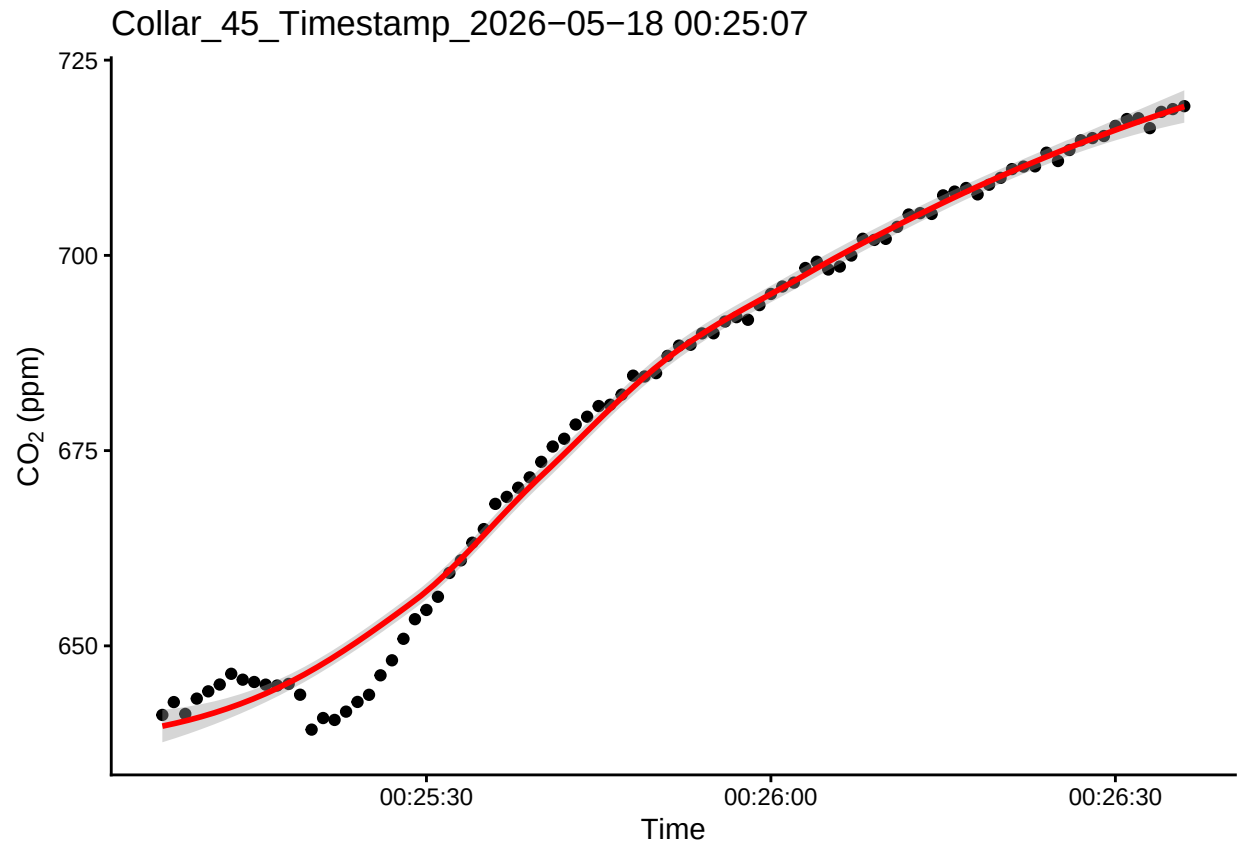
```
## 'geom_smooth()' using method = 'loess' and formula = 'y ~ x'
```



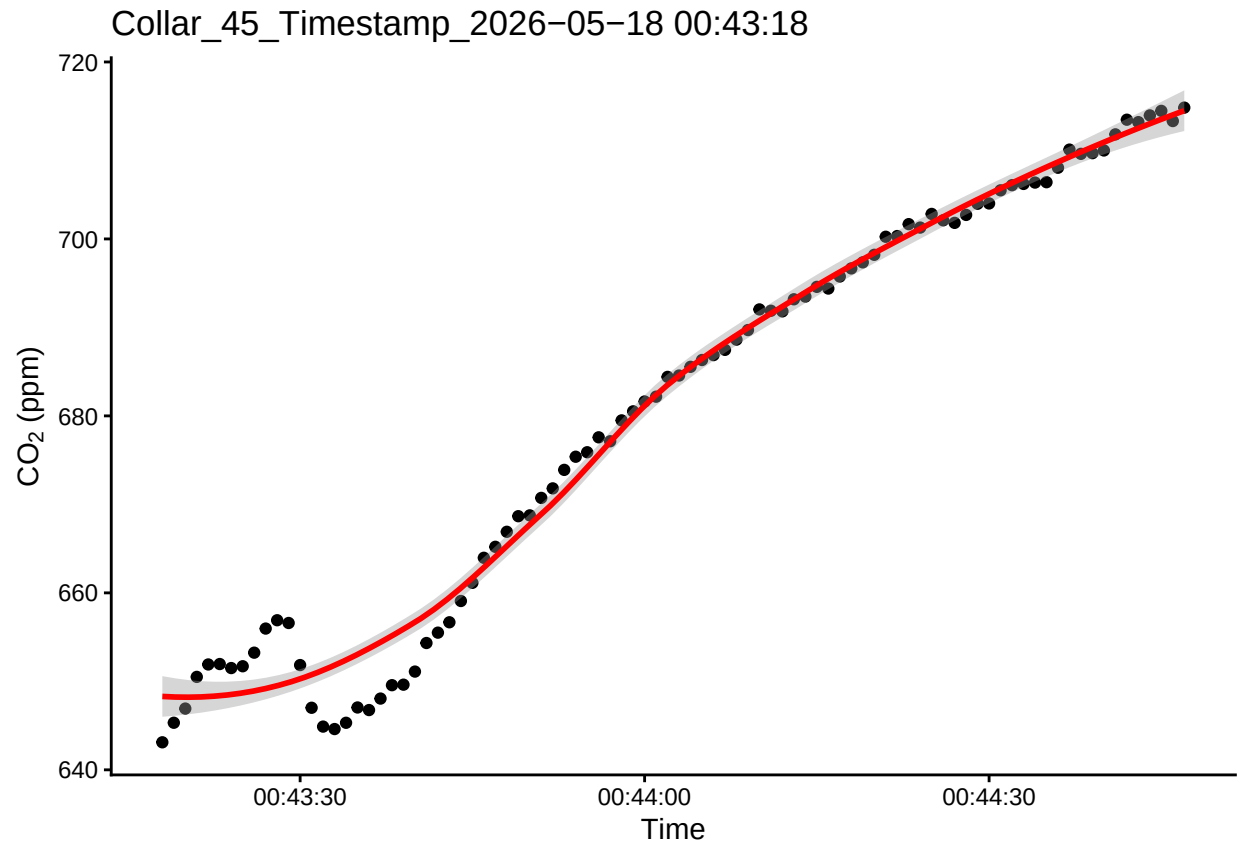
```
## 'geom_smooth()' using method = 'loess' and formula = 'y ~ x'
```



```
## 'geom_smooth()' using method = 'loess' and formula = 'y ~ x'
```

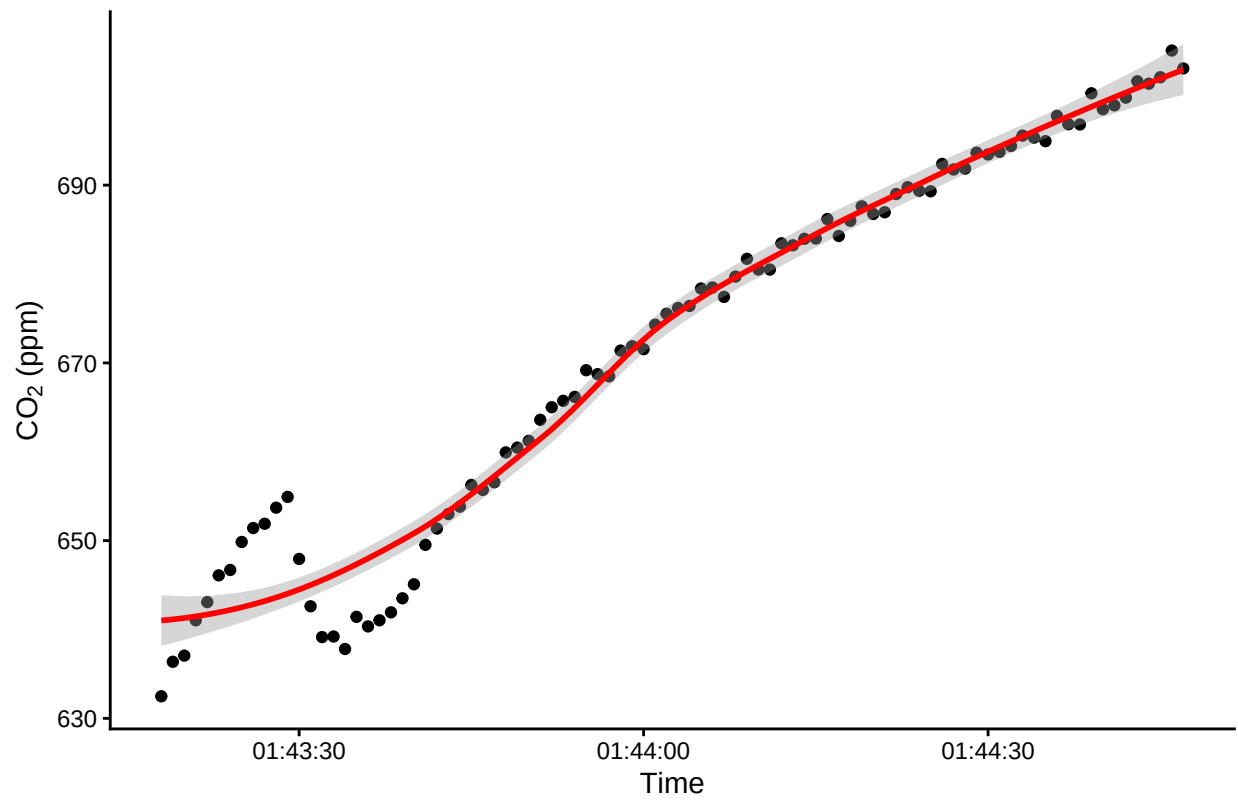


```
## 'geom_smooth()' using method = 'loess' and formula = 'y ~ x'
```

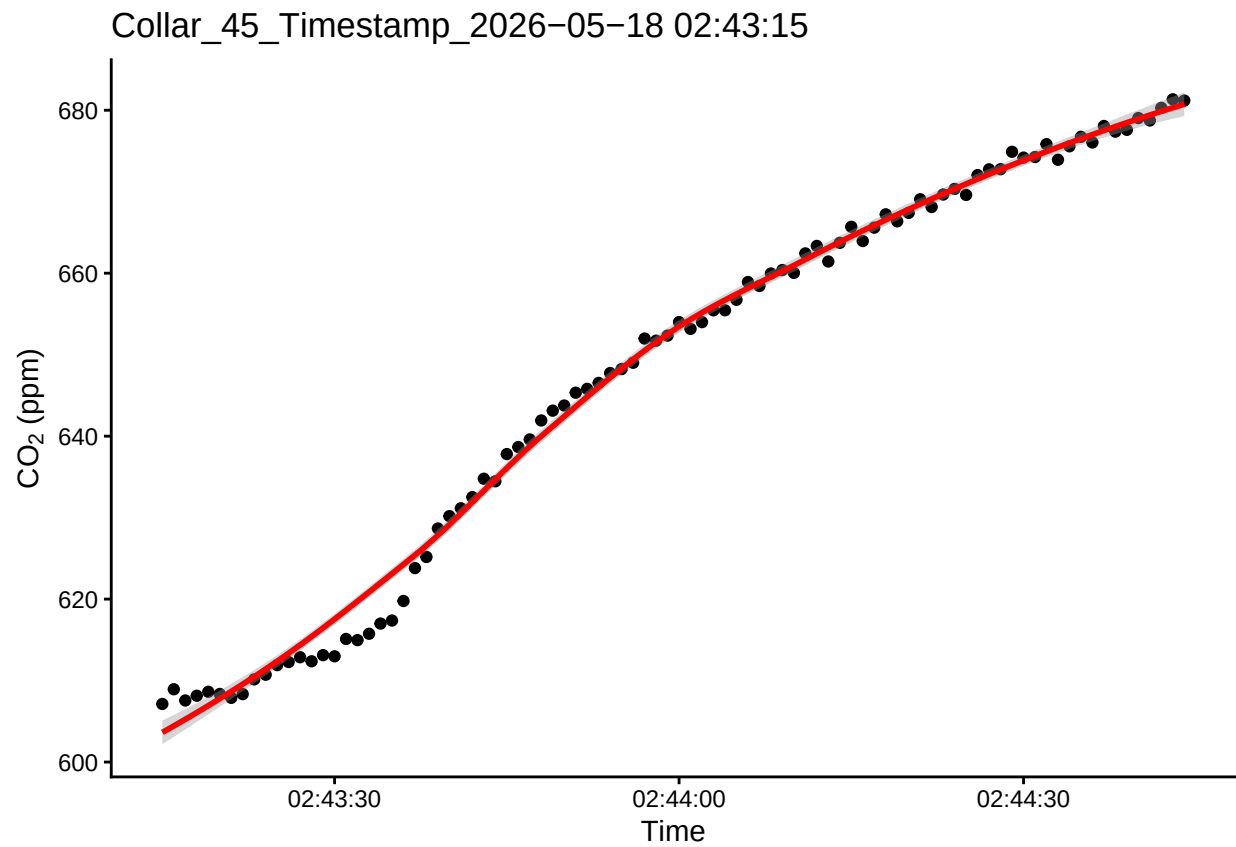


```
## 'geom_smooth()' using method = 'loess' and formula = 'y ~ x'
```

Collar_45_Timestamp_2026-05-18 01:43:18

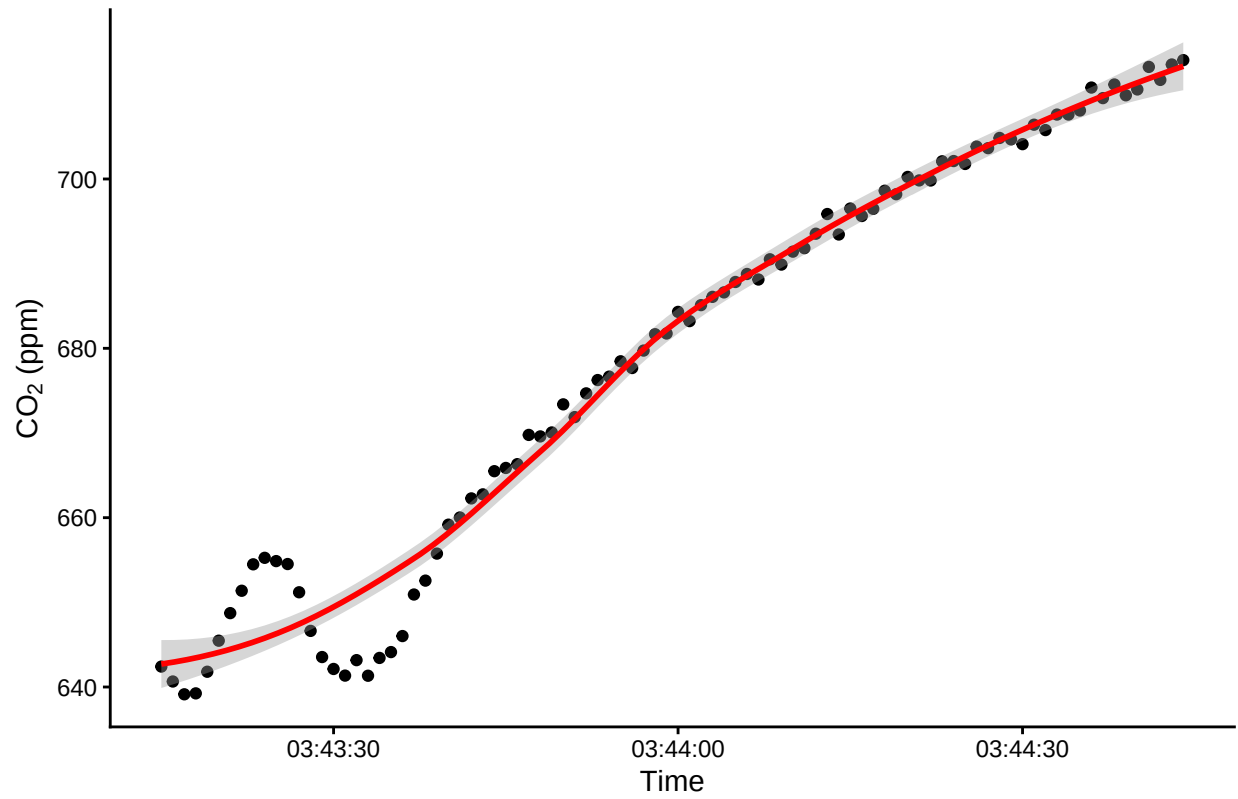


```
## 'geom_smooth()' using method = 'loess' and formula = 'y ~ x'
```

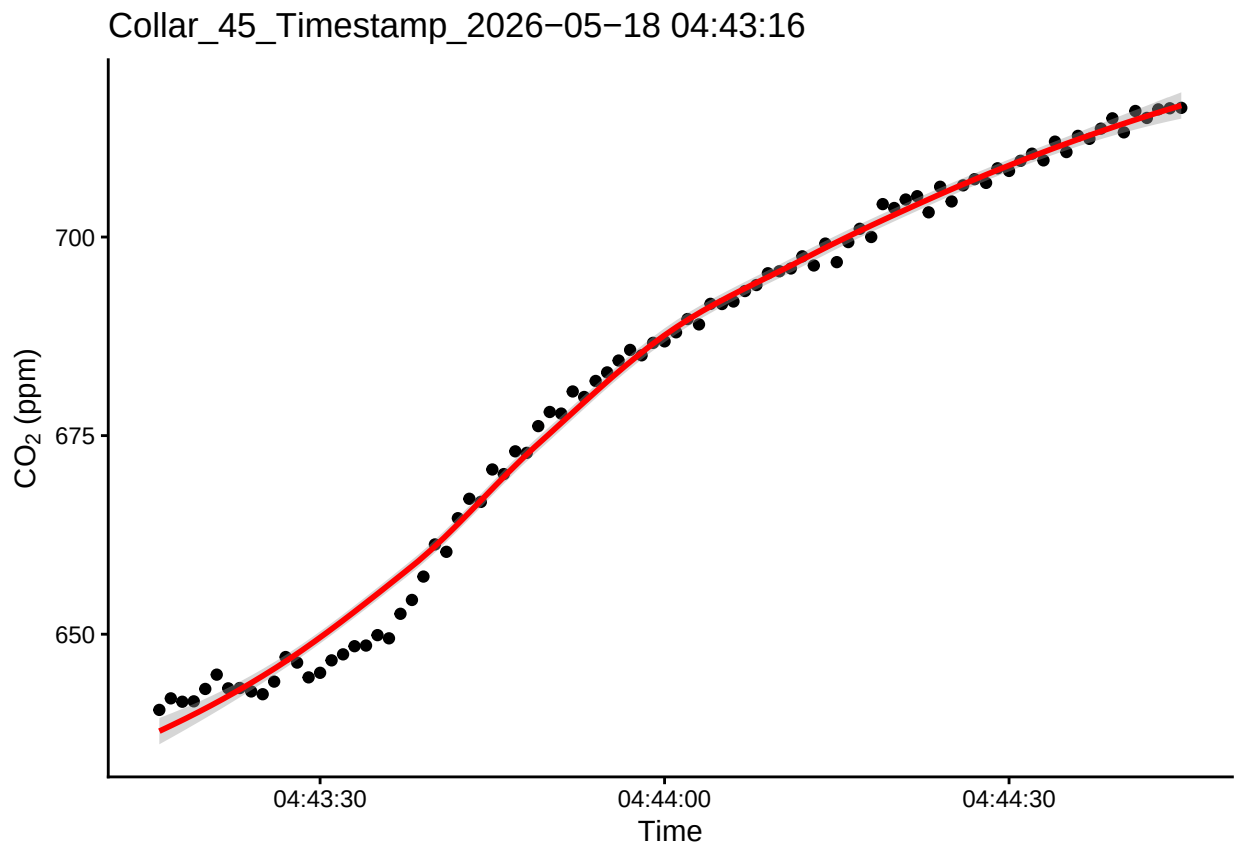



```
## 'geom_smooth()' using method = 'loess' and formula = 'y ~ x'
```

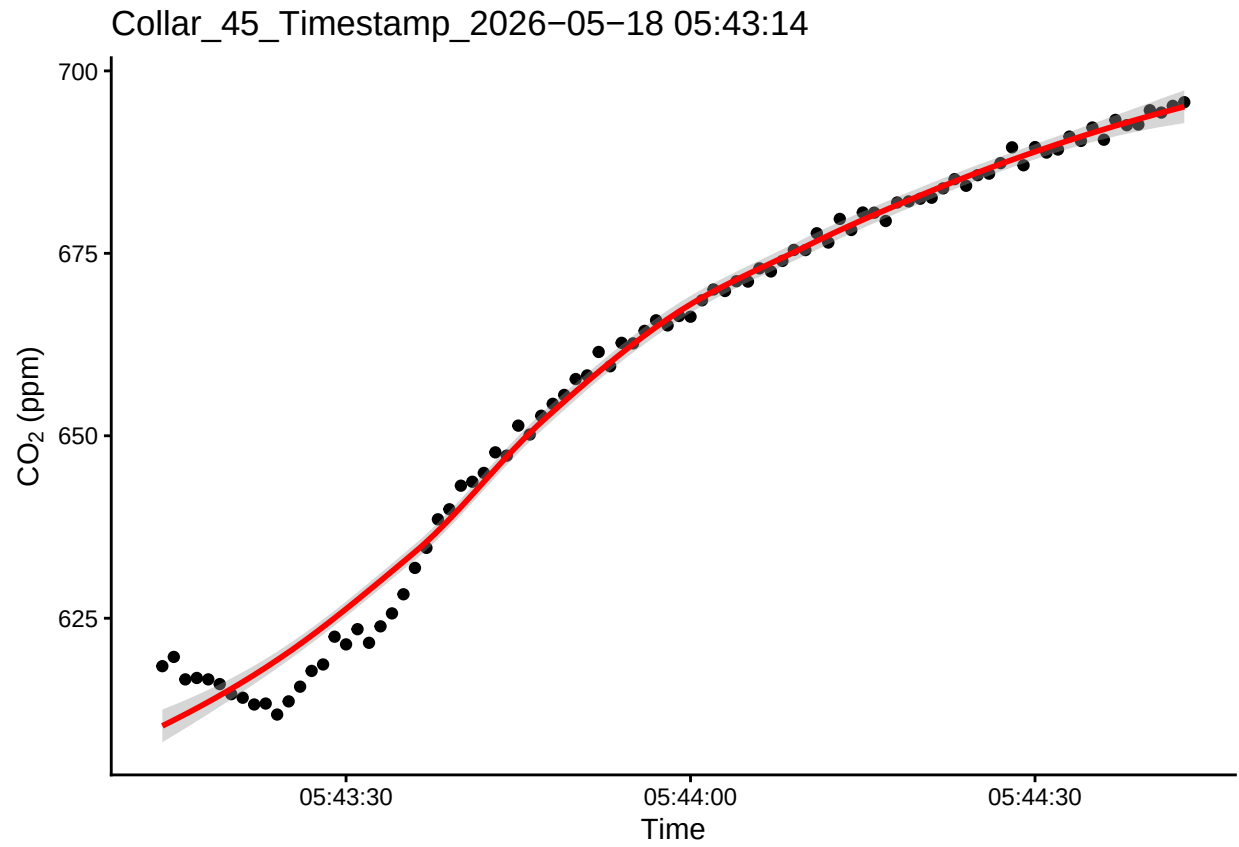
Collar_45_Timestamp_2026-05-18 03:43:15



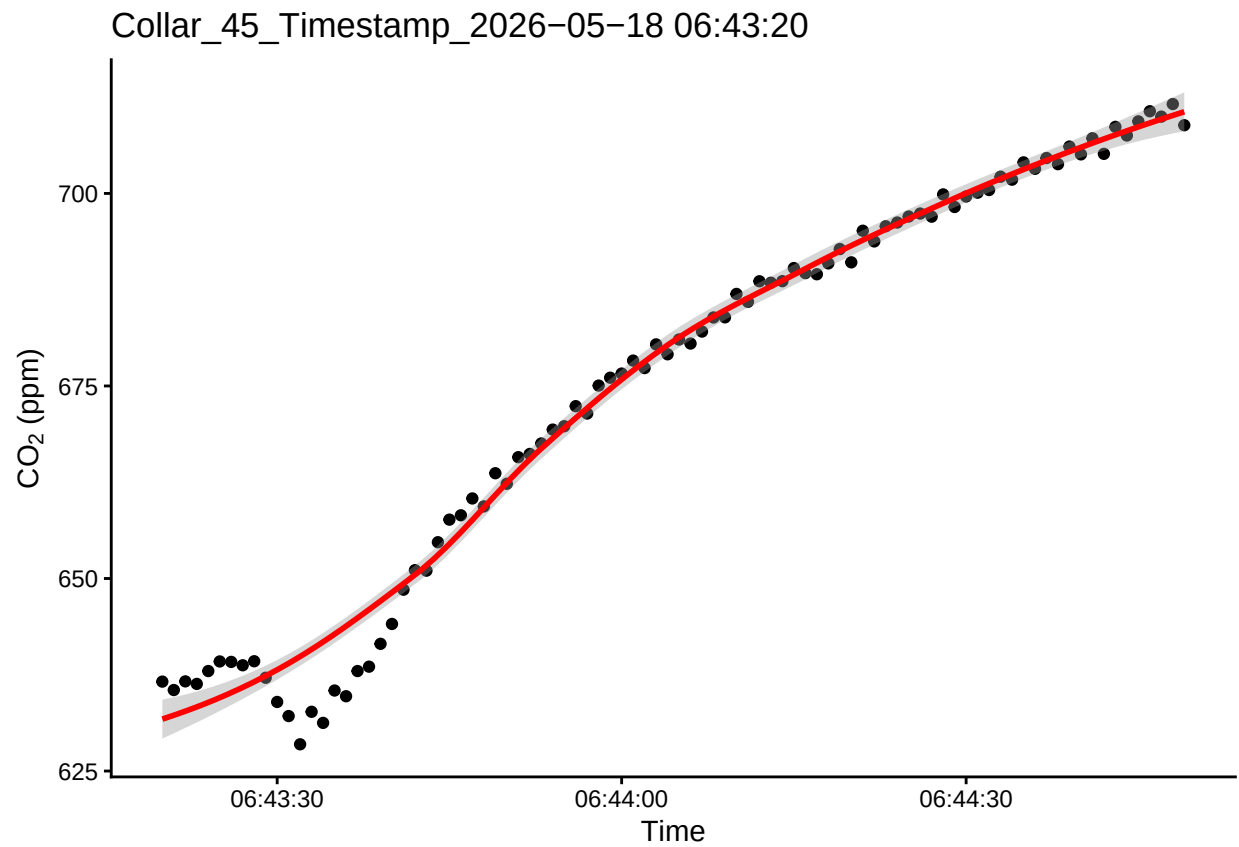
```
## 'geom_smooth()' using method = 'loess' and formula = 'y ~ x'
```



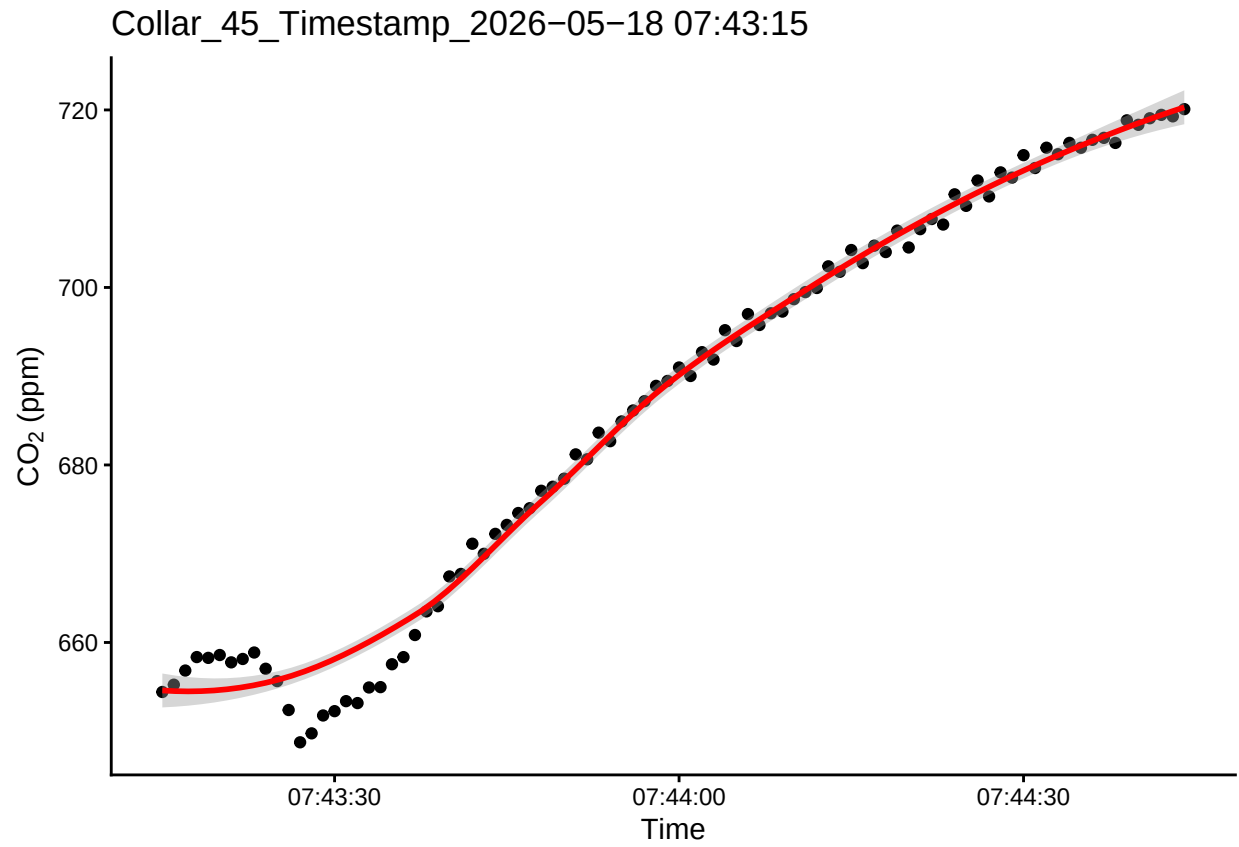
```
## 'geom_smooth()' using method = 'loess' and formula = 'y ~ x'
```



```
## 'geom_smooth()' using method = 'loess' and formula = 'y ~ x'
```

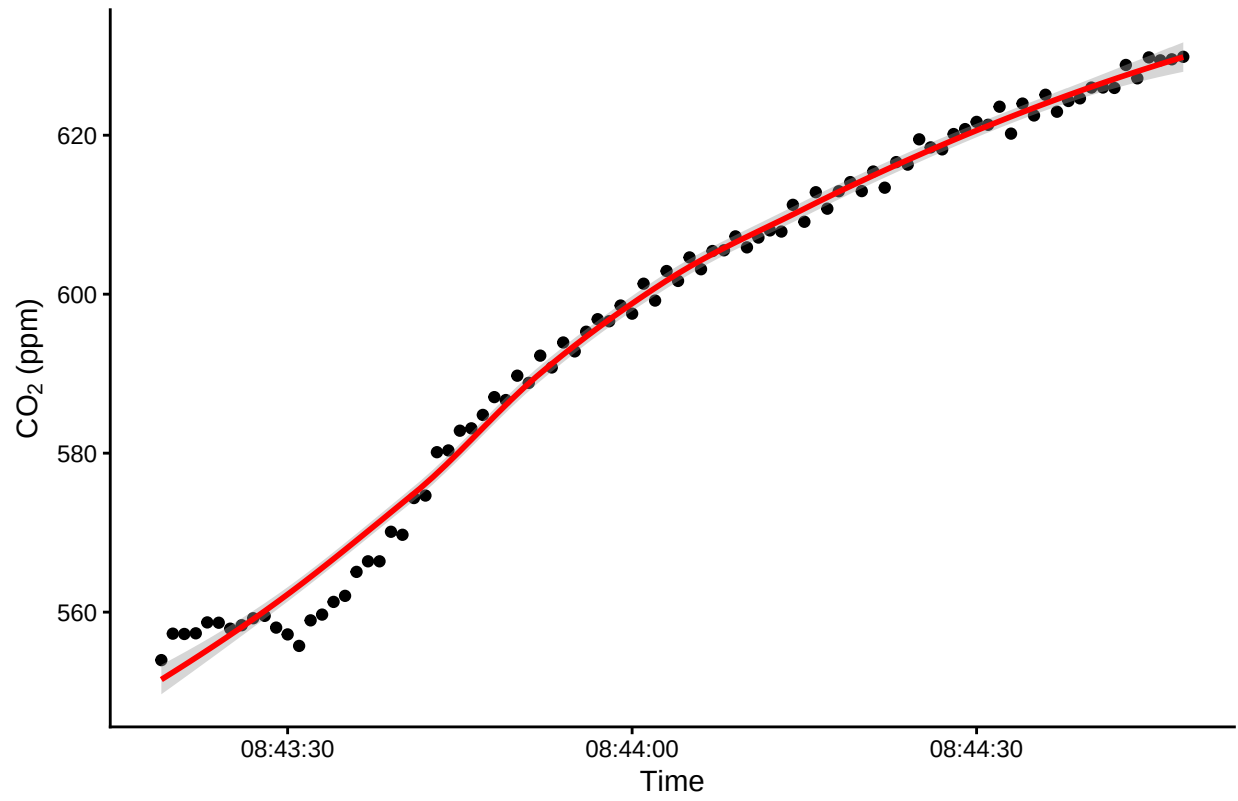


```
## 'geom_smooth()' using method = 'loess' and formula = 'y ~ x'
```

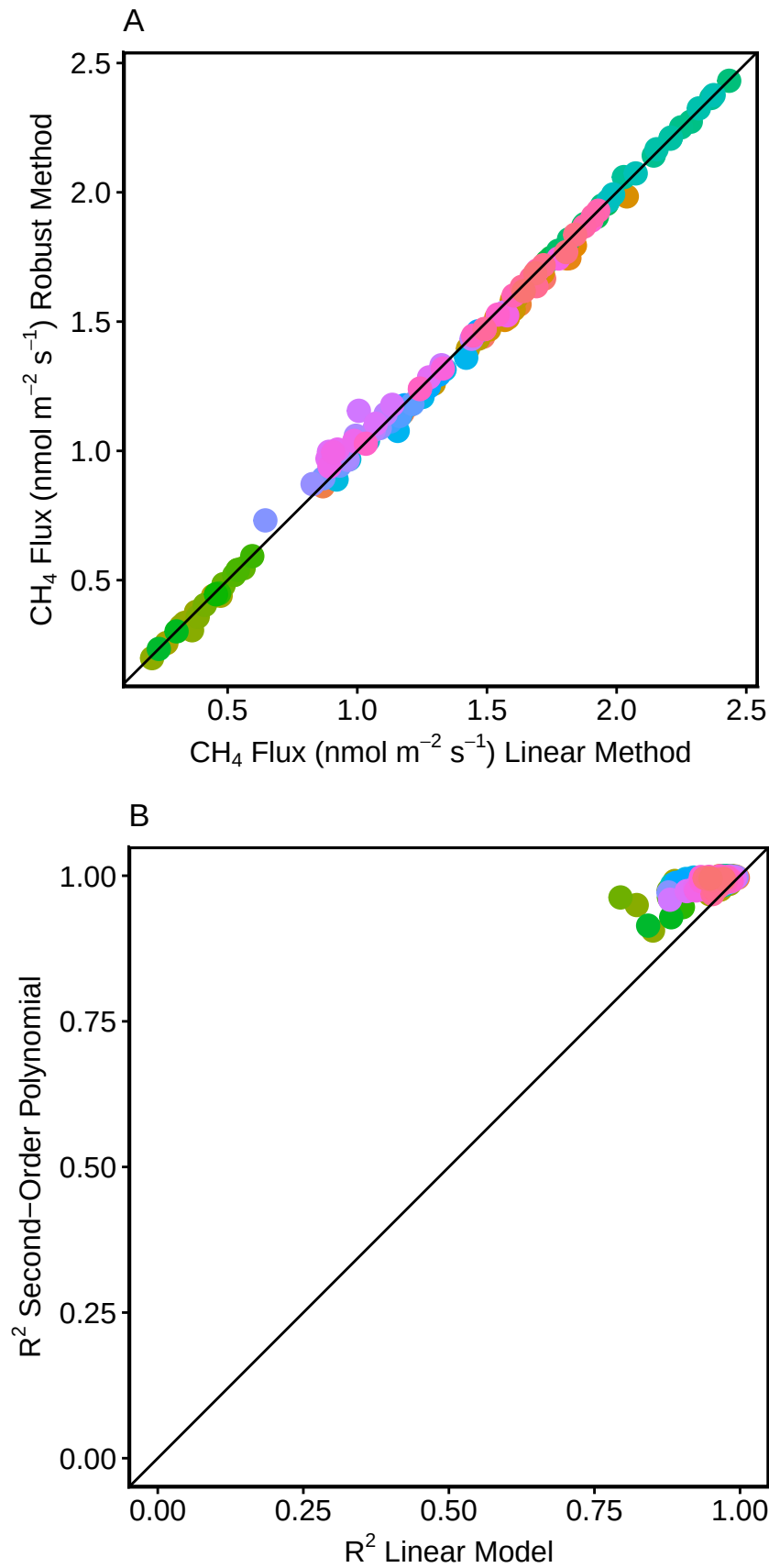


```
## 'geom_smooth()' using method = 'loess' and formula = 'y ~ x'
```

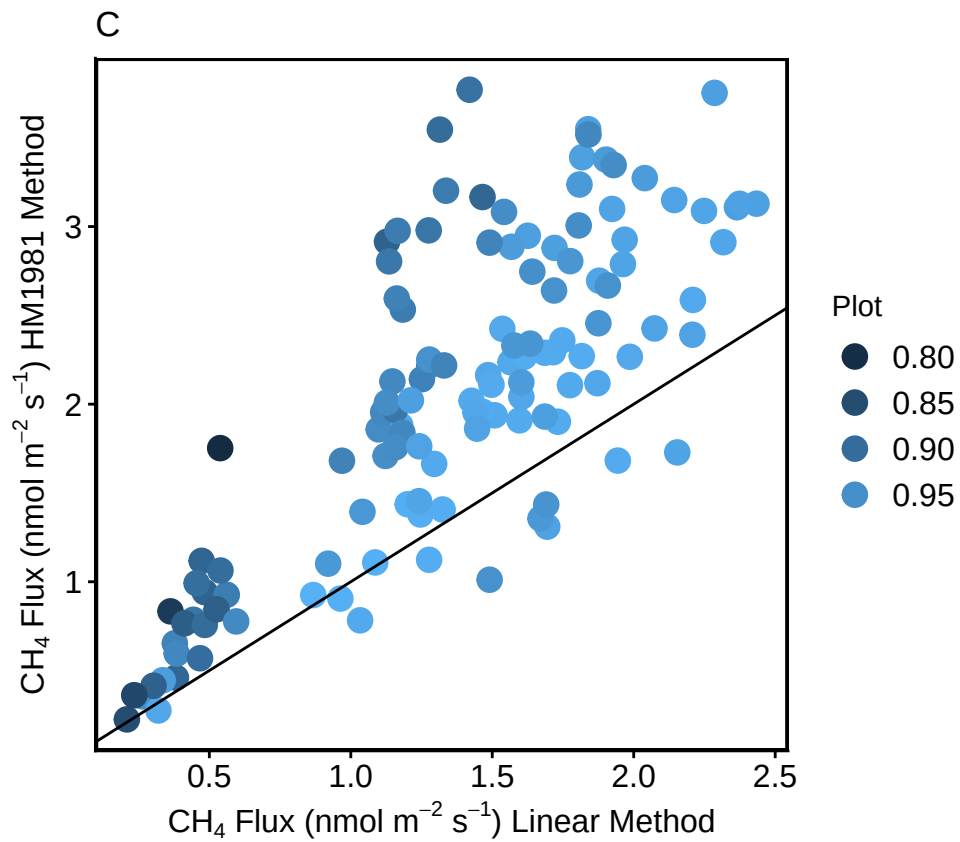
Collar_45_Timestamp_2026-05-18 08:43:19



Do some visualization of QAQC plots to assess fluxes




```
## Warning: Removed 20 rows containing missing values or values outside the scale range
## ('geom_point()').
```



Pull together dataframe and export for use elsewhere

```
##      NA label      NA      StartTime
## 1 Collar    38 Timestamp 2026-05-17 10:39:56
## 2 Collar    38 Timestamp 2026-05-17 11:39:55
## 3 Collar    38 Timestamp 2026-05-17 12:39:56
## 4 Collar    38 Timestamp 2026-05-17 13:39:55
## 5 Collar    38 Timestamp 2026-05-17 14:39:58
## 6 Collar    38 Timestamp 2026-05-17 15:39:57
```